

VU MEDICAL ZONE

Admin: Amaan Khan

Virology

BT601 Midterm Merge ppt

Lecture 1 to 107

Virology

VIRUSES

Definition and meaning

Viruses

MEANING

- Latin word
- Meaning **poison** or **venom**
- Given by Beijerinck (1897)



Viruses

DEFINITION

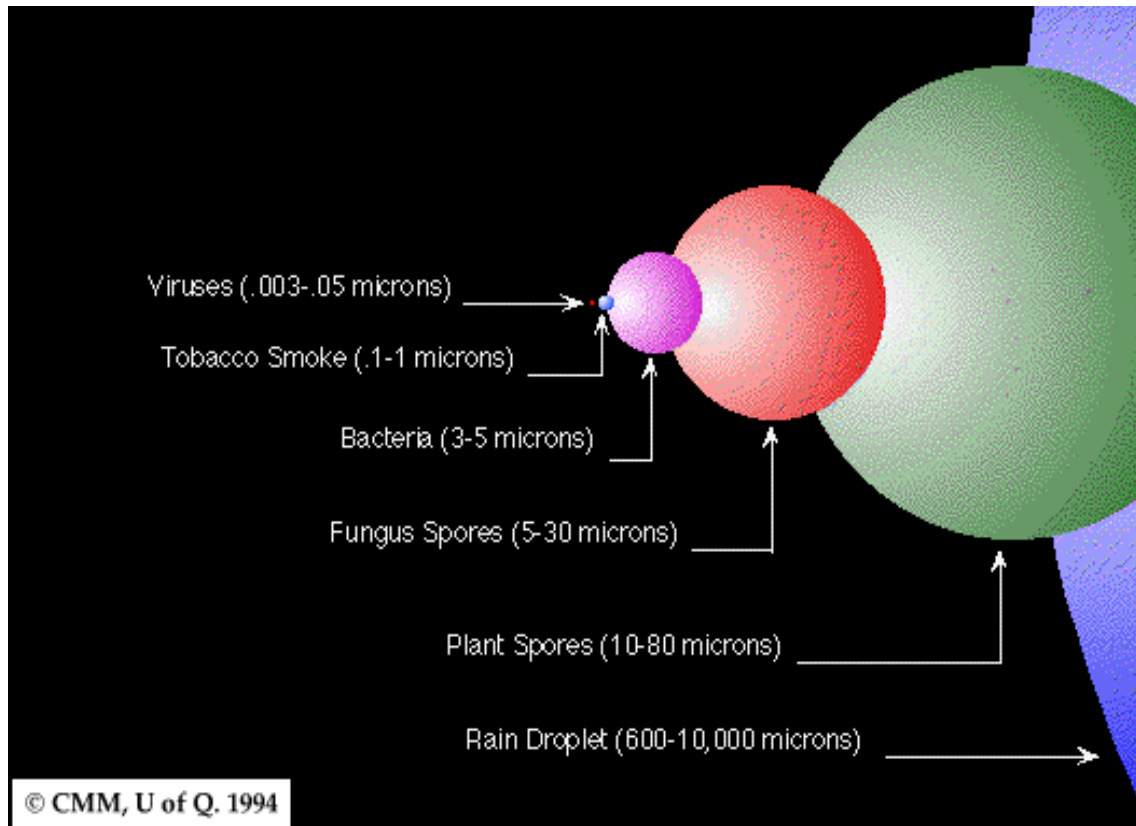


- Viruses are non-cellular particles made up of genetic material and proteins that can invade living cells
- Obligate intracellular parasites

Viruses

INTERESTING FACT

- Submicroscopic
- Smaller than bacteria
- Smaller than the smallest cell



Viruses

TAKE HOME QUESTION!

Can you find the size range
of viruses?



Virology

Discovery of viruses

Discovery of viruses

Ancient Times

- First written record
- From Egypt around 3700BC
- Signs of poliomyelitis in a priest



Discovery of viruses



Small pox

- Edward Jenner
(1796)
- Smallpox vaccine
using milder cowpox
virus
- No knowledge of viral
particles

Discovery of viruses



Iwanowski

- In 1892, Dmitri Iwanowski found out reason for disease in plants
- He thought the bacteria causing disease is small enough to pass the smallest filters
- He did not know the significance of his results

Discovery of viruses



The name “VIRUS”

- In 1898
- By Beijernick
- Studied filtered diseased plant juices and found the caused disease in healthy plants

Discovery of viruses

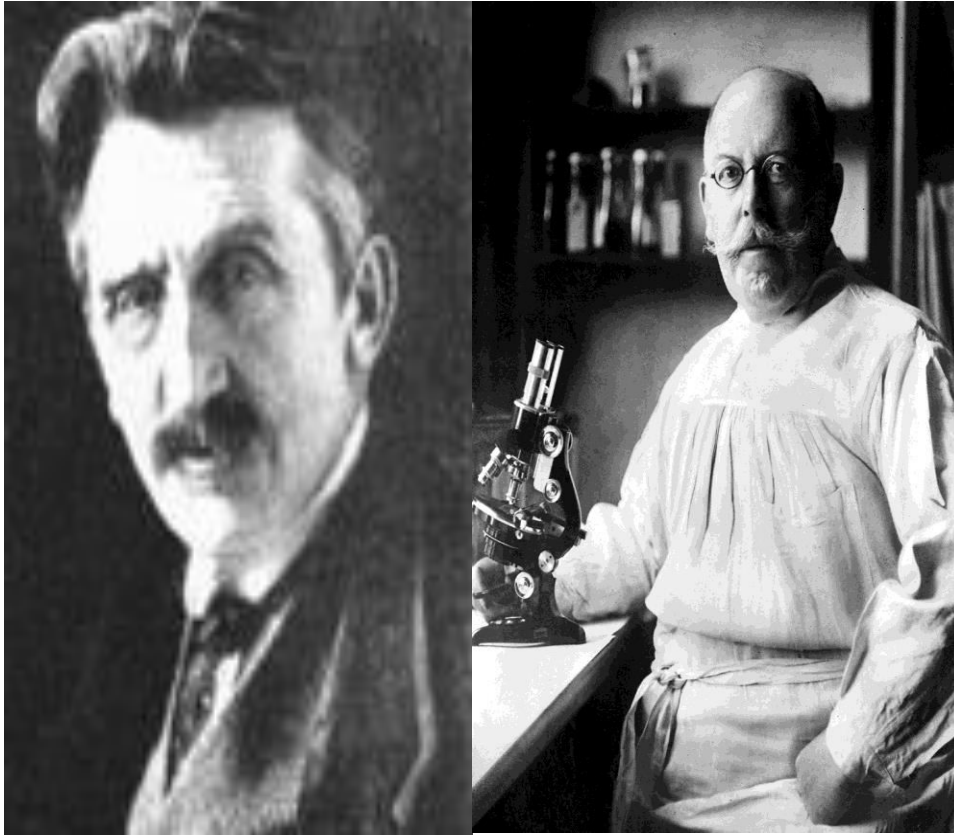
Tobacco Mosaic Virus³

- Wendell Stanley (1935) discovered that:
- Viruses are composed of protein and nucleic acid
- Studied Diseased tobacco plants
- Discovered isolates of virus
- Won Nobel award



Discovery of viruses

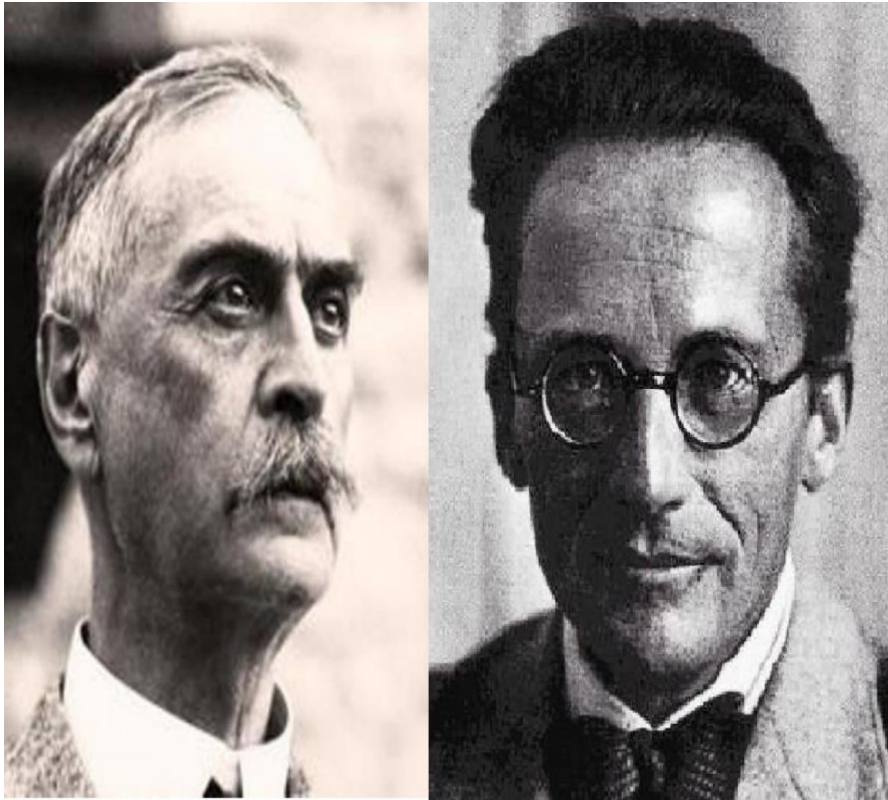
Foot and Mouth disease



- In 1898
- Recognized by
- Fredrich Loeffler
- Pal Frosch
- Disease in cattle

Discovery of viruses

Poliomyelitis



- 1909
- Karl Landsteiner and Erwin Popper
- First human virus to be found

Discovery of viruses

Bacteriophages

- Fredrick Twort (1915) and Felix D'Herelle (1917)
- Recognized viruses that infect bacteria
- Bacteriophages
(eaters of bacteria)

END slide



Virology

History Of Virology

History Of Virology

First written record

- Hieroglyph from Memphis (capital of ancient Egypt)
- 3700 BC
- Temple priest showing signs of paralytic poliomyelitis



History Of Virology

1196 BC



- Pharaoh Ramses V
- Dies in 1196 BC
- Well preserved mummified body kept in Cairo museum
- Believed to have small pox
- Pustular lesions on face

History Of Virology

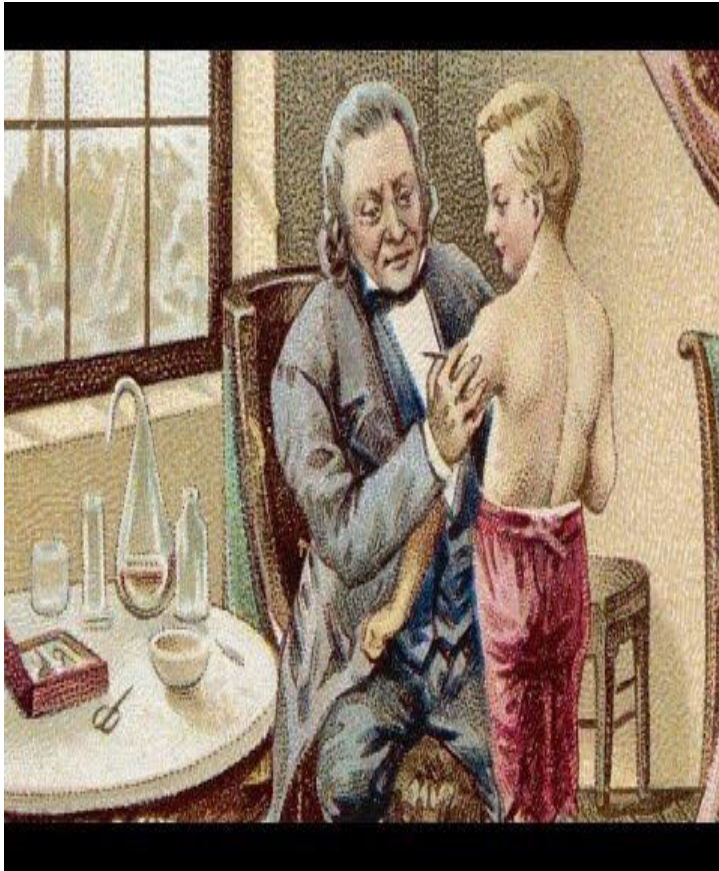
Endemic small pox



- Small pox endemic in China by 1000 BC
- Survivors did not suffer from re-infection
- Variolation was developed

History Of Virology

May 1796



- Edward Jenner introduced the concept of vaccination
- Cow pox against small pox
- Adopted worldwide in 19th century

History Of Virology

Martinus Beijerinck

- Tobacco Mosaic Virus
- First to develop modern idea of virus
- Referred as *contagium vivum fluidum* (soluble living germ)



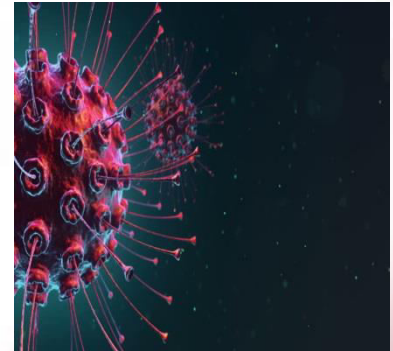
Virology

Importance of
Virology

Importance Of Virology

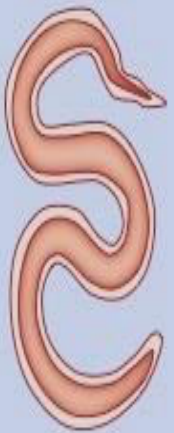
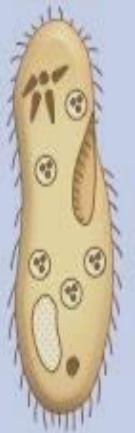
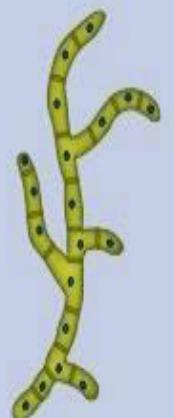
UNIQUE PROPERTIES

- Living as well as non-living characteristics
- Unable to cultivate in laboratory easily
- Simplest structure
- Complex interaction with host
- Replication within host



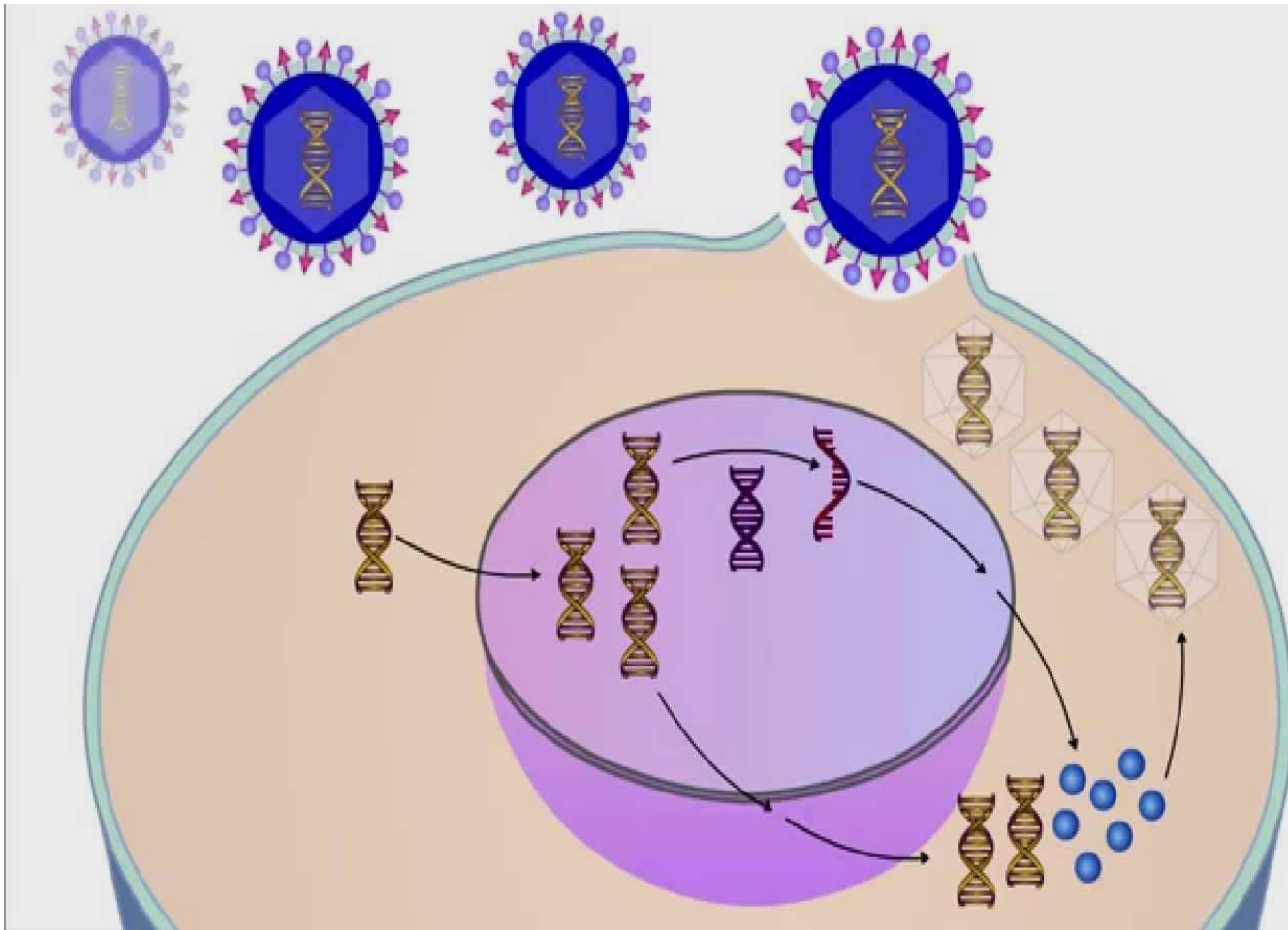
Importance Of Virology

1. To study an important class of pathogens

CELLULAR (LIVING)				ACELLULAR (NON-LIVING)	
					
Parasites (e.g. <i>helminthes</i>) ⇒ Tapeworm	Protozoa (e.g. <i>plasmodia</i>) ⇒ Malaria	Fungi (e.g. <i>tinea</i>) ⇒ Athlete's foot	Prokaryote (i.e. <i>bacteria</i>) ⇒ Leprosy	Virus (e.g. <i>HIV</i>) ⇒ AIDS	Prion ⇒ CJD

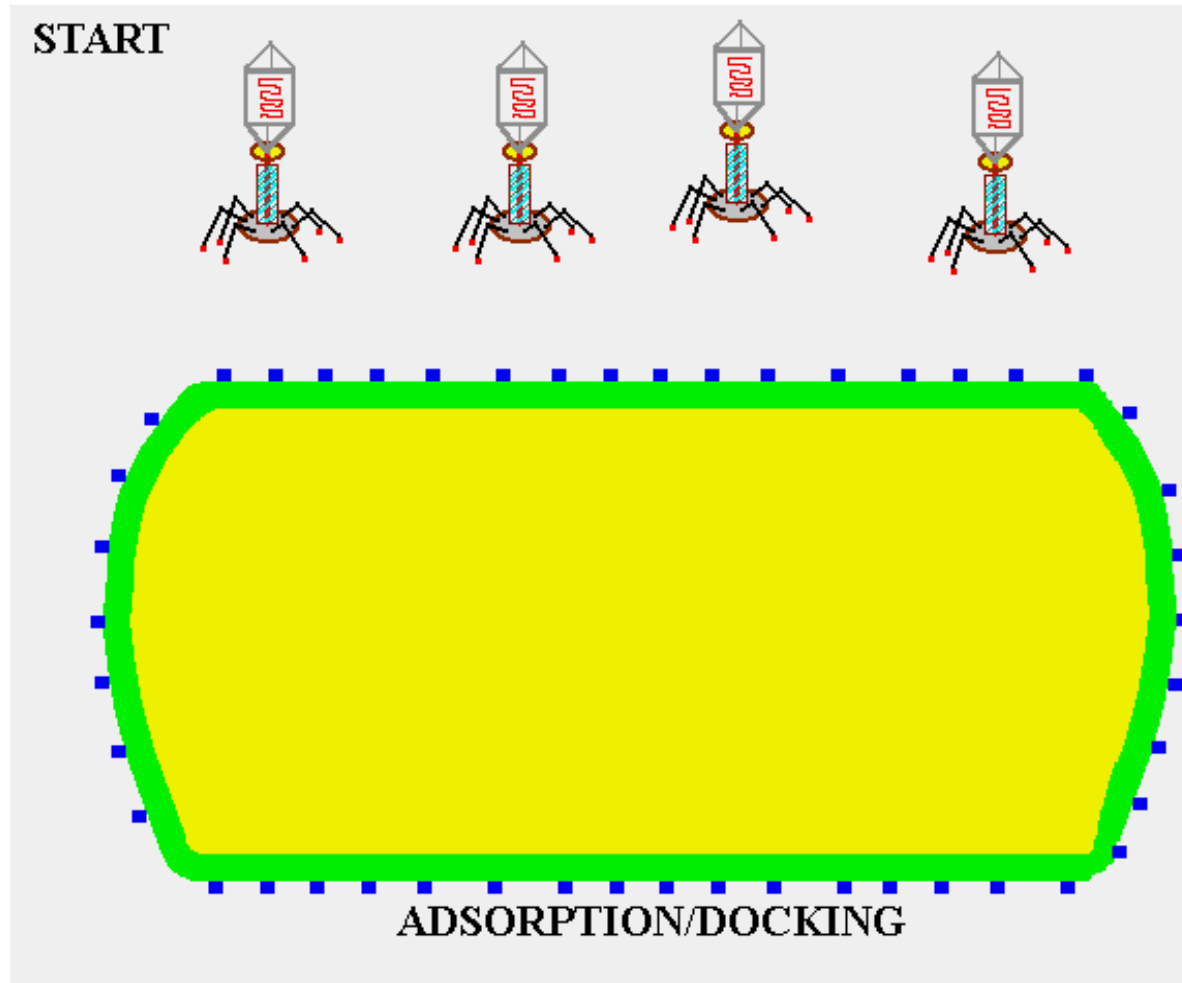
Importance Of Virology

2. To study pathogenesis of viruses



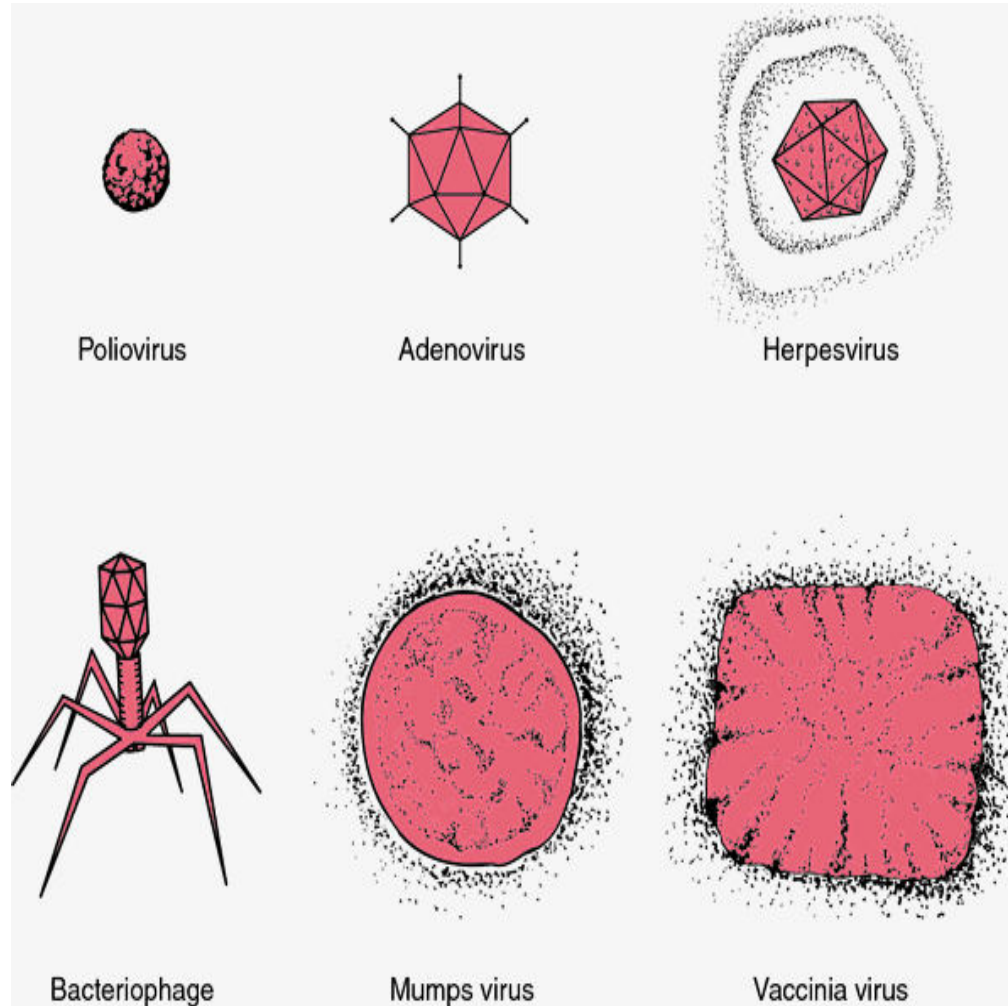
Importance Of Virology

3. To study lifecycle of viruses



Importance Of Virology

4. To study structure of viruses



Importance Of Virology

5. To study viral cultivation in lab

- a. Tissue culture
- b. Embryonated egg
- c. Live animal

Virology

VIRUSES

Living or non-living

Viruses Living or non-living

Are virus living entities?

Ambiguous question



Viruses Living or non-living

What is life?

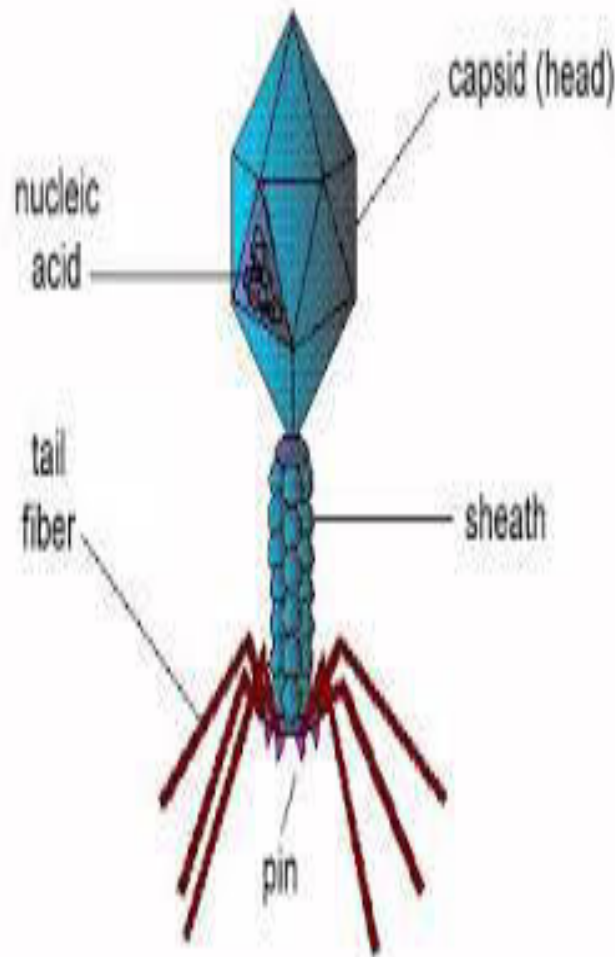
- Definition of life:
 - A complex set of processes resulting from the actions of proteins specified by nucleic acids
- Nucleic acids of living things are in action all the time

Viruses Living or non-living

What is Virus ?

- Viruses are complicated assemblies of molecules, including proteins, nucleic acids, lipids, and carbohydrates
- Viruses can only replicate themselves by infecting a host cell and therefore cannot reproduce on their own.

Viruses Living or non-living



CONCLUSION

- Viruses are:

“exceptionally complex aggregation of non-living chemicals”

OR

“exceptionally simple living organisms”

Virology

ORIGIN OF VIRUSES

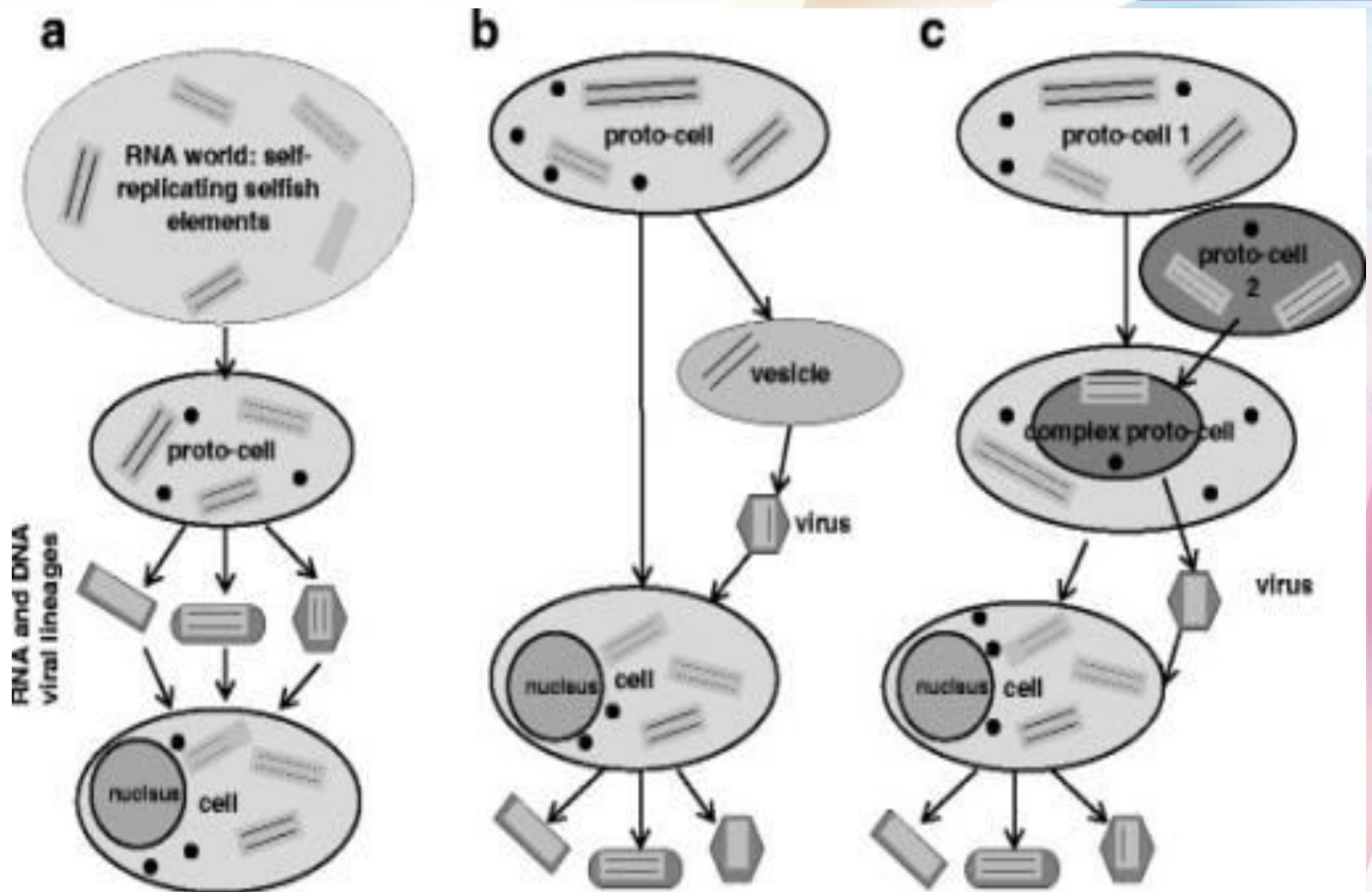
ORIGIN OF VIRUSES

THEORIES

Evolved to be:
Economic form of
microbial life

Most efficient
Three theories for
evolution

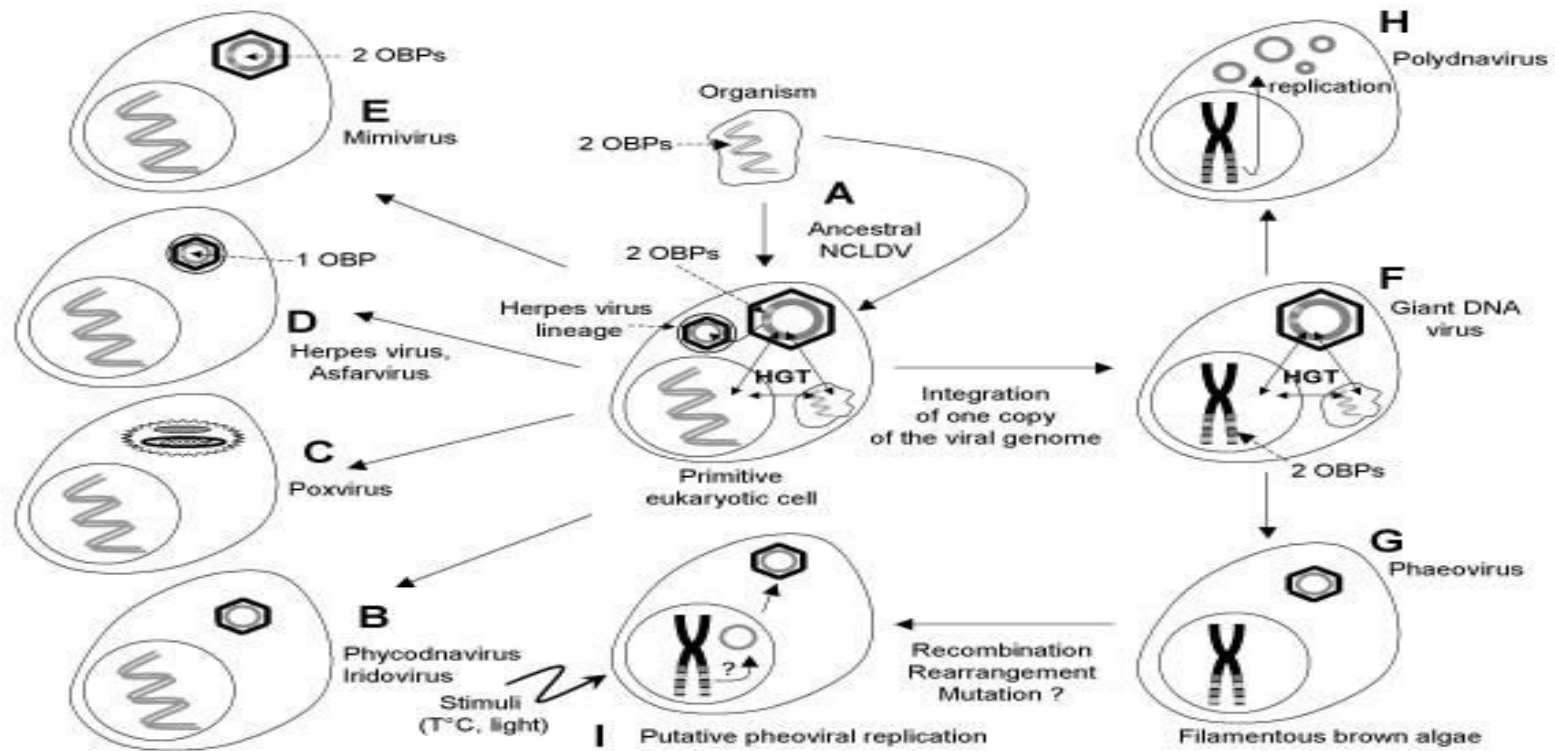
ORIGIN OF VIRUSES



ORIGIN OF VIRUSES

FIRST THEORY

They may have originated and evolved with the other life forms



ORIGIN OF VIRUSES

SECOND THEORY

They may have arisen from segments of the cellular nucleic acids which acquired the ability to replicate at the expense of host cell

ORIGIN OF VIRUSES

THIRD THEORY

- Regressive theory
- Viruses arose from free living organisms which gradually lost genetic information
- Until they became totally dependent on the biosynthetic pathways of their host cells

Virology

GENERAL PROPERTIES OF VIRUSES

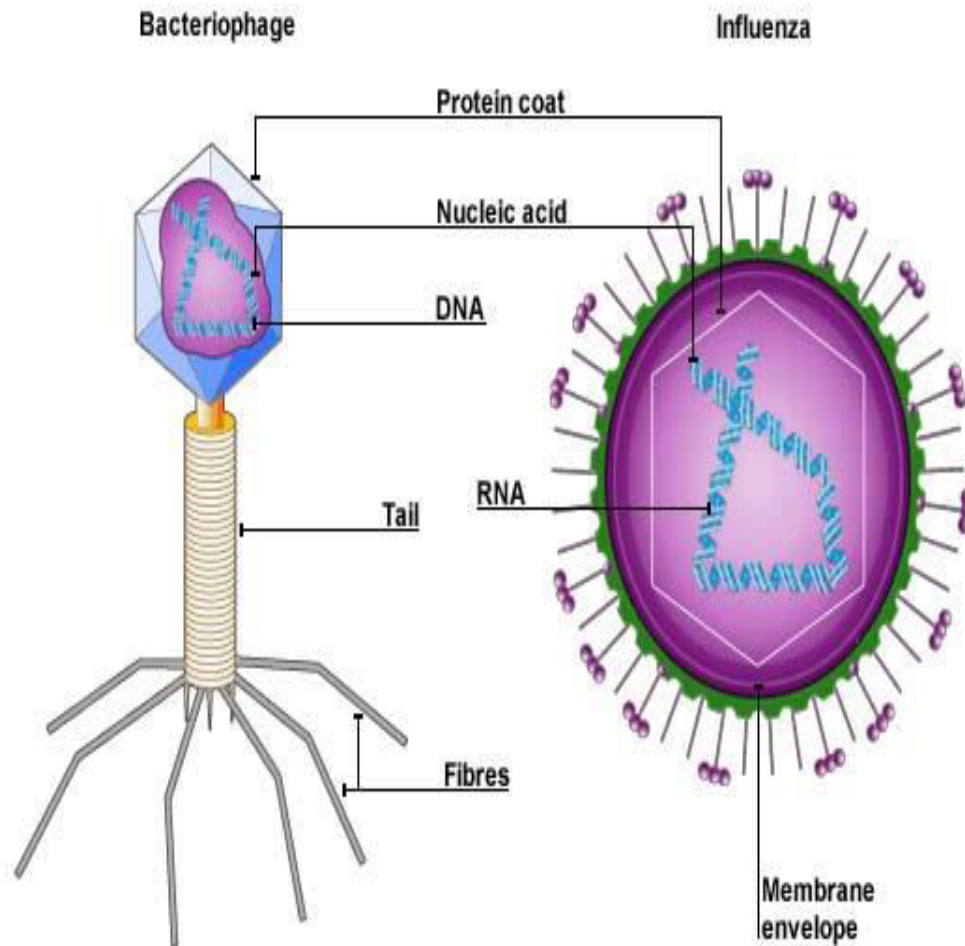
GENERAL PROPERTIES OF VIRUSES

VIRIONS

- Mature viruses
- Variable in size

VIRUS	SIZE
Foot and Mouth Disease virus	27nm
Pox virus	300nm

GENERAL PROPERTIES OF VIRUSES



STRUCTURE

1. Protein
2. Nucleic acid

Deoxyribonucleic
Acid (DNA)

Ribonucleic Acid
(RNA)

GENERAL PROPERTIES OF VIRUSES

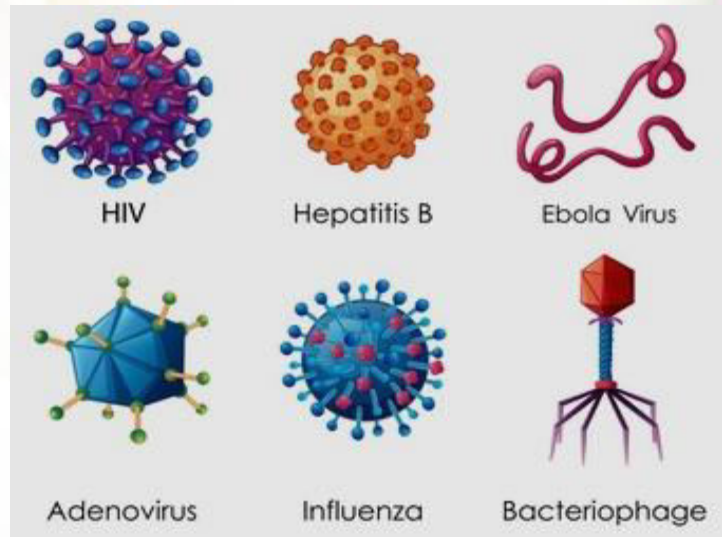
WHY VIRUSES NEED HOST?

- No enzymes of their own
- No metabolism
- Require host machinery

GENERAL PROPERTIES OF VIRUSES

COMPLEX VIRUSES

- Additionally contain:
 - Lipid
 - Polysaccharides
 - Trace elements



Virology

GENERAL PROPERTIES OF VIRUSES

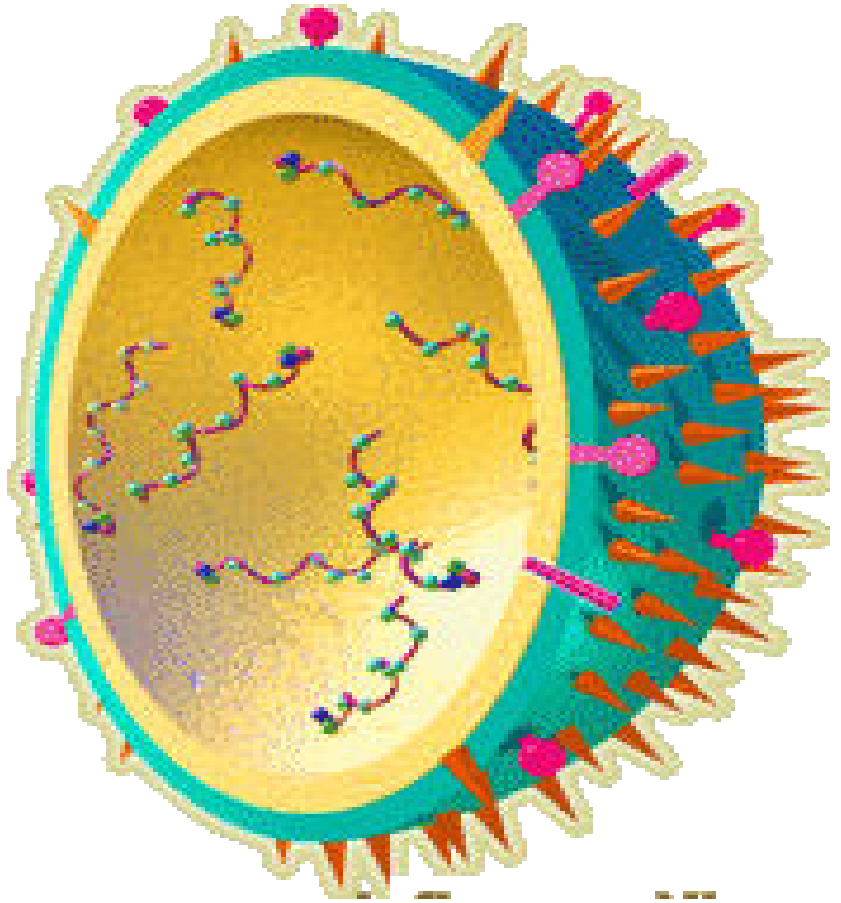
GENERAL PROPERTIES OF VIRUSES

- Non living structures
- Non-cellular
- Contain a protein coat called the capsid
- Have a nucleic acid core containing DNA or RNA (one or the other - not both)
- Capable of reproducing only when inside a HOST cell



GENERAL PROPERTIES OF VIRUSES

- Some viruses are enclosed in an protective envelope
- Some viruses may have spikes to help attach to the host cell
- Most viruses infect only specific host cells



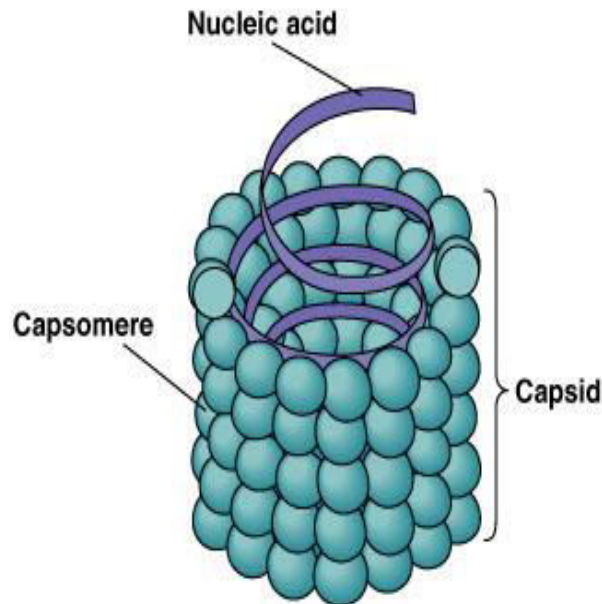
GENERAL PROPERTIES OF VIRUSES

MORPHOLOGY

- Viruses come in a variety of shapes
- Some may be helical shape like the Ebola virus
- Some may be polyhedral shapes like the influenza virus
- Others have more complex shapes like bacteriophages

GENERAL PROPERTIES OF VIRUSES

Types of Viruses: Helical Viruses



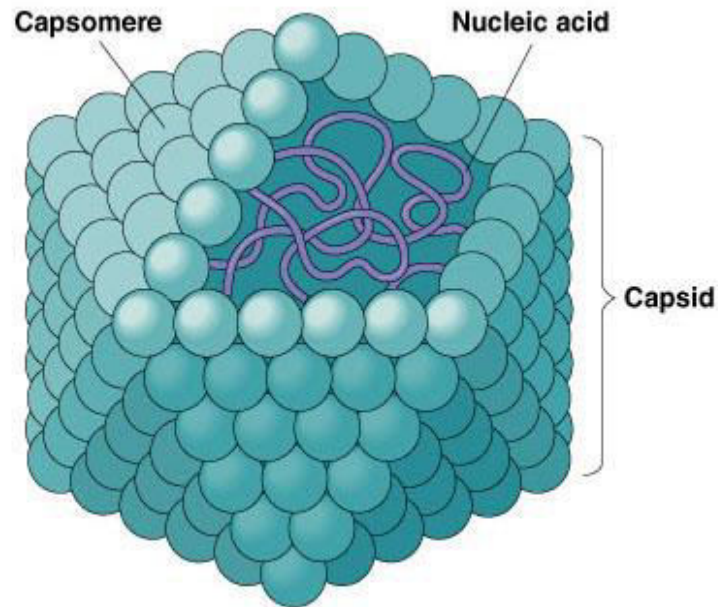
(a) A helical virus



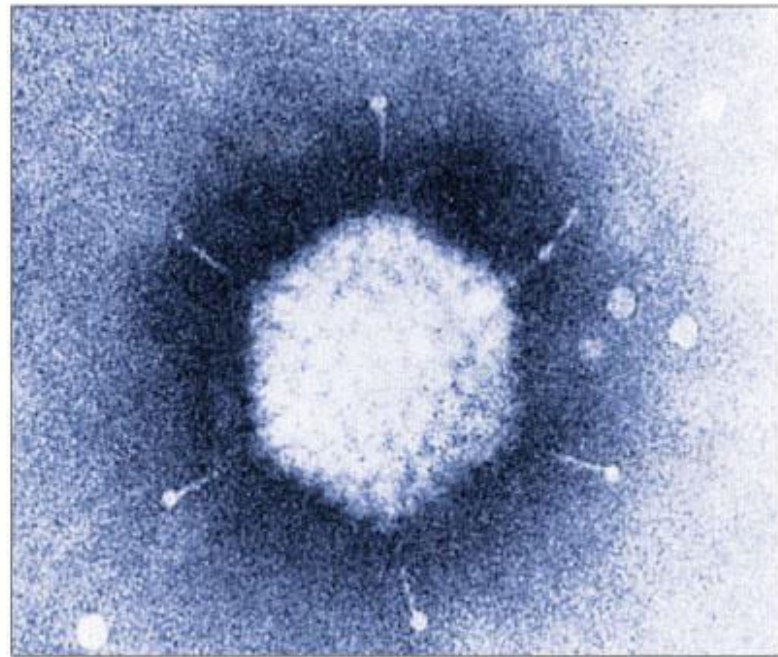
(b) Ebola virus

GENERAL PROPERTIES OF VIRUSES

Polyhedral Viruses



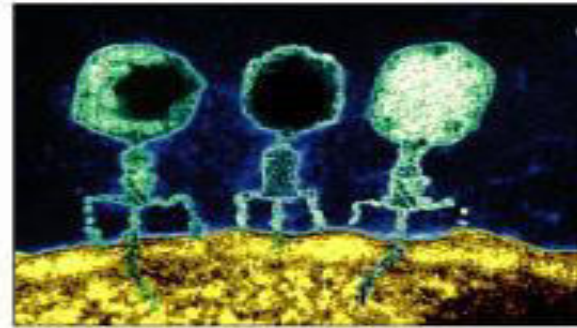
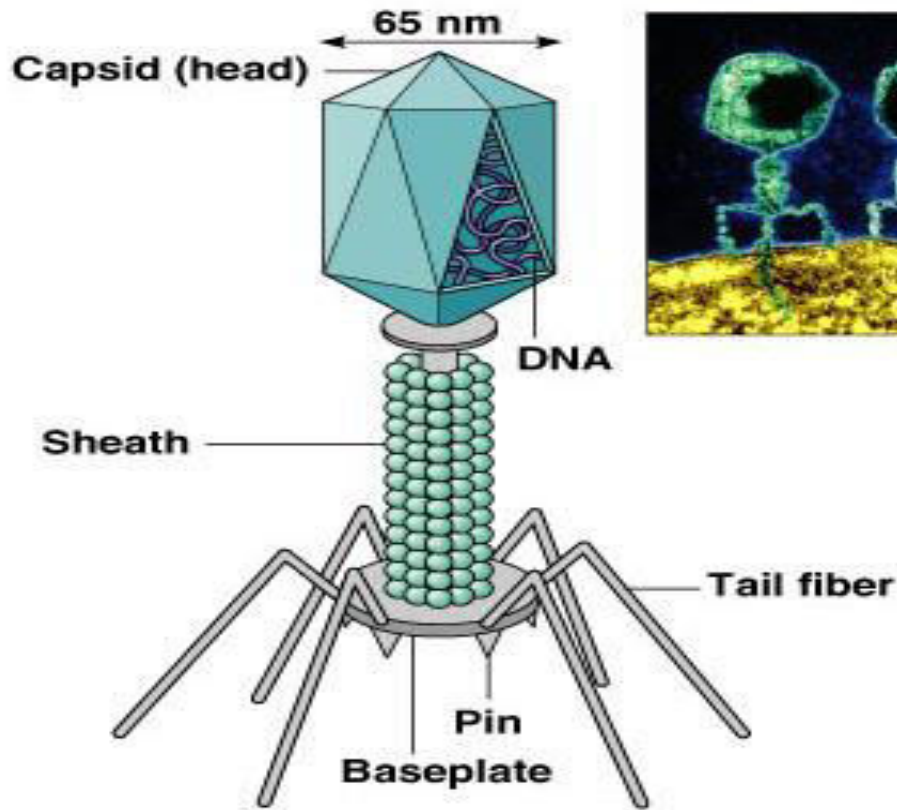
(a) A polyhedral virus



(b) A Mastadenovirus

GENERAL PROPERTIES OF VIRUSES

Complex Viruses

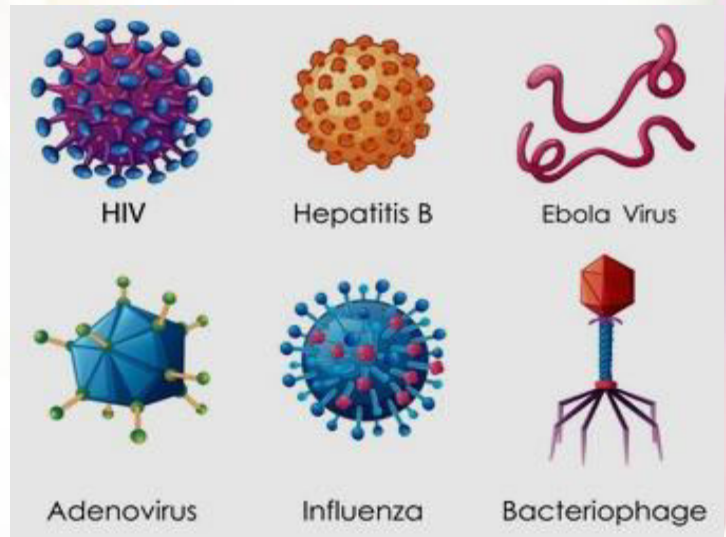


(a) A T-even bacteriophage

GENERAL PROPERTIES OF VIRUSES

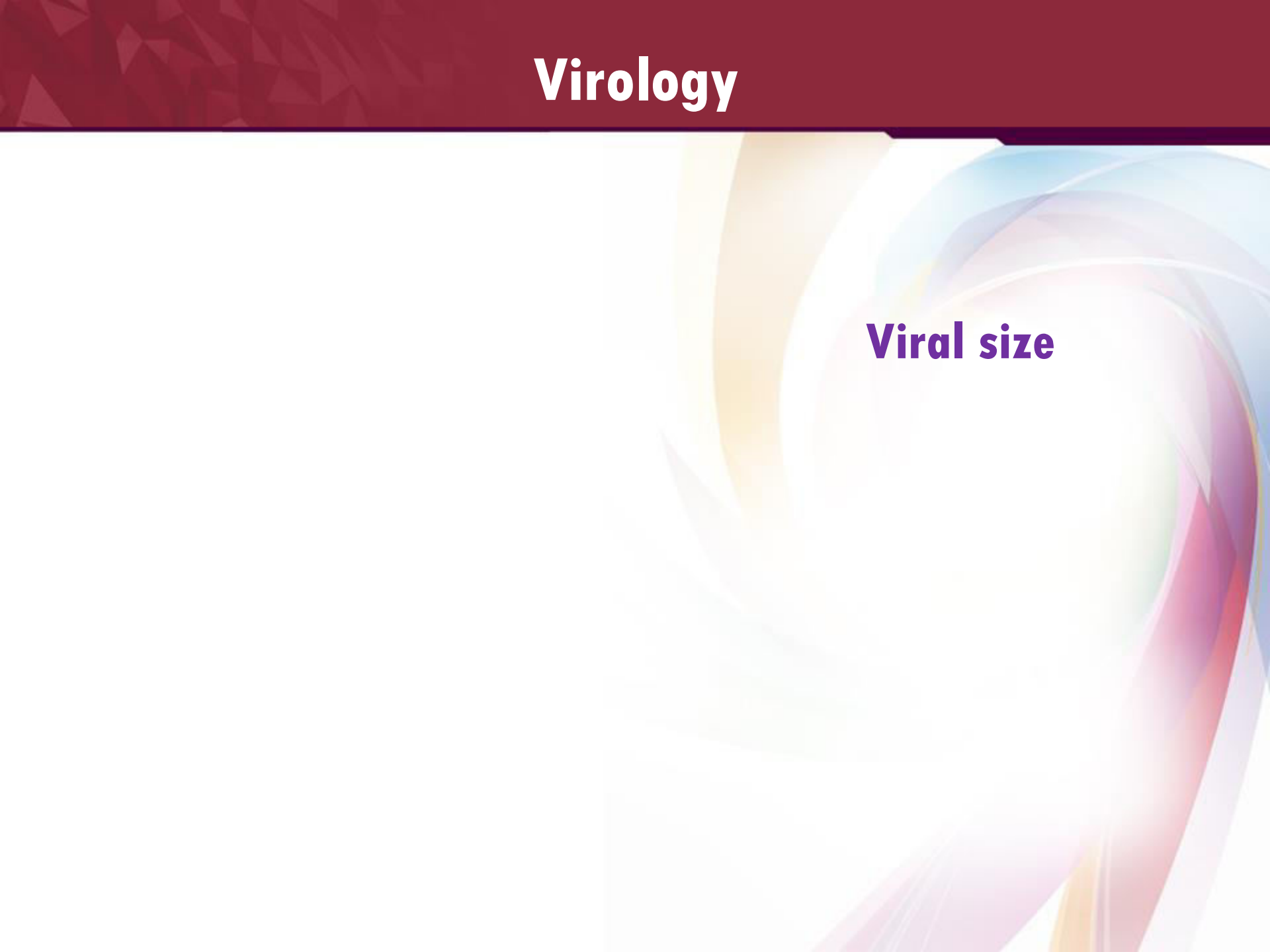
COMPLEX VIRUSES

- Additionally contain:
 - Lipid
 - Polysaccharides
 - Trace elements

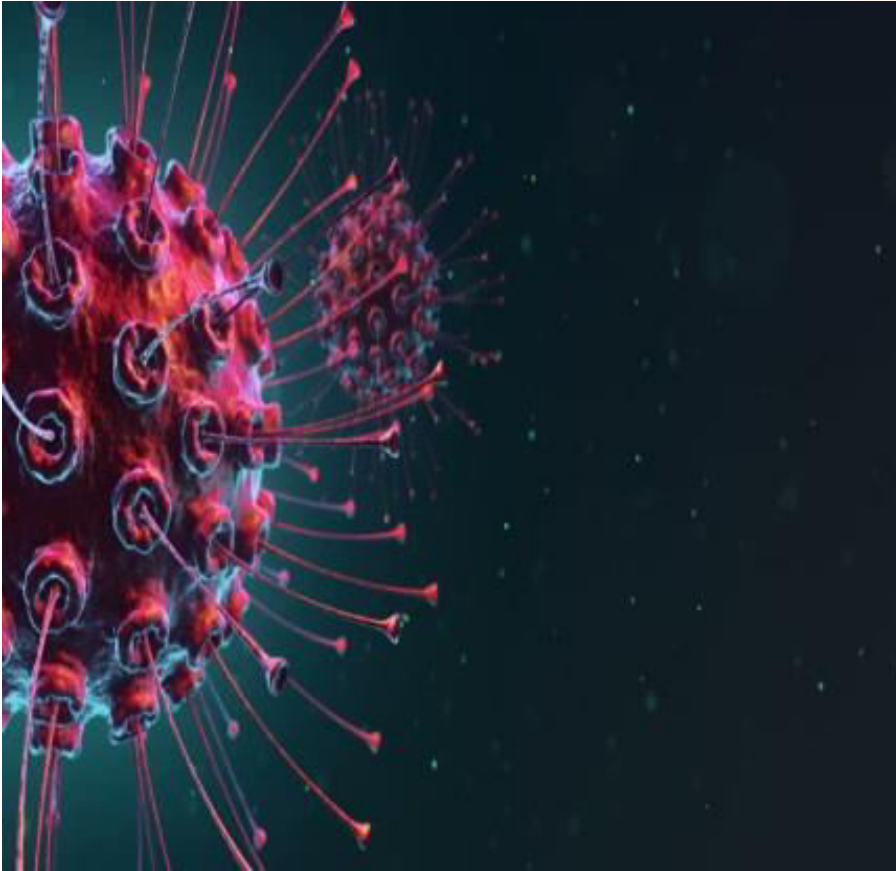


Virology

Viral size



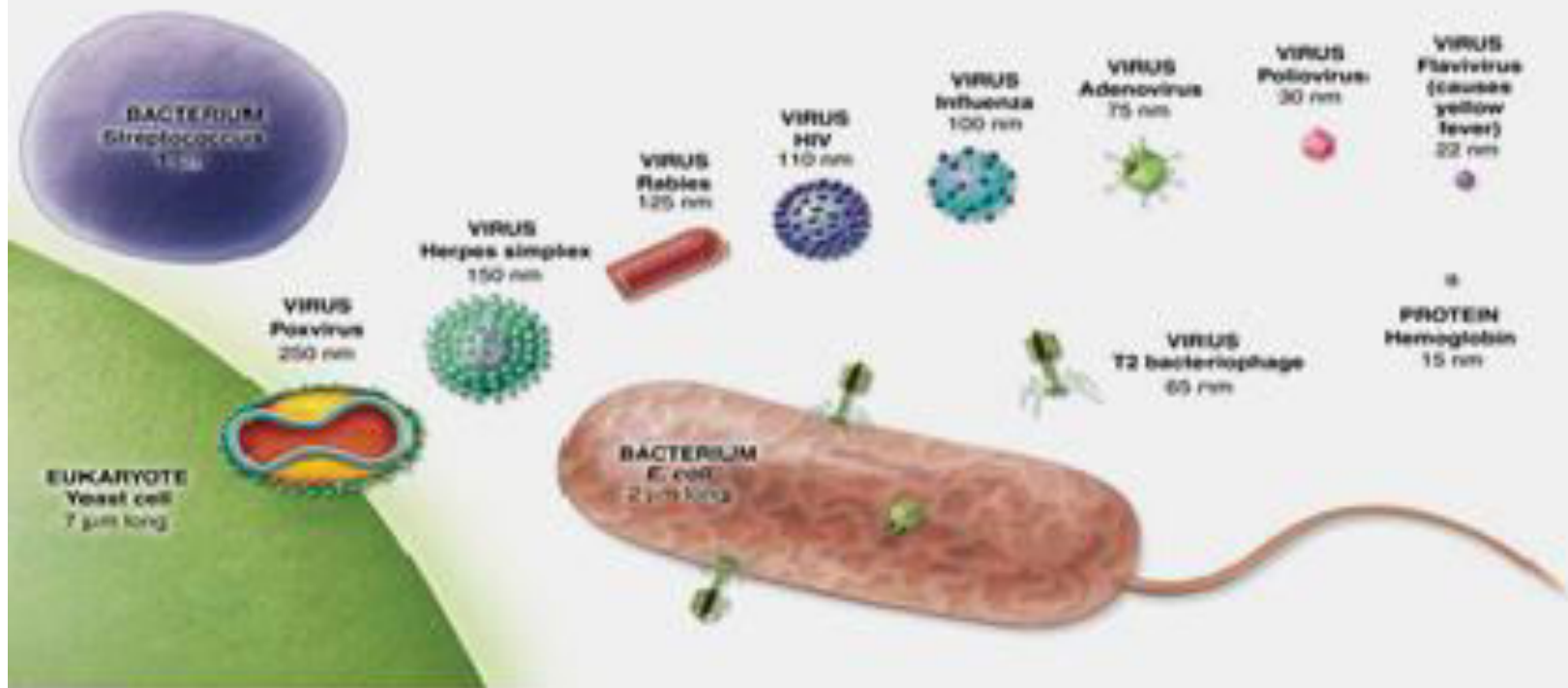
Viral size



- Determined by electron microscopy
- Vary in size
- Most are smaller than bacteria
- Some are as the same size of bacteria (vaccinia)

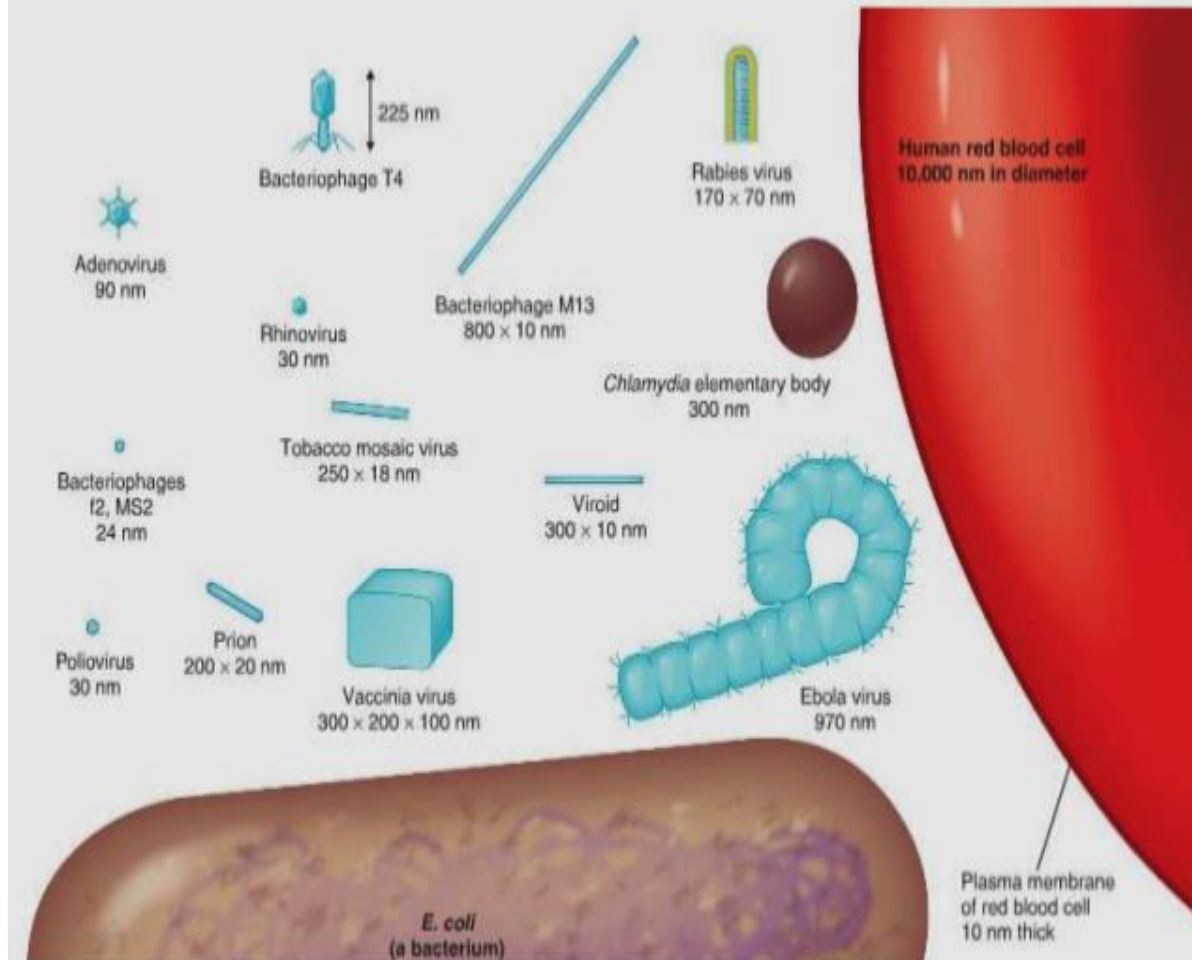
Viral size

Viruses vary in size, as well as in shape



Viral size

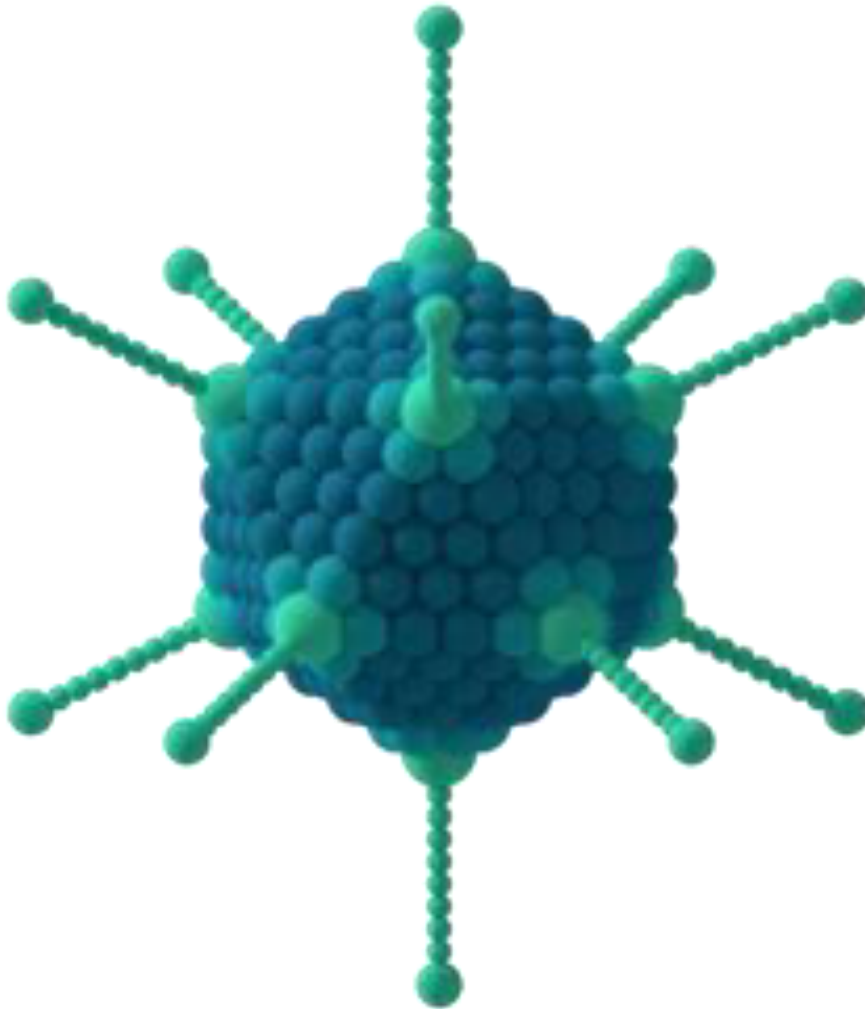
Size of Viruses



Virology

FUNCTIONAL CHEMISTRY OF VIRUSES

Function chemistry of viruses



CAPSID

- Outer protein coat
- Infectious component of virion
- Infectivity depends on break down of protein capsid (Capsomeres)
- Mostly host specific

Function chemistry of viruses

SPECIFICITY

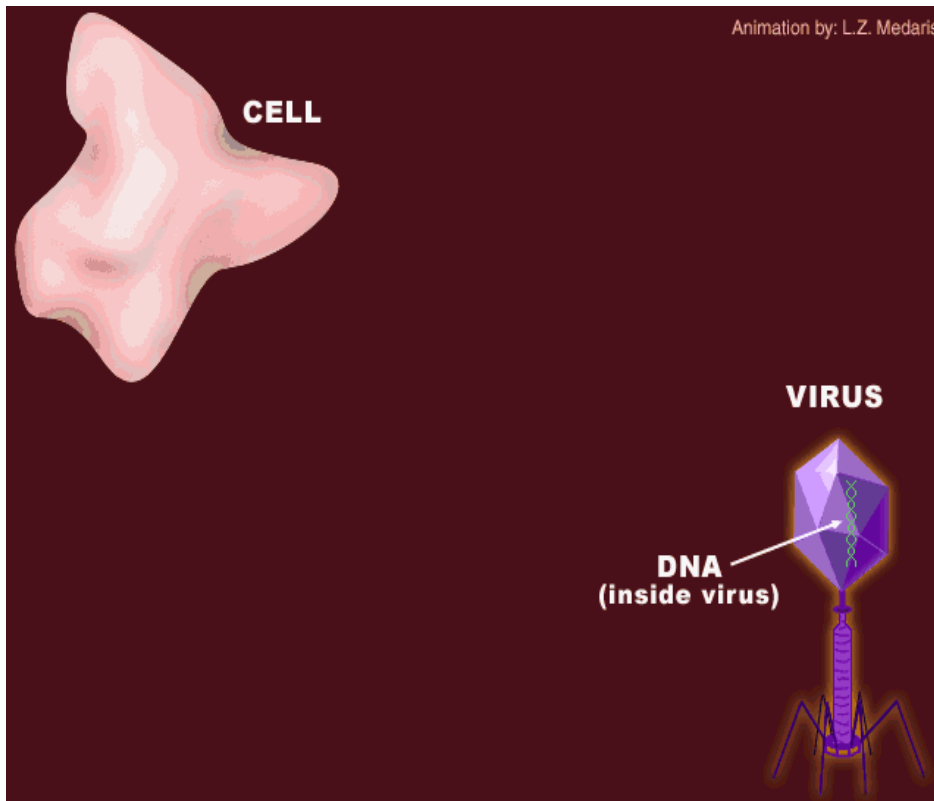
Most viruses infect specific host cells ie are host specific.

Host specificity is due to:

- **specific attachment sites on the host cells called receptors**
 - **Receptor sites for bacteriophage are found in bacterial cell walls or fimbriae or flagella**
 - **Animal cell membranes contain receptors for animal viruses**
- **availability of cellular factors required for viral multiplication in the host cells**

Function chemistry of viruses

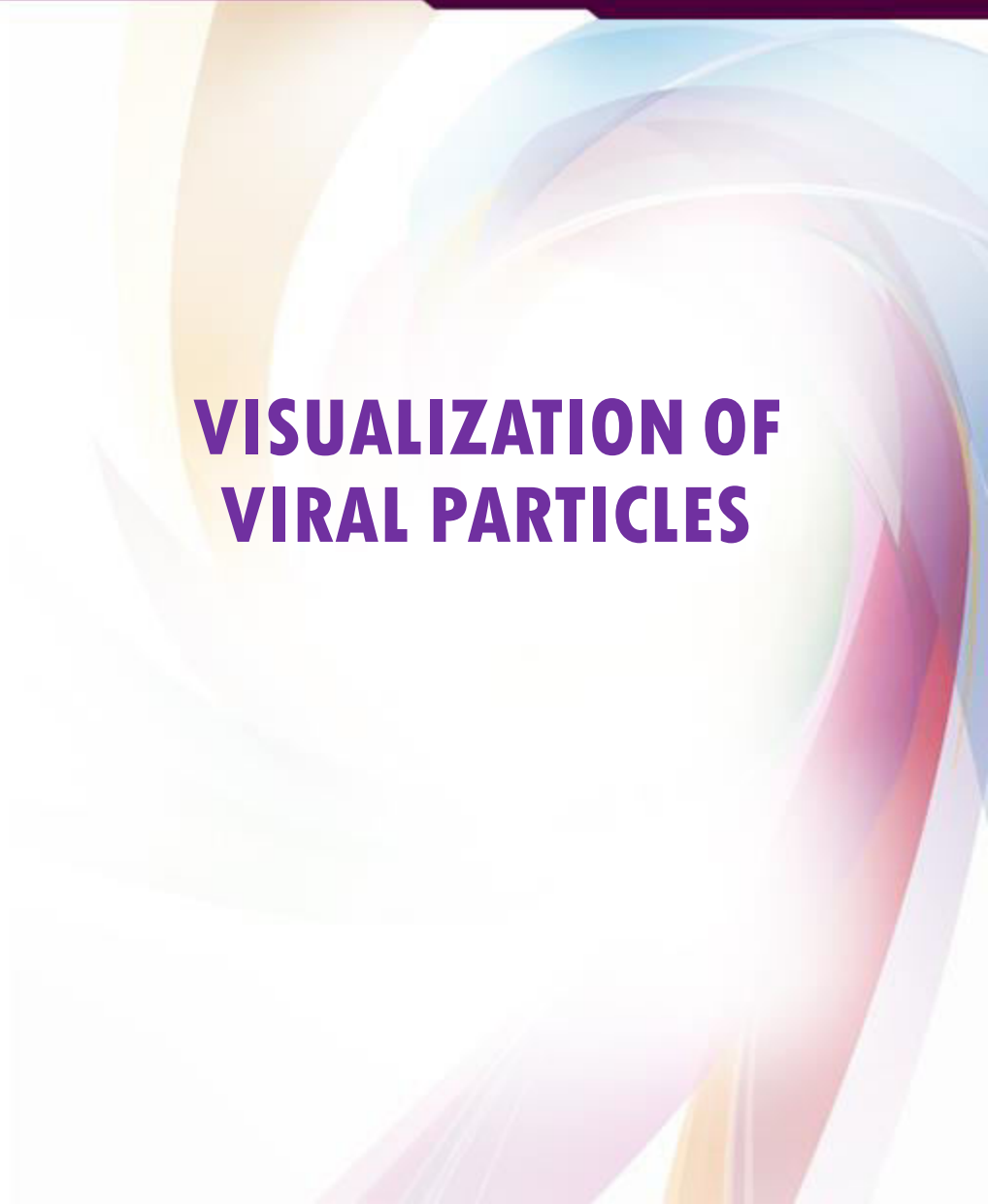
INFECTIVITY



- For infectivity in host cells:
 1. Viral nucleic acid
 2. Viral structural sub components (spikes or glycoproteins)
 3. Enzymes induced in host cells (polymerase enzymes)

Virology

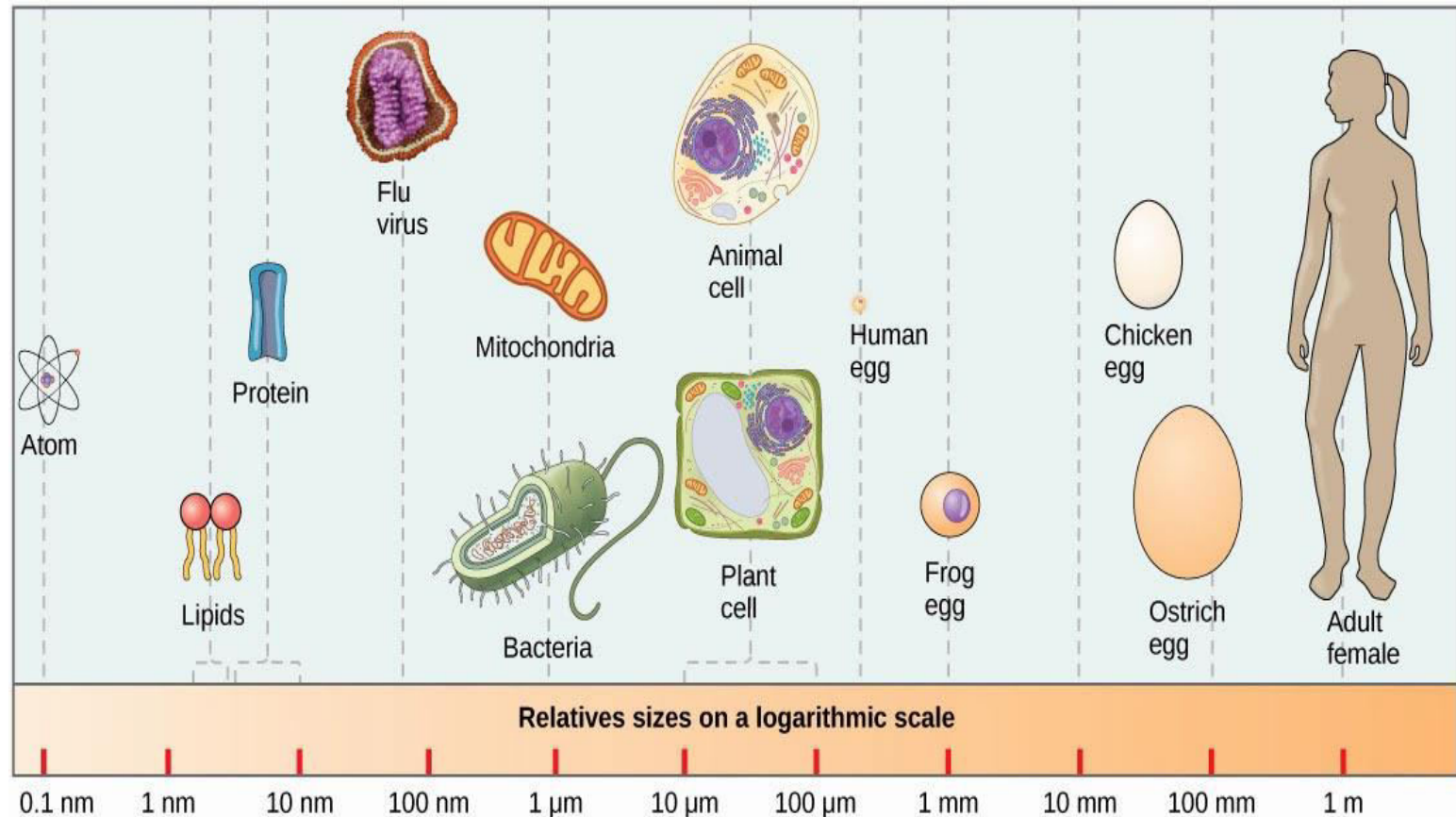
VISUALIZATION OF VIRAL PARTICLES



Visualization of Viral particles

Size

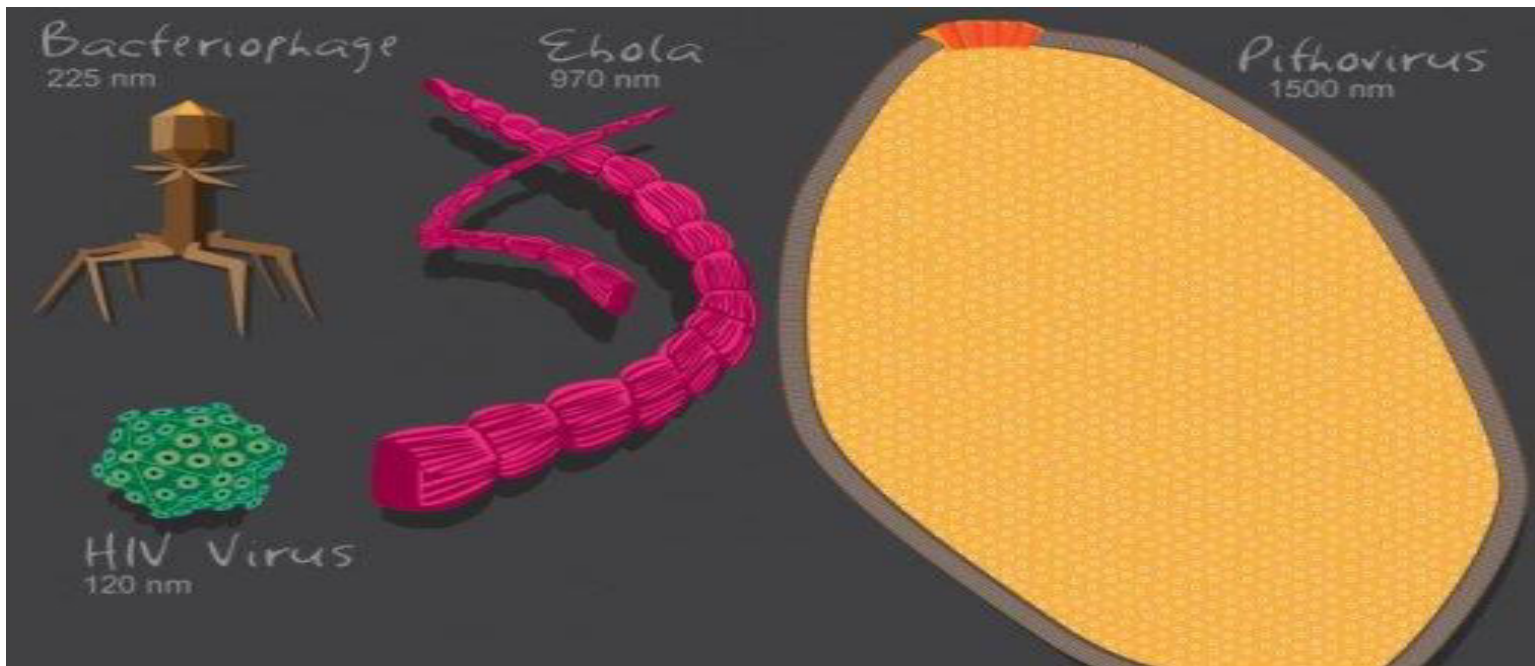
- Small size Not visible through simple microscope



Visualization of Viral particles

Variation

- Shape greatly varies
- Diversity
- Effects pathogenicity



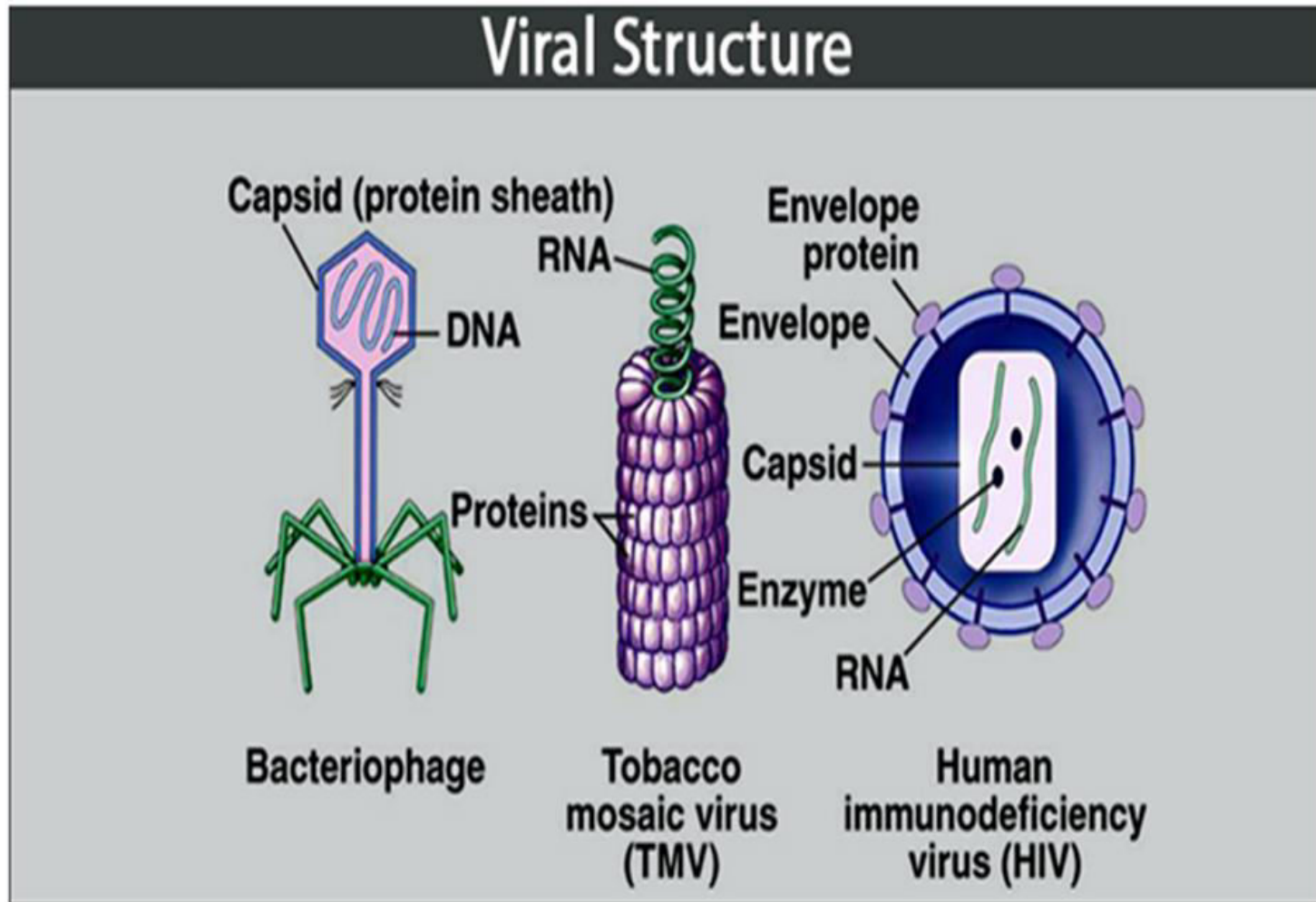
Visualization of Viral particles

General morphology

1. Icosahedral
2. Helical
3. Complex

Visualization of Viral particles

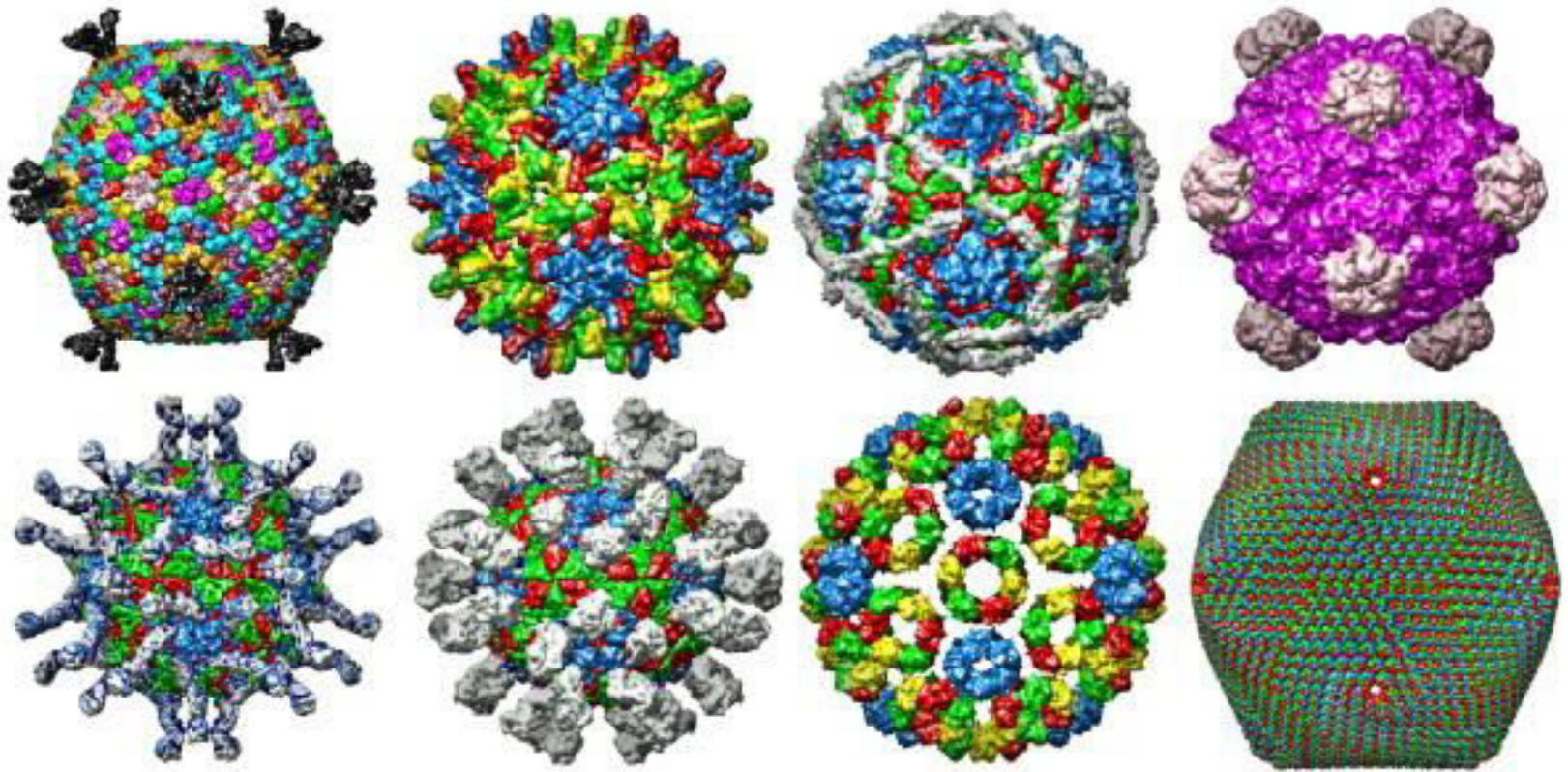
General morphology



Virology

BIOCHEMICAL AND CELLULAR BACKGROUND OF VIRUSES

BIOCHEMICAL AND CELLULAR BACKGROUND OF VIRUSES

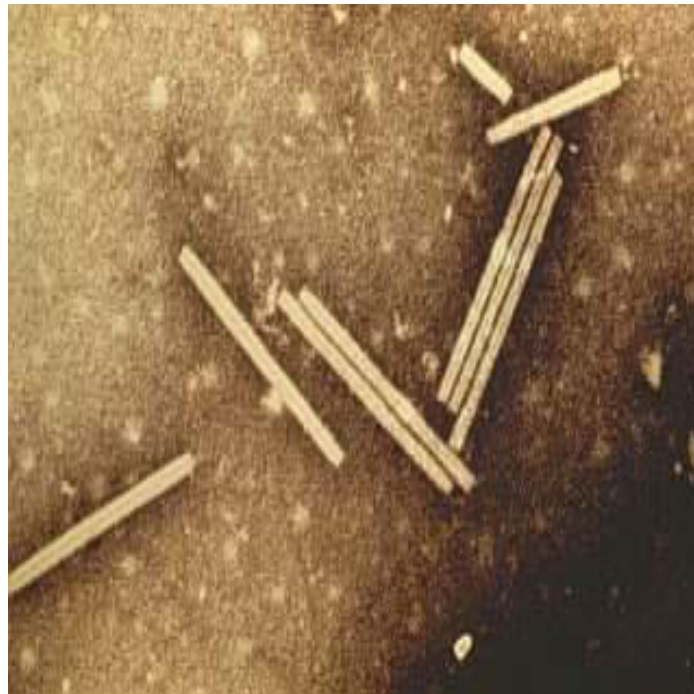


BIOCHEMICAL AND CELLULAR BACKGROUND OF VIRUSES

Wendell M. Stanley

1946 Nobel Prize in Chemistry

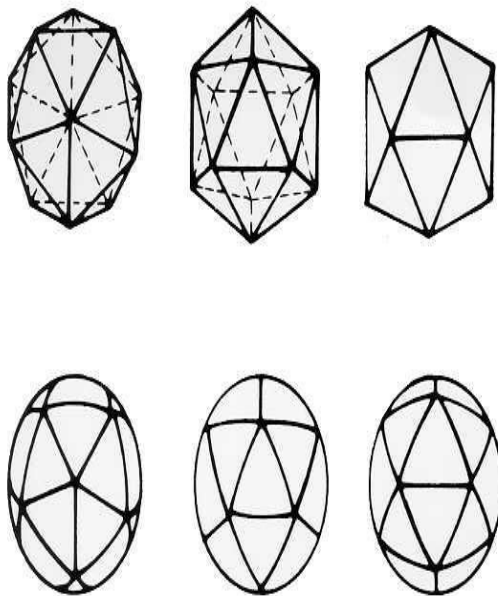
Crystallized Tobacco Mosaic Virus and
Systematically Investigated its Biochemistry



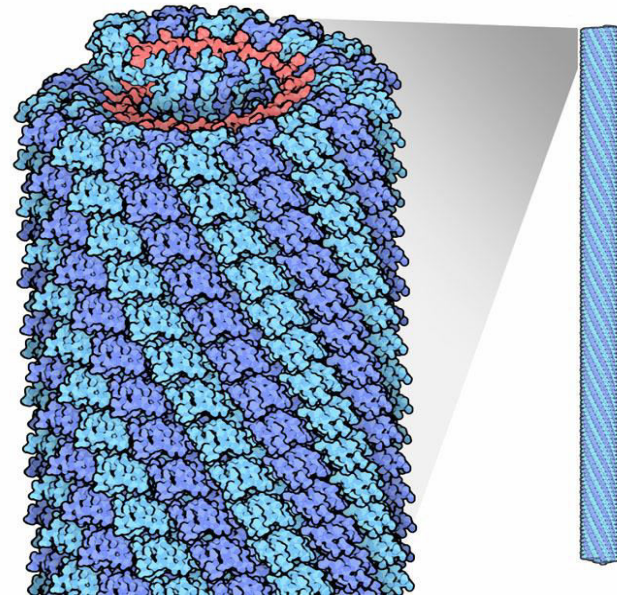
BIOCHEMICAL AND CELLULAR BACKGROUND OF VIRUSES

Symmetry of Viral Capsids

Icosahedral

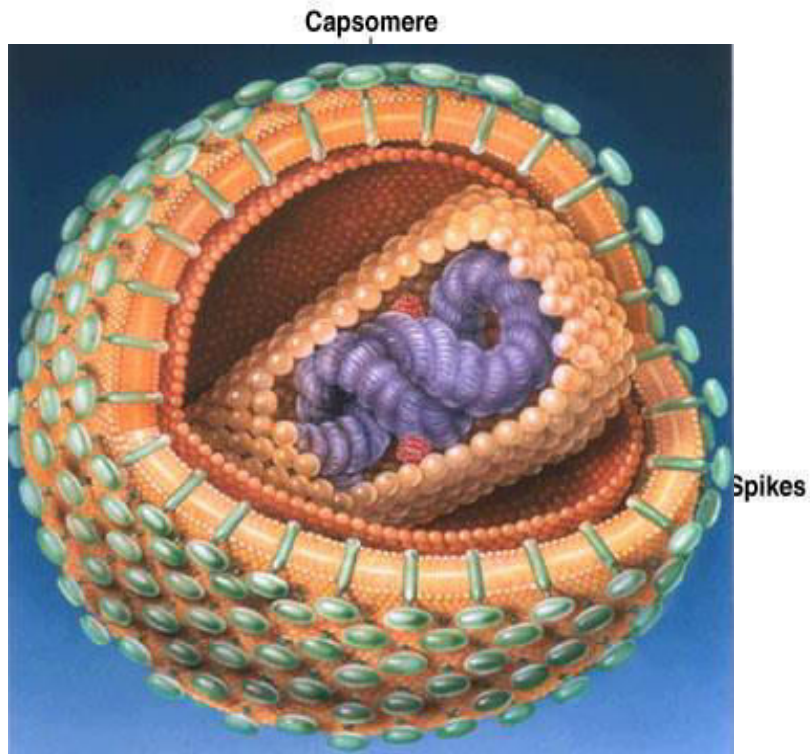


Helical



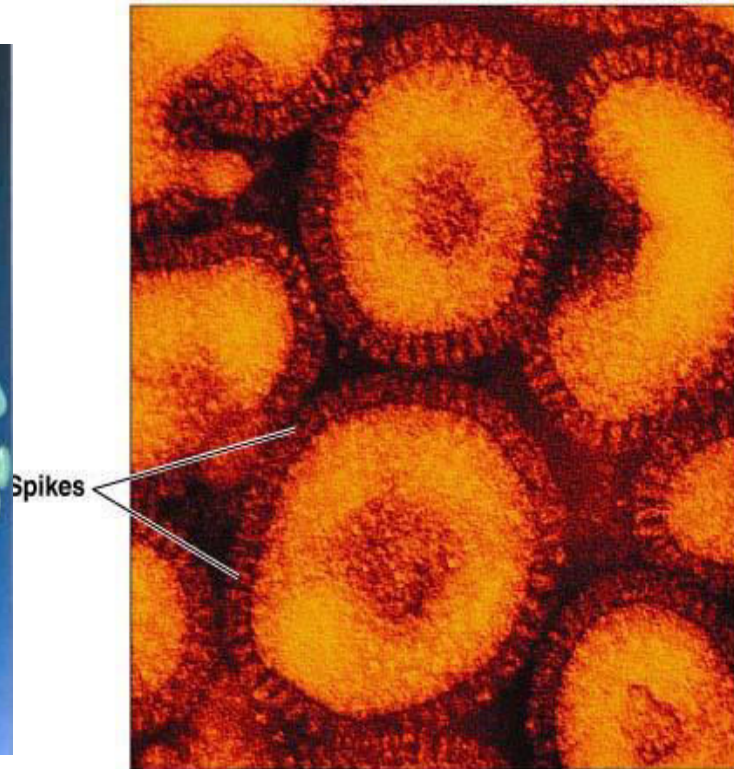
BIOCHEMICAL AND CELLULAR BACKGROUND OF VIRUSES

ENVELOPED VIRUSES



(a) An enveloped helical virus

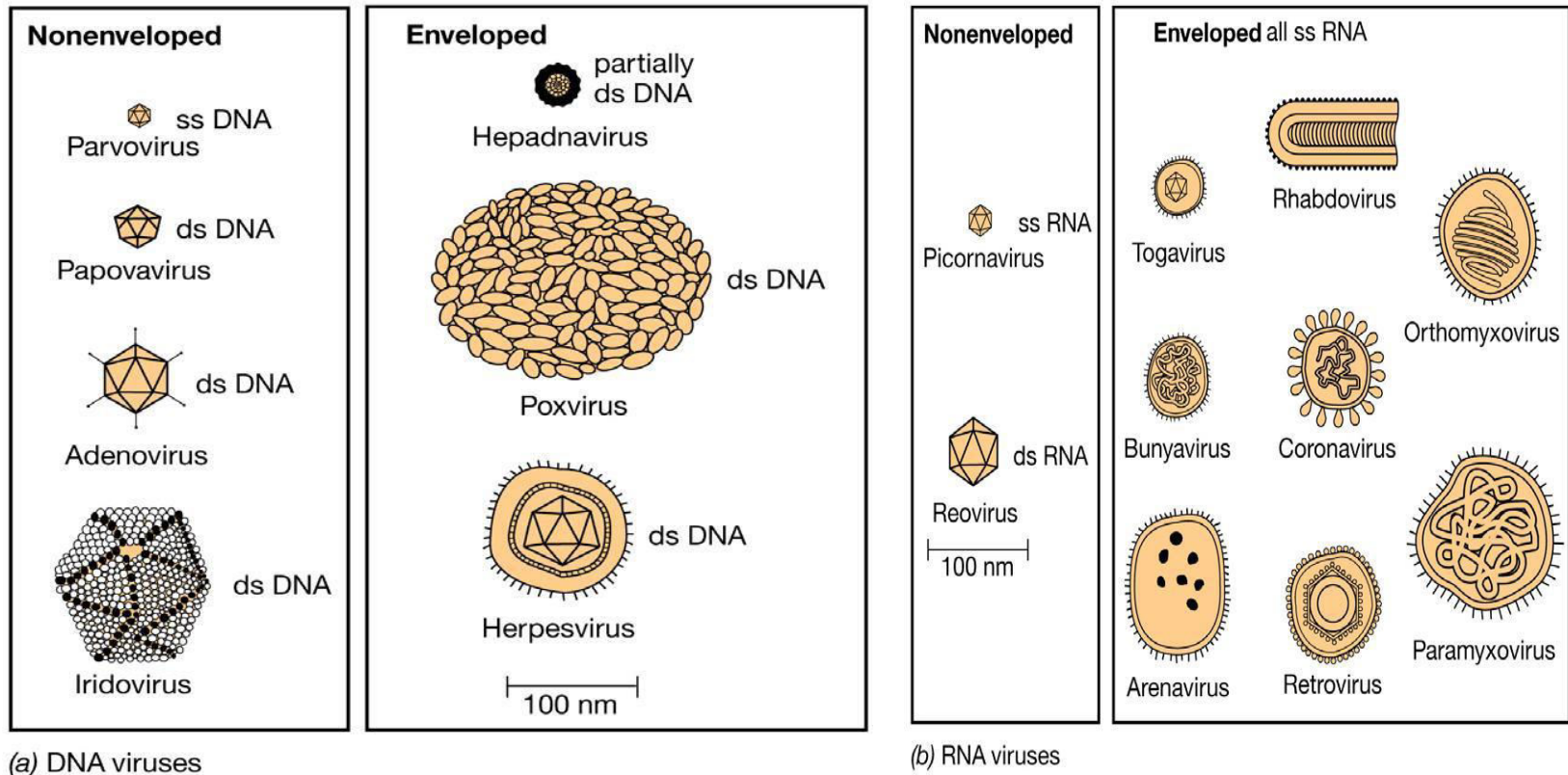
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(b) Influenzavirus

BIOCHEMICAL AND CELLULAR BACKGROUND OF VIRUSES

SURVEY OF HUMAN VIRUSES



Virology

**Living characters
in host cells**

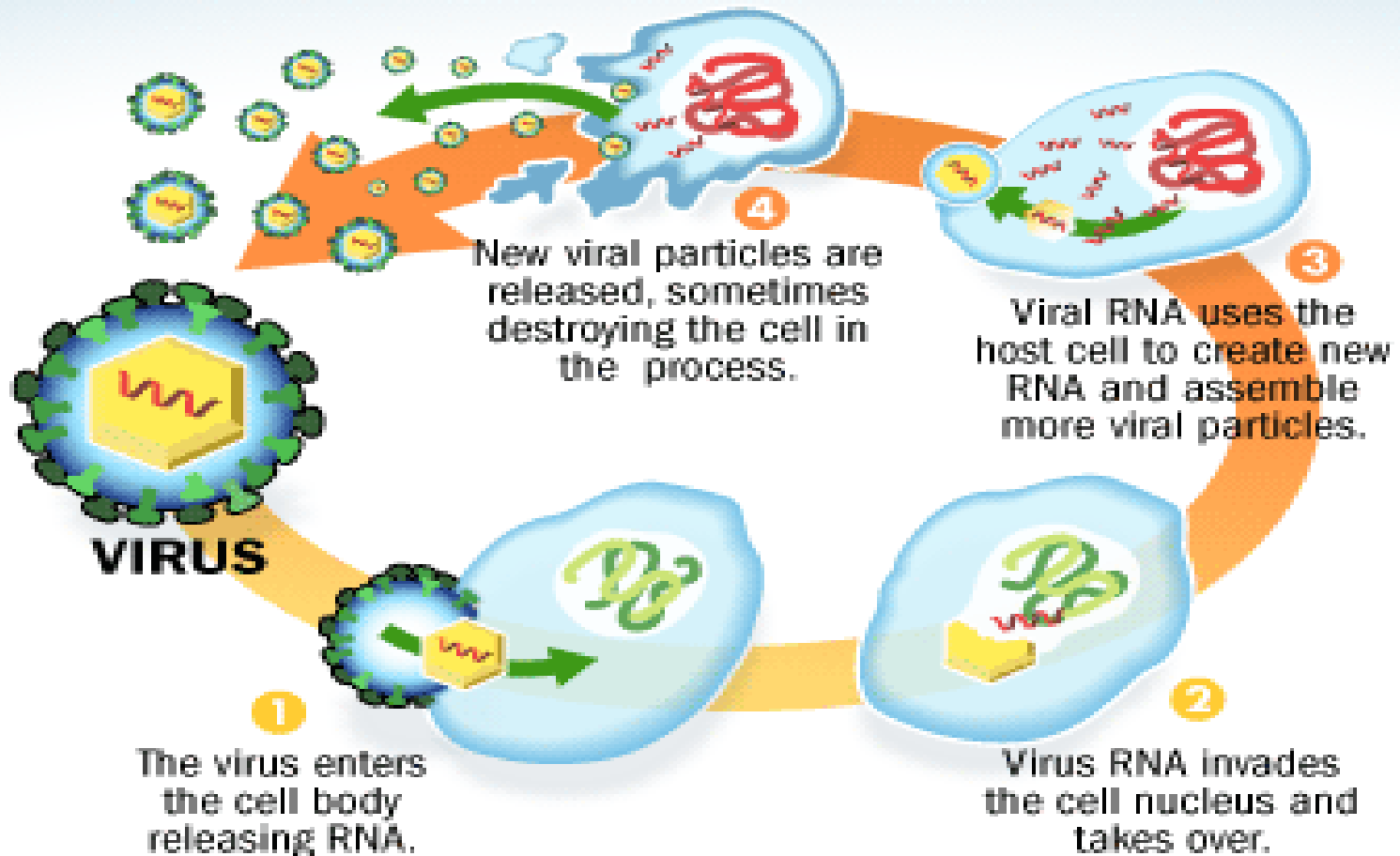
Living characters in host cells

- Molecular hijackers
- Biologically inert
- Dependent on host cells for replication and protein synthesis
- Lack in their own enzymes or organelles etc
- Sometimes alter the host genome

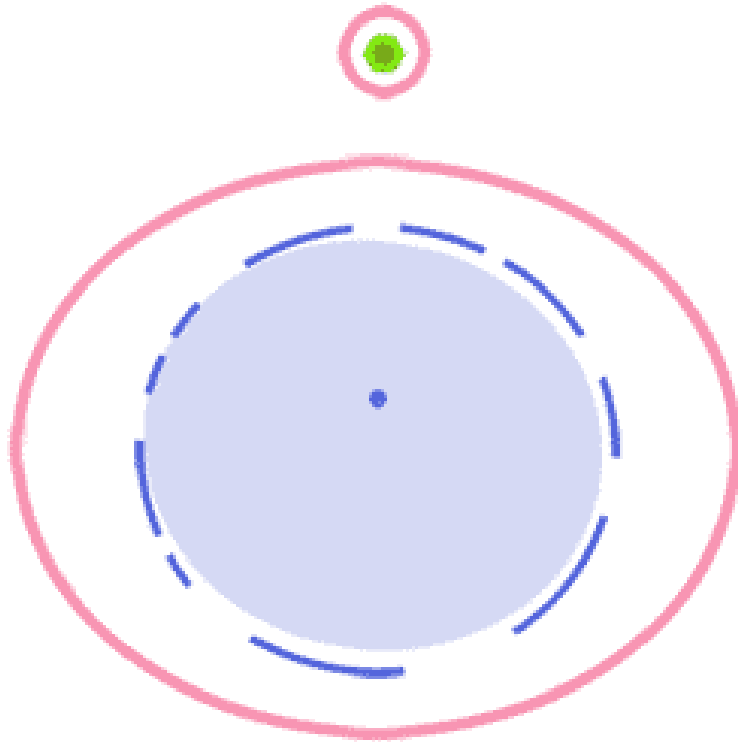
Living characters in host cells

How a Virus Works

©2010 HowStuffWorks



Living characters in host cells



Replication

- Using host machinery
- Force replication in host
- Increase in number
- Leave the cell simply or either by bursting the host cell

Living characters in host cells

Effects on host

- None
- Genetic alterations
- Physical damage
- Oncogenesis

Shelter

- Take shelter in specific host cells
- Take nourishment
- Complete lifecycles
- Continuity of viral life

Virology

HOST RANGE

Host Range

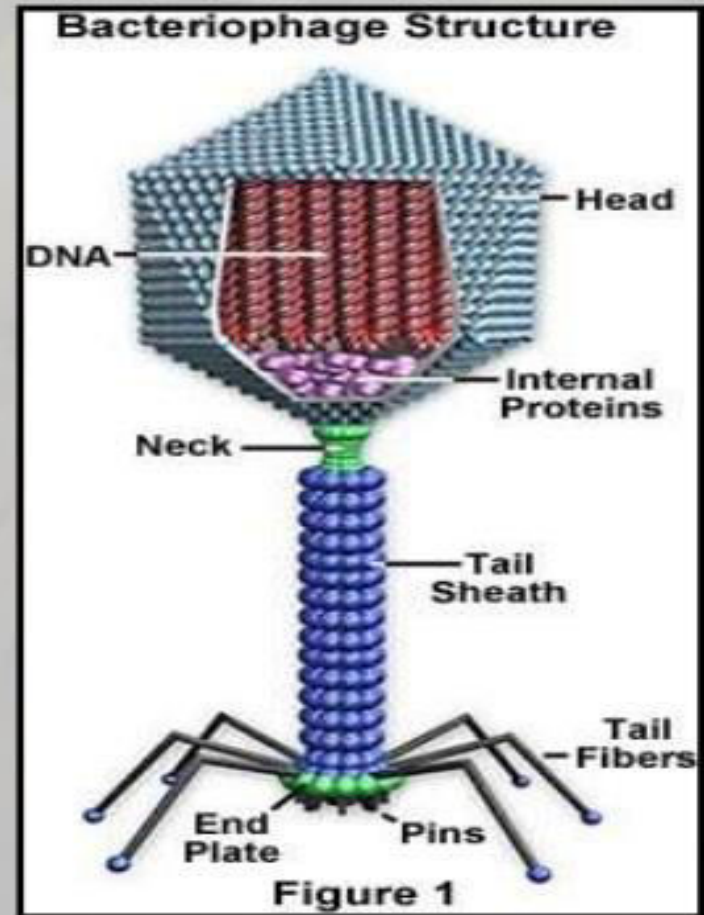
What is a host range?

- The spectrum of host cells that the virus can infect.

Host Range

Host Range

1. Host range refers to the spectrum of host cells in which a virus can multiply.
2. Most viruses infect only specific types of cells in one host species.
3. Host range is determined by the specific attachment site on the host cell's surface and the availability of host cellular factors.
4. Viruses that infect bacteria are called bacteriophages, or phages.



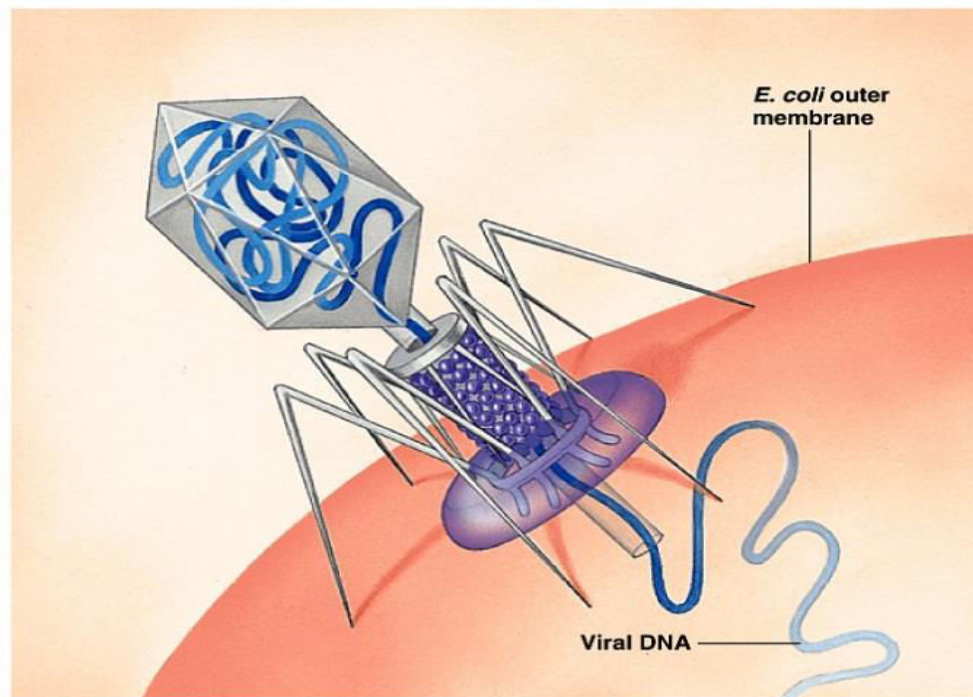
Host Range

Viral host range

- Plants
- Vertebrates
- Invertebrates
- Protists
- Bacteria
- Fungi

Host Range

- Most viruses are host specific
- Bind to specific receptors on cells
- Animal viruses: receptors on plasma membrane
- Form hydrogen bond with viral surface



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Virology

Viral specificity



Viral Specificity

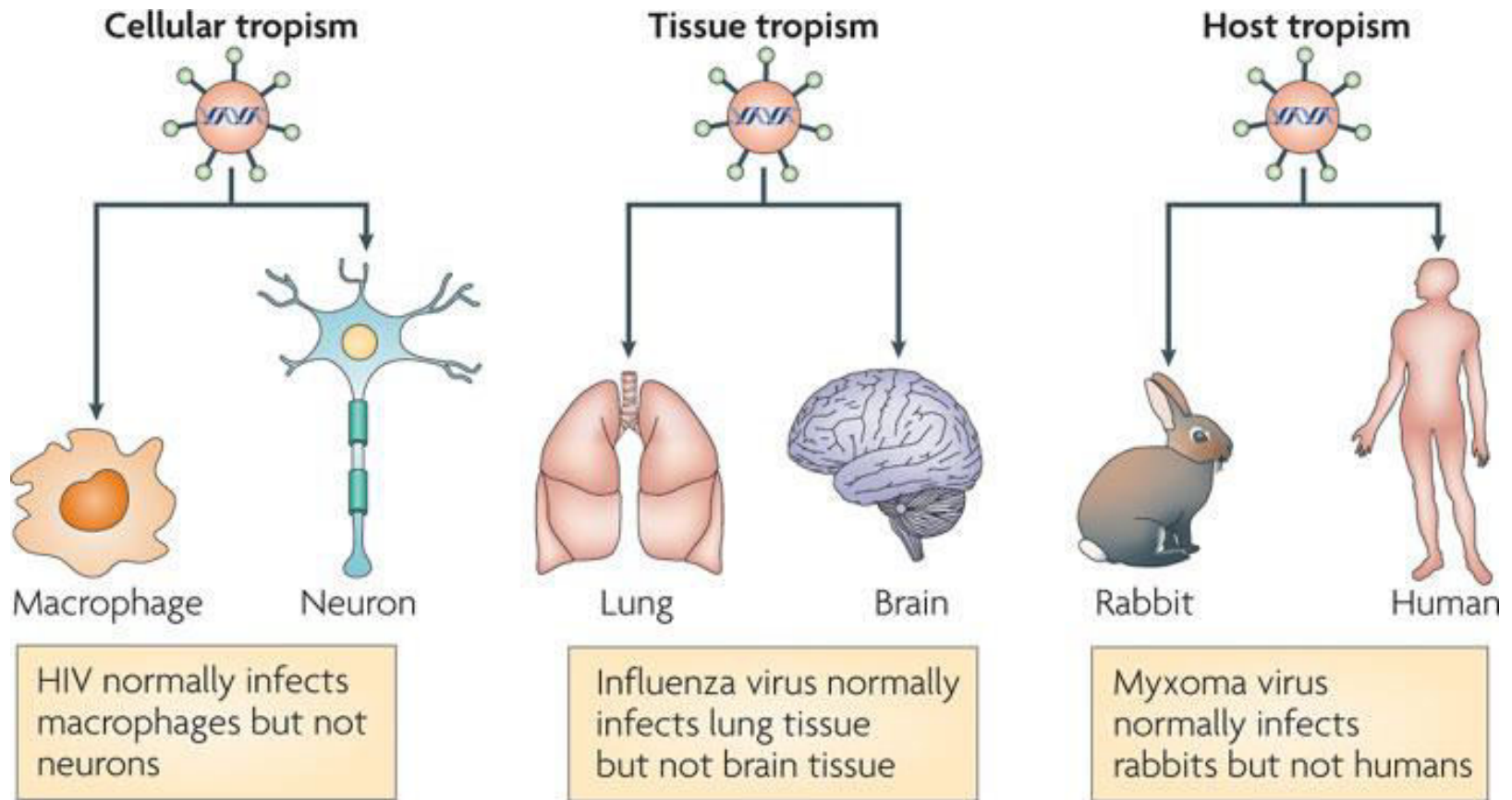
- All kingdoms can be infected by viruses
- Viruses that infect one kingdom are unable to infect another
- Viruses might infect closely related organisms
- Usually species specific

Viral Specificity

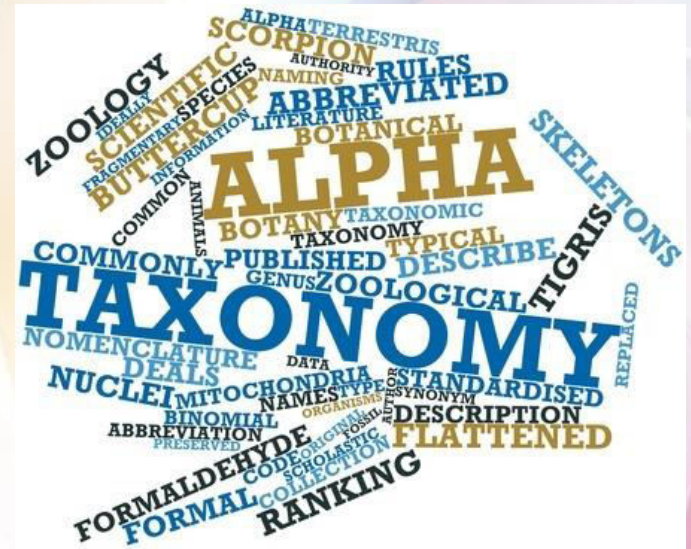
Examples

VIRUS	HOST
Tobacco Mosaic Virus	Tobacco plant
Rabies	Mammals
Swine flu	Pigs and humans
Foot and Mouth disease	Hooved animals
Bird Flu	Birds and human
Myxoma virus	Rabbits

Viral Specificity



Virology



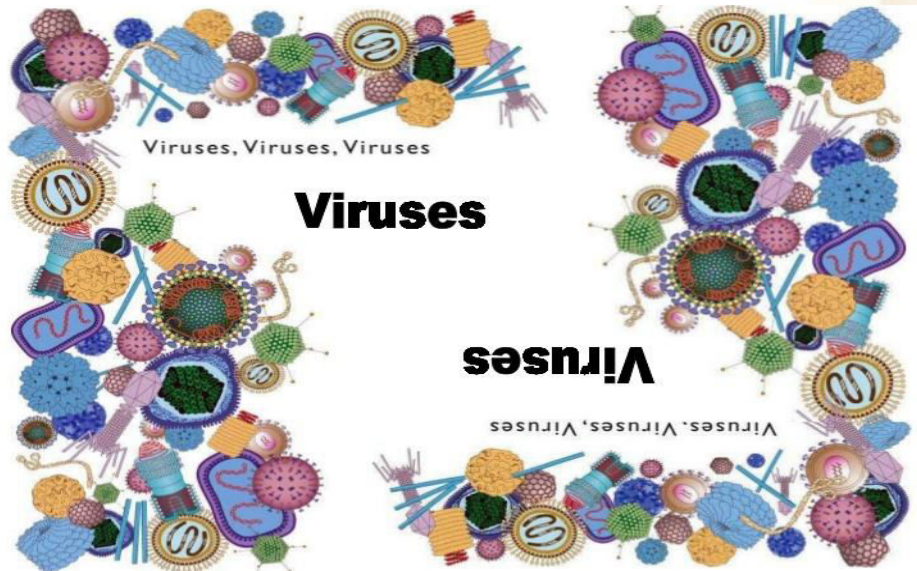
TAXONOMY I

Taxonomy

WHAT IS

- Naming of something
- OR
- Scheme of classification and nomenclature of organisms
- Viral taxonomy is the classification of viruses based on viral nature

Taxonomy

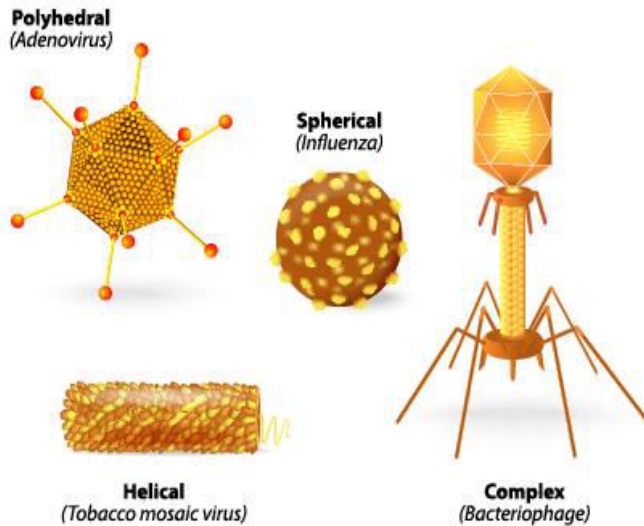


SCHEMES FOR CLASSIFICATION

- International Committee on Taxonomy of Viruses (ICTV) 1966
- Baltimore system of classification 1971

Taxonomy

VIRAL SHAPES



BASED ON PHENOTYPE

Morphology

Structure of capsid

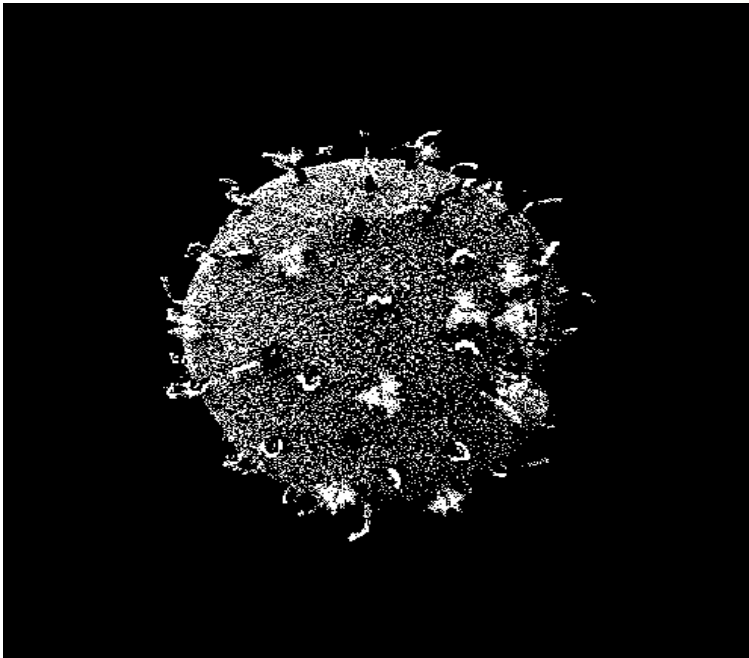
Presence of envelope

Type of nucleic acid

Mode of replication

- Host organism
- Diseases caused

Taxonomy



ICTV CLASSIFICATION

•A class where members have several properties in common e.g.

- Host
- Ecological niche
- Pathogenicity
- Mode of transmission
- Habitat

Taxonomy

LEVEL OF CLASSIFICATION

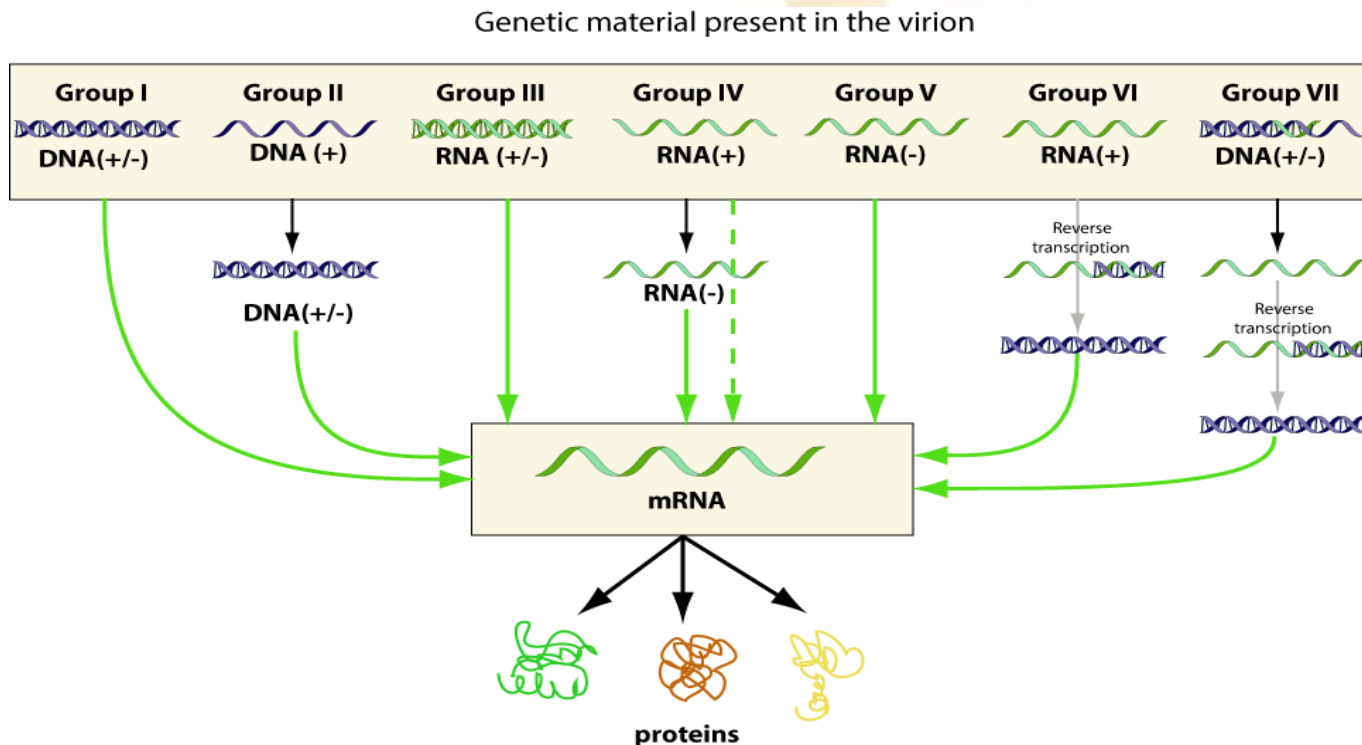
ORDER	FAMILY	SUBFAMILY	GENUS	SPECIES
• -virales	• -viridae	• -virinae	• -virus	

Virology

TAXONOMY-II

Taxonomy

- **BALTIMORE CLASSIFICATION**
- David Baltimore
- 1971

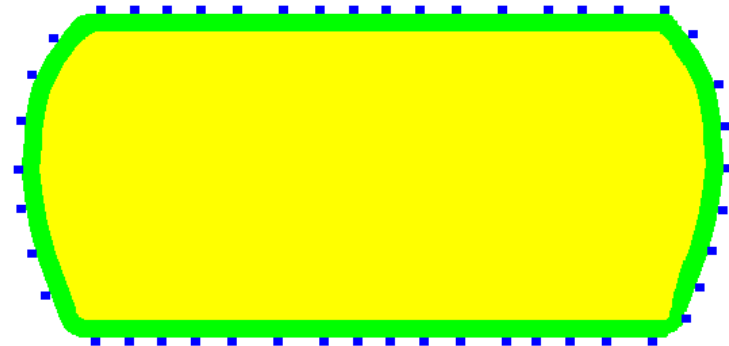
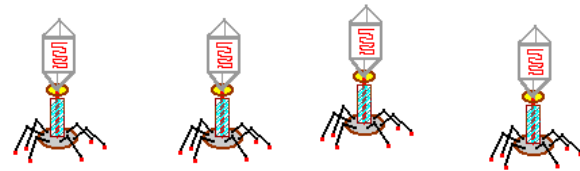


Taxonomy

DEPENDS ON

1. Expression of nucleic acid
2. Stranded
3. Segmented/ Sense
4. Method of Replication

START



ADSORPTION/DOCKING

Taxonomy

7 CLASSES:

GROUP I: double stranded DNA viruses (Adeno, Herpes)

GROUP II: single stranded DNA viruses (Parvo)

GROUP III: double stranded RNA viruses (Reo)

GROUP IV: positive sense single stranded RNA viruses (Polio, Picorna)

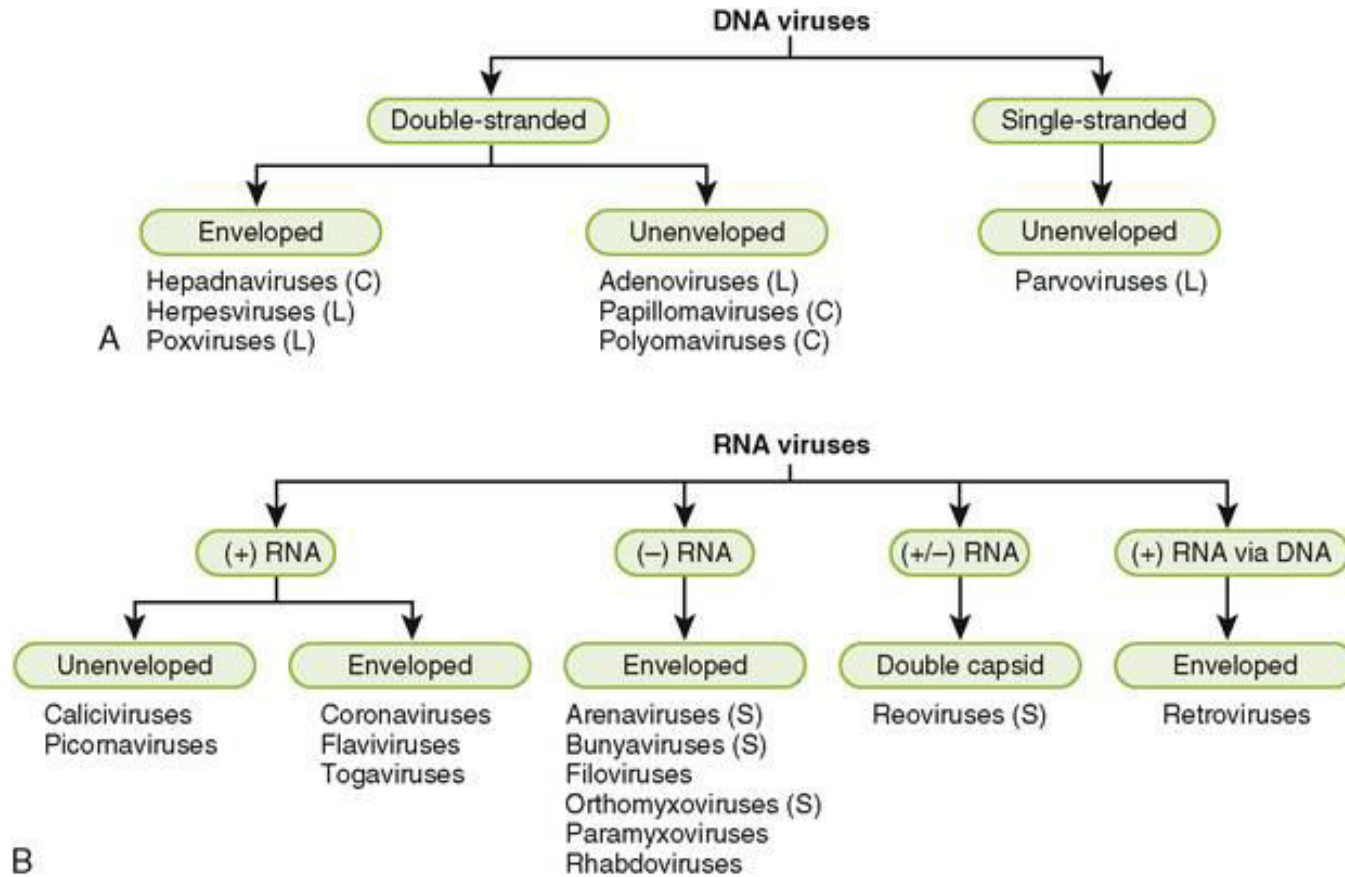
GROUP V: negative sense single stranded RNA viruses (ortho)

GROUP VI: Single stranded RNA viruses (Retro)

GROUP VII: Double stranded DNA viruses (Hepadna)

Taxonomy

VIRAL CLASSIFICATION



Virology

ARTIFICIAL TRANSMISSION OF VIRUSES IN PLANTS

Artificial transmission of viruses in plants

BARRIERS

- External surface
- Structural differentiation of host
- Internal barriers
- Direct transmission only through wounds or poorly protected tissues



Artificial transmission of viruses in plants

MECHANICAL TRANSMISSION



- Few plant viruses
- TMV and WMV by root inoculation
- Potato viruses by infected tuber plantation

Artificial transmission of viruses in plants

- **SEED TRANSMISSION**
- Rare
- But exists

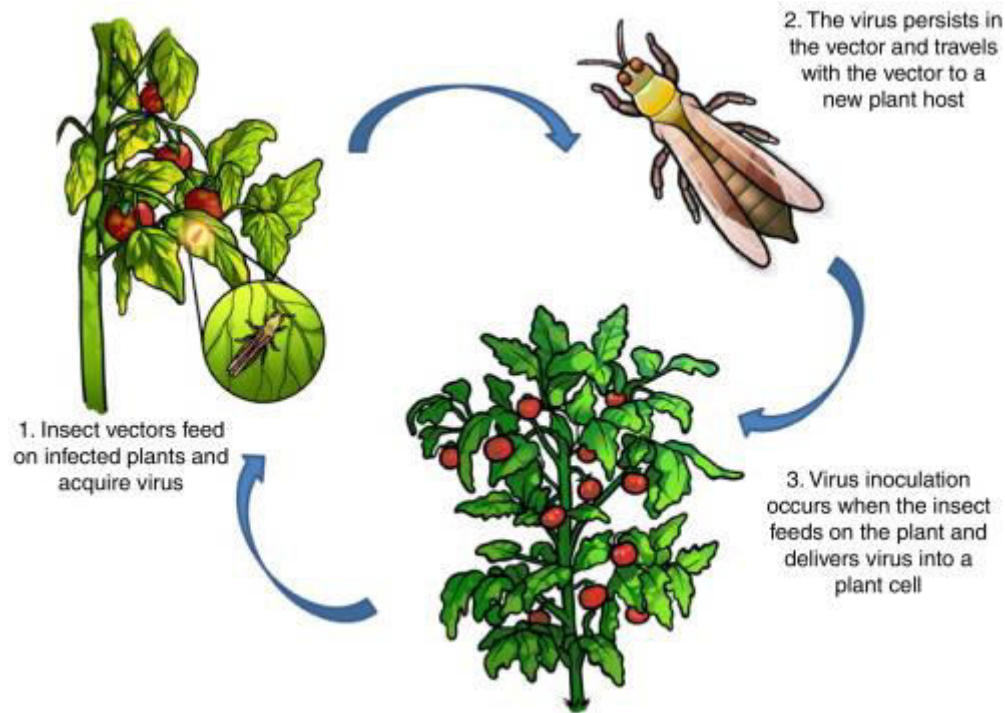


Artificial transmission of viruses in plants

• VECTOR

TRANSMISSION

- Most common
- Grass hopper
- Aphids
- Leaf hoppers
- White flies
- Mites
- Beetles



Artificial transmission of viruses in plants

DISEASES

- Rice dwarf virus
- Maize streak
- Tomato spoiled wilt virus
- Pigeon pea sterility virus

Virology

ARTIFICIAL TRANSMISSION OF VIRUSES IN ANIMALS

Artificial transmission of viruses in animals

REASON

- Viruses that cannot be cultivated on cell culture or embryonated eggs
- To experiment on animals rather human beings



Artificial transmission of viruses in animals

ANIMAL INOCULATION



- Route:
Subcutaneous,
intracerebral,
intraperitoneal,
Intradermal or
intraocular
- Virus multiplies in
host and disease
develops

Artificial transmission of viruses in animals

ANIMALS USED

- Monkey
- Rabbit
- Guinea pig
- Rat
- Hamsters mice



Artificial transmission of viruses in animals

GROWTH CONFIRMATION

- Pathogenesis
- Lesions
- Immune response
- Oncogenesis
- Death

Artificial transmission of viruses in animals

ADVANTAGES

- Clinical manifestations are determined
- Primary isolation of viruses
- Used for study of immune response

Virology

MICROSCOPY



Microscopy

MICROSCOPE

- An instrument used to see objects that are too small for the naked eye.



Microscopy

MICROSCOPY

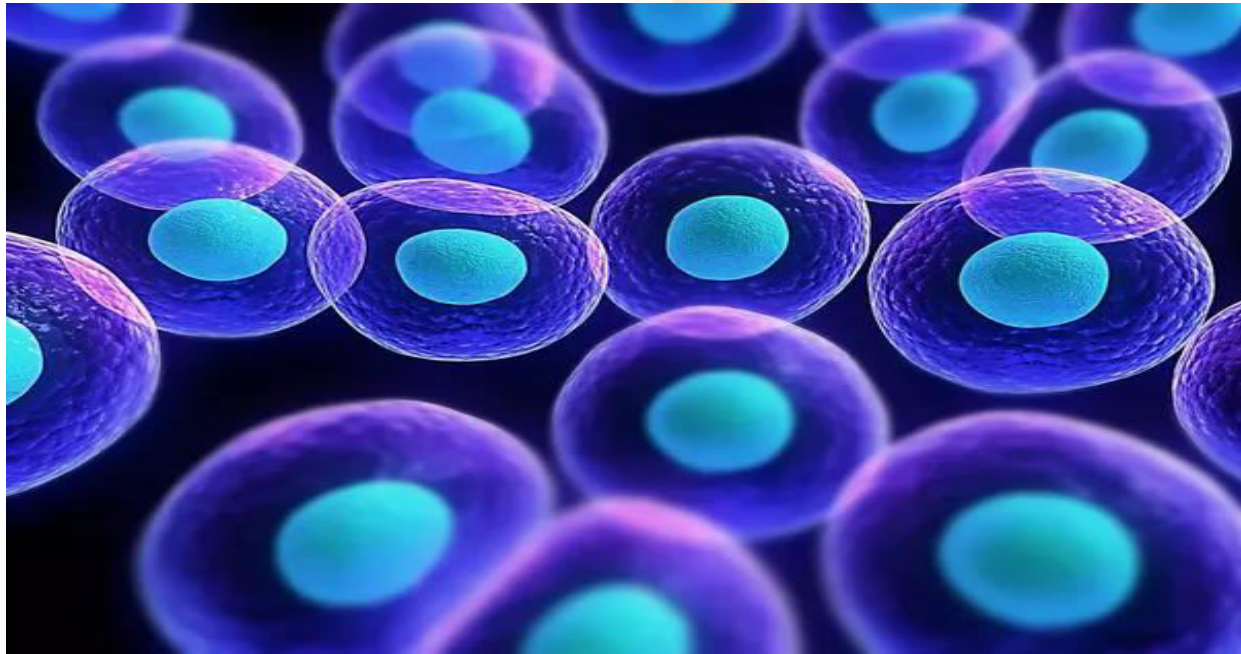
- The science of investigating small objects using such an instrument is called microscopy



Microscopy

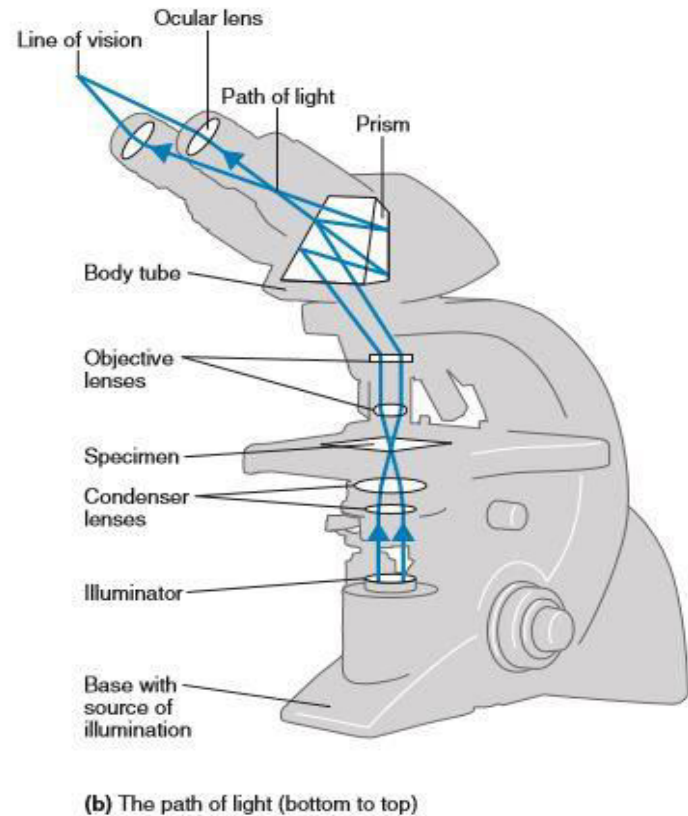
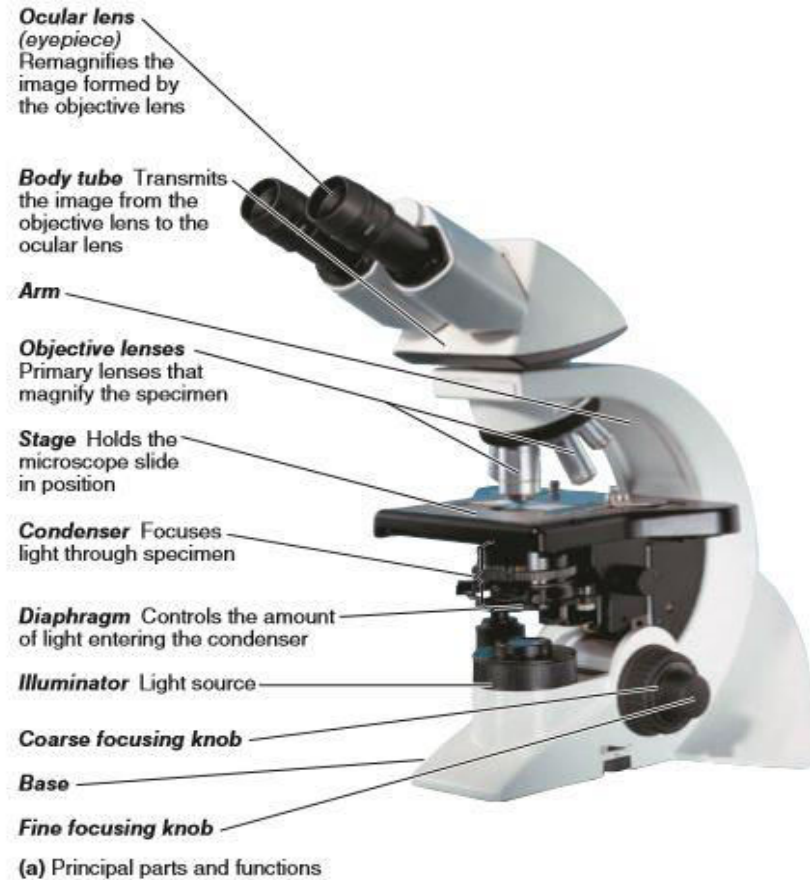
MICROSCOPIC

- Microscopic means invisible to the eye unless aided by a microscope



Microscopy

PARTS OF MICROSCOPE



Microscopy & its Types

Amina Mustafa

Department of Molecular Biology

Virtual University of Pakistan

Introduction

micro- = “small”; *-scope* = “to look at”) to study them.

A **microscope** is an instrument that magnifies objects otherwise too small to be seen, producing an image in which the object appears larger. Most photographs of cells are taken using a microscope, and these pictures can also be called **micrographs**.

Microscopy

Invented by :

Antonie van Leeuwenhoek
(Father of Microbiology)



Von Leeuwenhoek
Microscope
(circa Late 1600s)

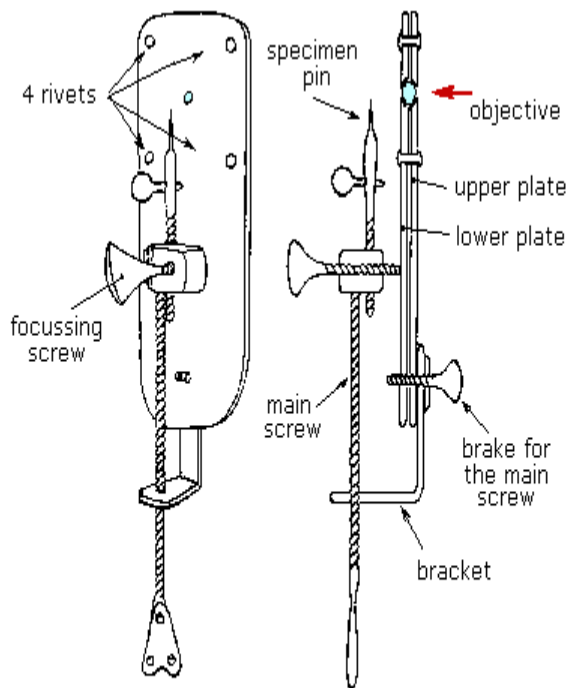


Figure 1 - Diagram of the microscope constructed by Antoni van Leeuwenhoek in the XVII century

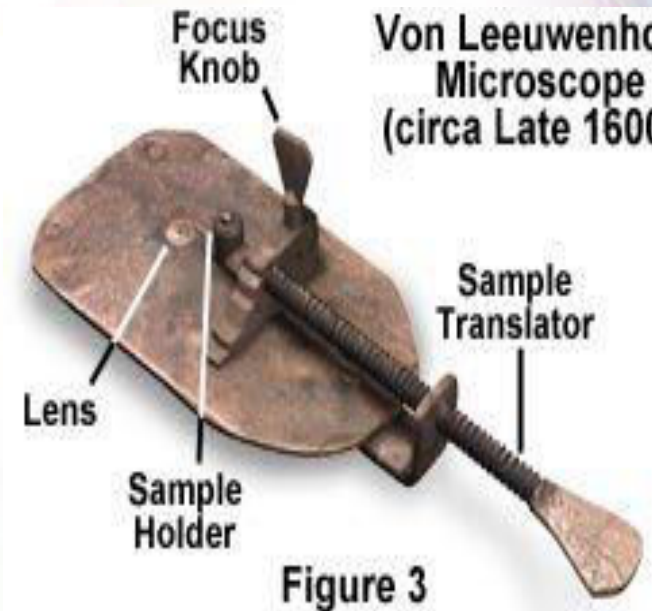


Figure 3

Microscopy

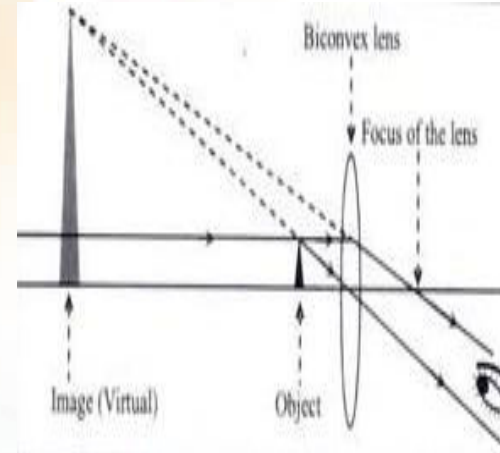
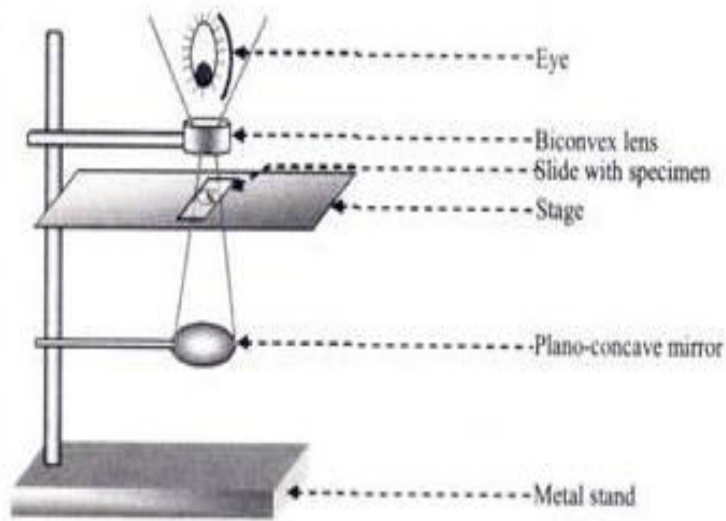
USES:

- Diagnostics
- Identification
- Histology – Tissue analysis
- Examining Forensic Evidence
- Studying cells
- Studying Micro-organisms
- Scientific Research

Types of Microscopy

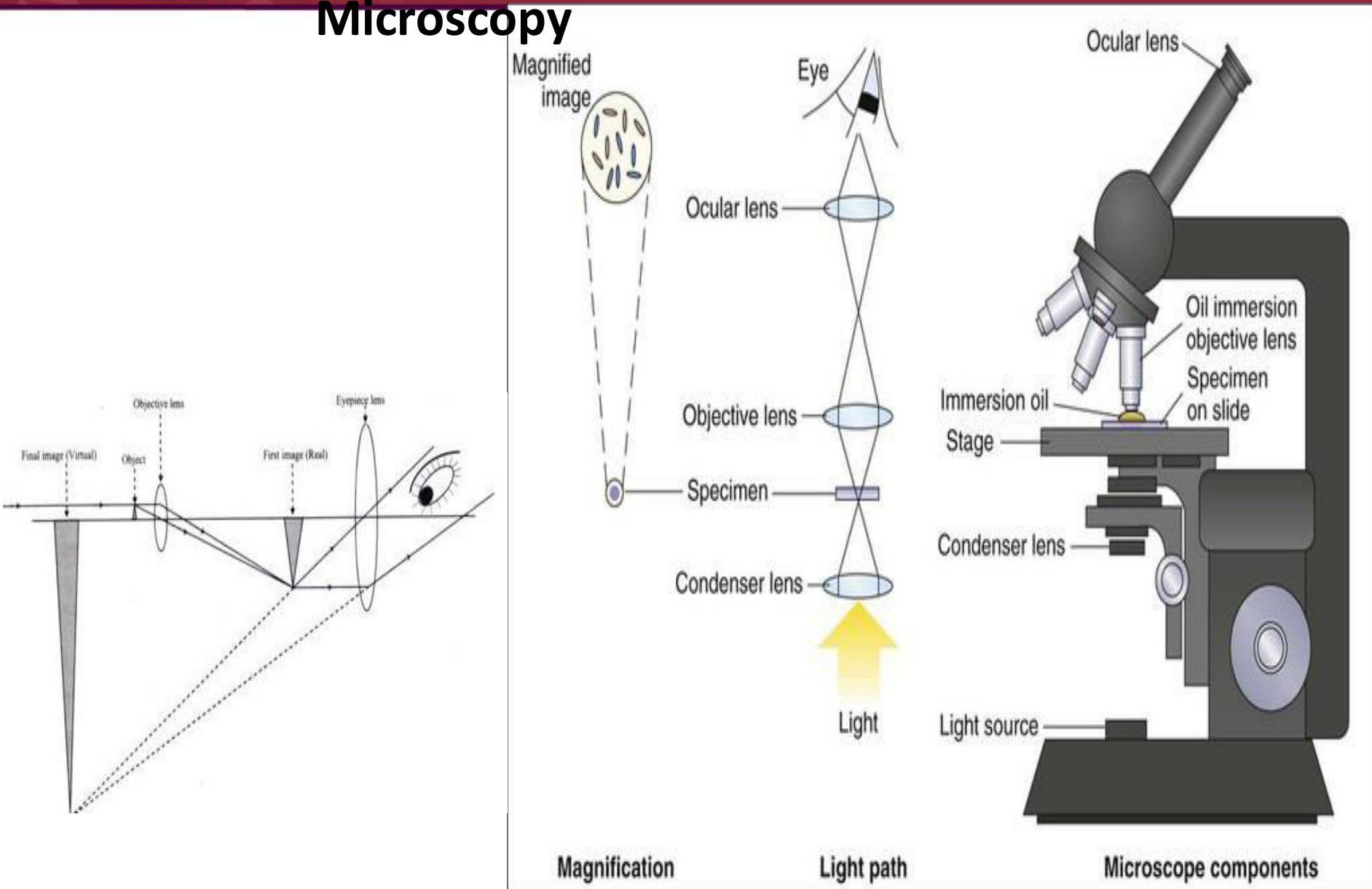
- Simple Microscope
- Compound Microscope
- Dark Field Microscope
- Phase Contrast Microscope
- Flourescent Microscope
- Interference Microscope
- Electron Microscope
- Atomic Force Microscope
- Polarization Microscope

Simple Microscope



USES: Study of microscopic algae, fungi and biological specimen

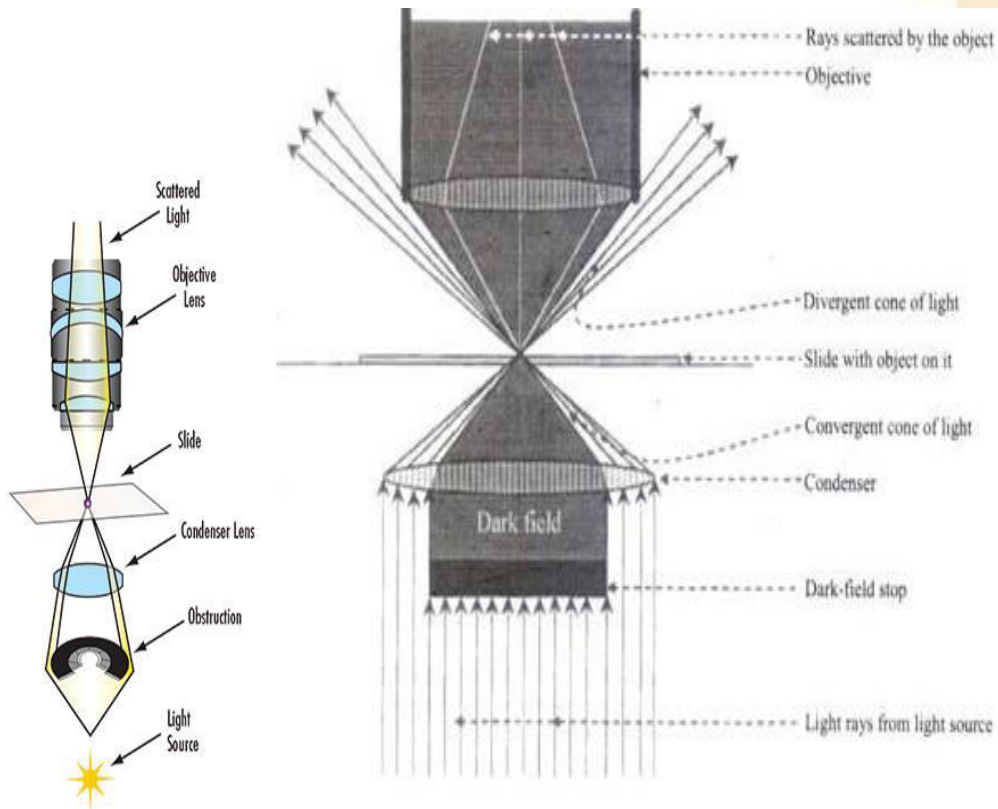
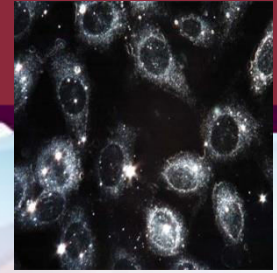
Compound/Bright-Field (Light) Microscopy



S.N.	Simple Microscope	Compound Microscope
1	Single lens	3 to 5 objective lenses
2	Has only one lens for magnifying objects.	Has two sets of lenses for magnifying objects: eyepiece lens and objective lenses
3	Low Magnification	Higher magnification.
4	Condenser lens is absent	Condenser lens is present which is used to adjust the intensity of light for magnification of object.
5	Light source is natural	Illuminator is a source of light
6	Has only one adjustment screw that is used to move the limb up and down for focusing an object.	Has coarse adjustment screw (for rapid focusing an object) and fine adjustment screw (for fine and sharp focusing).
7	Can only be used in simple ways such as enlarging small letters while reading.	Has a wide range of use such as in studying the structure of different objects

Dark-Field Microscope

- Examination of unstained microbes
- Special kind of condenser, with a dark-field stop, which is an opaque disc obstructing the path of light from the light source centrally, but allowing a peripheral ring of light



Phase Contrast Microscope

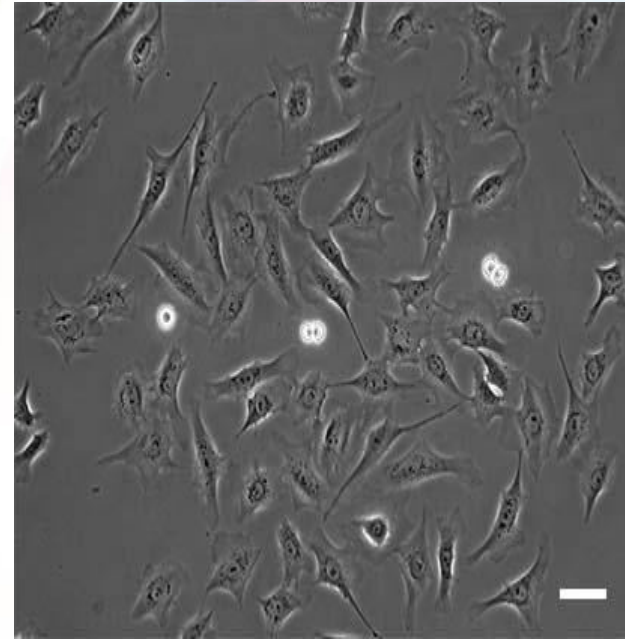
Applications:

- Cell culture
- Live cell imaging

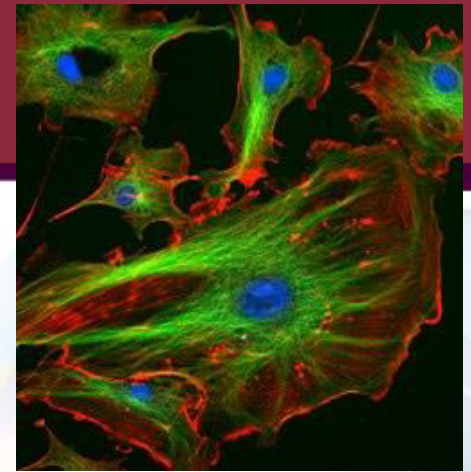
Living cells can be observed in their natural state without previous fixation or labeling.

Principle:

When light passes through cells, small phase shifts occur. These phase shifts are converted into changes in amplitude, which can be observed as differences in image contrast.

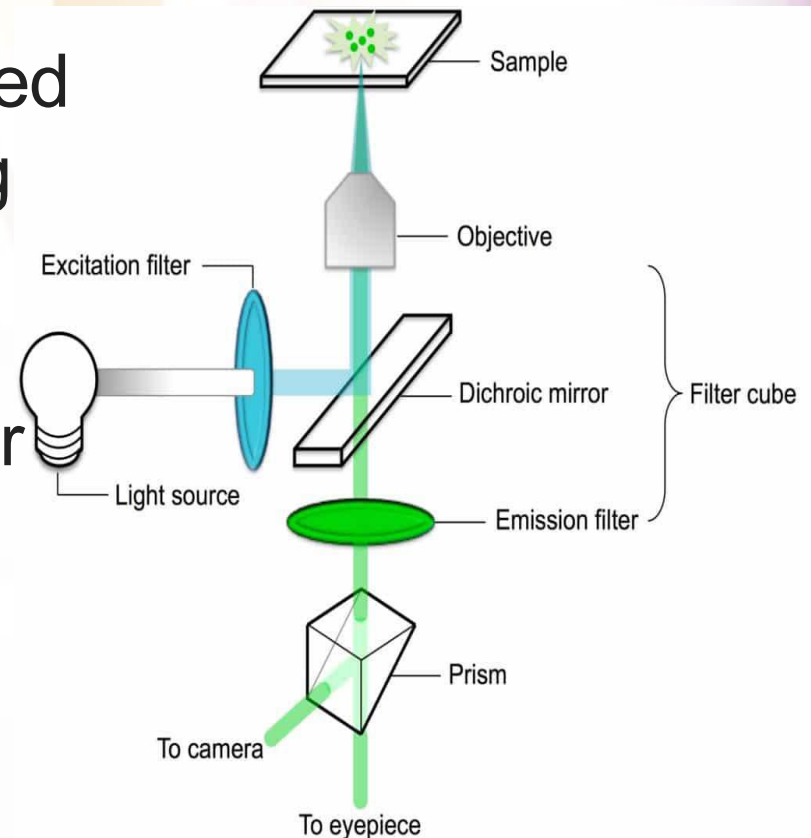


Flourescent Microscope



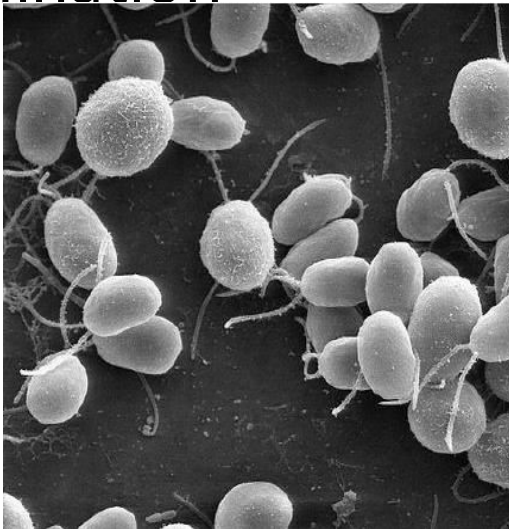
Principle.

- The specimen is illuminated with light of a specific wavelength which is absorbed by the fluorophores, causing them to emit light of longer wavelengths
- Florescent dyes are used for specimen staining
- **Examples of Flourescent stains:** [DAPI](#) and [Hoechst](#)



Electron Microscope

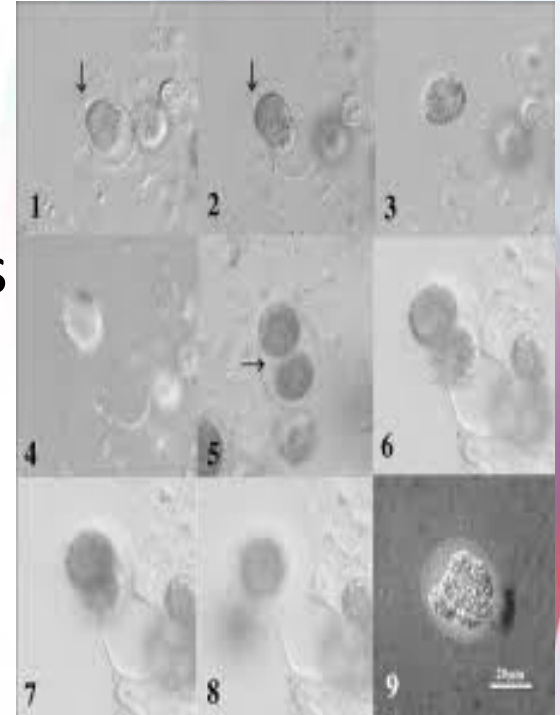
- Microscope with high magnification and resolution
- Uses a beam of accelerated electrons as a source of illumination



Interference Microscope

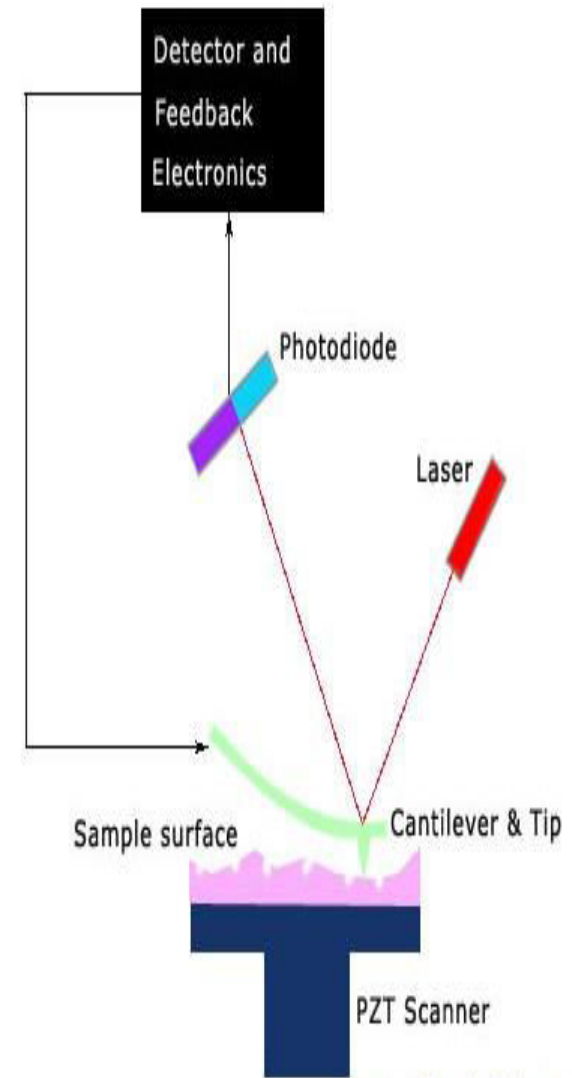
Principle:

A prism is used to split light into two slightly diverging beams that then pass through the specimen. It is thus based on measuring the differences in refractive index upon recombining the two beams. **Interference** occurs when a light beam is retarded or advanced relative to the other



Atomic Force Microscope

- High resolution
- Measure in fractions of a nanometer
- Uses a cantilever with a sharp probe that scans the surface of the specimen



Polarization Microscope

- **Used** mainly in geological studies for geological specimens but also in medicine and biology.
- Uses a **polarized** light, in which the light waves vibrate in one direction.

Organism Group	Bright-Field Microscopy	Fluorescence Microscopy	Dark-Field Microscopy	Electron Microscopy
Bacteria	+	+	±	—
Fungi	+	+	—	—
Parasites	+	+	—	±
Viruses	—	+	—	±

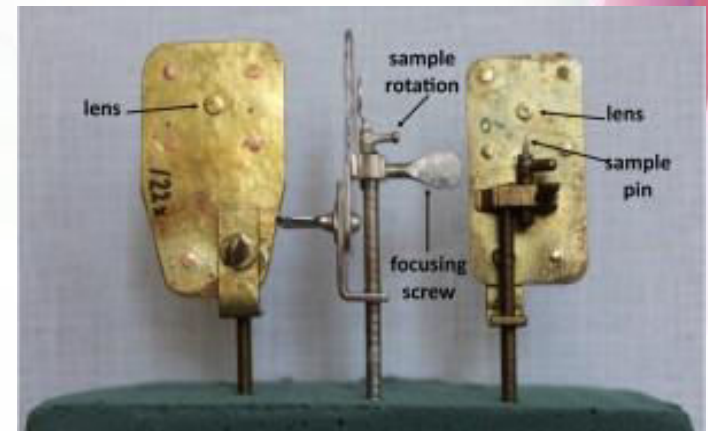
Virology

HISTORY OF MICROSCOPY

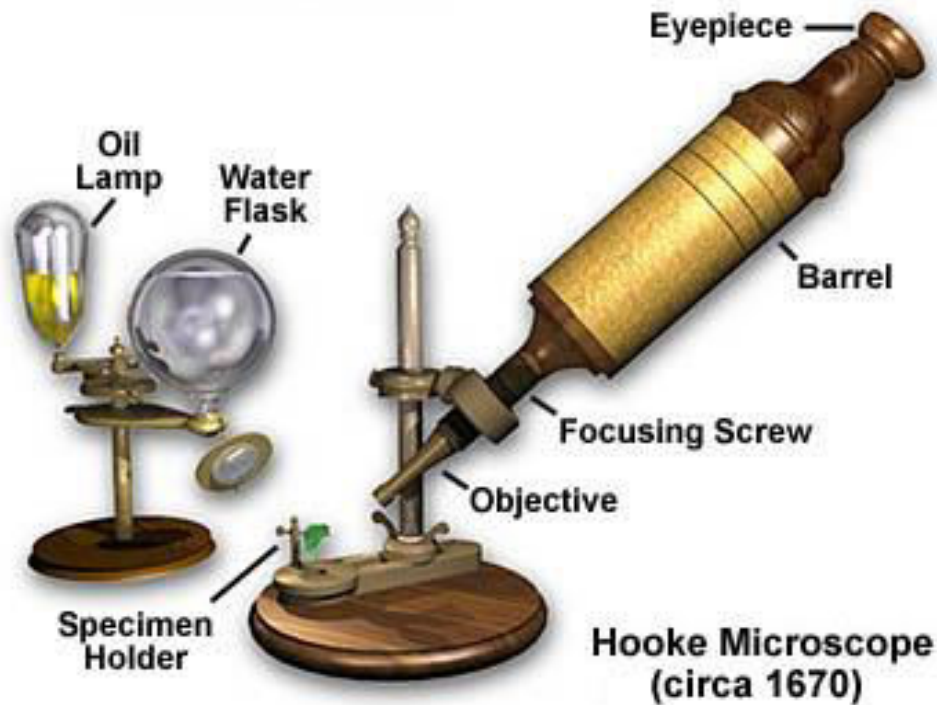
History of Microscopy

FIRST MICROSCOPE

- First simple microscope by Leeuwenhoek
- 17th century



History of Microscopy



ROBERT HOOKE

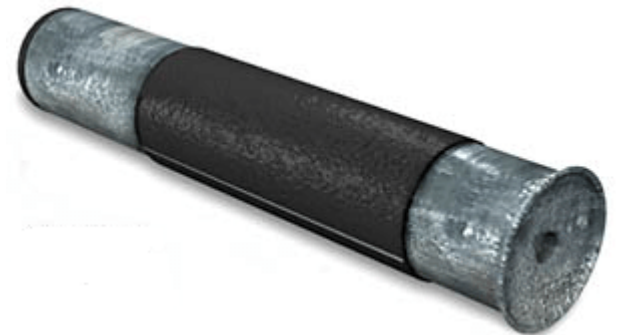
- Robert Hooke in 1665 observed cork cells

History of Microscopy

COMPOUND

MICROSCOPE

- Zaccharias Janssen made the first compound microscope



History of Microscopy

BETTER MICROSCOPE

- In 1830 a better microscope introduced by Joseph Lister



History of Microscopy



**Leeuwenhoek
Microscope
(circa late 1600s)**



**British
Microscope
(circa 1865)**



**Hand-Held Microscope
(circa early 1700s)**



**Winkel-ZEISS
Dissecting
Microscope
(circa 1927)**



**ZEISS
Primo Star
(circa 2010)**

Microscopes Over the Ages

Figure 1

Virology

LIGHT MICROSCOPY



Light Microscopy

LIGHT MICROSCOPY

- Light microscopy refers to the use of any kind of microscope that uses visible light to observe specimens.

Light Microscopy

TYPES:

- Compound light microscopy
- Dark field Microscopy
- Phase-contrast Microscopy
- Differential interference contrast (DIC) Microscopy
- Fluorescence Microscopy
- Confocal Microscopy

Light Microscopy

COMPOUND LIGHT MICROSCOPY

- Light rays from an illuminator (the light source)
- Pass through a condenser, which has lenses that direct the light rays through the specimen
- From here, light rays pass into the objective lenses
- The image of the specimen is magnified again by the ocular lens (eyepiece)

Light Microscopy

MAGNIFICATION

- Objective lens X Eyepiece lens
- It is the ratio of the size of an object seen under microscope to the actual size observed with unaided eye.

Light Microscopy

RESOLUTION:

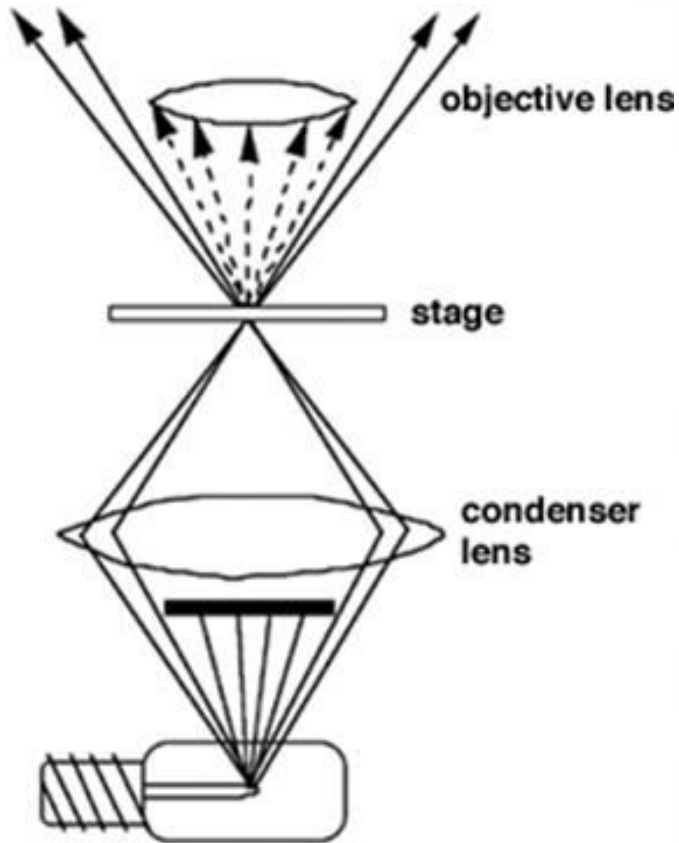
- refers to the ability of the lenses to distinguish two points that are a specified distance apart
- The resolving power of human eye is 0.25 mm
- The light microscope can separate dots that are 0.25 μ m apart.
- The electron microscope can separate dots that are 0.5nm apart.

Virology

DARK FIELD MICROSCOPY



Dark Field Microscopy



DARK FIELD MICROSCOPY

- Refers to the use of microscope that excludes the unscattered beam from the image
- As a result, the field around the specimen (i.e., where there is no specimen to scatter the beam) is generally dark

Dark Field Microscopy



USE:

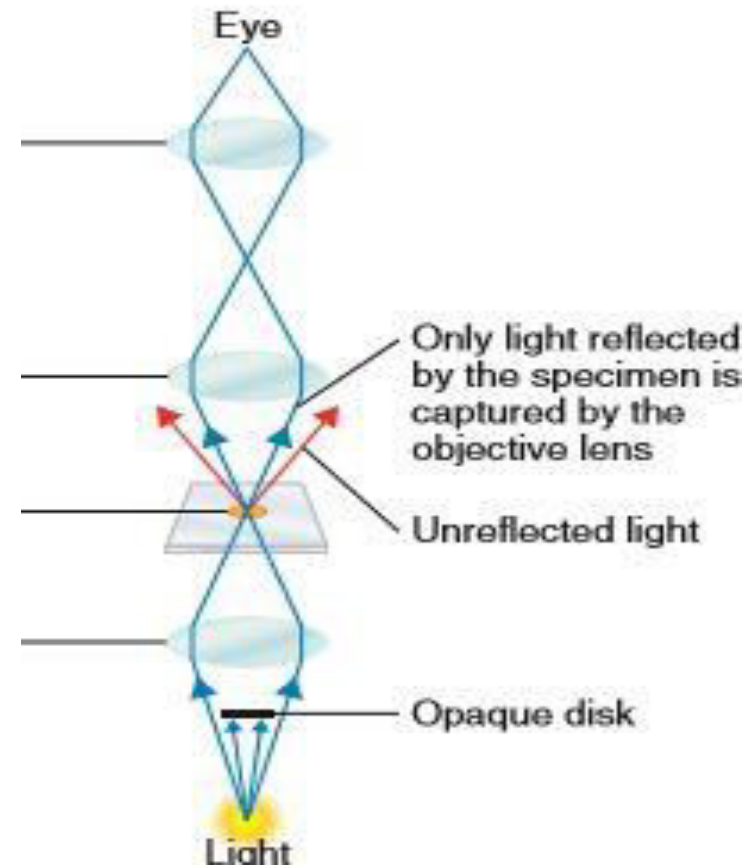
- Used for live microorganisms that either are invisible in the ordinary light microscope, cannot be stained by standard methods.

Dark Field Microscopy

DARK FIELD MICROSCOPY

- Uses a darkfield condenser containing an opaque disk
- The disk blocks light that would enter the objective lens
- Only light that is reflected off (turned away from) the specimen enters the objective lens
- Because there is no direct background light, the specimen appears light against a black background—the dark field.

Dark Field Microscopy



Virology

DIFFERENTIAL INTERFERENCE CONTRAST MICROSCOPY



DIC Microscopy

DIC MICROSCOPY

- Also known as Nomarski interference contrast microscopy
- used to enhance the contrast in unstained, transparent samples
- DIC works on the principle of interferometry to gain information about the optical path length

DIC Microscopy



WORKING:

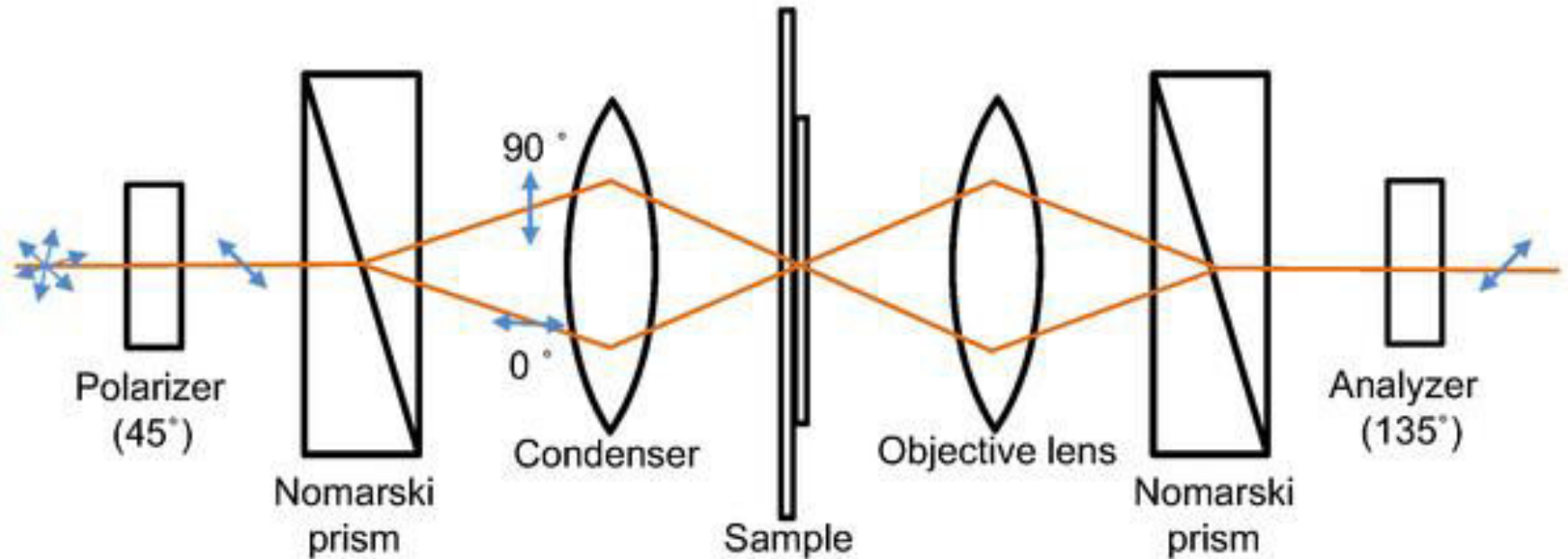
- DIC microscope uses two beams of light instead of one
- prisms split each light beam, adding contrasting colors to the specimen
- image is brightly colored and appears nearly three-dimensional

DIC Microscopy

LIGHT PATH

- Unpolarized light enters and becomes polarized
- Then it enters the prism and gets separated into two beams
- These rays are focused by condenser
- Rays travel through specimen
- Then enter objective lens
- Second prism recombines the rays forming an image

DIC Microscopy



Virology

FLUORESCENCE MICROSCOPY



Fluorescence Microscopy

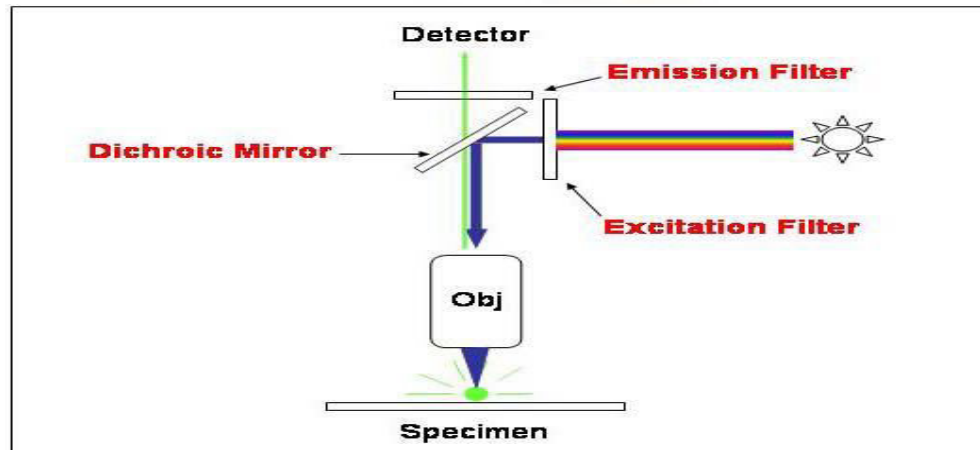
FLUORESCENCE MICROSCOPE

- A microscope that uses fluorescence and phosphorescence instead of/ or in addition to reflection and absorption to study properties of a microbe

Fluorescence Microscopy

PRINCIPLE:

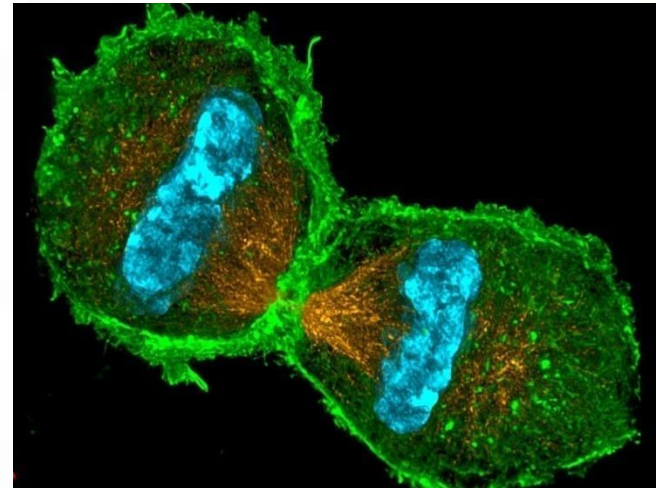
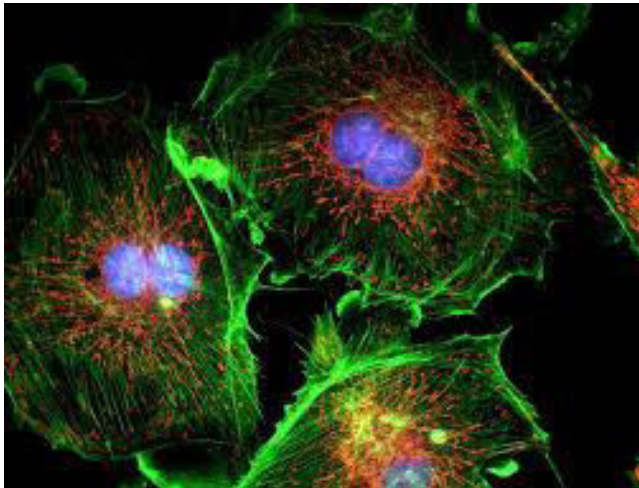
- Fluorescence microscopy takes advantage of fluorescence, the ability of substances to absorb short wavelengths of light (ultraviolet) and give off light at a longer wavelength (visible).



Fluorescence Microscopy

IMAGE:

- When microorganisms stained with a fluorochrome are examined under a fluorescence microscope with an ultraviolet or near-ultraviolet light source, they appear as luminescent, bright objects against a dark background.



Fluorescence Microscopy

LIGHT SOURCE:

- Monochromatic light source
- Four types:
 1. Xenon arc lamps
 2. Mercury vapor lamps
 3. Super-continuum source
 4. High power LED

Virology

CONFOCAL MICROSCOPY



Confocal Microscopy

CONFOCAL MICROSCOPY

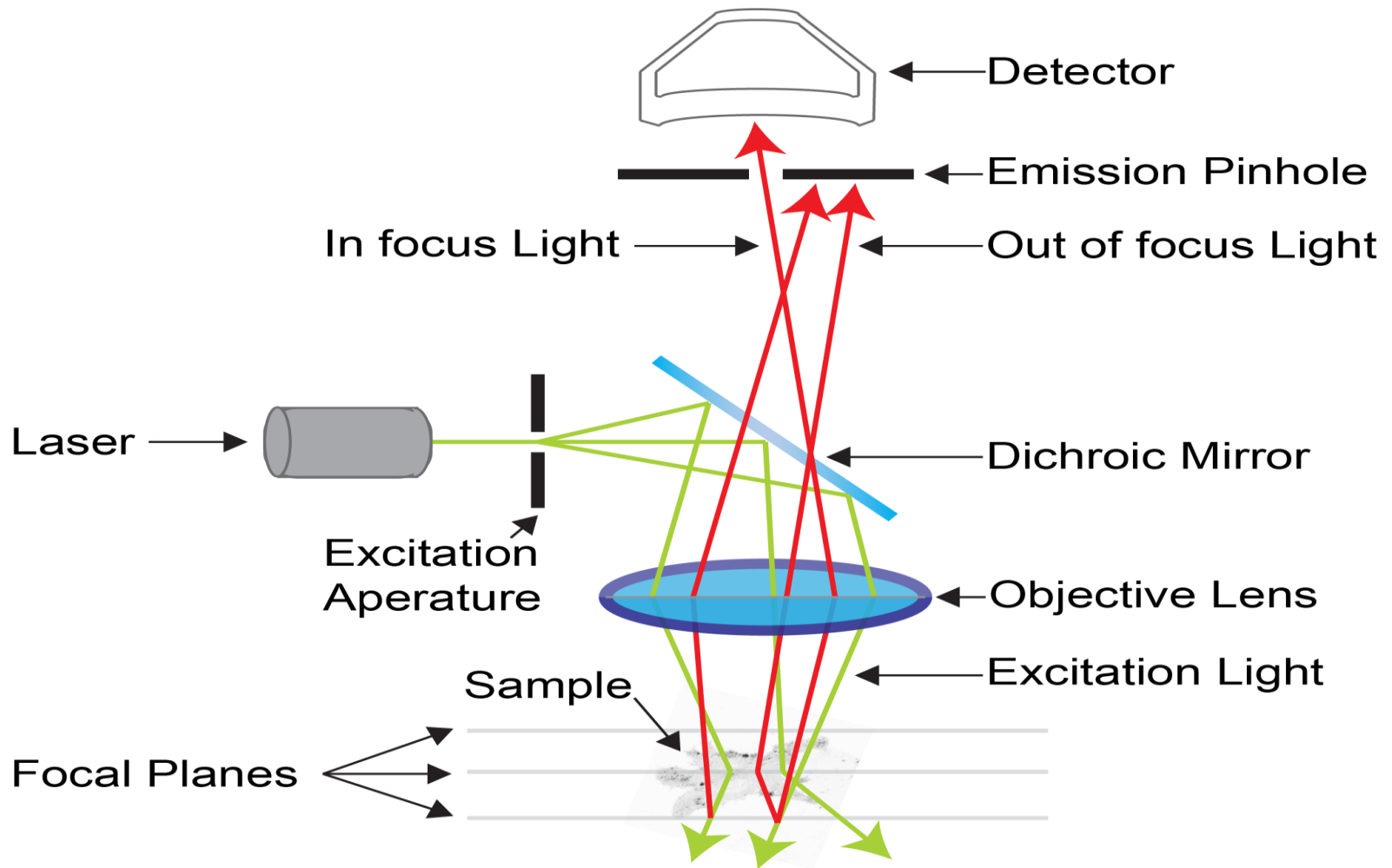
- An optical imaging technique for increasing optical resolution and contrast of a micrograph
- by using a spatial pinhole to block out-of-focus light in image formation

Confocal Microscopy

PRINCIPLE:

- used to reconstruct three-dimensional images
- Like fluorescent microscopy, specimens are stained with fluorochromes so they will emit or return light
- But instead of illuminating the entire field, one plane of a small region of a specimen is illuminated with a short-wavelength (blue) light which passes the returned light through an aperture aligned with the illuminated region

Confocal Microscopy



Confocal Microscopy

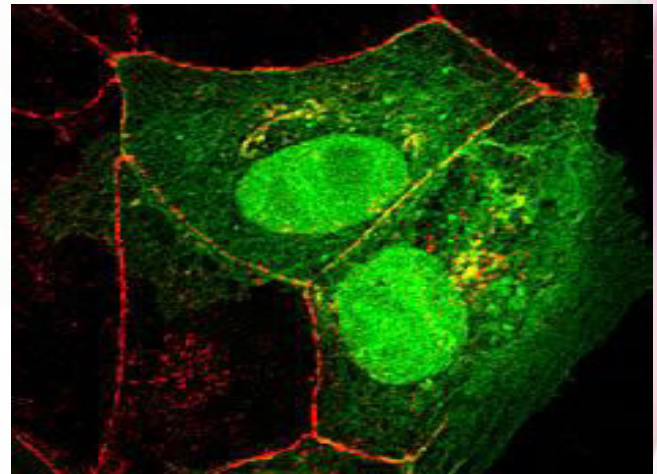
PROPERTIES:

- Each plane corresponds to an image of a fine slice that has been physically cut from a specimen
- Successive planes and regions are illuminated until the entire specimen has been scanned
- Confocal microscopy can image cells in detail only to a depth of less than 100 mm

Confocal Microscopy

IMAGE

- Point laser source
- 3D dimensional fraction pattern
- Better resolution of Wide field illumination



Confocal Microscopy

TYPES:

1. Confocal laser scanning microscope
2. Spinning disk confocal microscope
3. Microlens enhanced or dual spinning disk confocal microscope
4. Programmable array microscope (PAM)

Virology

TWO PHOTON MICROSCOPY



Two Photo Microscopy

TWO PHOTON MICROSCOPY

- It is a fluorescence imaging technique that allows imaging of living tissue up to about one millimeter in depth.

Two Photo Microscopy

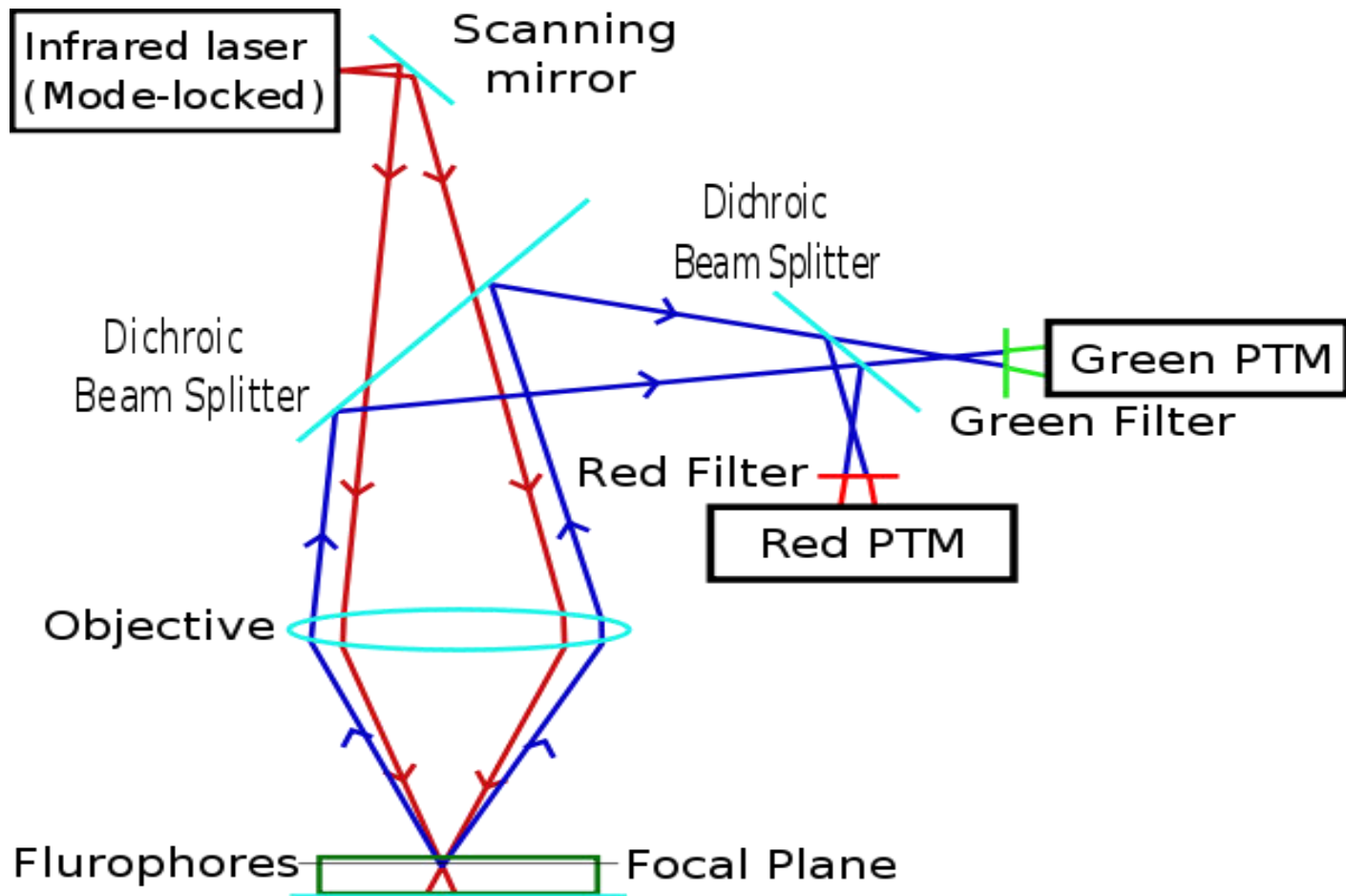
PRINCIPLE:

Specimens are stained with a fluorochrome
Two-photon microscopy uses long-wavelength (red) light, and therefore two photons, instead of one, are needed to excite the fluorochrome to emit light. The longer wavelength allows imaging of living cells in tissues up to 1 mm (1000 mm) deep



Two Photon Microscopy

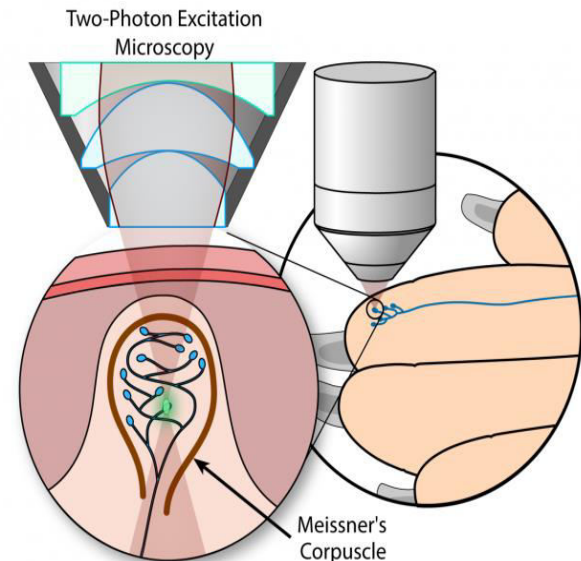
TWO PHOTON MICROSCOPY ILLUSTRATION



Two Photo Microscopy

APPLICATIONS:

- Applicable for very thin transparent cells (skin)
- Non-invasive optical biopsy



Virology

SCANNING ACOUSTIC MICROSCOPY



Scanning Acoustic Microscopy

SCANNING ACOUSTIC MICROSCOPY

- A device that uses focused sound to image an object
- Applicable in biological and medical research
- Developed in 1974

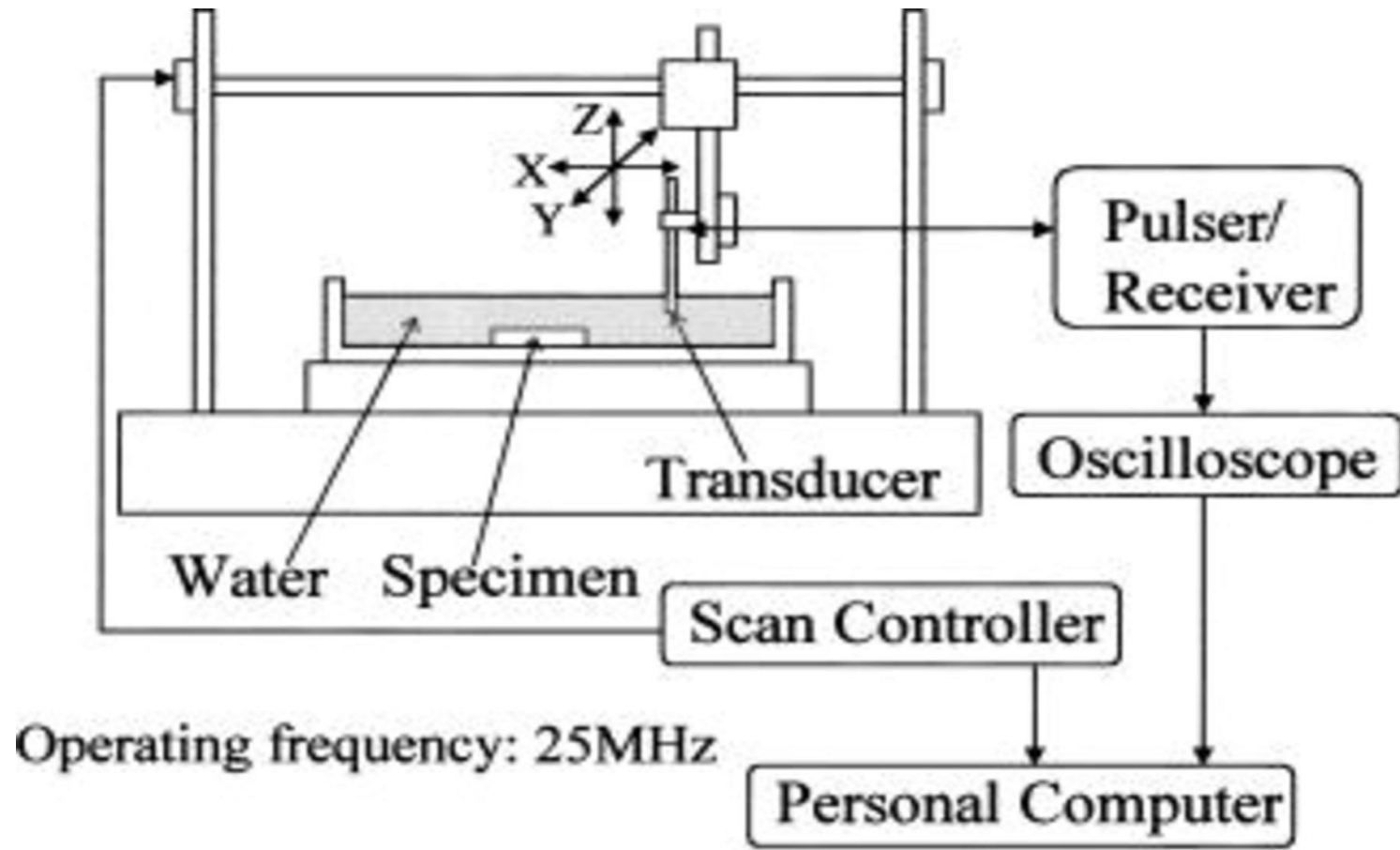
Scanning Acoustic Microscopy

PRINCIPLE:

- Scanning acoustic microscopy (SAM) basically consists of interpreting the action of a sound wave sent through a specimen
- A sound wave of a specific frequency travels through the specimen, and a portion of it is reflected back every time it hits an interface within the material

Scanning Acoustic Microscopy

SAM ILLUSTRATION:



Scanning Acoustic Microscopy

IMAGE

- Resolution= 1 μm
- Both 2D and 3D image



Scanning Acoustic Microscopy

USE:

SAM is used to study living cells attached to another surface, such as cancer cells, artery plaque, and bacterial biofilms that foul equipment

Virology

TRANSMISSION ELECTRON MICROSCOPY



Transmission Electron Microscopy

TEM MICROSCOPY

A technique that uses a beam of electrons

- Transmitted through the specimen
- Ultra-thin section of specimen
- Image on imaging device

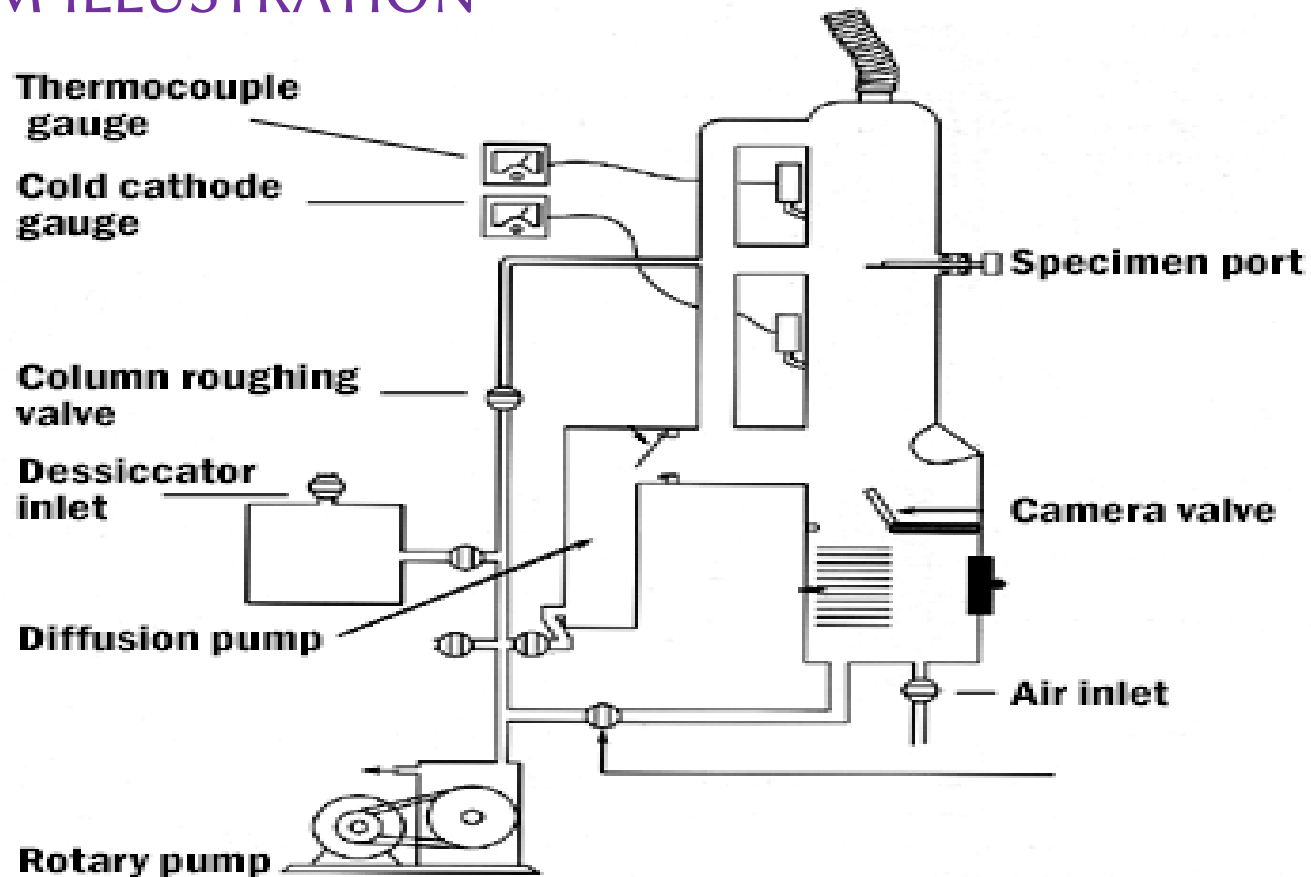
Transmission Electron Microscopy

PRINCIPLE:

- Focused beam of electrons from an electron gun passes through ultrathin section of the specimen
- Beam gets focused on a small area of the specimen by an electromagnetic condenser lens
- Instead of being placed on a glass slide, the specimen is usually placed on a copper mesh grid

Transmission Electron Microscopy

TEM ILLUSTRATION



Transmission Electron Microscopy

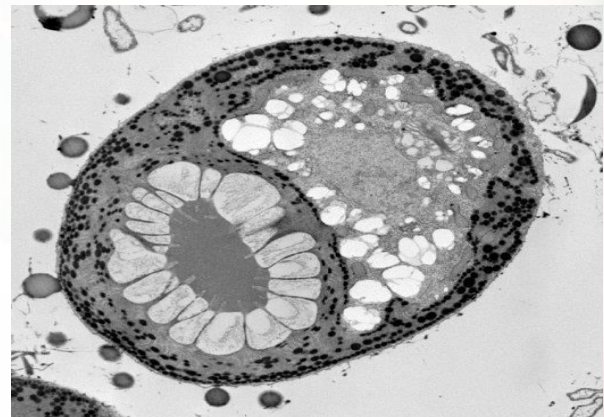
IMAGE FORMATION:

- Electrons pass through the specimen and then through an electromagnetic objective lens, which magnifies the image
- The electrons are focused by an electromagnetic projector lens onto a fluorescent screen or photographic plate
- The final image, called transmission electron micrograph, appears as many light and dark areas, depending on the number of electrons absorbed by different areas of the specimen

Transmission Electron Microscopy

IMAGE

- The transmission electron microscope can resolve objects as close together as 10 pm
- The objects are generally magnified 10,000 to 100,000*



Transmission Electron Microscopy

DYE:

- Contrast can be greatly enhanced by using a “dye” that absorbs electrons and produces a darker image in the stained region
- Lead, osmium, tungsten, and uranium (Stains)
- These metals can be fixed onto the specimen (positive staining)
- or used to increase the electron opacity of the surrounding field (negative staining)

Transmission Electron Microscopy

LIMITATIONS:

- Only a very thin section of a specimen (about 100 nm) can be studied effectively
- Specimens must be fixed, dehydrated, and viewed under a high vacuum to prevent electron scattering
- These treatments not only kill the specimen, but also cause shrinkage and distortion



Virology

SCANNING ELECTRON MICROSCOPY



Scanning Electron Microscopy

SEM MICROSCOPY

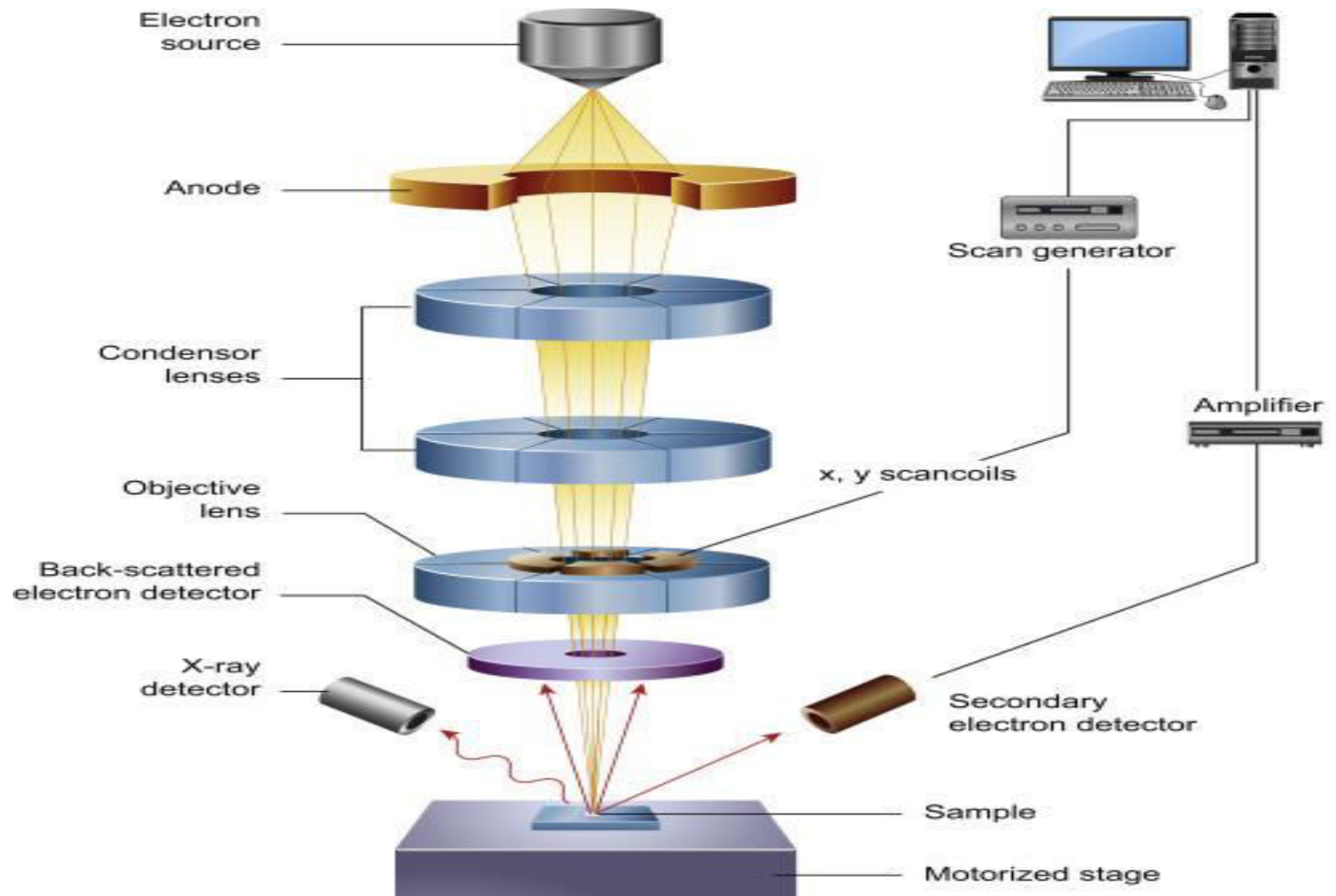
- A microscope that produces images of a sample by scanning the surface with a focused beam of electrons
- Produces various signals that contain information about the surface topography and composition of the sample

Scanning Electron Microscopy

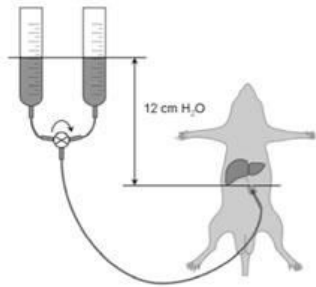
PRINCIPLE:

- An electron gun produces a finely focused beam of electrons called the primary electron beam
- These electrons pass through electromagnetic lenses and are directed over the surface of the specimen
- The primary electron beam knocks electrons out of the surface of the specimen, and the secondary electrons thus produced are transmitted to an electron collector, amplified, and used to produce an image on a viewing screen or photographic plate. The image is called a scanning electron micrograph

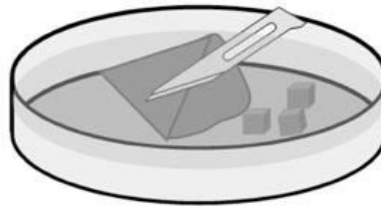
Scanning Electron Microscopy



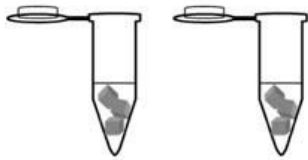
Scanning Electron Microscopy



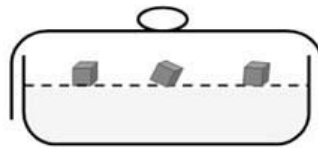
Step I (2-10 min)



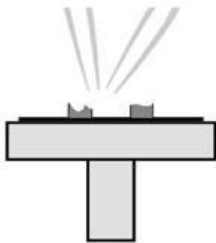
Step II (10 min)



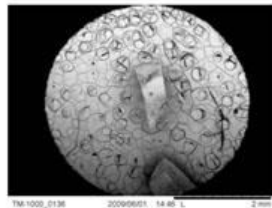
Step III (5×2 min & 10 min)



Step IV (10 min)



Step V (10 min)



Step VI

SAMPLE PREPARATION:

- Prepared in presence of vacuum and beam of electrons
- Mounted on specimen holder
- Specimens get conductive by beam of electrons

Scanning Electron Microscopy

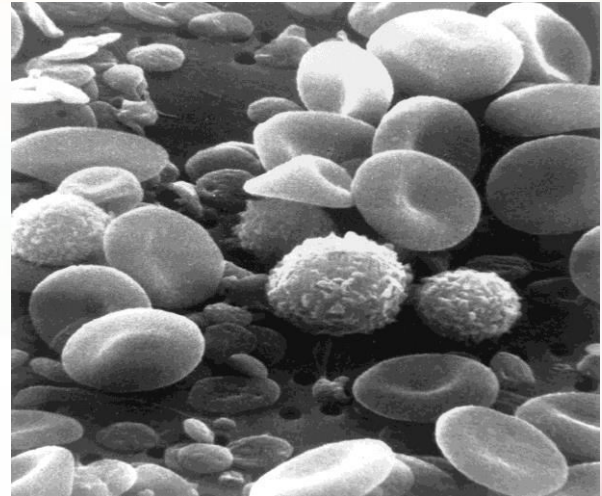
IMAGE

- Provides striking three-dimensional views of specimens
- it can resolve objects as close together as 10 nm, and objects are generally magnified 1000 to 10,000

Scanning Electron Microscopy

USES:

- useful in studying the surface structures of intact cells and viruses
- Also used for fractography



Virology

SCANNED PROBE MICROSCOPY



Scanned Probe Microscopy

SCANNED PROBE MICROSCOPY

- Forms images of surfaces using physical probe that scans the specimen
- Founded in 1981

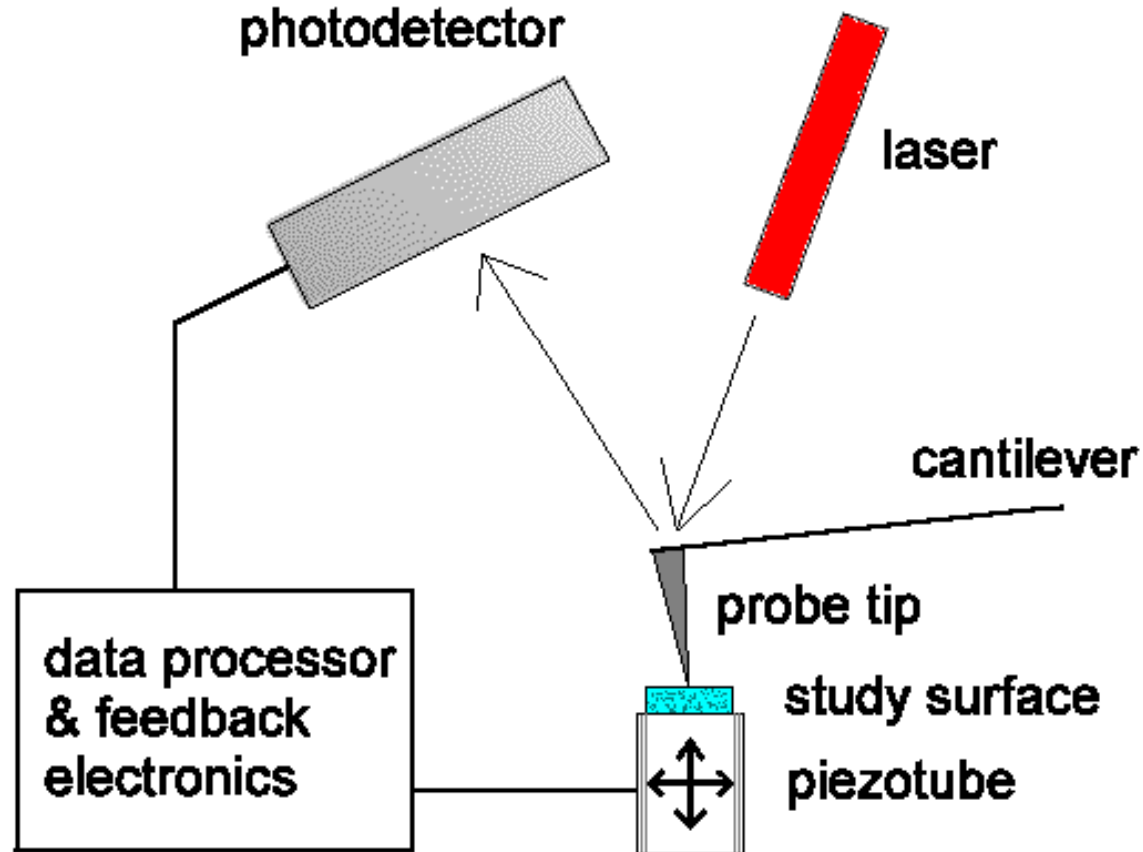
Scanned Probe Microscopy

PRINCIPLE:

- The interaction between the sharp probe and surface provides 3D topographic image of surface at the atomic scale
- Two most popular modes:
 1. Scanning Tunneling Microscopy (STM)
 2. Atomic Force Microscopy (AFM)

Scanned Probe Microscopy

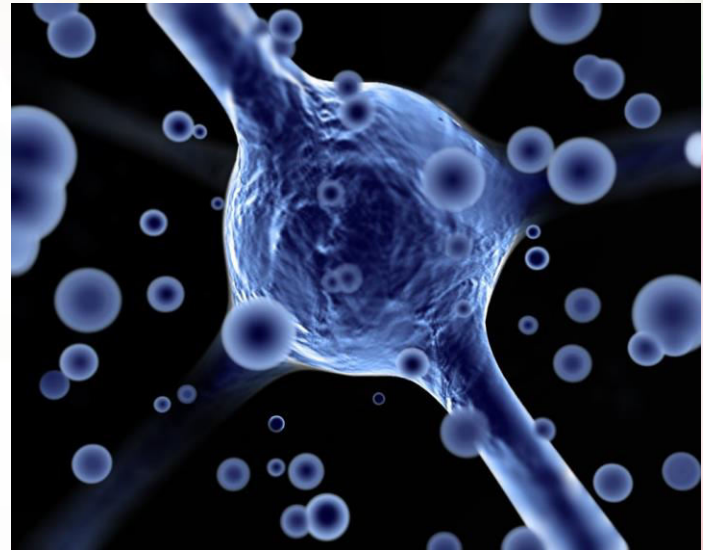
SPM ILLUSTRATION:



Scanned Probe Microscopy

IMAGE FORMATION:

- Scans tip over surface
- A value is recorded
- Displayed as heat map



Scanned Probe Microscopy

LIMITATIONS:

- Detailed shape of probe is difficult to determine sometimes
- Slower in image acquiring
- Maximum image size is smaller
- Cannot examine buried solid-solid or liquid-liquid interfaces

Virology

STRUCTURE AND FORMATION OF VIRAL PARTICLES

Structure and Formation of Viral particles

Viral Capsids and Envelopes

Structure and Formation of Viral particles

CAPSID

Virus particles contain the viral genome packaged in a protein coat called the capsid.

Structure and Formation of Viral particles

ENVELOPE

Surrounding the capsid

composed of lipid bilayer that contains viral proteins

derived from the host cell membranes.

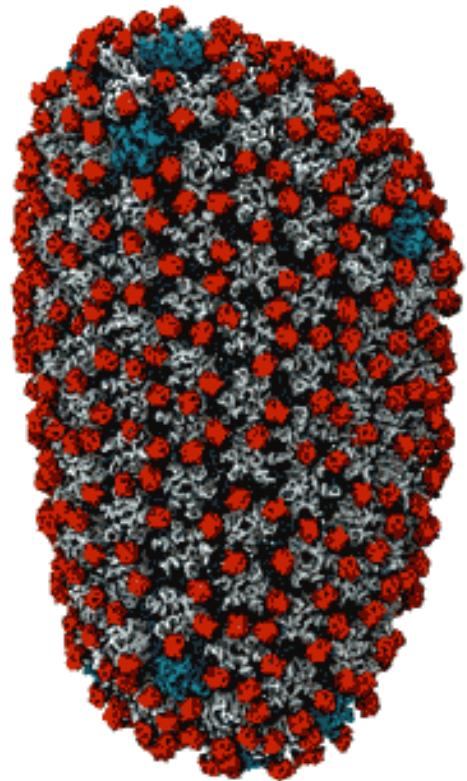
Structure and Formation of Viral particles

FUNCTIONS

The capsid and envelope play many roles in viral infection, including virus attachment to cells, entry into cells, release of the capsid contents into the cells, and packaging of newly formed viral particles. The capsid and envelope are also responsible for transfer of the viral genetic material from one cell to another. These structures also determine the stability characteristics of the virus particle, such as resistance to chemical or physical inactivation.

Virology

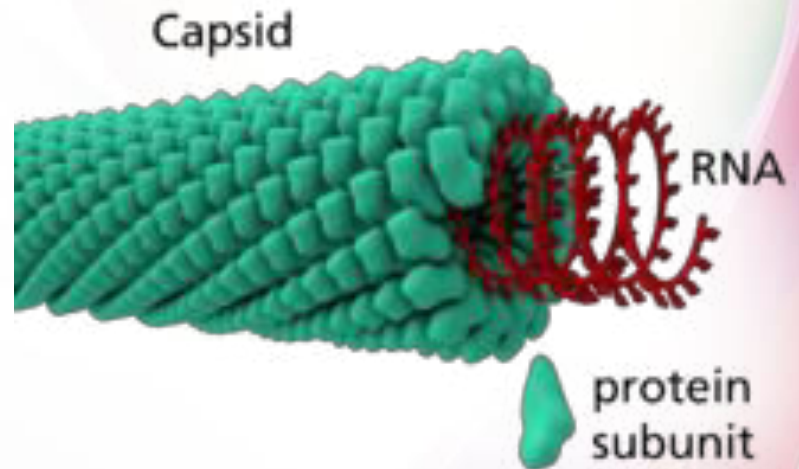
VIRAL CAPSID



Viral Capsid

MEANING

- Protein shell of virus
- Made up of protein sub-units (protomers)
- Encloses genome



Viral Capsid

FUNCTION

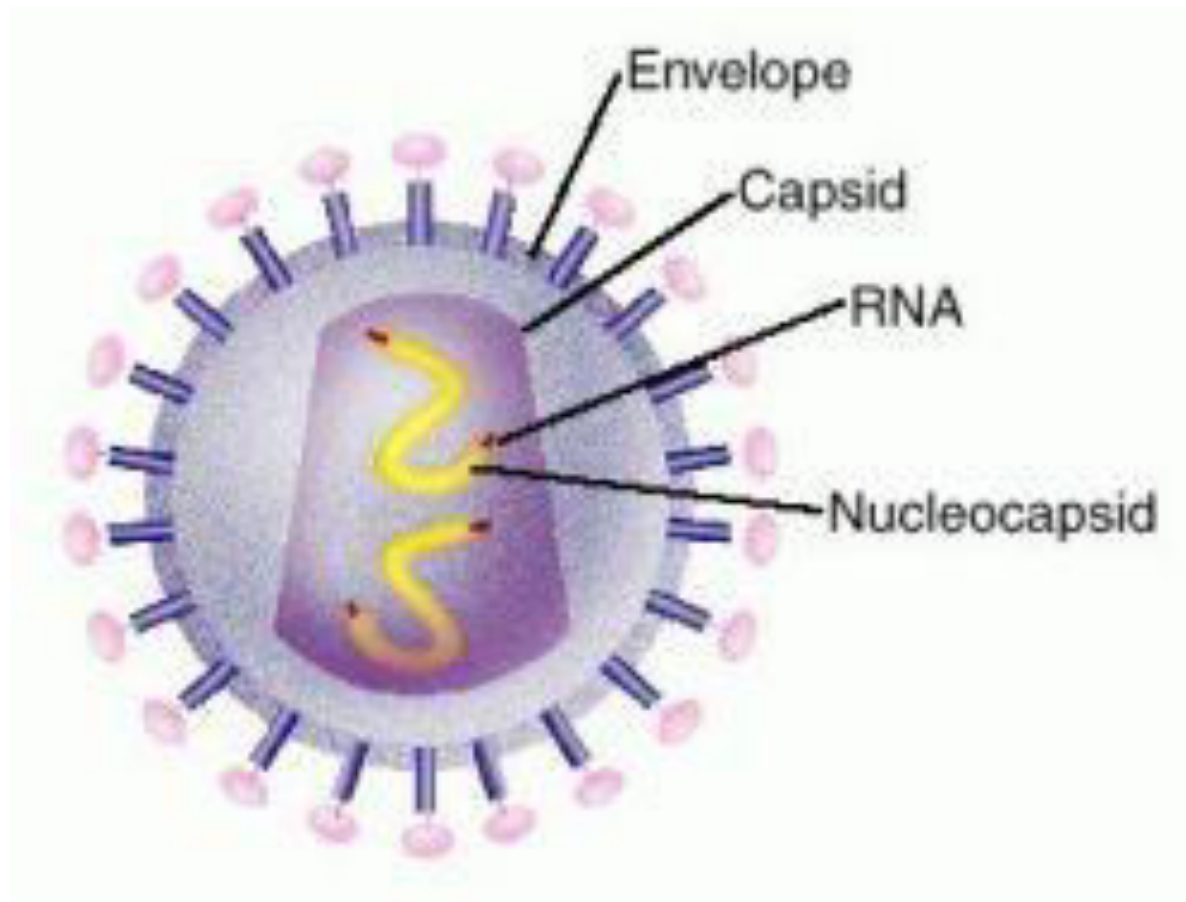
- Protection of genome
- Prevent digestion by cellular enzymes
- In cytoplasm
Helps to introduce genome in host cell
- Bind to cell surface

Viral Capsid

NUCLEOCAPSID

Genome plus capsid

Collective term



Viral Capsid

TYPES

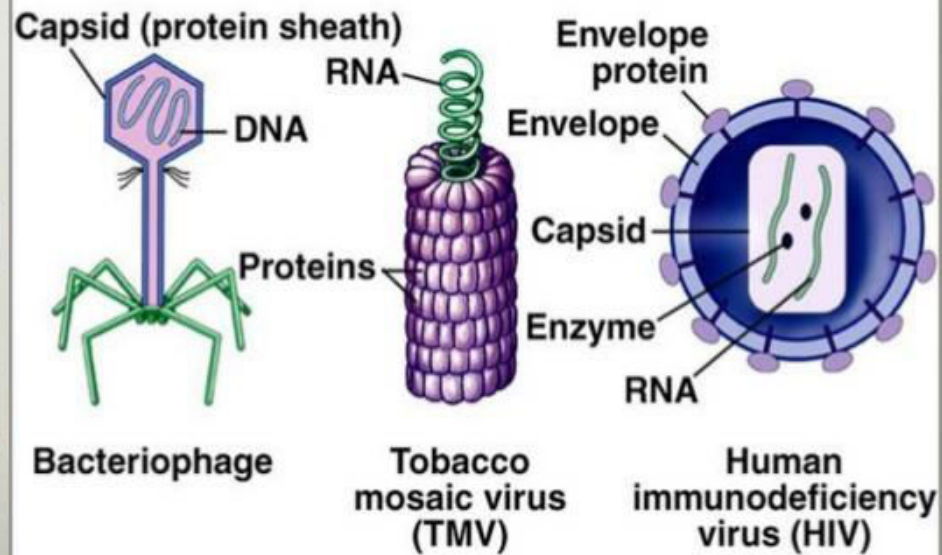
THE CAPSID & VIRAL SYMMETRY

COMPLEX

HELICAL

ICOSAHEDRAL

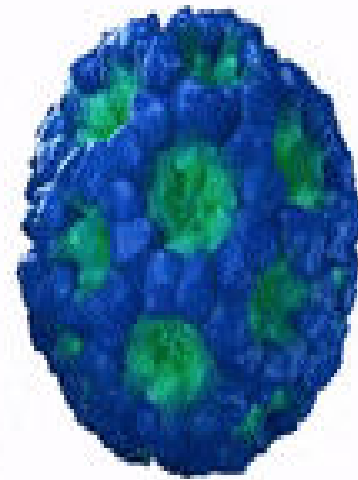
Viral Structure



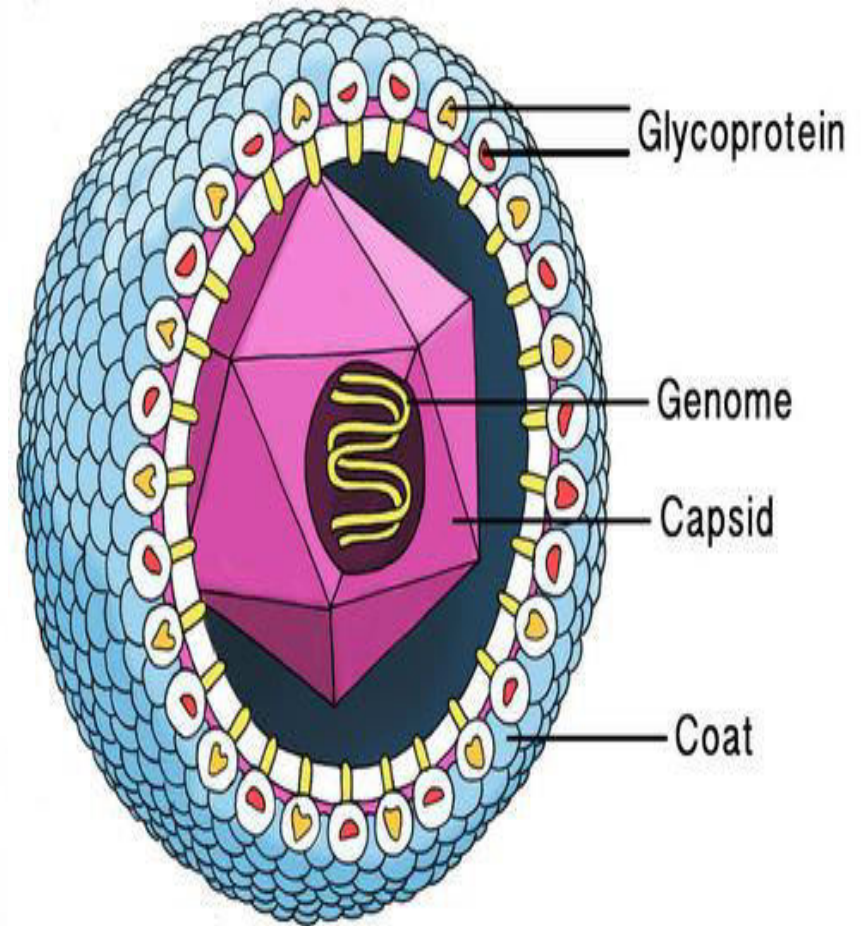
Virology

CAPSID

Symmetry and Architecture



Capsid; Symmetry and Architecture



Capsid; Symmetry and Architecture

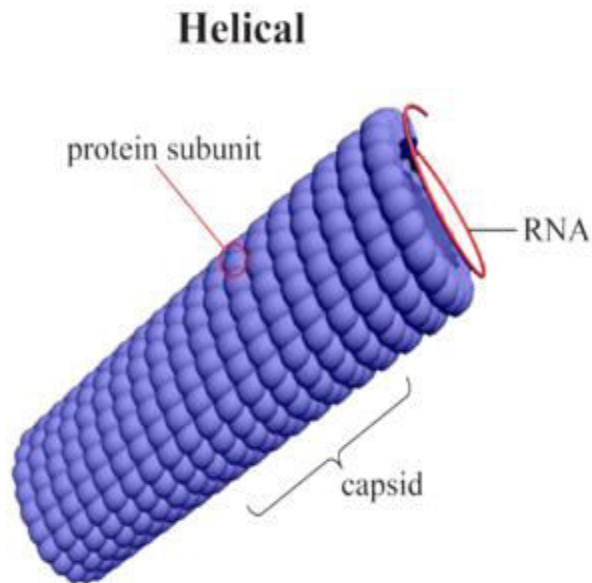
COMPOSITION

- Proteinic nature
- Composed of protomers and/or
- capsomers

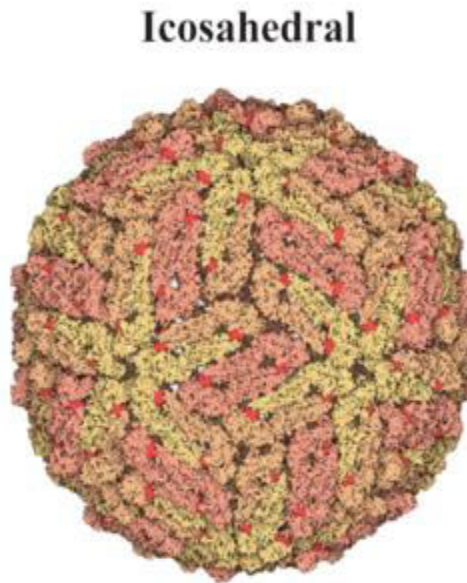
Capsid; Symmetry and Architecture

TYPES

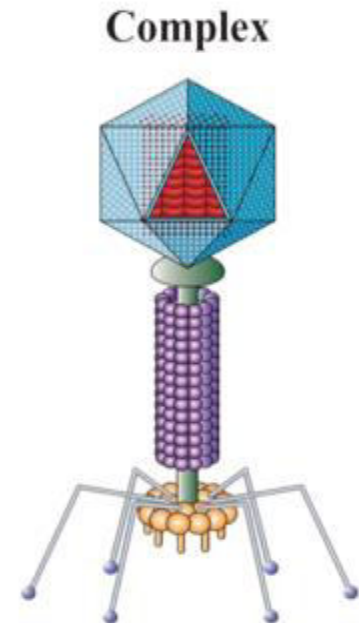
Helical



Icosahedral

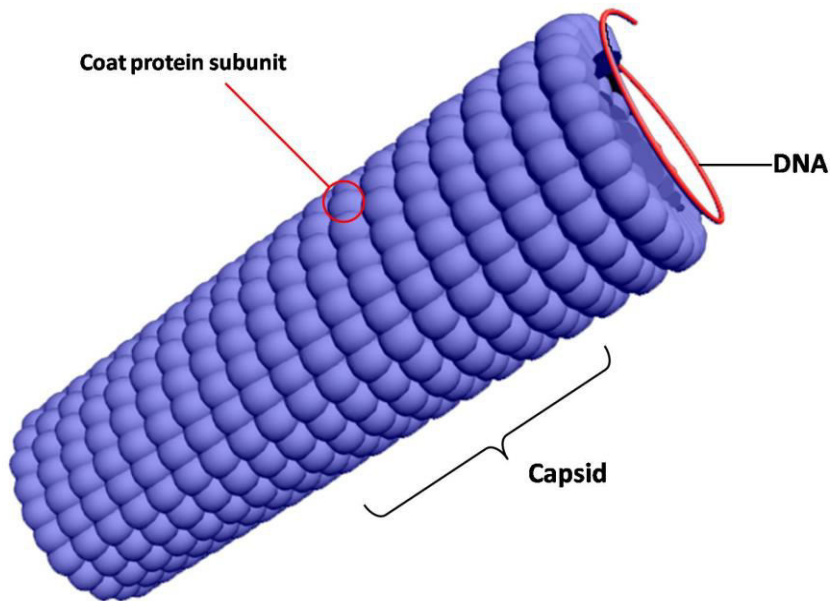


Polyhedral



Capsid; Symmetry and Architecture

HELICAL SYMMETRY



- Rod like structure
- Genome in the centre of helix
- Helix made by repeating units in a spiral
- Examples,
Tobacco Mosaic
Virus

Capsid; Symmetry and Architecture

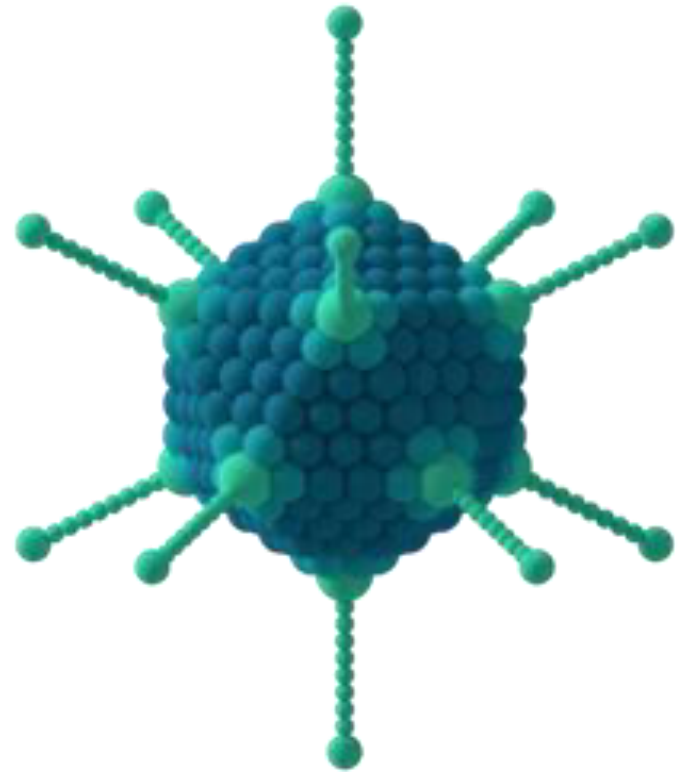
HELICAL SYMMETRY

- Virus particles elongated or pleomorphic (not spherical)
- Nucleic acid also spiral
- Capsomeres arranged around Nucleic acid

Capsid; Symmetry and Architecture

ICOSAHEDRAL SYMMETRY

- Characteristic of many spherical viruses
- In adenovirus (icosahedral) composition consists of : Hexon, Pentone base, and fibre



Capsid; Symmetry and Architecture

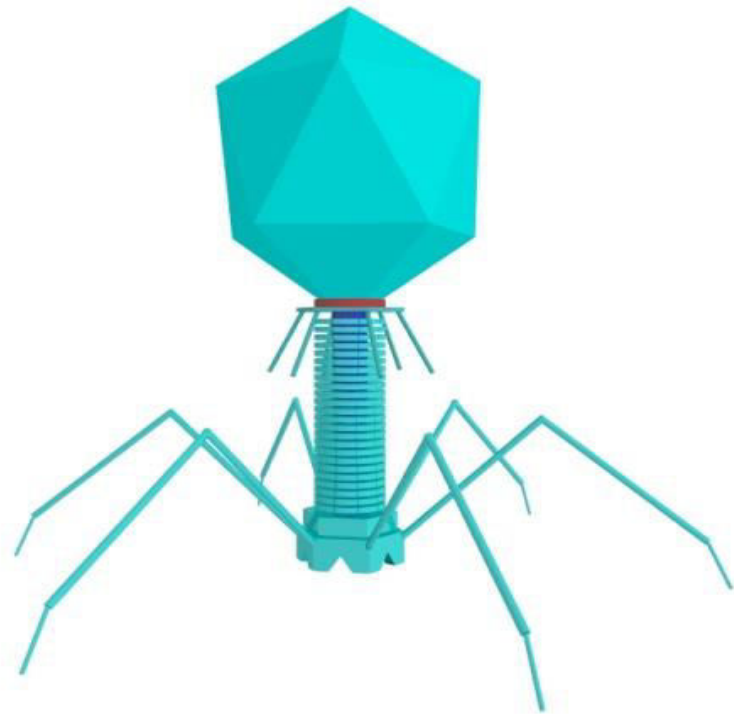
ICOSAHEREDRAL SYMMETRY

- Also called cubic capsid
- Mostly spherical
- Particles composed of 20 equilateral triangles, 12 vertices and 2,3,5 rotational symmetry

Capsid; Symmetry and Architecture

POLYHEDERAL SYMMETRY

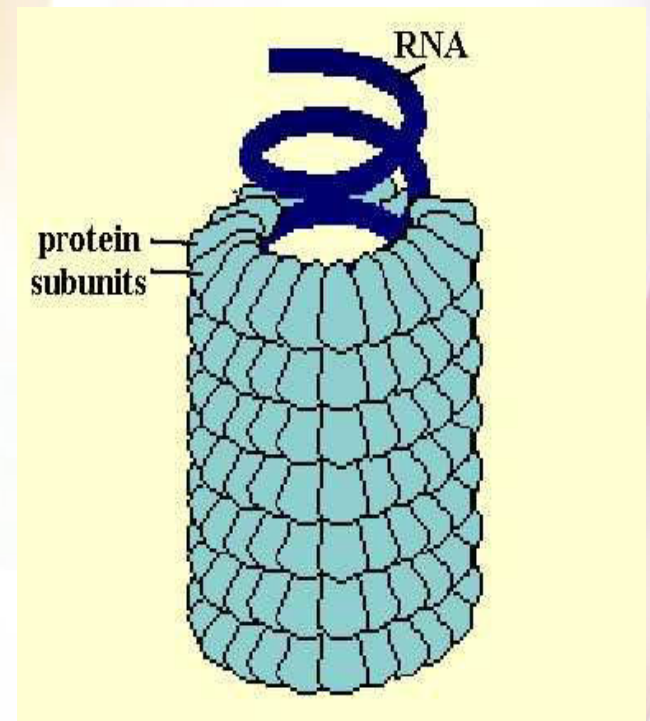
- Also called complex
- Neither helical nor icosahedral symmetry
- E.g Bacteriophage



Virology

CAPSID SYMMETRY

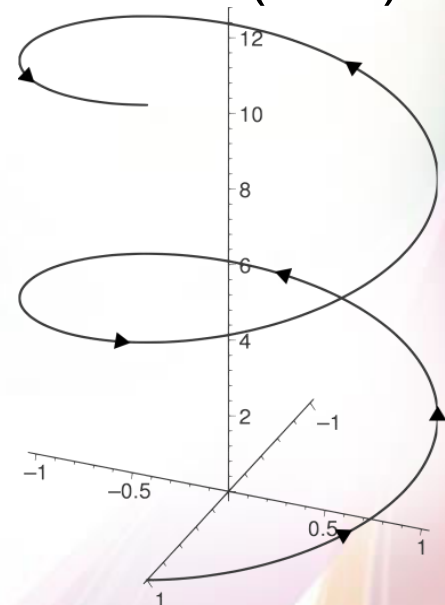
Helical



Helical Symmetry

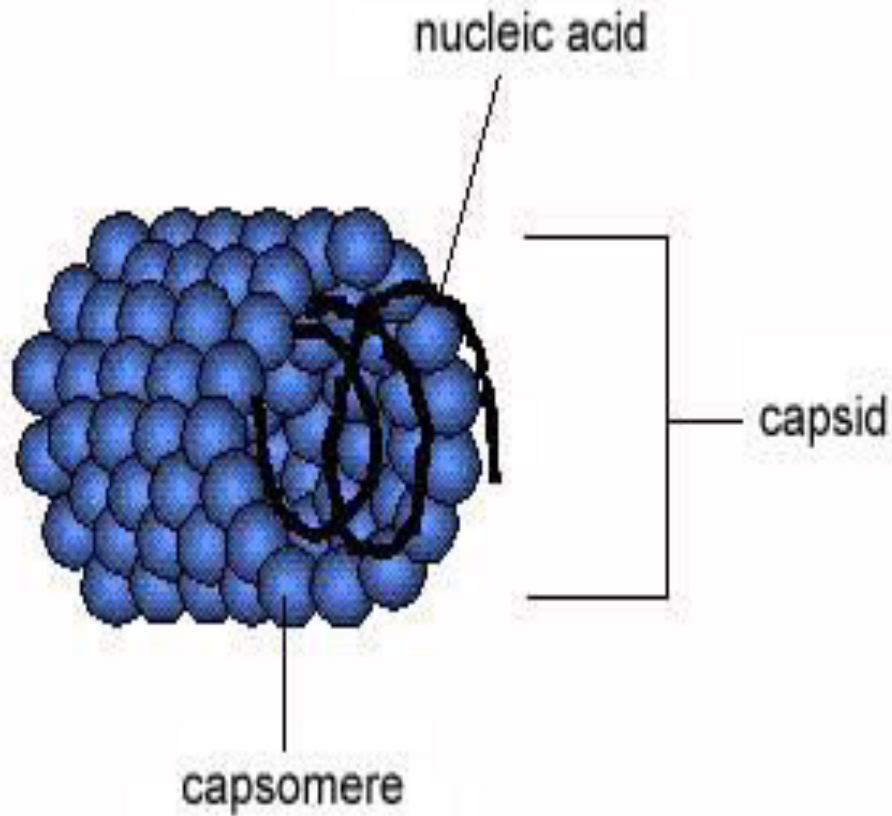
WHAT IS HELICAL SHAPE?

- Property that a tangent line at any point makes a constant angle with a fixed line (axis)



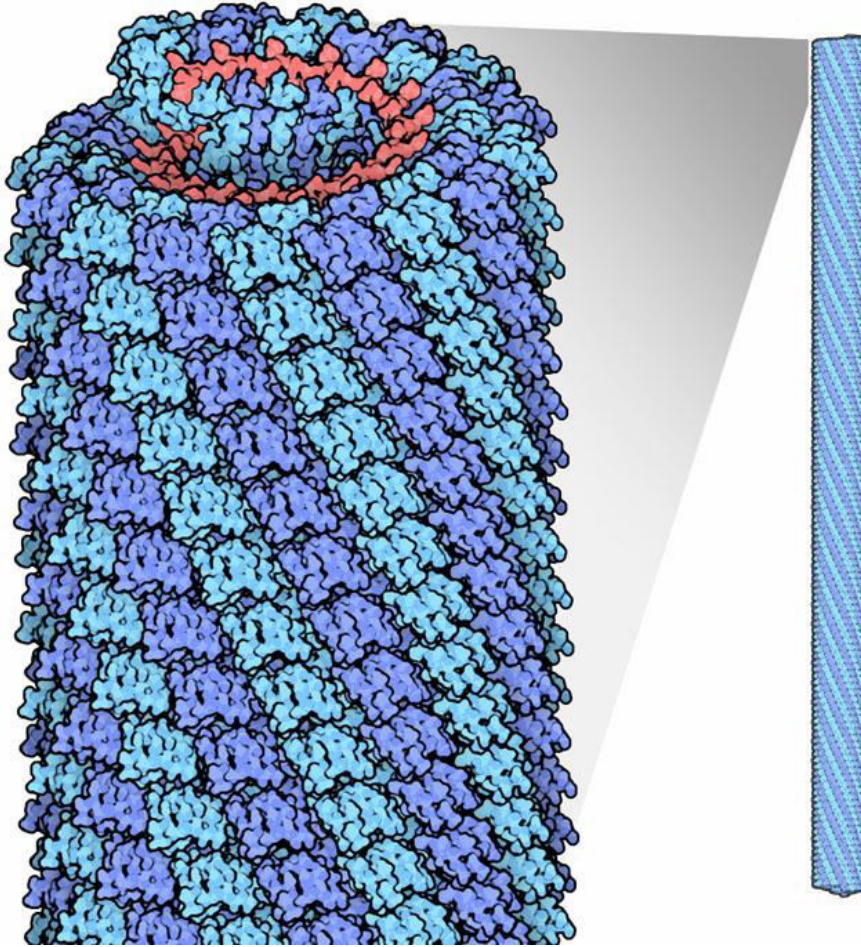
Helical Symmetry

HELICAL VIRUS



- Resembles long rods
- Ribbon-like structure
- Hollow tubes with protein walls

Helical Symmetry

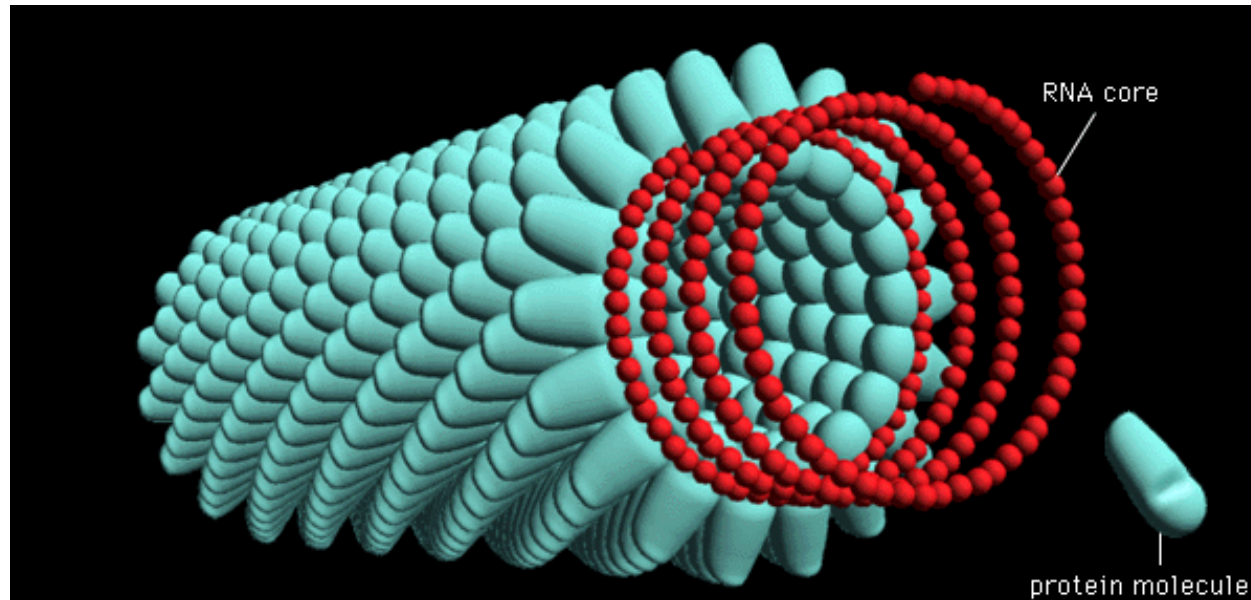


- A single protomer associates together in a helical arrangement to produce a long rigid tube

Helical Symmetry

GENOME

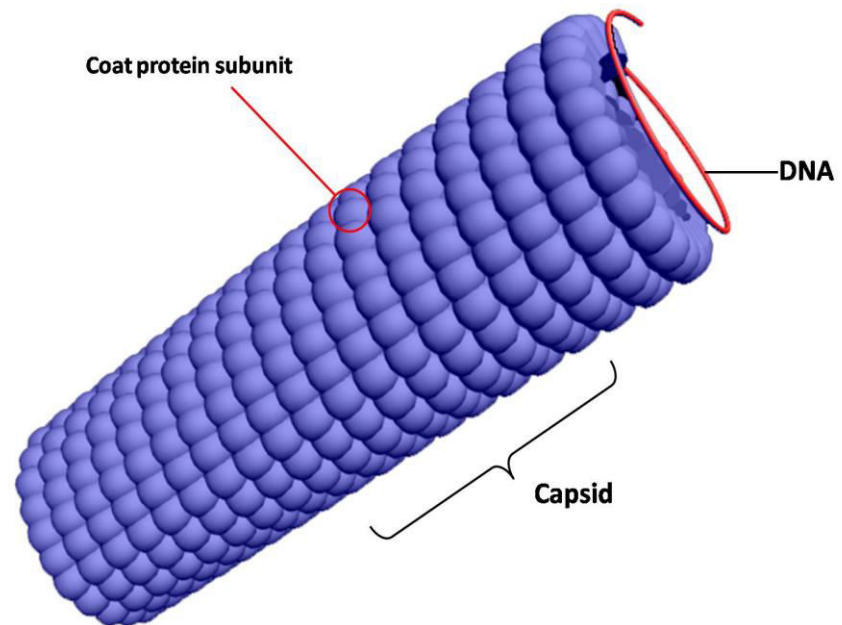
- Genome is wound in a spiral
- Inside the capsid
- Within a groove



Helical Symmetry

LENGTH

- The Nucleic acid determines the length of the helix
- Capsid doesn't extend much beyond the genome end



Helical Symmetry

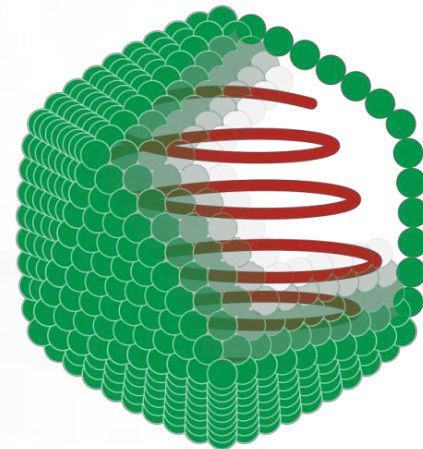
EXAMPLES

- Most are enveloped
- All are RNA viruses
- Tobacco Mosaic Virus (TMV)
- Influenza virus

Virology

CAPSID SYMMETRY

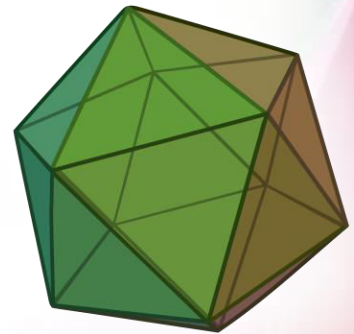
Icosahedral



Icosahedral Symmetry

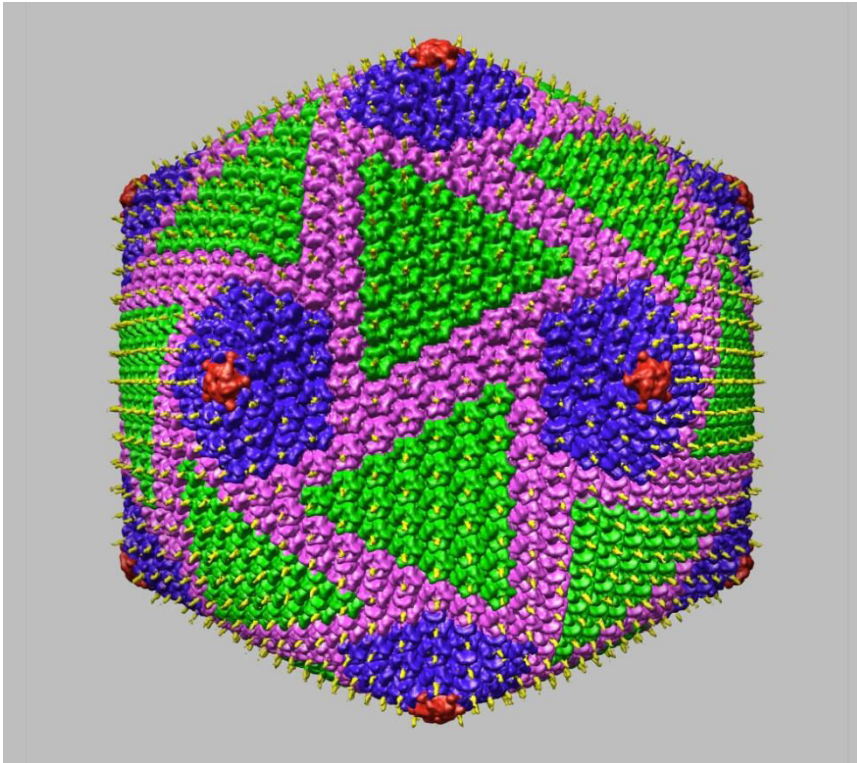
WHAT IS ICOSAHEREDRAL SHAPE?

- An icosahedron is a polyhedron with 12 vertices (corner), 20 faces (sides) and 30 edges



Icosahedral Symmetry

WHY ICOSAHEREDRAL?

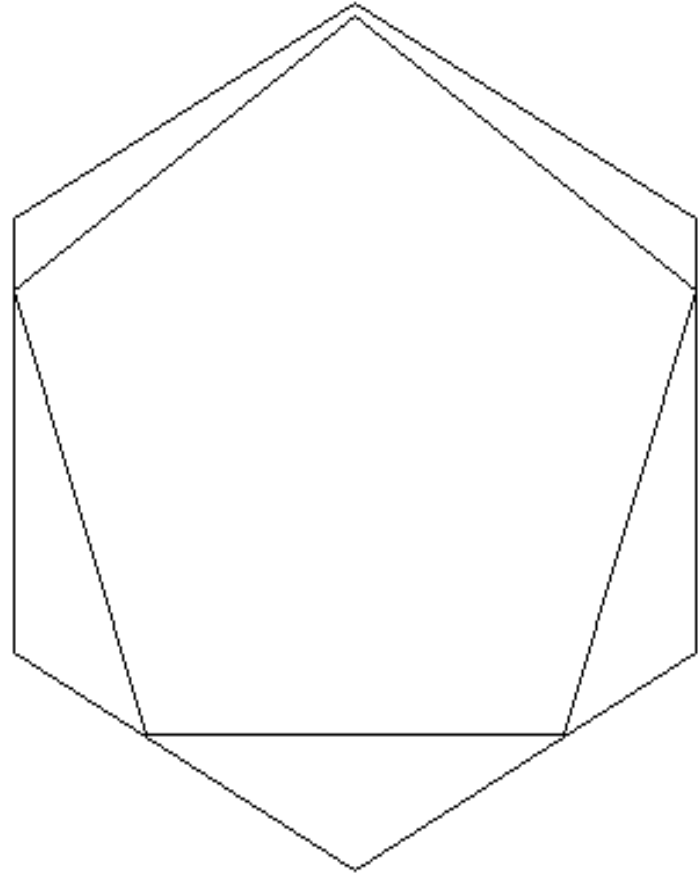


- Most efficient way to enclose a space
- small number of linear genes can specify large structure

Icosahedral Symmetry

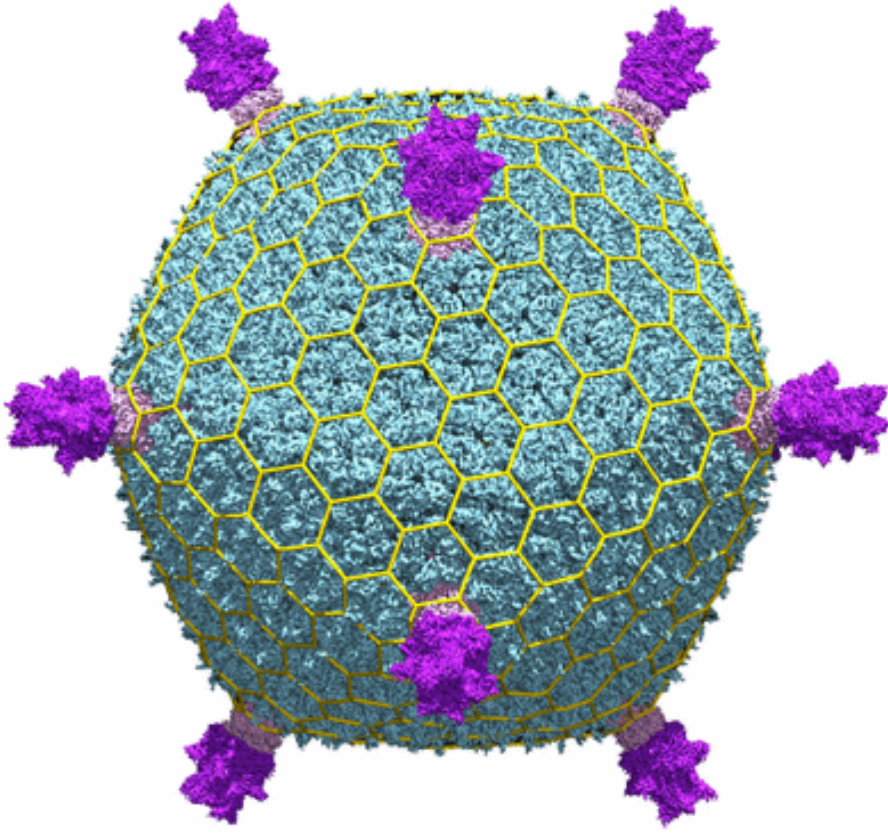
TYPES

- Hexamers (6 units)
- Pentamers (5 units)



Icosahedral Symmetry

CAPSOMERE

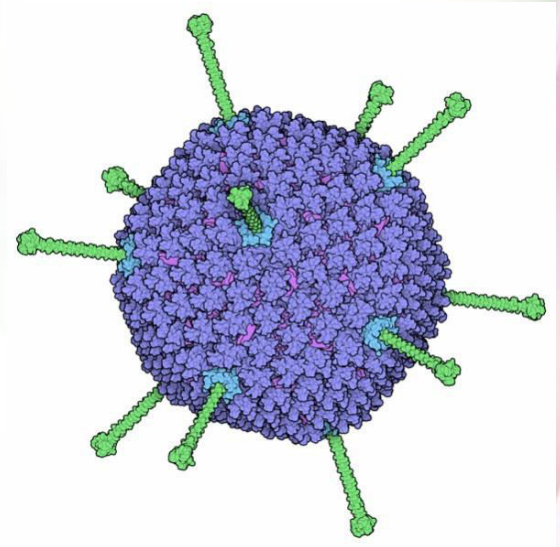


- Composed of knob-shaped structures
- Called capsomeres
- Composed of 5 or 6 protomers

Icosahedral Symmetry

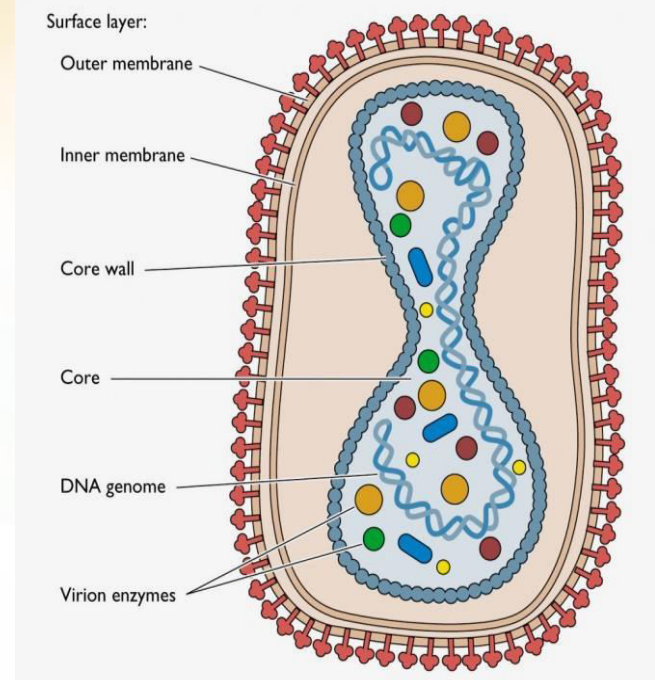
EXAMPLES

- most stable
- Adenovirus
- Picornavirus



Virology

CAPSID SYMMETRY Complex

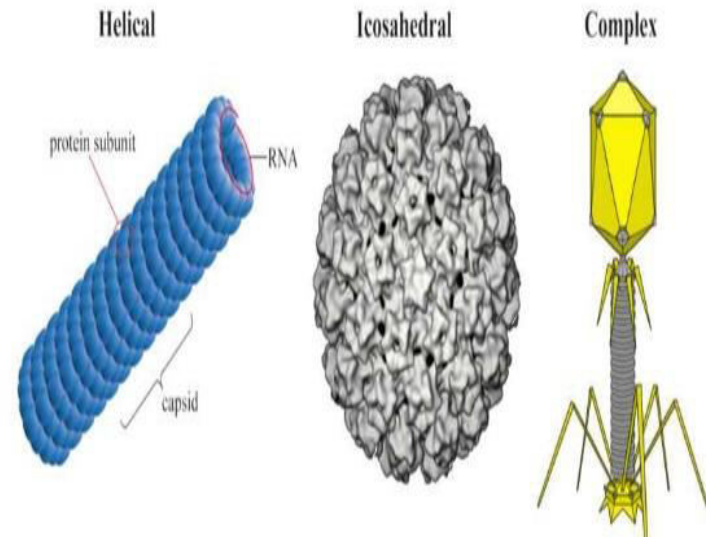


Complex Symmetry

WHAT IS COMPLEX SHAPE?

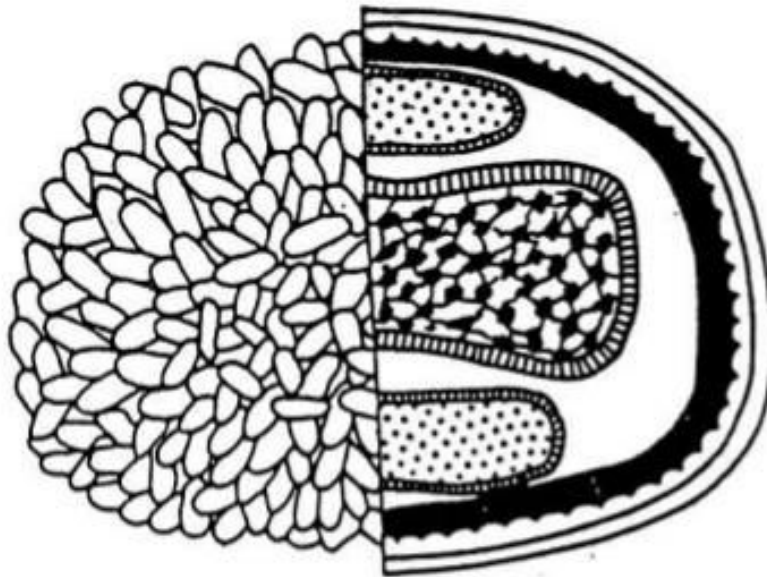
- Neither helical nor icosahedral

Symmetry of virus



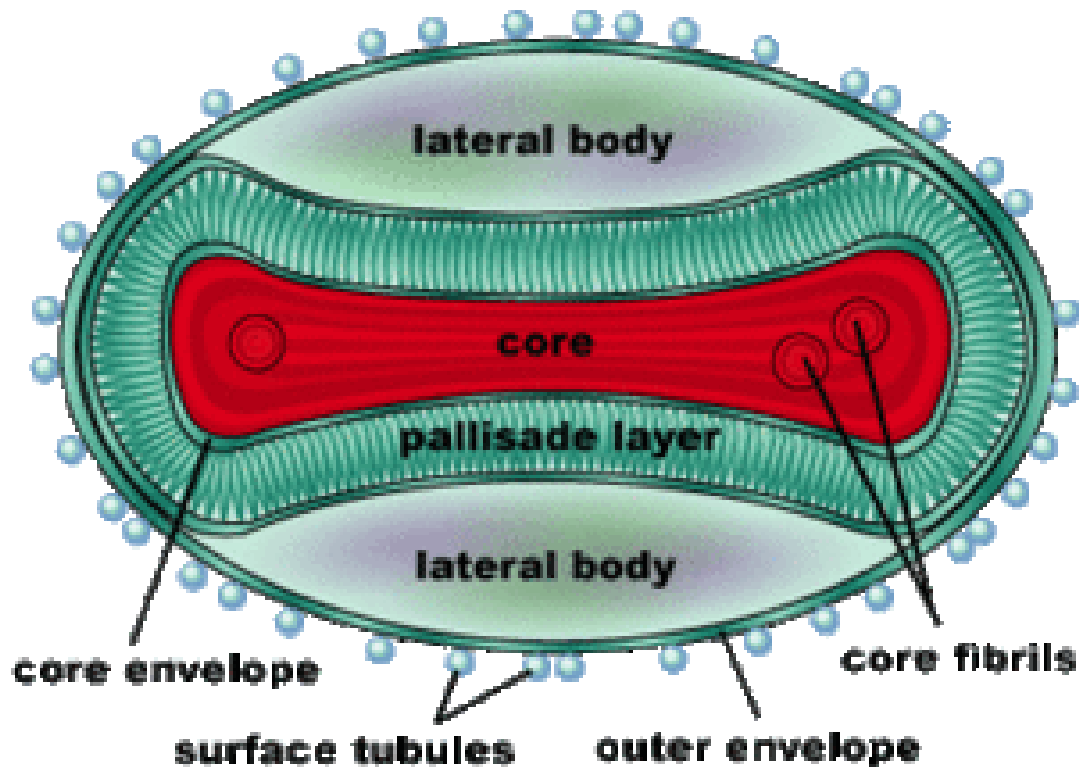
Complex Symmetry

HELICAL VIRUS



- Complex
- Ovoid interior
- Brick-shaped exterior

Complex Symmetry

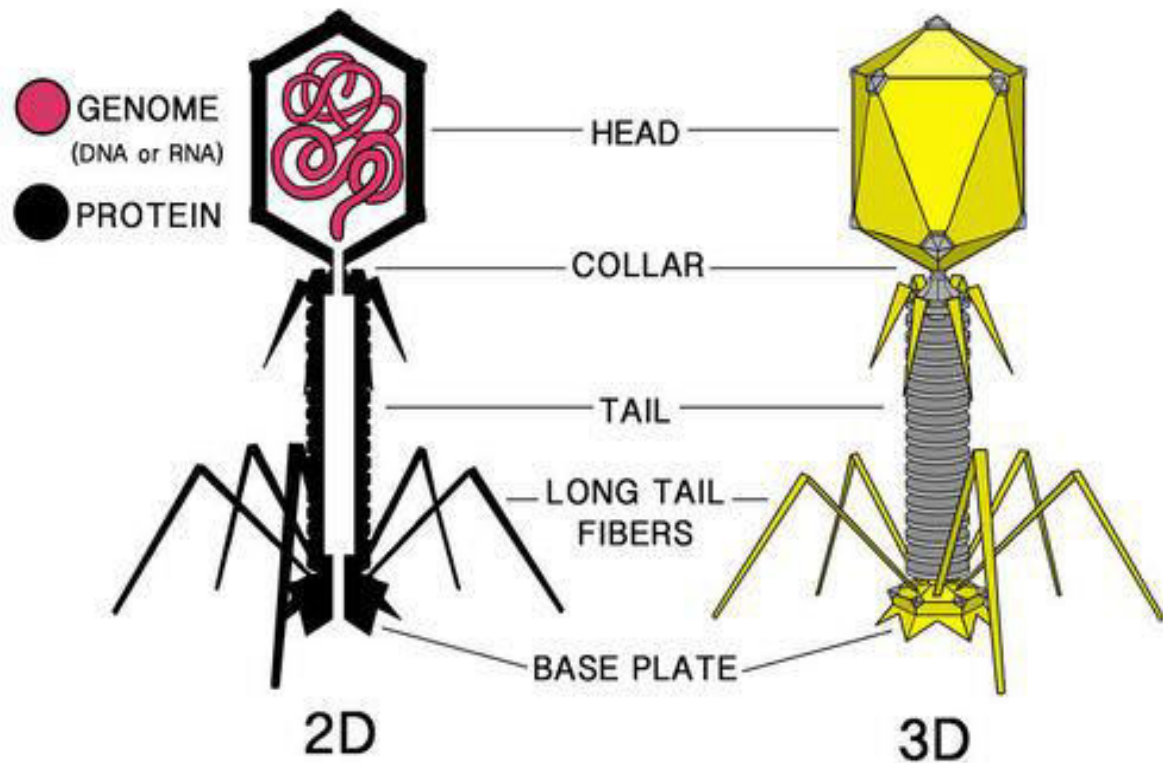


Several
separate
capsomeres

Complex Symmetry

GENOME

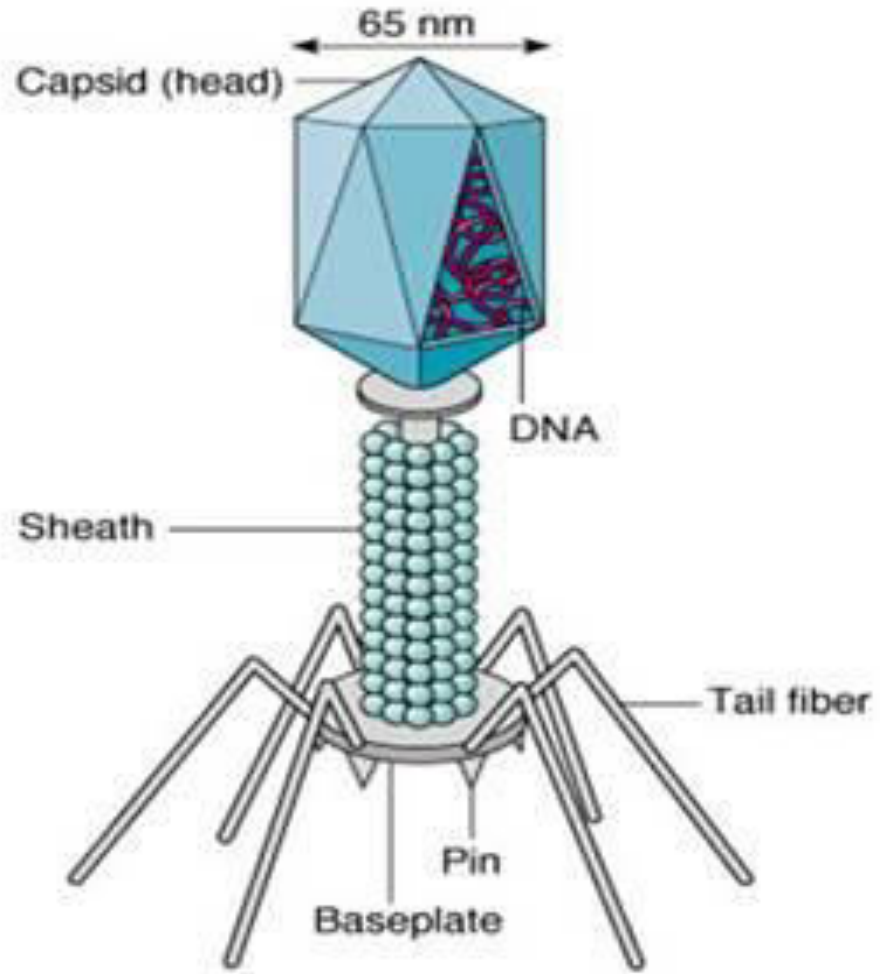
- Genome associated with proteins
- Contained in nucleoid



Complex Symmetry

EXAMPLES

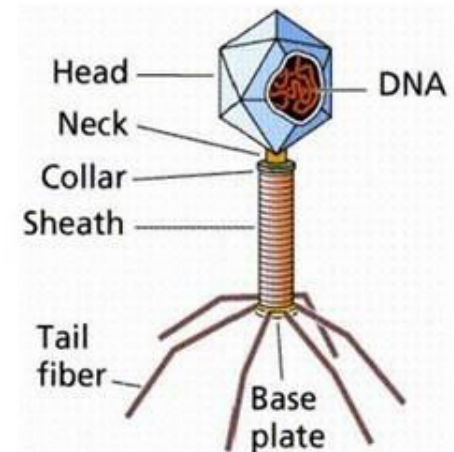
- Bacteriophages
- Poxviruses



Complex Symmetry

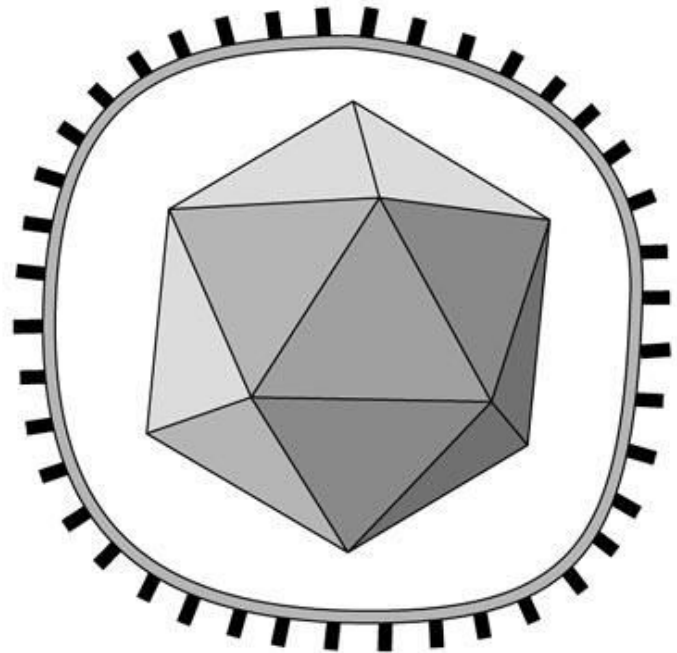
T-PHAGES

- Head and tail
- Called binal symmetry
- Icosahedral head and helical tail symmetry



Virology

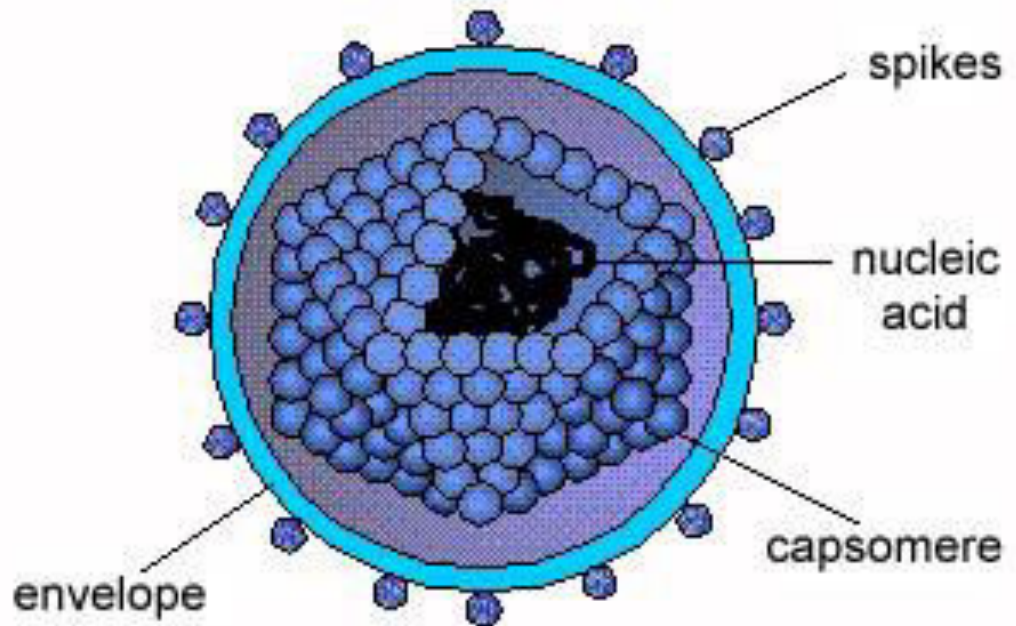
ENVELOPED VIRUSES



Enveloped Viruses

EXAMPLES

- Many animal viruses
- HIV
- Influenza virus
- Herpesvirus
- Adenovirus
- Hepatitis B virus
- Parvovirus



Enveloped Viruses

CONCLUSION

- Surrounds capsid
- Composed of glycoproteins and lipid bilayer

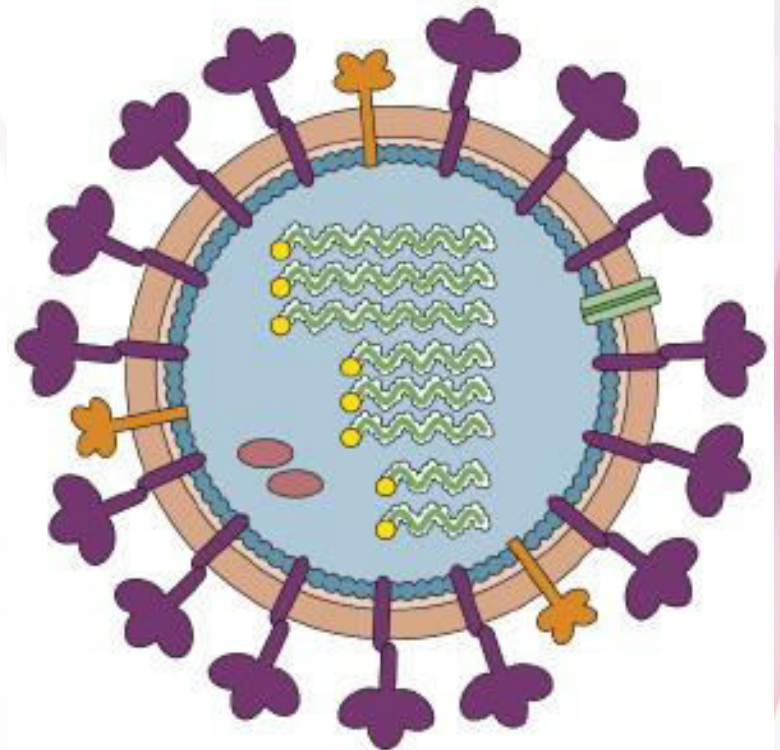
Enveloped Viruses

CONCLUSION

- Helps virus to enter host
- Not all viruses have it (Most animal viruses)

Virology

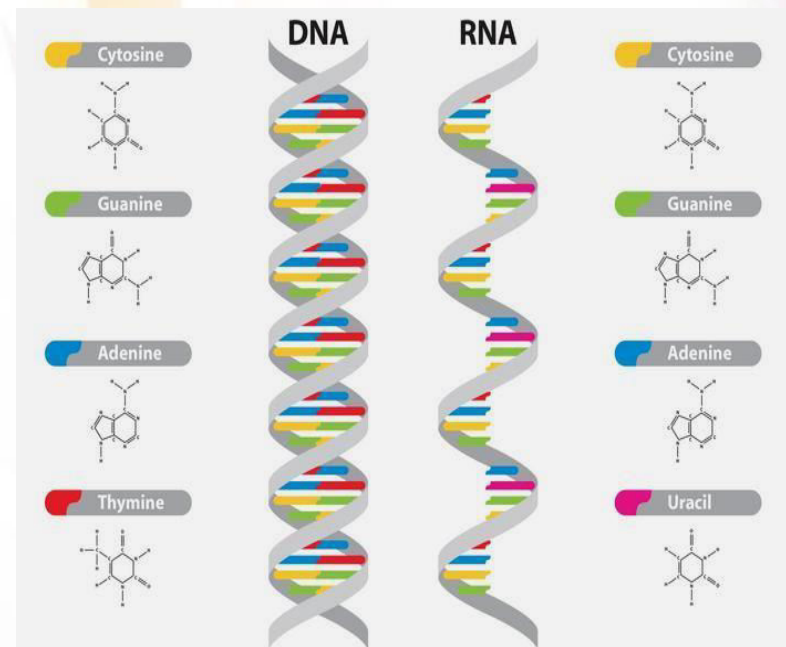
VIRAL GENOME



Viral Genome

NUCLEIC ACID

- Either DNA
- Or RNA



Viral Genome

STRANDEDNESS

- Single stranded
- Double stranded



Viral Genome

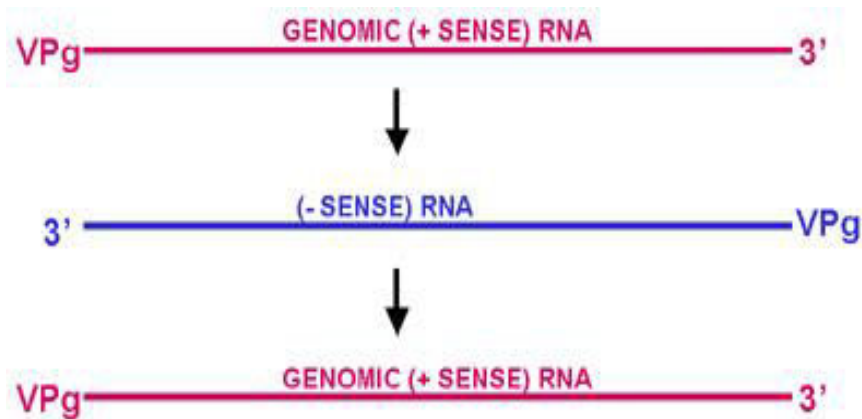
SHAPE

- Circular
- Linear
- Segmented



Viral Genome

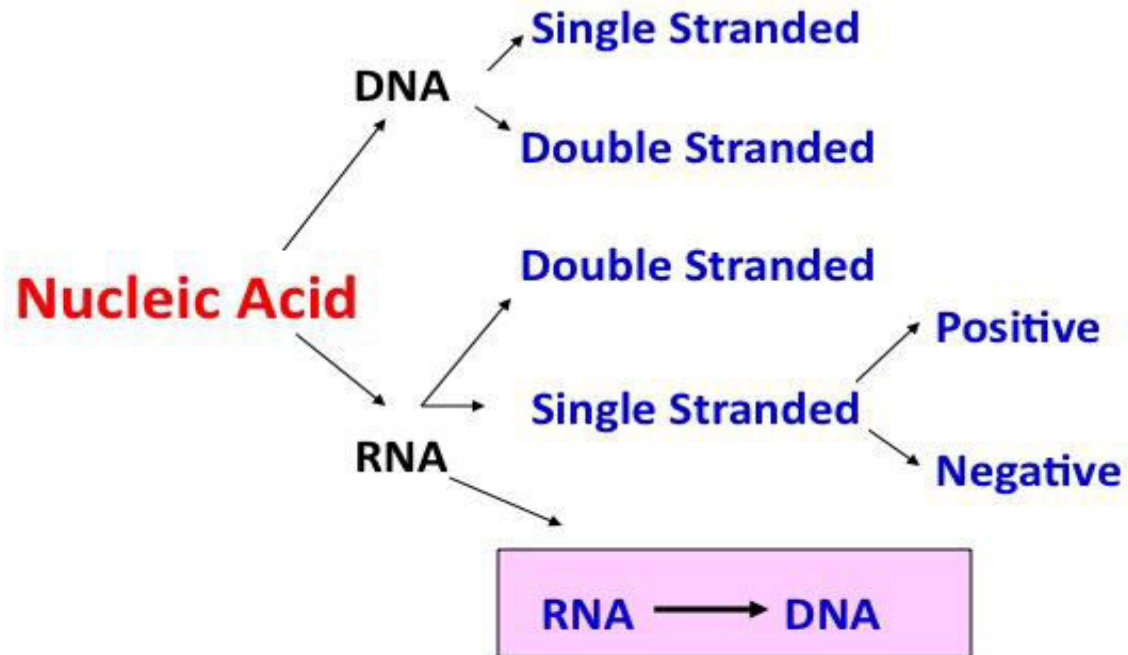
SENSE



- Positive sense (+)
- Negative sense (-)
- Ambisense (+ / -)

Viral Genome

Viral Genomes



Viral Genome

PLANT VIRUSES

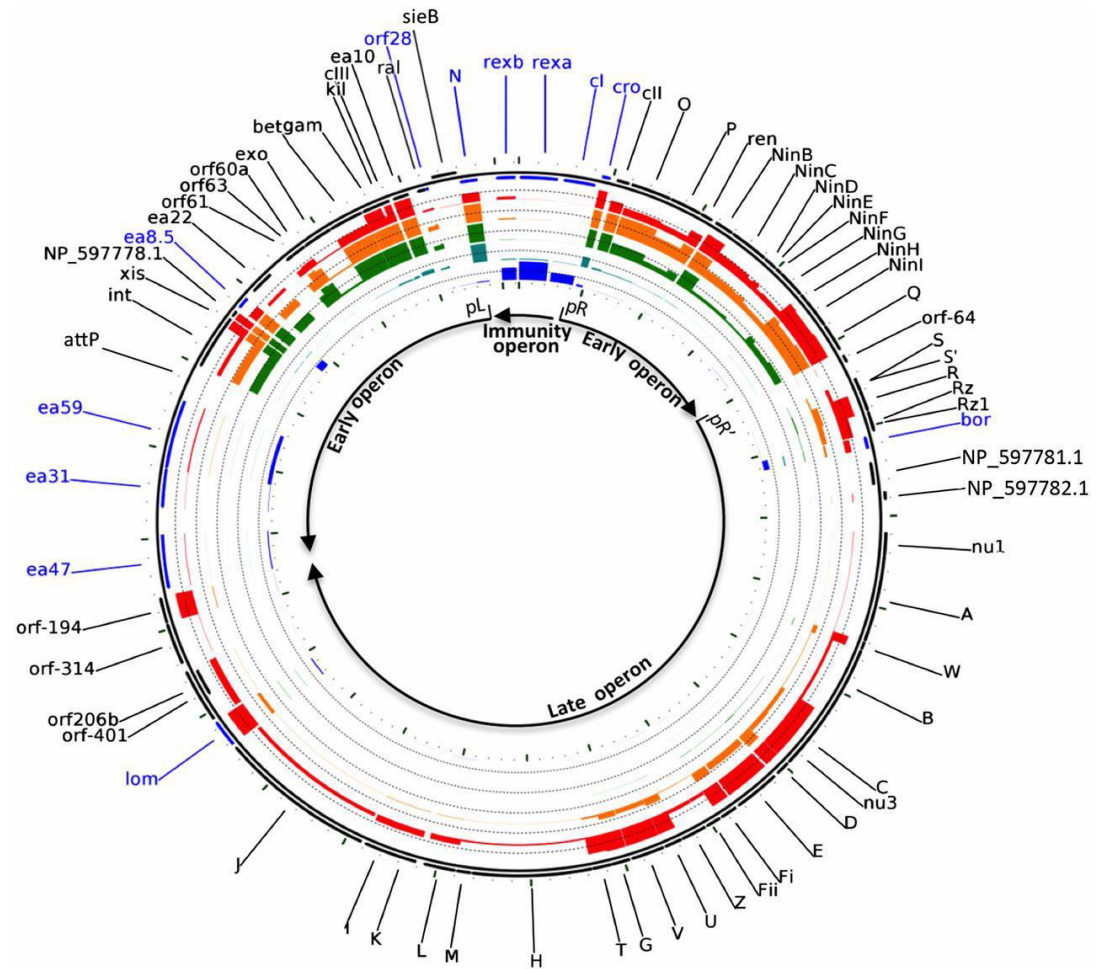
- Single stranded
- RNA genome



Viral Genome

BACTERIOPHAGES

- Double stranded
- DNA genome



Viral Genome

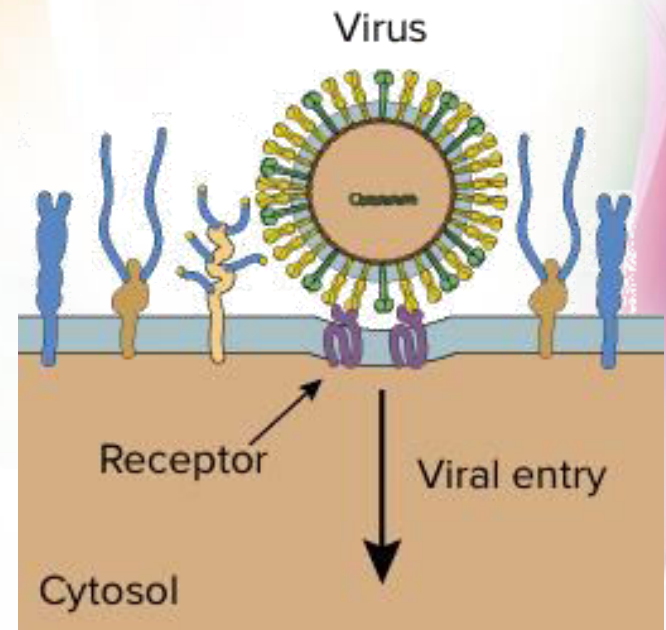
GENOME SIZE

- Varies
- Smallest genome of *Circovirus* (2 kilobases)
- Largest *Pandoravirus* (2 Megabases)

Virology

VIRUS RECEPTORS

Recognition and Binding



Virus Receptors

ENTRANCE

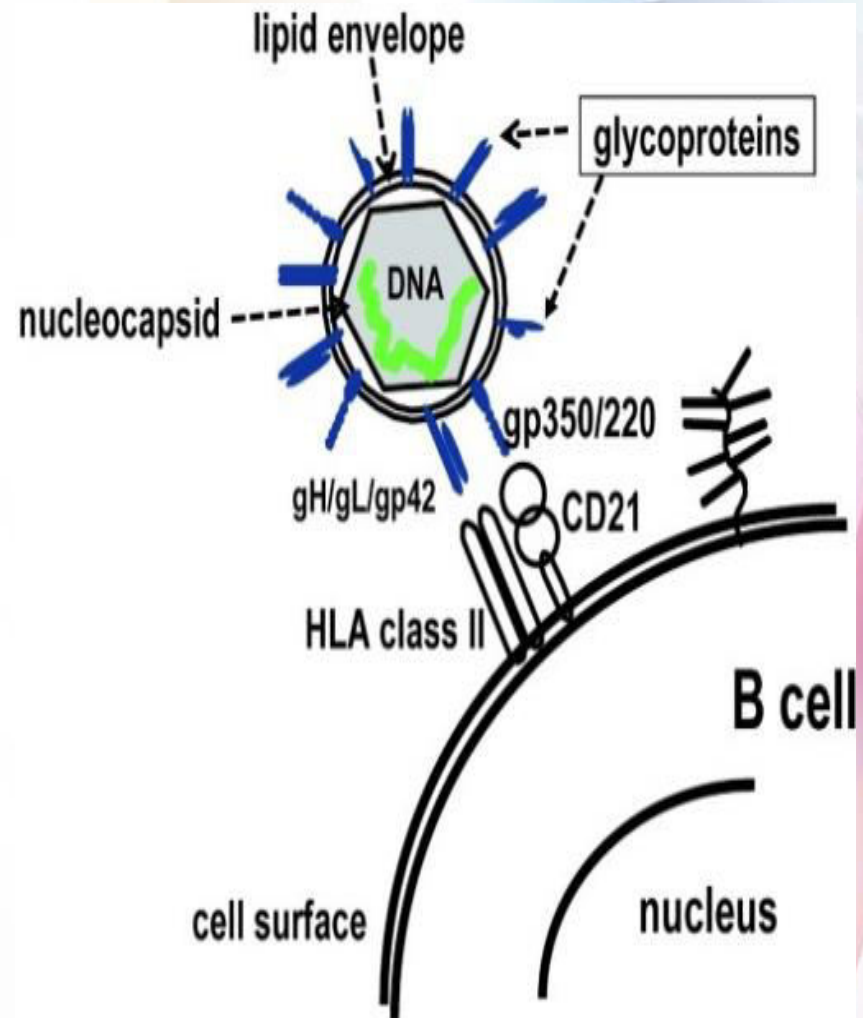
- Virus interacts with host receptors
- Cascade of events
- Leading entry of viruses
- Same for enveloped and nonenveloped

Virus Receptors

RECEPTORS

- Host and virus interaction through receptors
- Receptors can be protein, lipids or carbohydrates

Virus Receptors

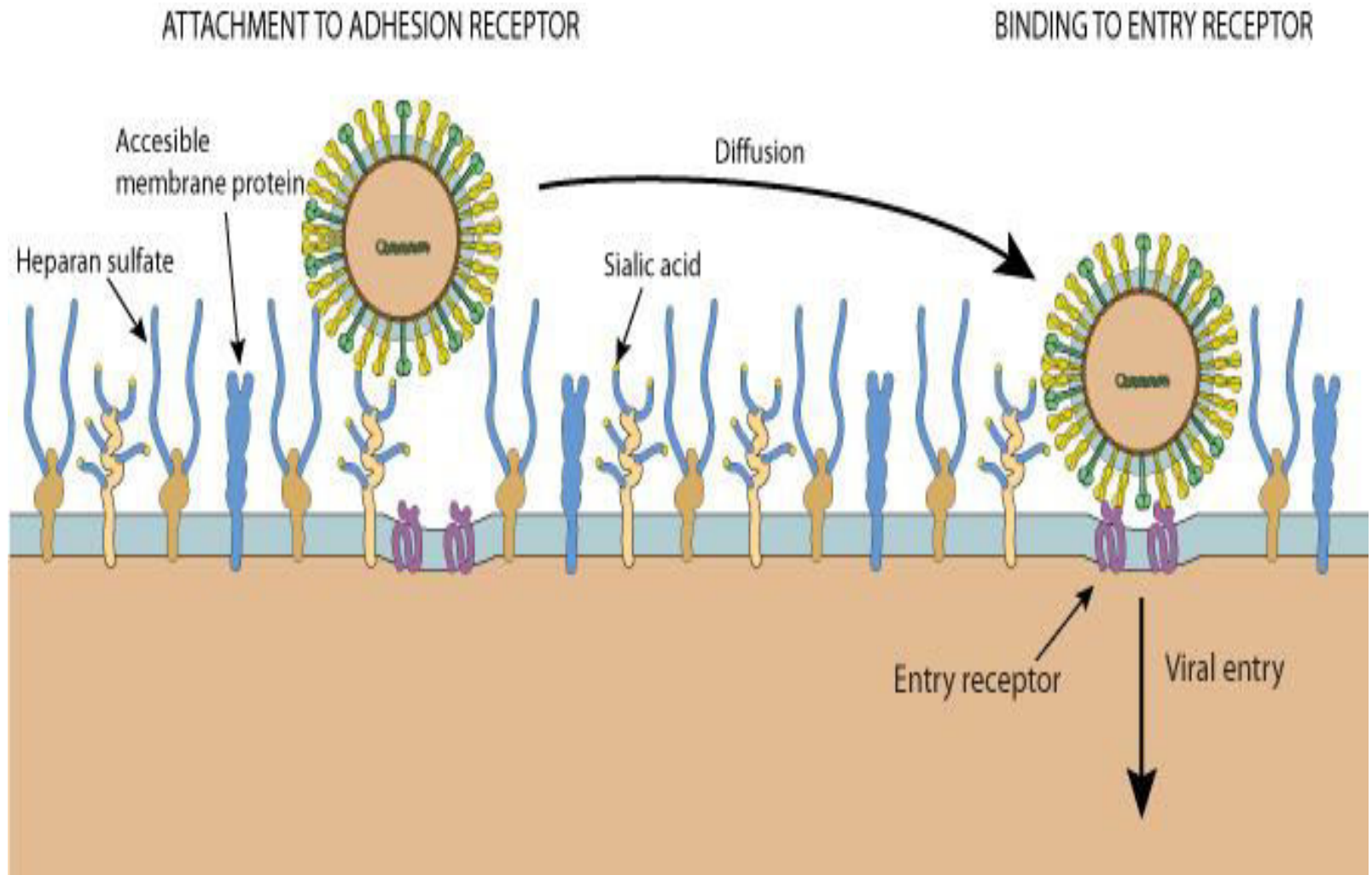


Virus Receptors

RECEPTOR RECOGNITION

- Viruses have various receptor recognition structures on surface
- Frequently have spikes
- Bind on the receptors on host cell surface

Virus Receptors



Virus Receptors

ENTRANCE

Variable

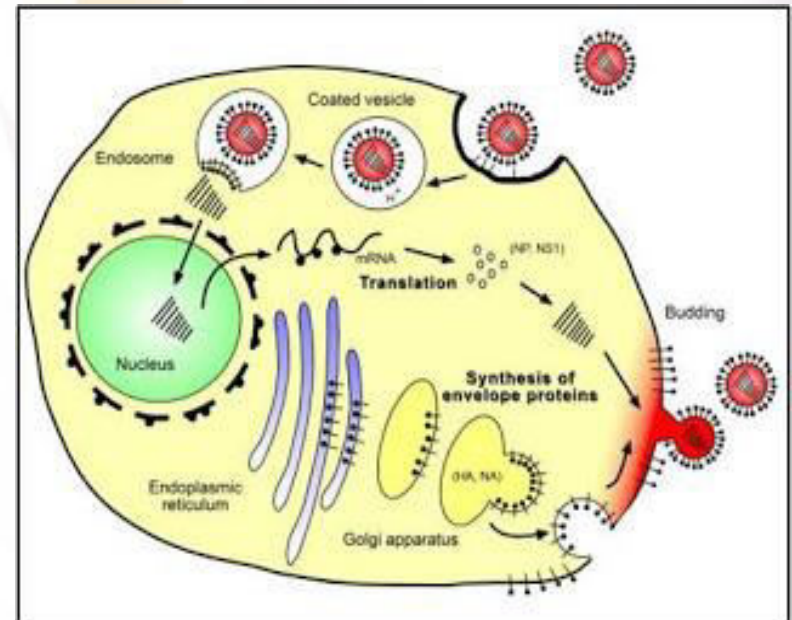
- Pinocytosis/endocytosis

Membrane fusion

- Viral Penetration

Virology

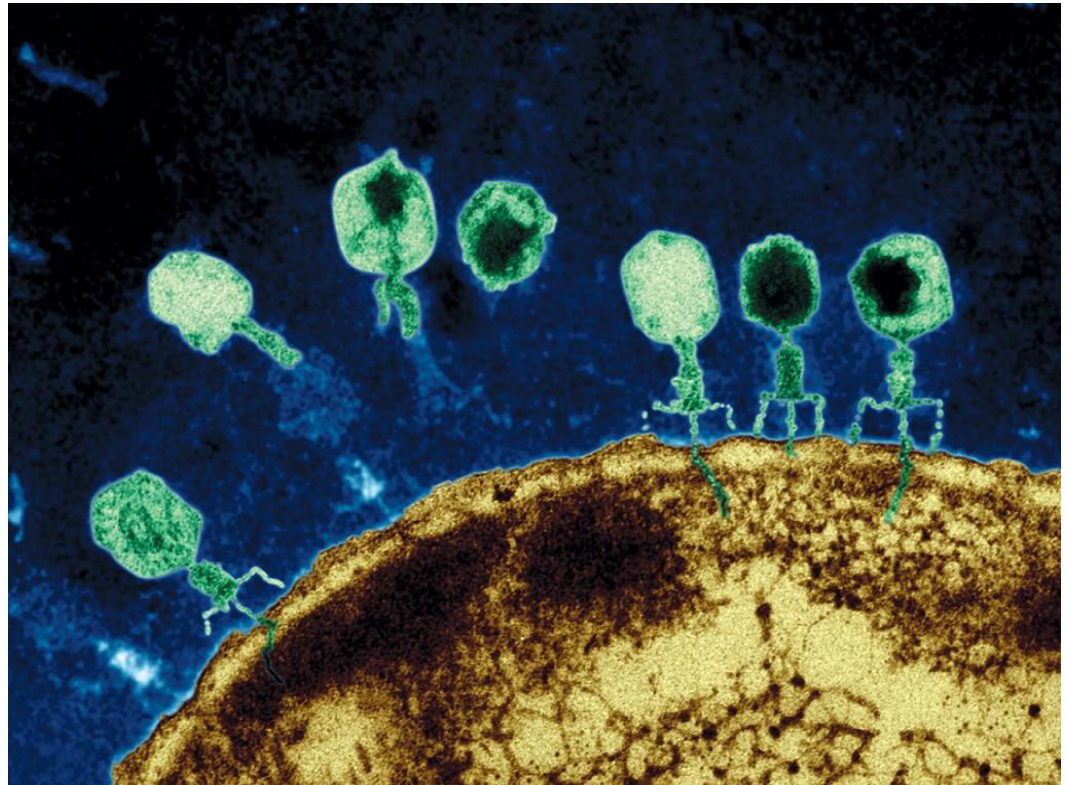
Interaction of Virus and Host Cells



Interaction of Virus and Host Cells

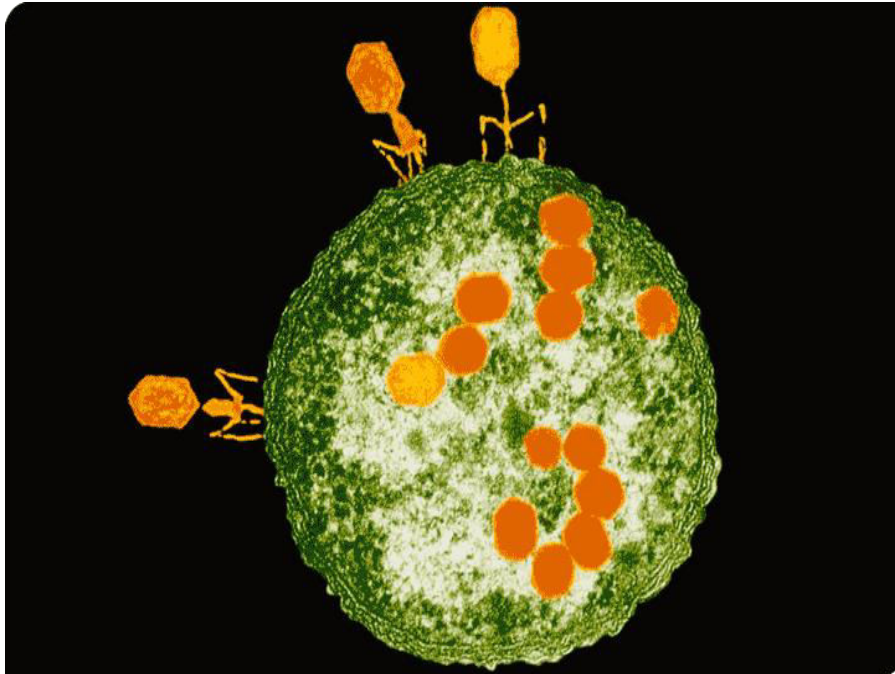
VIRUS-HOST

- Can cause
 - No damage
 - Cell damage
 - Cell death
 - Oncogenesis



Interaction of Virus and Host Cells

NON-IMMUNOGENIC RESPONSE



- Phagocytosis
- Raised body temp
- Hormones
- Malnutrition

Interaction of Virus and Host Cells

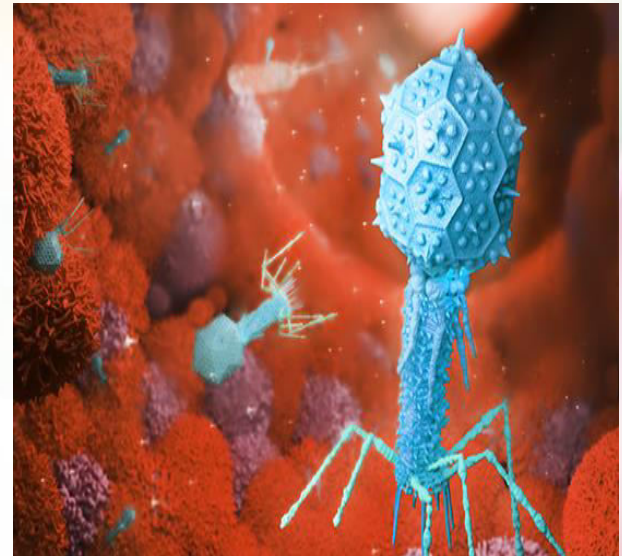
BARRIERS

- Skin
- Lack of membrane receptors
- Mucus
- Epithelium
- Low pH
- Immune system

Interaction of Virus and Host Cells

VIRAL INVASION

- Virus can escape immune system
- Cause generalized immuno-suppression



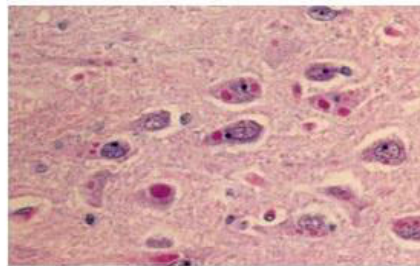
Interaction of Virus and Host Cells

FORMATION OF INCLUSION BODIES

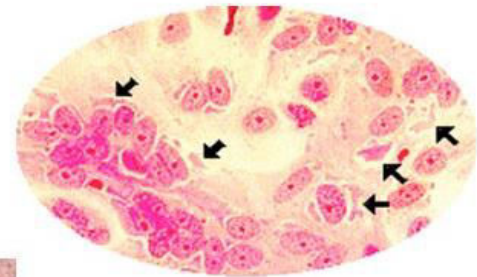
- Nuclear or cytoplasmic aggregates
- Stainable
- Sites of viral multiplication



Adenovirus
nuclear inclusions



Rabies virus
cytoplasmic inclusions
Negri bodies

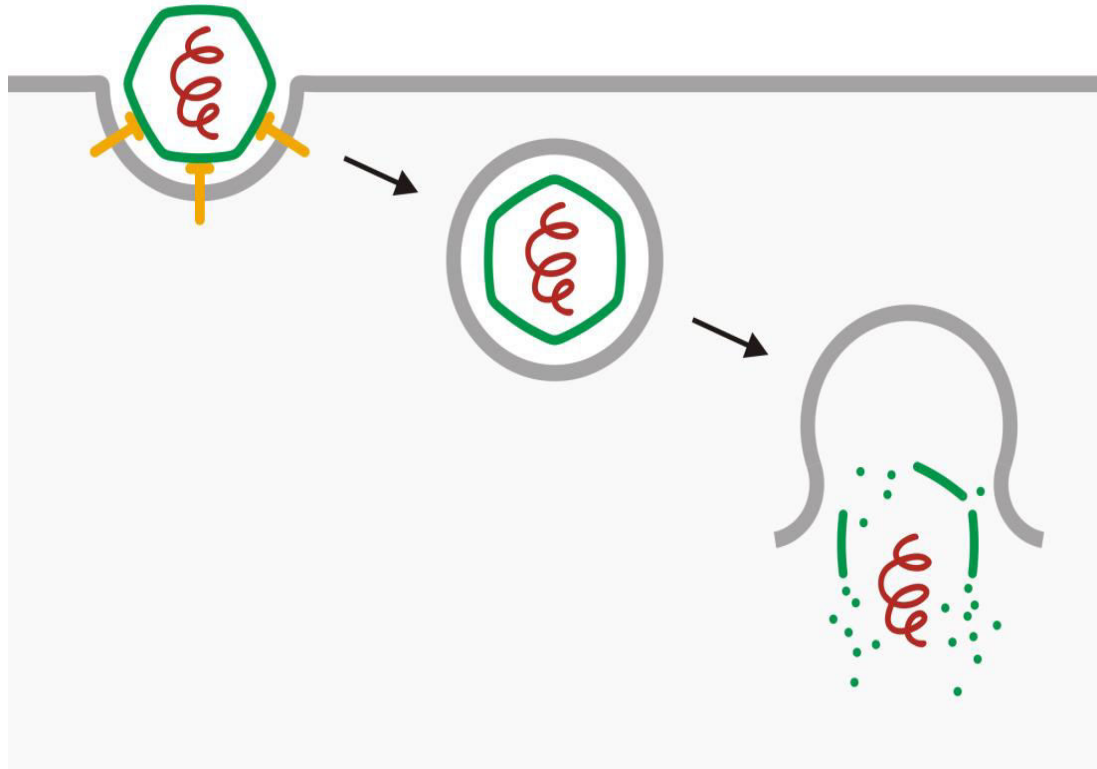


RSV
cytoplasmic inclusions

Interaction of Virus and Host Cells

PATHOGENESIS

- Inapparent
- Apparent
- Acute
- Subacute
- Chronic
- Latency



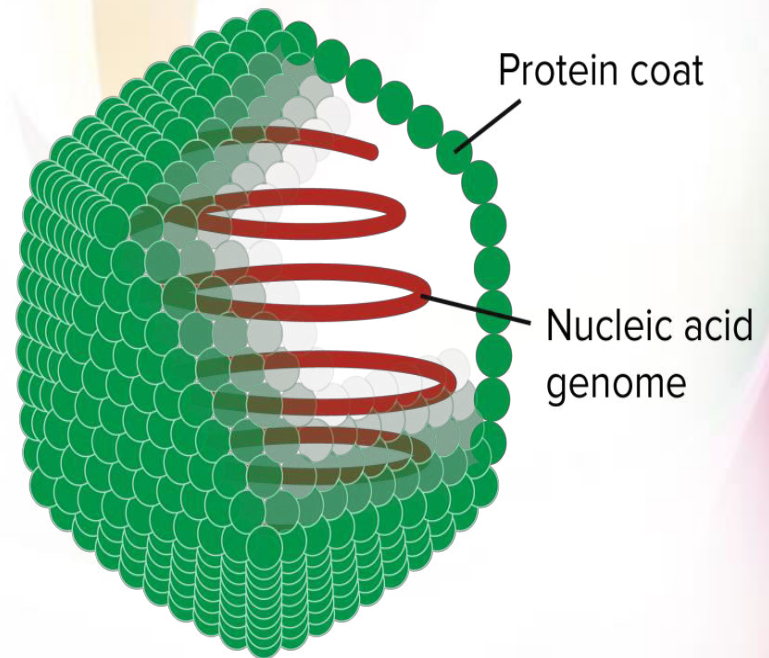
Interaction of Virus and Host Cells

ENTRANCE

- Respiratory tract
- Digestive tract
- Skin
- Conjunctivae
- Sexual contact
- Vectors

Virology

Protein-Nucleic acid Interactions



Protein-Nucleic acid Interactions

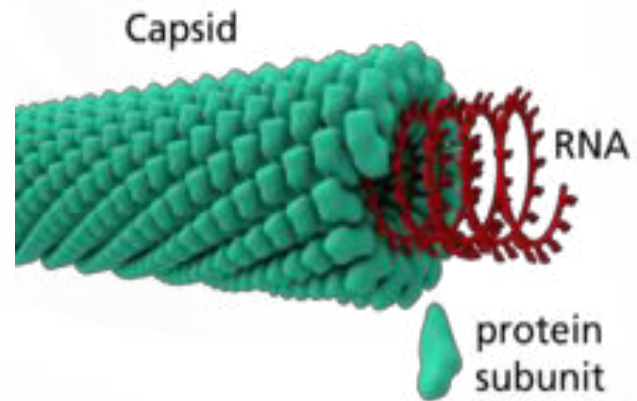
HISTORY

- Late 19th century, Microscopic observation
- Since then researchers have used variety of procedure to understand the relation
- Play a crucial role in biological processes

Protein-Nucleic acid Interactions

PRETECTION OF GENOME

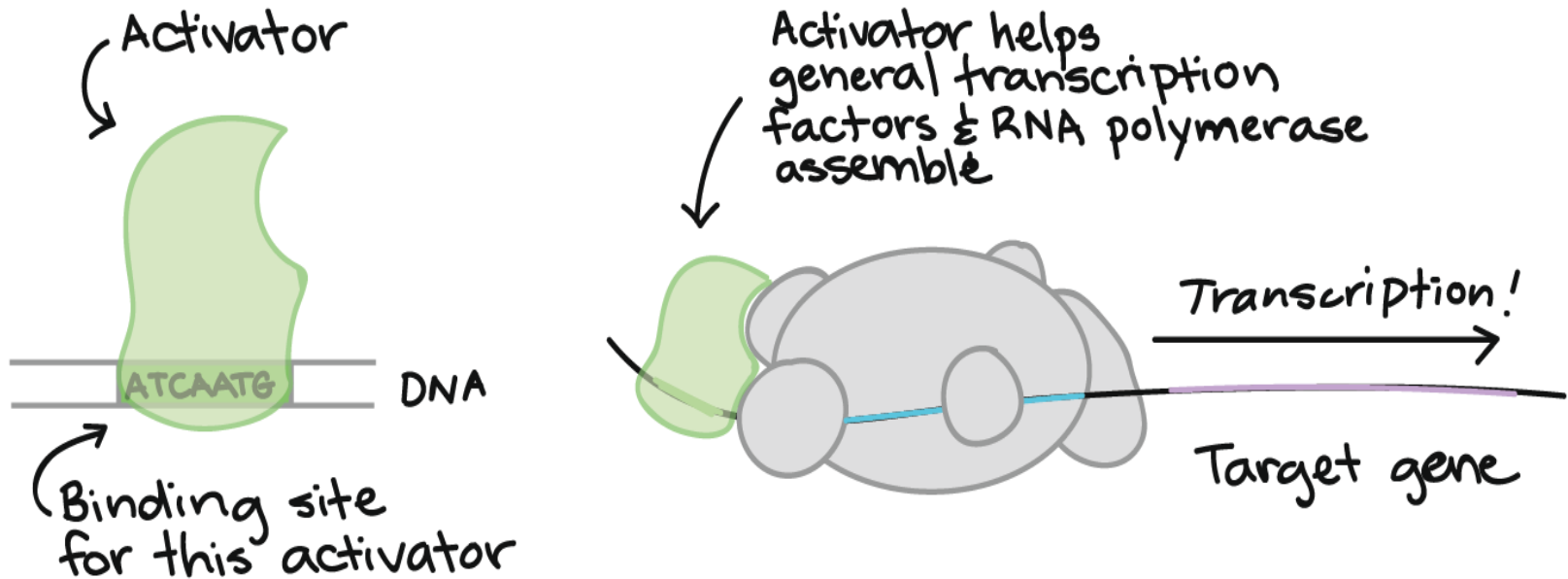
- Primary function of virus
- Capsid proteins protect the genome



Protein-Nucleic acid Interactions

REGULATION OF BIOLOGICAL FUNCTIONS

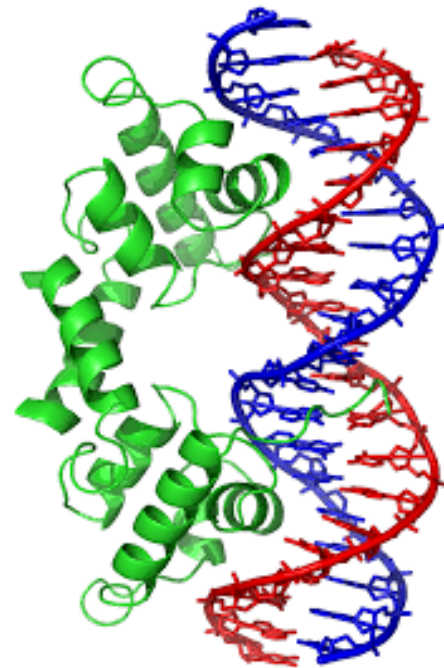
- Usually expression of gene
- Usually transcription factors
- Gene regulation



Protein-Nucleic acid Interactions

STRUCTURE

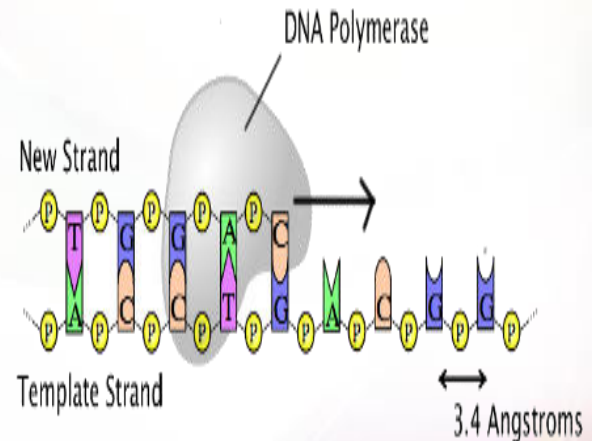
- DNA binding motif
- With the help of different ions



Protein-Nucleic acid Interactions

ENZYMATIC ROLE

- Polymerases and Restriction Endonucleases
- For replication and translation of proteins



Virology

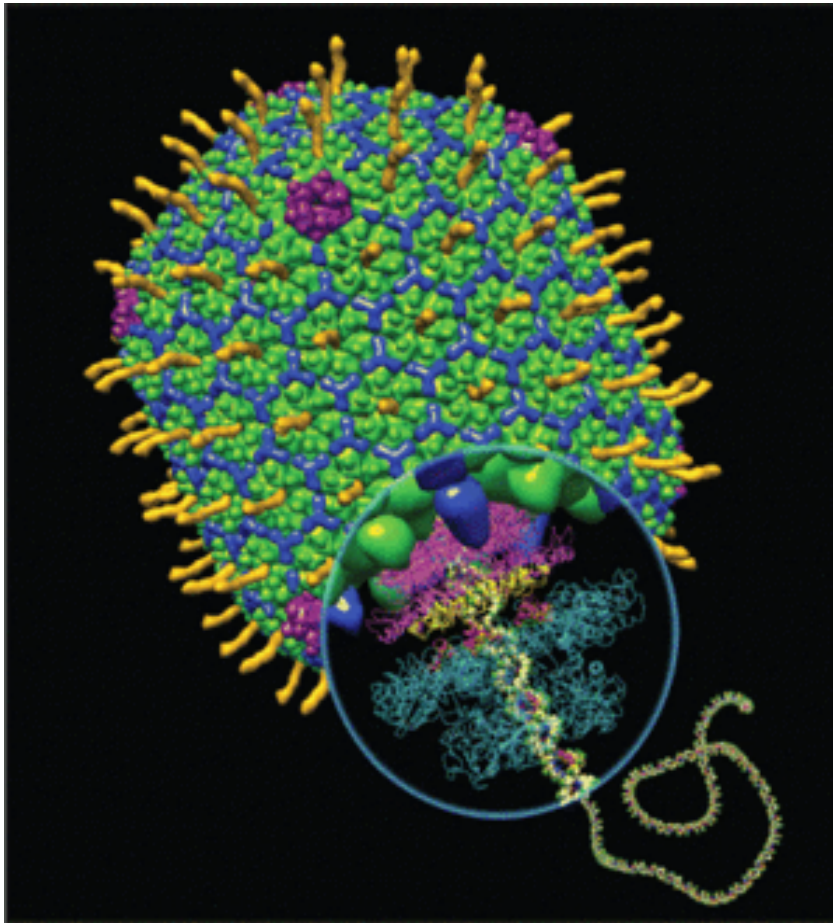
Genome Packaging

Genome Packaging

- Fundamental step in viral lifecycle
- Usually preformed through capsids
- Capsids are reformed sometimes

Genome Packaging

PACKAGING MOTOR PROTEIN



- Driven by hydrolysis of ATP
- Condenses genome into a confined place

Genome Packaging

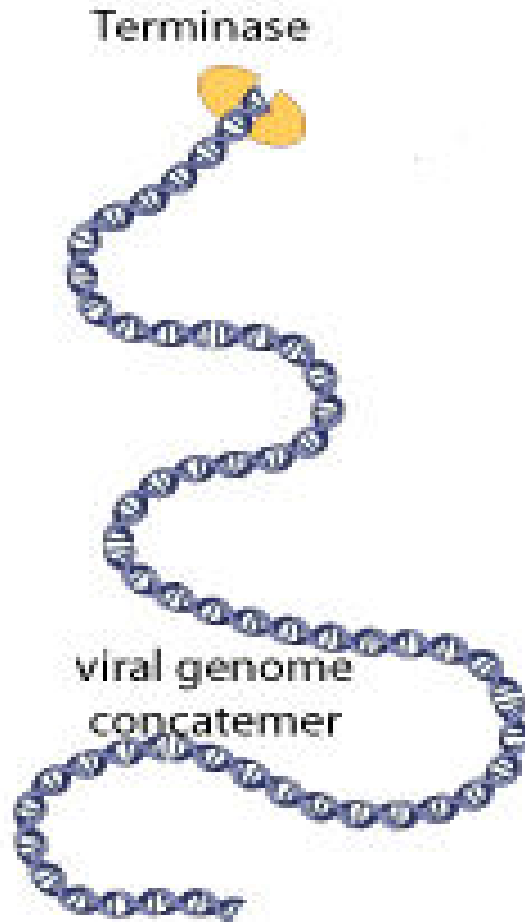
STEPS

1. Binding of Terminase
2. Binding to Procapsid portal
3. DNA translocation
4. Contamer Cleavage

Genome Packaging

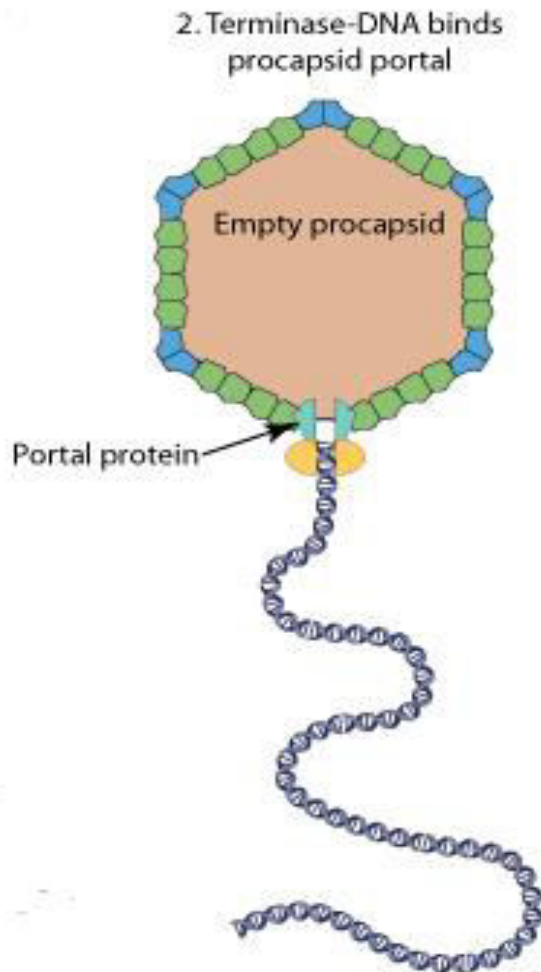
BINDING OF TERMINASE

- First step
- Terminase binds to viral genome



Genome Packaging

TERMINASE-DNA BINDS TO PROCAPSID PORTAL

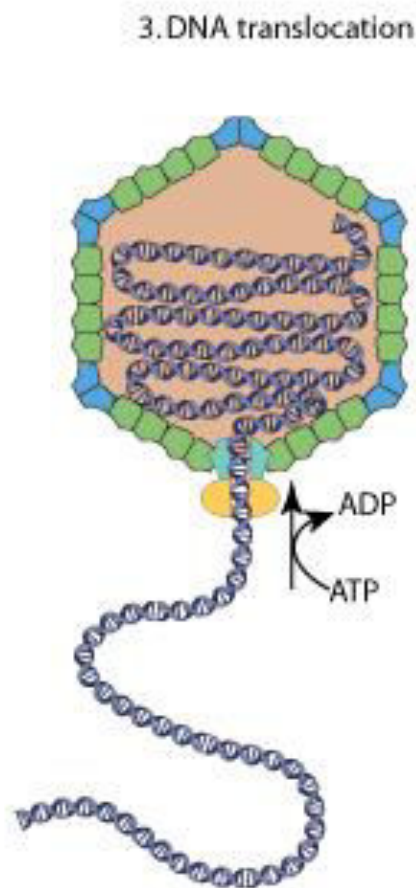


- Preassembled capsid or procapsid usually
- Terminase-DNA complex binds to the portal of entry on the capsid

Genome Packaging

DNA TRANSLOCATION

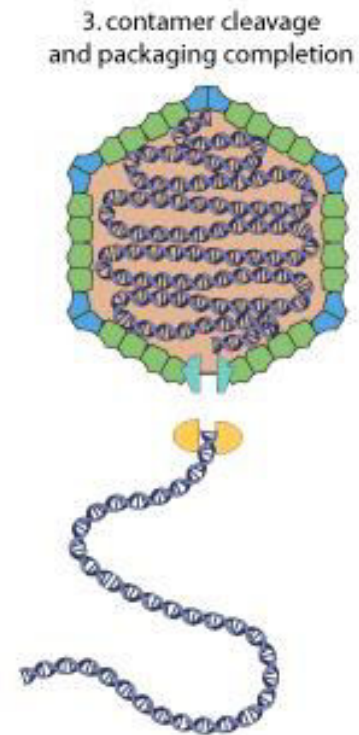
- Portal structure provides entry to DNA
- Packaging motor protein
- Large terminase



Genome Packaging

CONTAMER CLEAVAGE

- DNA cleavage
- Completion of genome packaging



Virology

An Introduction to Viral Genomics

An Introduction to Viral Genomics

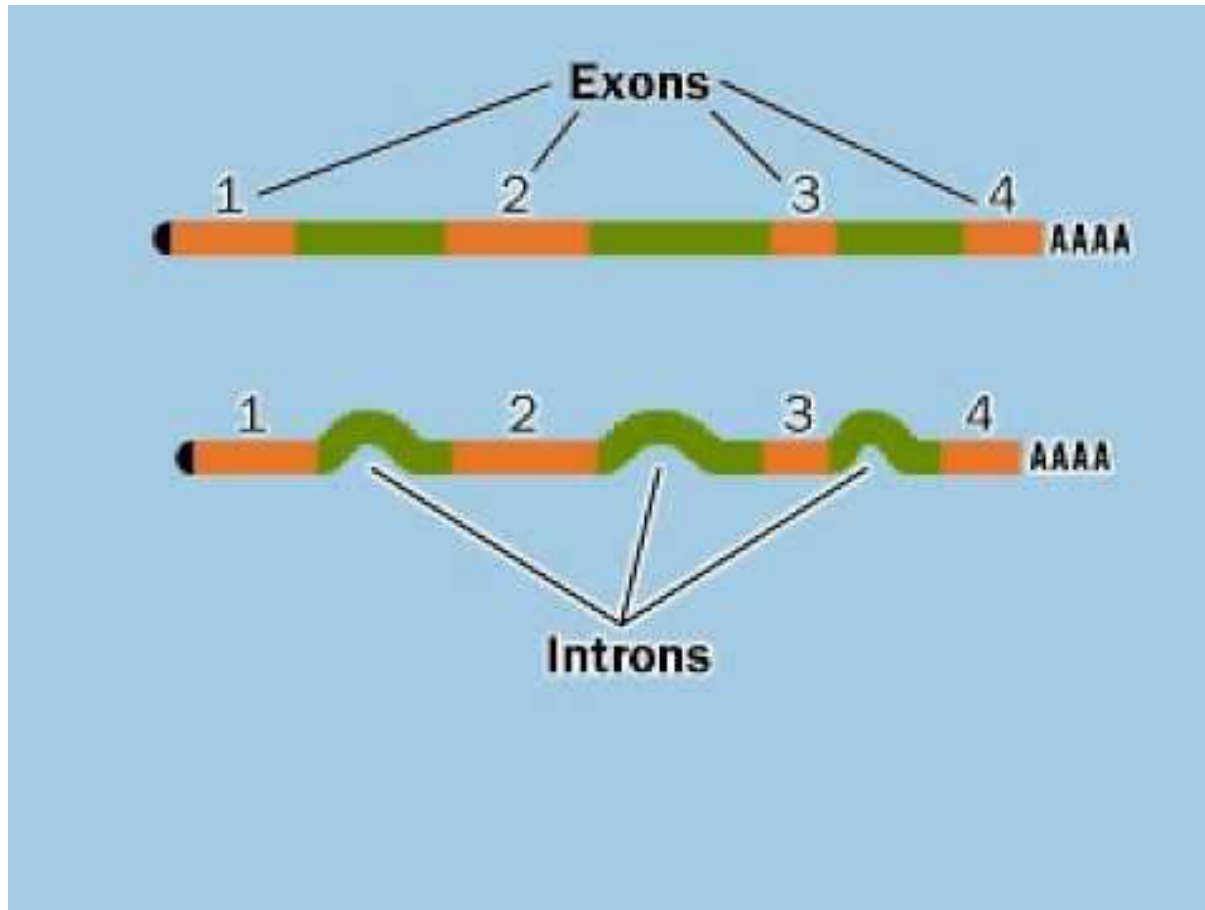
VIRAL GENOME

- More complex than any other species
- DNA or RNA
- Single stranded or double stranded
- Positive sense
- Negative sense

An Introduction to Viral Genomics

INTRONS

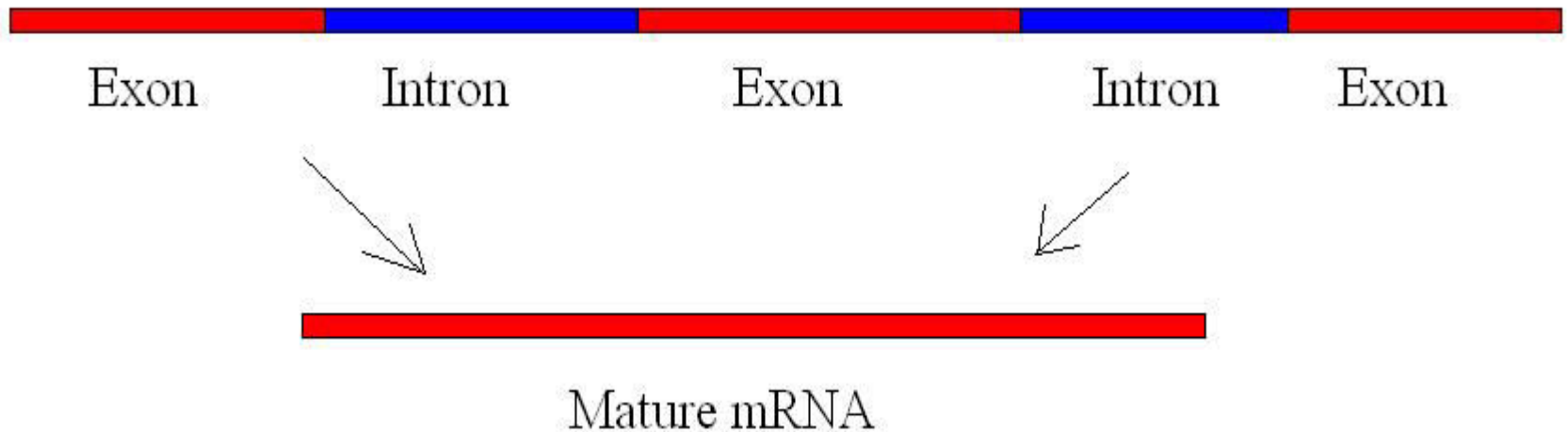
- Non-coding regions on viral genome



An Introduction to Viral Genomics

EXONS

- Coding regions on viral genome



An Introduction to Viral Genomics

MAIN GENES

- All viruses have these:
- Replicase – an enzyme that replicates genome
- Capsid – a protein that protects the genome

An Introduction to Viral Genomics

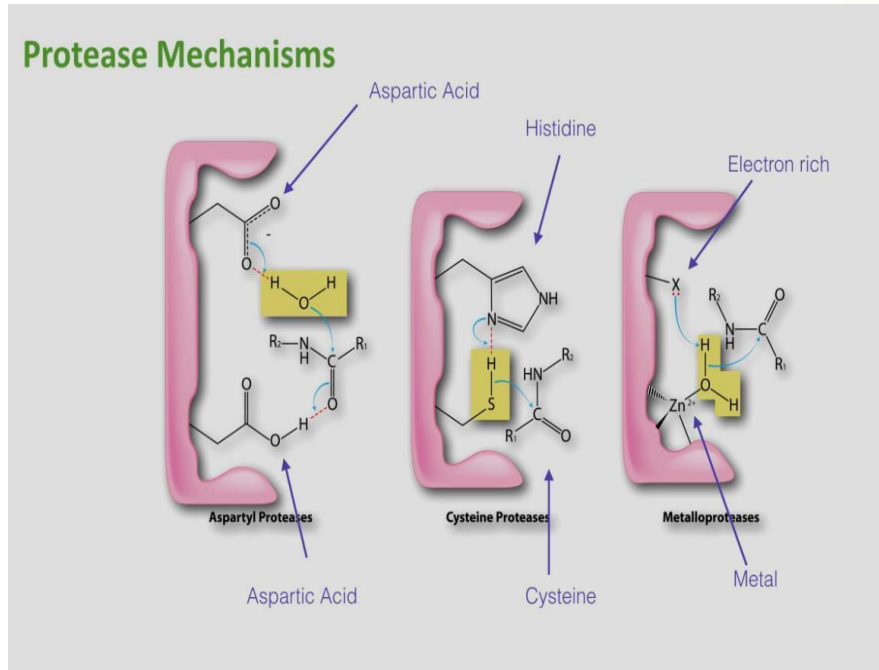
OTHER PROTEIN GENES

- Protease
- Glycoprotein
- Host-shut-off proteins
- Anti-host defence proteins

An Introduction to Viral Genomics

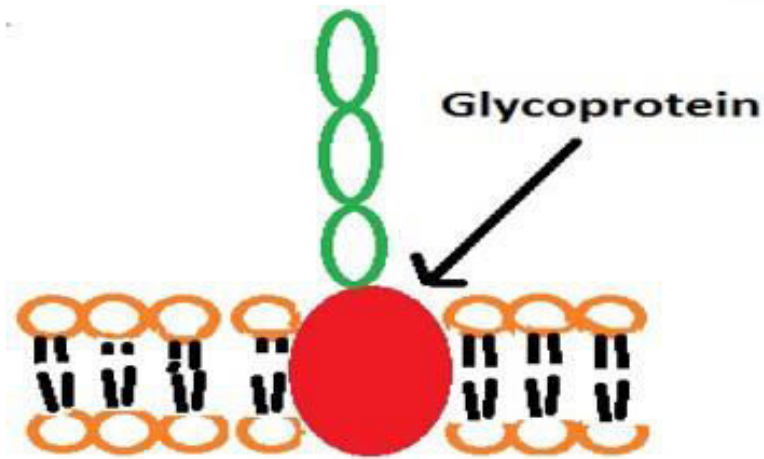
PROTEASE

- Processes viral proteins
- Allows assembly
- Maturation to infect other cells



An Introduction to Viral Genomics

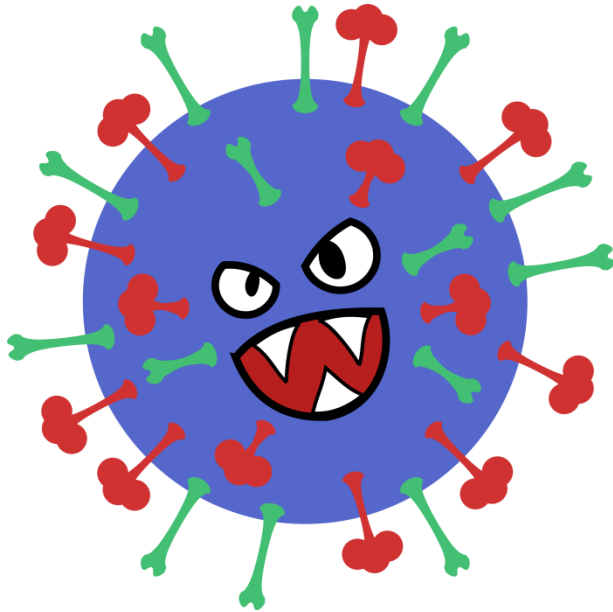
GLYCOPROTEIN



- Only in enveloped viruses
- Allows viral entry into host cell
- Targets specific type of host cells
- Aids in viral assembly

An Introduction to Viral Genomics

HOST-SHUT-OFF PROTEINS



- As name indicates
- Shut off host activities
- Only viral genes are processed

An Introduction to Viral Genomics

ANTI-HOST DEFENCE PROTEINS

- Number and type of these genes vary
- Prevent host defense mechanism from stopping viral replication

Virology

Terminologies used in Viral Genetics

Terminologies used in Viral Genetics

GENE

A segment of DNA molecule that encodes for synthesis of a single polypeptide



Terminologies used in Viral Genetics

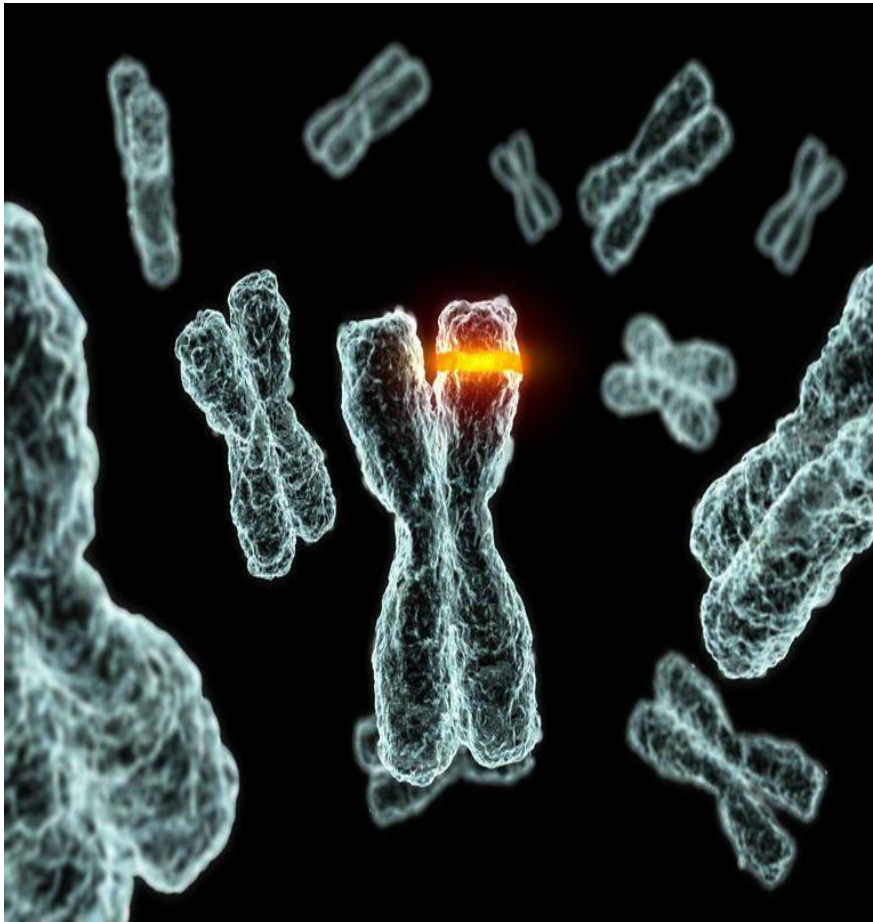
GENOME

- The entire DNA sequence of an individual/species



Terminologies used in Viral Genetics

MUTATION



- A permanent inheritable change in a single gene that results in the existence of two or more alleles occurring at the same locus

Terminologies used in Viral Genetics

PSEUDOTYPES

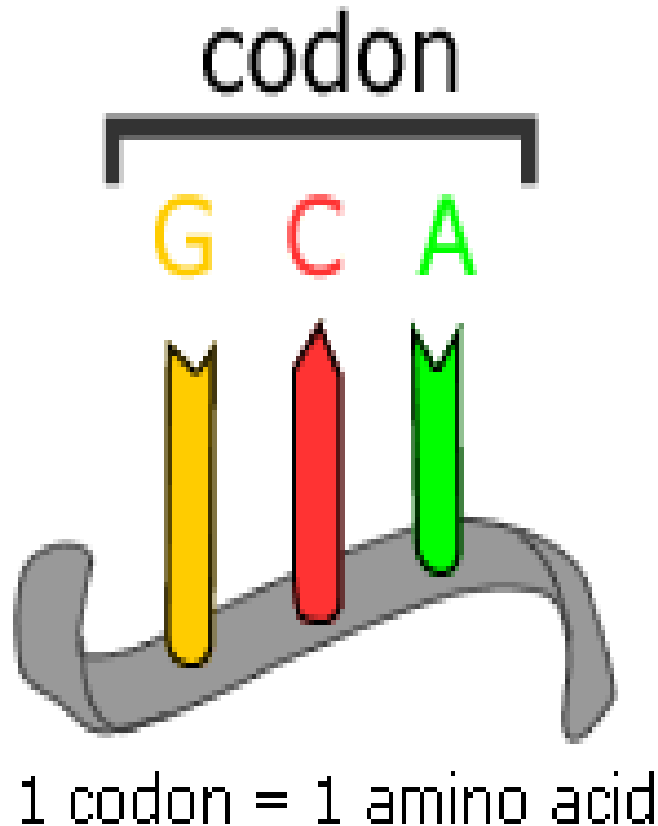
- viruses that have acquired the genotype of one parent and phenotype of another

Terminologies used in Viral Genetics

RECOMBINANTS

- A new combination of genes not found together on the chromosome in either parent
- A recombinant results from crossing over

Terminologies used in Viral Genetics



CODON

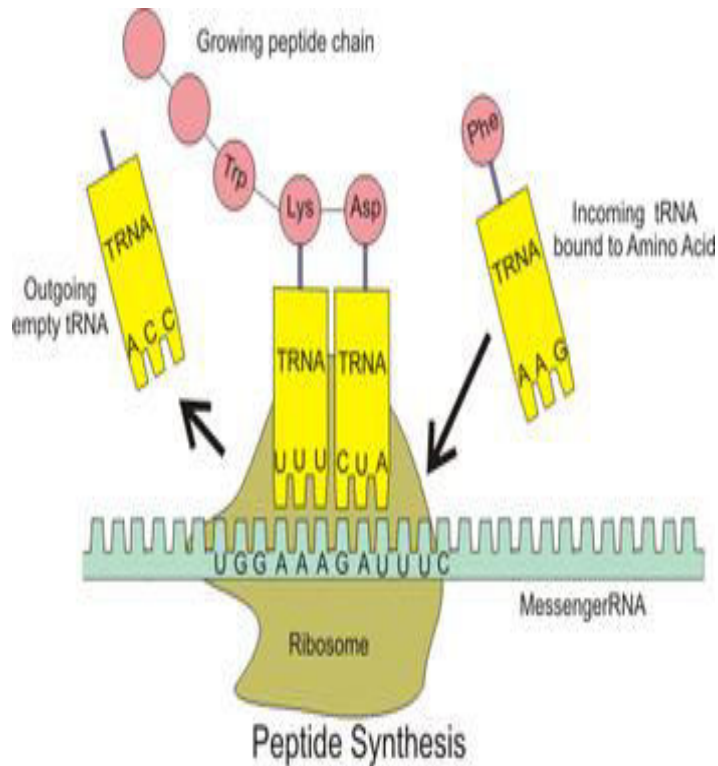
- A sequence of three bases in DNA or RNA that codes for a single amino acid
- Enables specific proteins to be made by specific genes

Terminologies used in Viral Genetics

ANTI-CODON

- A sequence of three bases in tRNA that is complementary to a codon in mRNA
- Enables tRNA to sequence amino acids in the order specified by mRNA

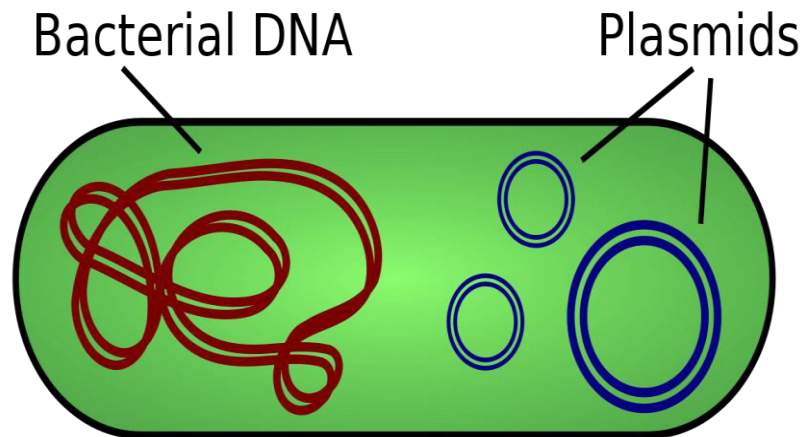
Terminologies used in Viral Genetics



RIBOSOMES

- Complexes of rRNA and protein in cytoplasm that serve as platforms for translation for mRNA into protein

Terminologies used in Viral Genetics



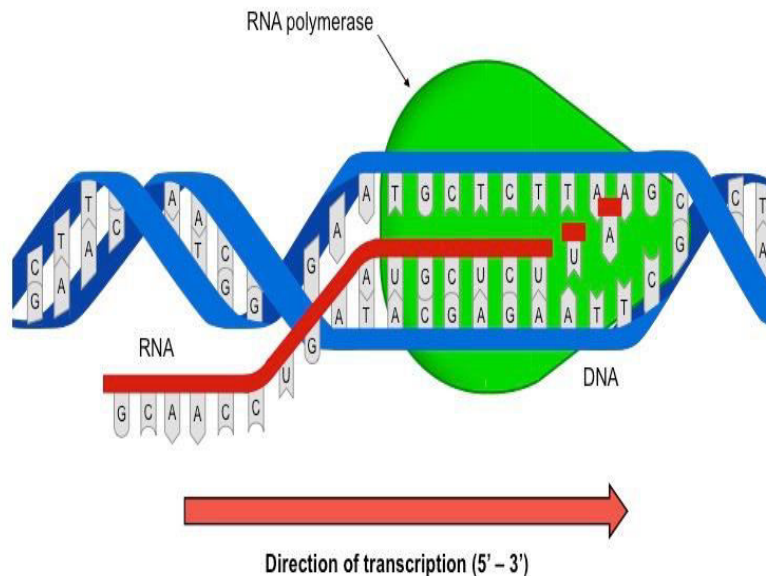
PLASMID

- Extrachromosomal circular DNA in bacteria
- Independently replicate and encode a product

Terminologies used in Viral Genetics

TRANSCRIPTION

- Synthesis of single-stranded RNA by RNA polymerase using DNA as a template
- The process in the nucleus whereby DNA is transcribed into mRNA



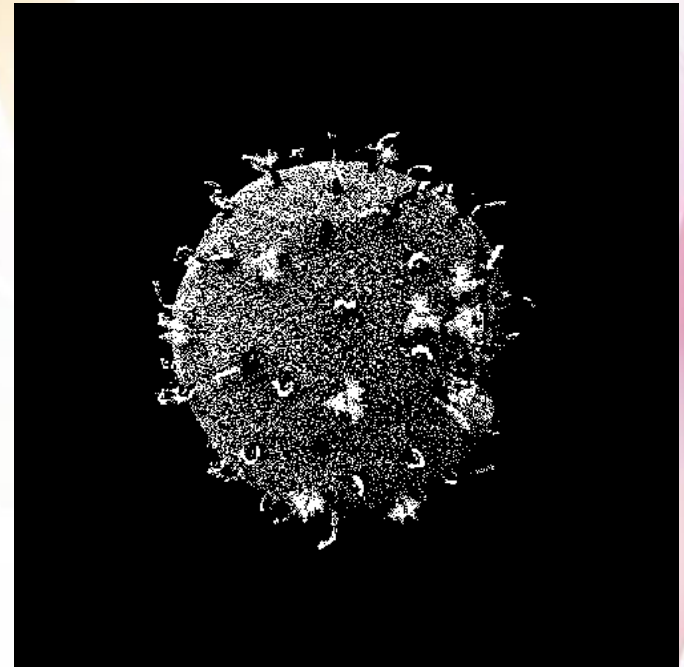
Terminologies used in Viral Genetics

TRANSLATION

- The process of translating the codon sequence in mRNA into polypeptides with the help of tRNA and ribosomes

Virology

Molecular Genetics of Viruses



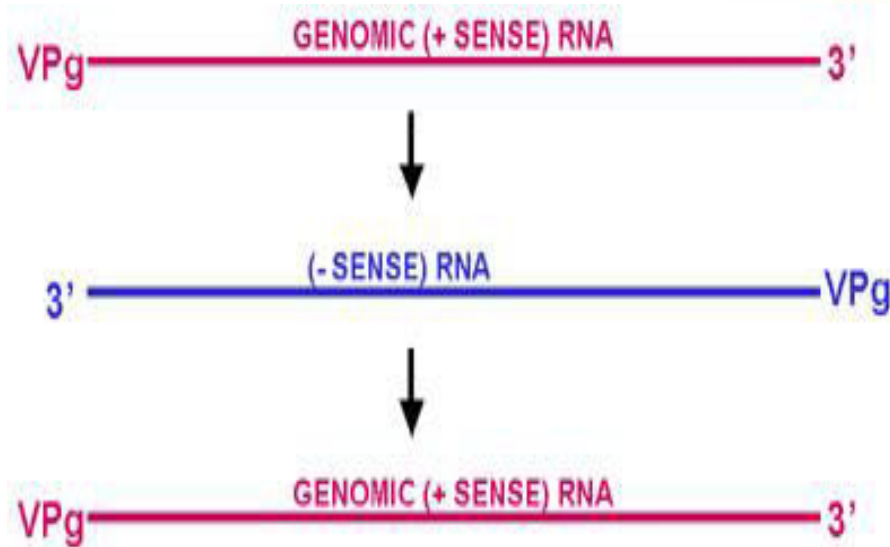
Molecular Genetics of Viruses

VIRAL GENETICS

- Viruses store their genetic info in six different types of genome
- Based on how nucleic acid is transcribed into viral mRNA

Molecular Genetics of Viruses

POSITIVE RNA STRAND



- RNA dependent RNA polymerase produces single stranded negative strand of RNA
- Then it produces positive strand of RNA
- Togavirus, Picornavirus etc.

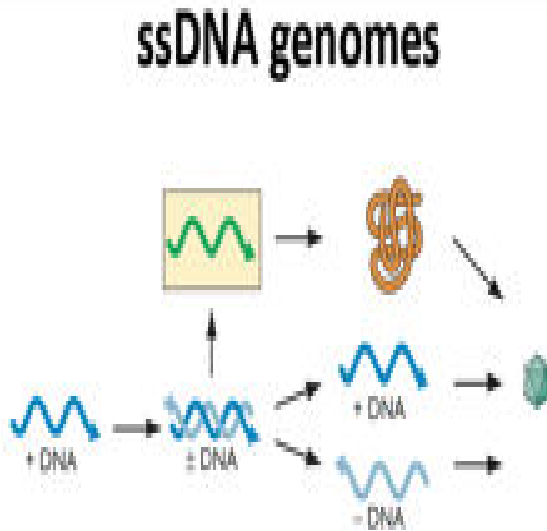
Molecular Genetics of Viruses

POSITIVE/ NEGATIVE DNA STRANDS

- Transcribed into mRNA
- DNA dependent DNA polymerase copies the strand to DNA strands
- RNA dependent RNA polymerase converts RNA into mRNA
- Then translated into viral proteins

Molecular Genetics of Viruses

POSITIVE SINGLE STRANDED DNA



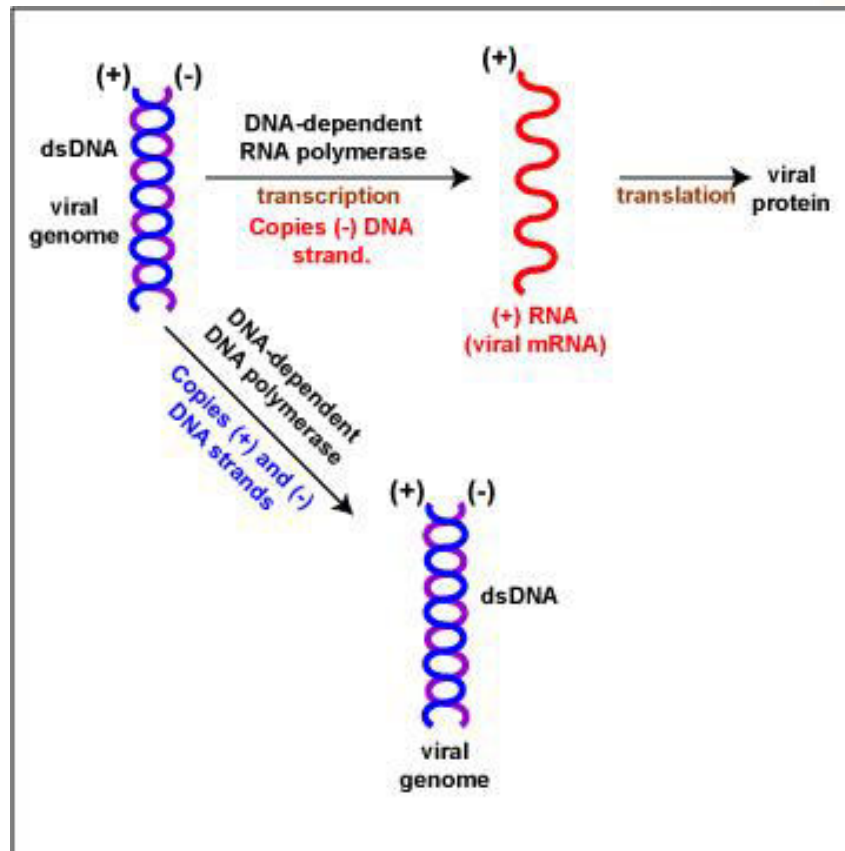
- DNA dependent DNA polymerases forms a double stranded DNA strand

DNA dependent RNA polymerase forms the mRNA

- Example, Parvovirus

Molecular Genetics of Viruses

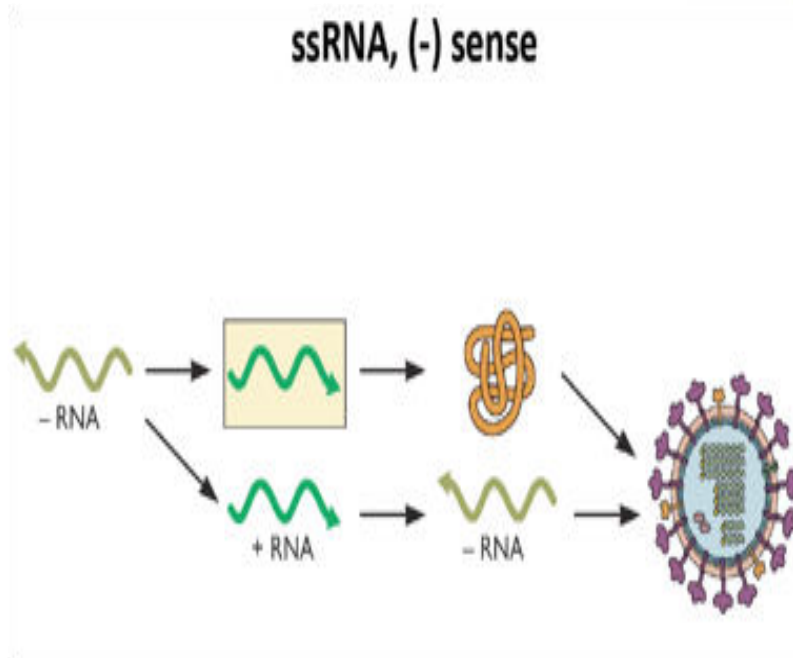
POSITIVE/ NEGATIVE DOUBLE STRANDED DNA



- RNA dependent RNA polymerases forms positive and negative RNA strand
- RNA dependent RNA polymerase forms the mRNA
- Translated into proteins

Molecular Genetics of Viruses

NEGATIVE SINGLE STRANDED DNA



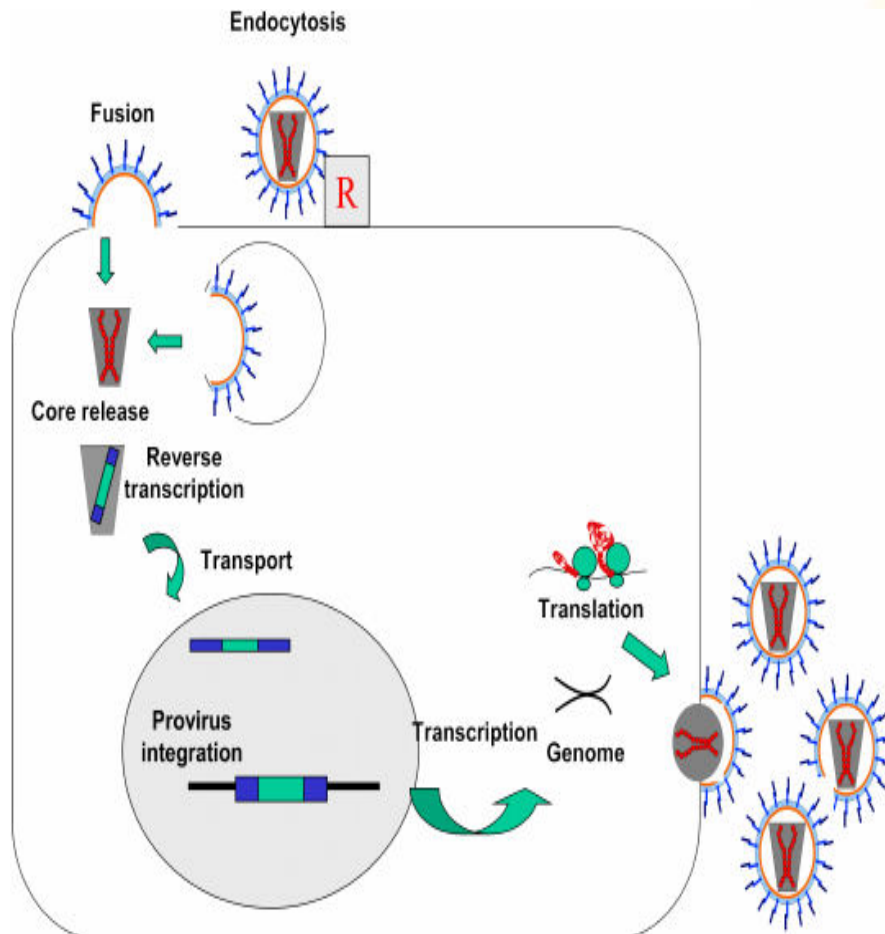
- RNA dependent RNA polymerases forms positive RNA strand
- RNA dependent RNA polymerase forms the mRNA
- Translated into proteins

Molecular Genetics of Viruses

POSITIVE RNA RETEROVIRUSES

- Reverse transcriptase enzyme
- Produces single stranded negative sense DNA
- DNA dependent DNA Polymerase copies to form double stranded DNA

Molecular Genetics of Viruses

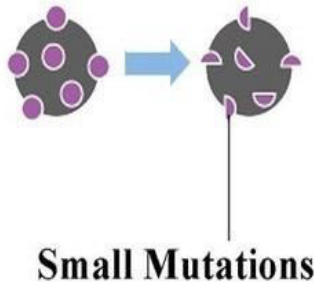


- DNA dependent RNA polymerase produces single stranded positive RNA genomes (mRNA)
- Examples, HIV-1, HIV-2, HTLV-1

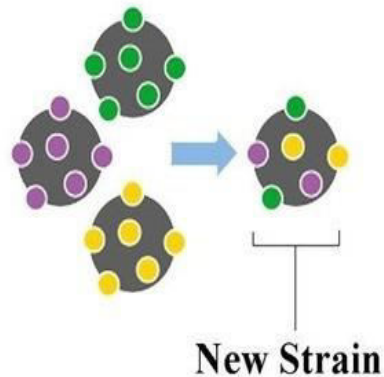
Molecular Genetics of Viruses

VIRAL GENETICS

Mutation
Antigenic drift



Antigenic shift

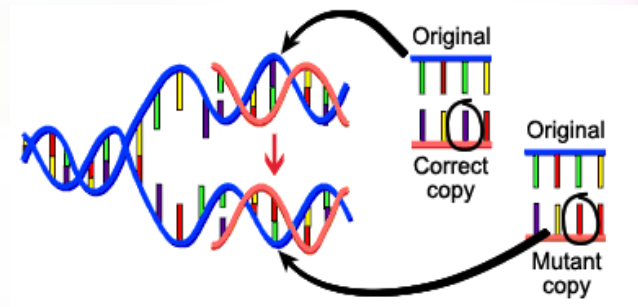


- Viruses usually grow rapidly, producing more chance of mutation
- Genetic changes by various mechanisms:
 - Genetic drift (Bases change)
 - Antigenic Shift (Recombination)

Molecular Genetics of Viruses

MUTATION

- Spontaneous Mutation that arise naturally during replication
- Induced Mutation that is caused by physical or chemical means



Virology

Genetic Techniques in Virology



Genetic Techniques in Virology

TYPES

- In-vitro (Physical analysis)
- In-vivo (viral genotype and phenotype)



Genetic Techniques in Virology

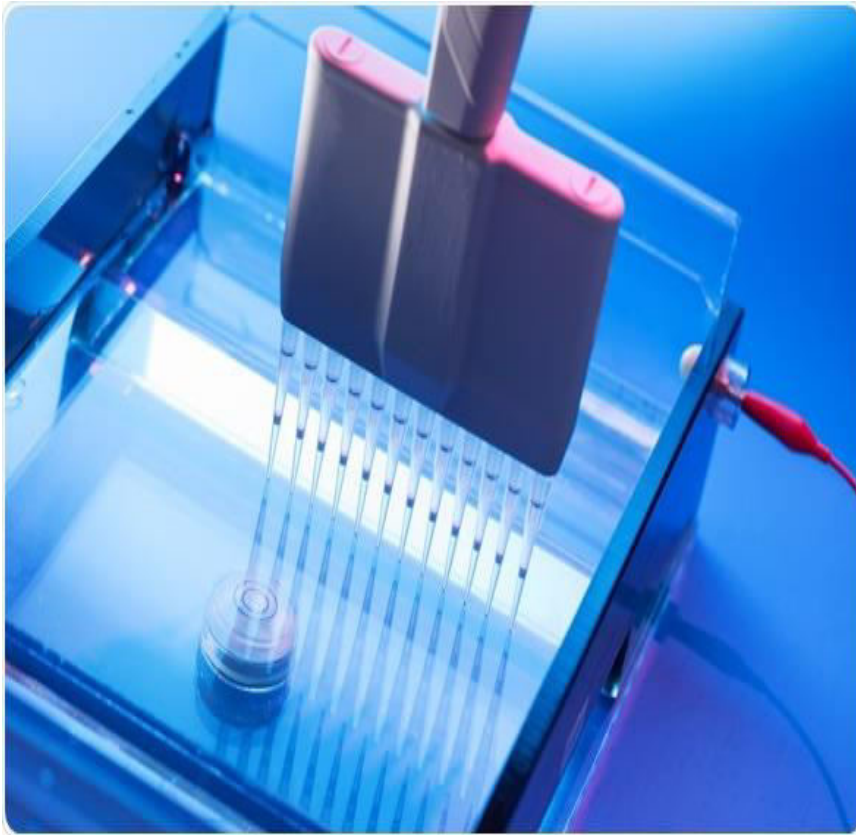
IN-VITRO



- Physical analysis
- Isolation of nucleic acids from virus preparations
- Direct analysis of DNA using restriction endonucleases enzyme
- Use of vectors in DNA sequencing
- Electron microscopy
- RNA sequence analysis using Reverse transcriptase

Genetic Techniques in Virology

MOLECULAR ANALYSIS



- Molecular cloning
- Gel electrophoresis
- Hybridization of complementary nucleotide sequences

Genetic Techniques in Virology

PHENOTYPIC ANALYSIS

- Standard technique in virology
- Cell-free extracts to translate mRNAs
- Transfection tests
- In-vitro protein formation
- In-vivo protein replication

Virology

LATENT VIRUSES

Latent viruses

MEANING

- The pathogenic viruses having the ability to lie dormant (or latent) within host cell
- Called lysogenic part of lifecycle

Latent viruses

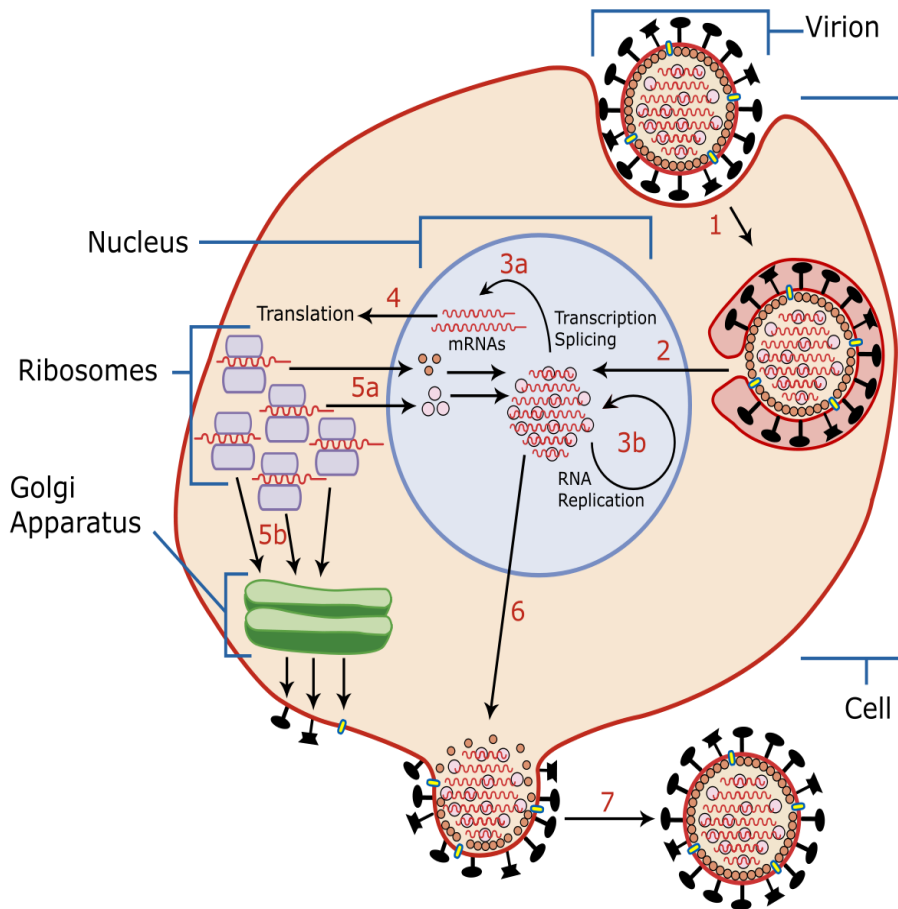
LATENCY



- The phase in which viral proliferation ceases after initial infection
- Viral genome not fully eradicated

Latent viruses

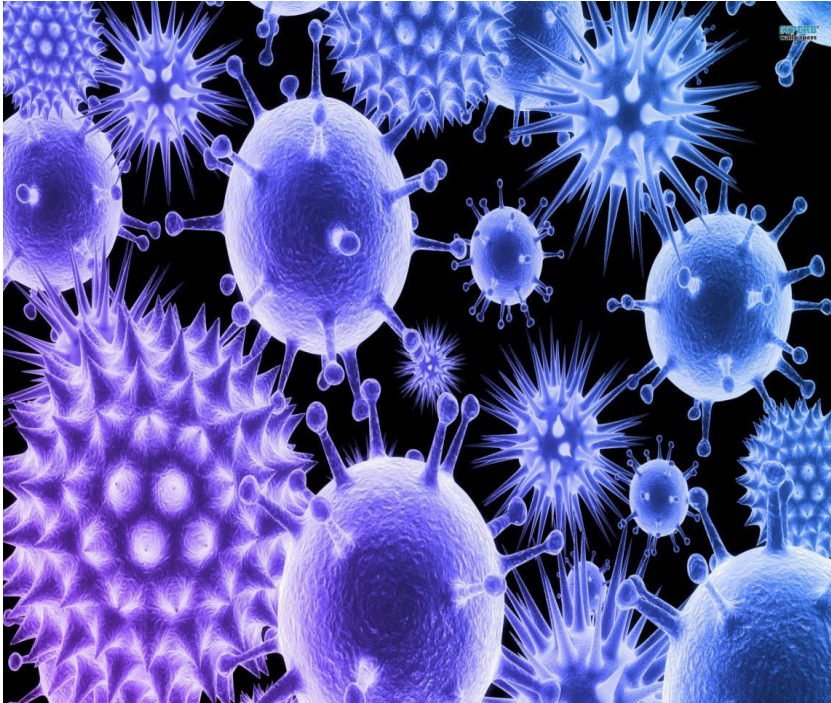
LYTIC PHASE



- The virus activates
- Continues to multiply increases its progeny
- Clinical signs appear

Latent viruses

MECHANISMS



1. Episomal Latency
2. Proviral Latency
3. Maintaining Latency

Latent viruses

EPISEMAL LATENCY

- Use of genetic episomes
- Viral genes float in cytoplasm or nucleus
- More vulnerable to host immune system
- Example, chicken pox and herpes simplex viruses

Latent viruses

PROVIRAL LATENCY

- Virus genome is integrated into the DNA of a host cell
- Automatic host cell division during viral replication
- Viral genome enters nucleus
- Example; HIV

Latent viruses

MAINTAINING LATENCY

- For continuity of infection
- By genes expressed during latency
- To evade host immune system
- To protect virus genome to be digested by cellular enzymes

Virology

EXAMPLE OF LATENT VIRUS

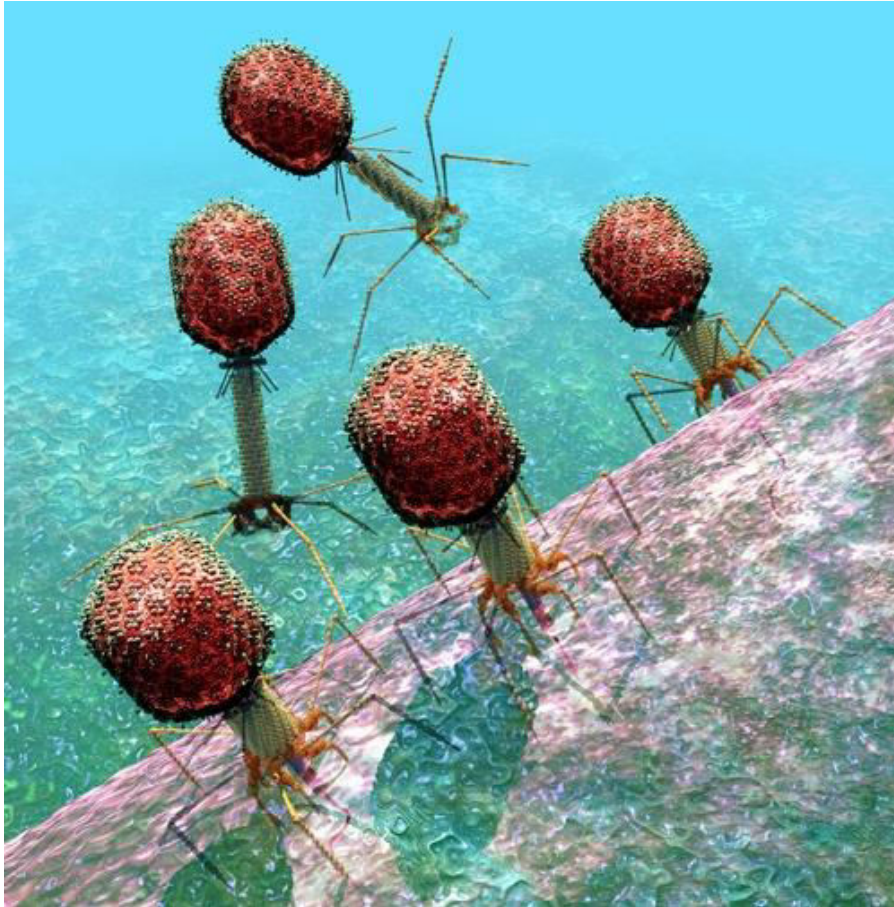
Example of Latent viruses

LATENT VIRAL INFECTIONS

1. Chronic persistent infections
2. Latent Occult Viral infections

Example of Latent viruses

CHRONIC PERSISTENT INFECTIONS



- Enveloped viruses
- e.g. *Paramyxoviruses*, *Retroviruses*, *Arenavirus*
- No disruption of cellular functions
- Promotes shredding of viral Antigens from the cell surface

Example of Latent viruses

LATENT OCCULT VIRAL INFECTIONS

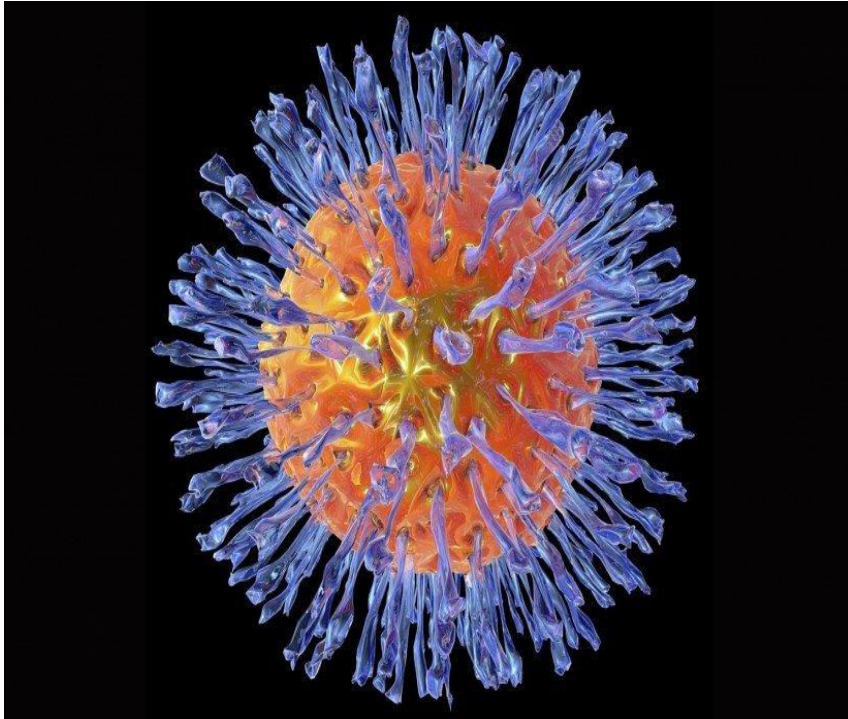
Some viruses reappear to cause acute disease

Three examples

- i. HSV
- ii. Adenovirus
- iii. SSPE

Example of Latent viruses

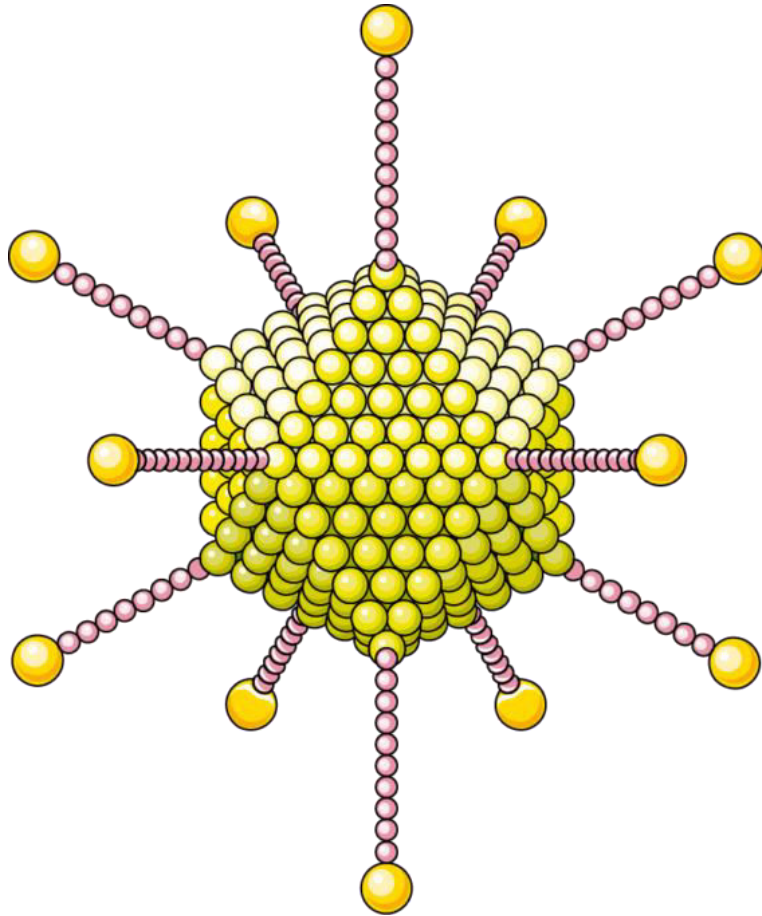
HSV



- Herpes Simplex Virus
- Primary infection in early age
- Found only during recurrent acute episodes
- Virus can be isolated from tissue homogenates

Example of Latent viruses

ADENOVIRUS



- Self-limiting
- Persistent infection of tonsils and adenoids
- Failure in viral isolation
- Absence of mature virion
- DNA in linear episomal form

Example of Latent viruses

MEASLES VIRUS

- Incomplete viral production
- Immature nucleocapsids produced
- Disease depends on immune response
- May cause tumorigenesis

Virology

VIRAL MUTANTS

Introduction

Viral Mutants

MUTANT

- A mutant is an organism or a new genetic character arising or resulting from an instance of mutation, which is generally an alteration of the DNA/RNA sequence of the genome or chromosome of an organism

Viral Mutants



RELEVANT TERMS

- Strain
- Type
- Variant
- Mutant

Viral Mutants



DIFFERENCE IN RELEVANT TERMS

- Strain: lines or isolate of same virus
- Type: serotypes of same virus
- Variant: virus, whose phenotype varies from original virus
- Mutant: virus showing genetic alteration

Viral Mutants

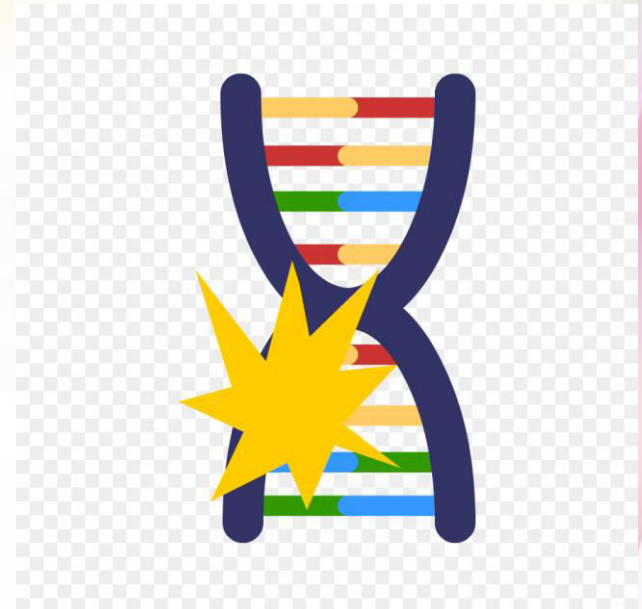
ORIGIN OF MUTATION

- **Spontaneous mutation:**
 - Very high to low rate of mutation
 - Due to difference in type of genome
 - Error rate higher in RNA dependent RNA polymerase
- **Induced Mutation**
 - In-vivo
 - or in-vitro mutagens

Viral Mutants

MAIN REASON

- Replication Error
- Chemicals
- Radiations



Viral Mutants

TYPES OF MUTANT VIRUSES

- Biochemical markers: drug-resistance mutations
- Deletions: similar to non-sense mutations
- Host range: refers to whole host animal
- Non-sense mutation: alteration of coding sequence
- Plaque morphology
- Temperature sensitive
- Cold sensitive
- Reverents

Virology

VIRAL MUTANTS

Spontaneous Mutation



Spontaneous Mutation

RATE

- High to low
- Very high in HIV
- Very low in Herpesvirus
- Resulting mutation in cellular DNA

Spontaneous Mutation

REASON

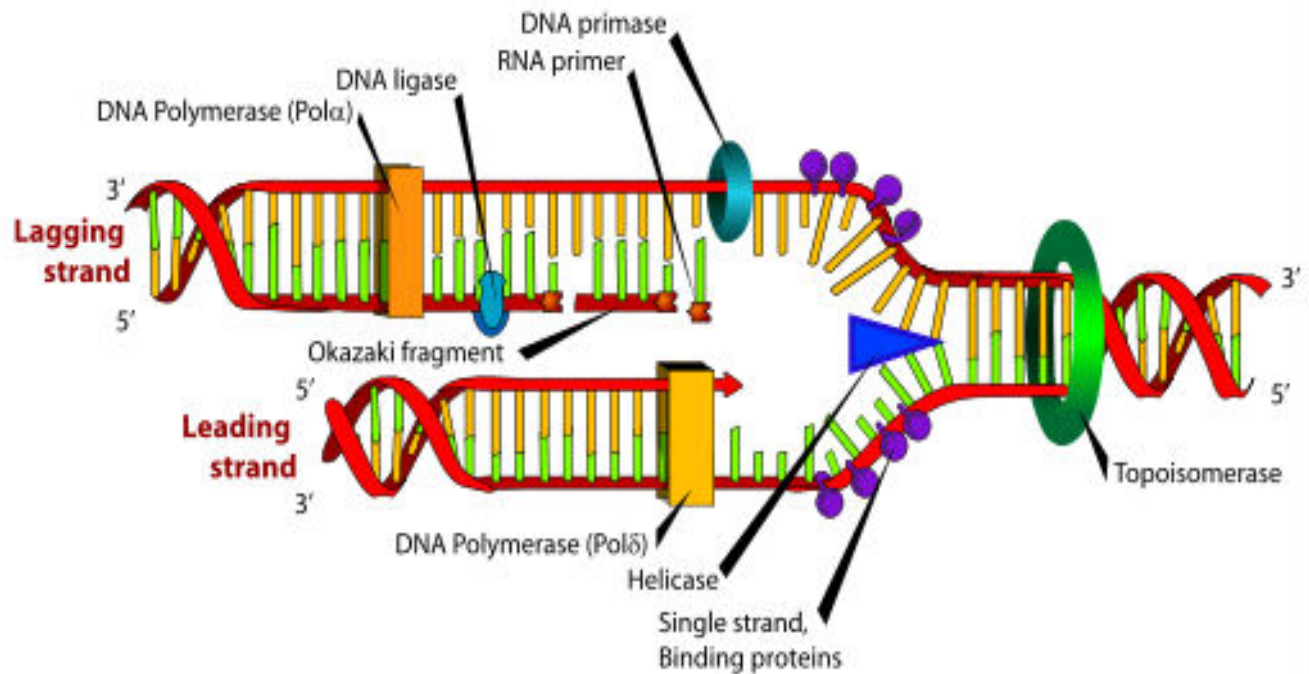
- Mechanism of genome replication
- RNA dependent RNA polymerase shows higher error rate than DNA dependent DNA polymerase



Spontaneous Mutation

DO RNA VIRUSES HAVE PROOF-READING?

- Yes!
- Yet they show higher rates of mutation than DNA viruses



Spontaneous Mutation

MUTATION: GOOD OR BAD?

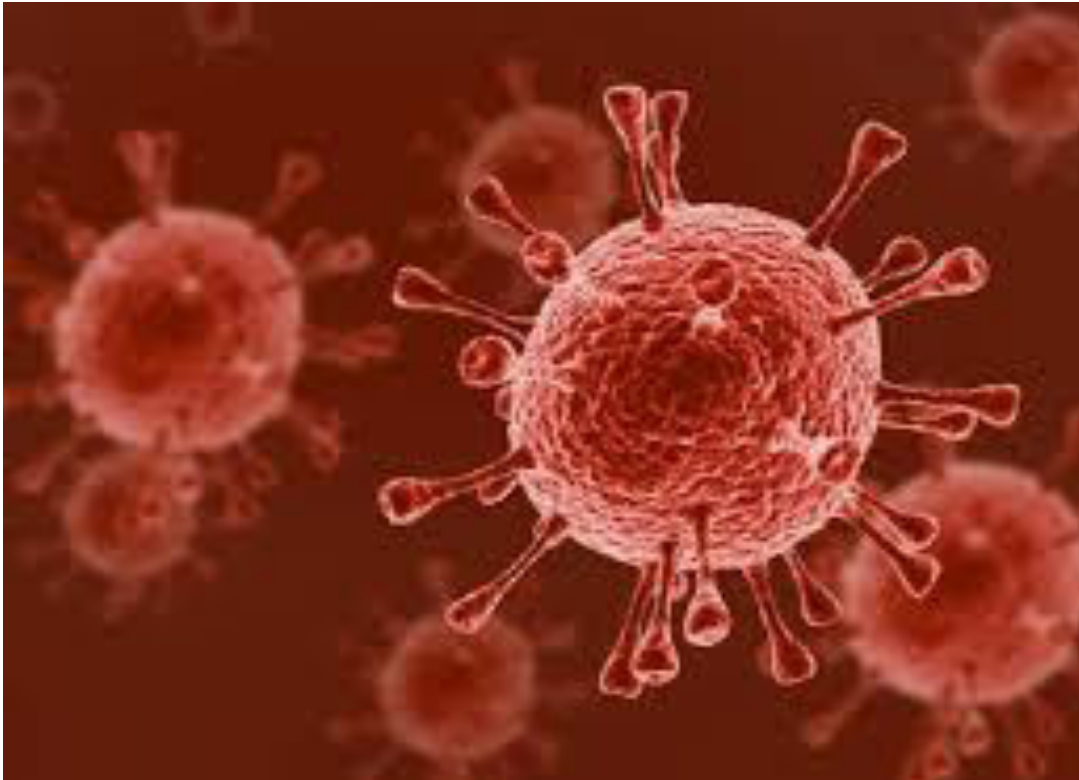
- Sometimes good for the virus
- Helps escape the immune system
- But some mutations may be deleterious
- Due to production of defective particles



Spontaneous Mutation

HIV

- Genome is 9.7kb long
- Mutation rate can be 0.9 to 9.7 mutations in every genome copied



Spontaneous Mutation

QUASISPECIES

- Mixture of molecular variants
- Usually the wild-type virus dominates
- Mostly non-infectious
- Important in viral evolution

Spontaneous Mutation

REASON?

- Tautomerism: base change due to repositioning of Hydrogen atom
- De-purination: Loss of purine base
- De-amination and hydrolysis: change of normal base to atypical base
- Slipped strand mis-pairing: Denaturation of new strand
- Replication slipping: error during replication process

Virology

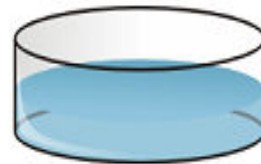
VIRAL MUTANTS

Induced Mutation

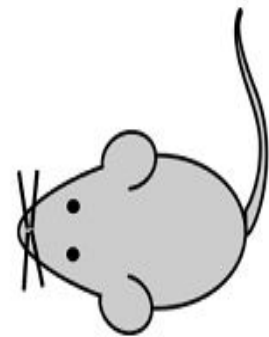
Induced Mutation

TYPES:

- In-vitro
- In-vivo



In Vitro



In Vivo

Induced Mutation



IN-VITRO

- Chemically modify nucleic acids
- Do not require replication
- E.g: nitrous acid, hydroxylamine and alkylating agents like nitrosoguanidine

Induced Mutation

IN-VIVO

- Requires Metabolically active nucleic acid that is replicating
- Compounds incorporated into newly synthesized nucleic acids
- E.g: 5-bromouracil resulting erroneous base pairing, intercalating bases that stack between bases, UV irradiation

Induced Mutation

CHEMICAL REASON

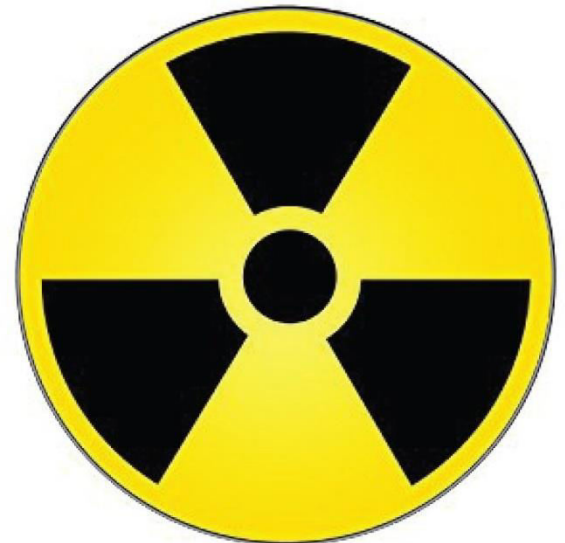


- Hydroxylamine
- Base analogs
- Alkylating agents
- DNA intercalating substances
- DNA cross linkers
- Oxidative damage
- Nitrous acid

Induced Mutation

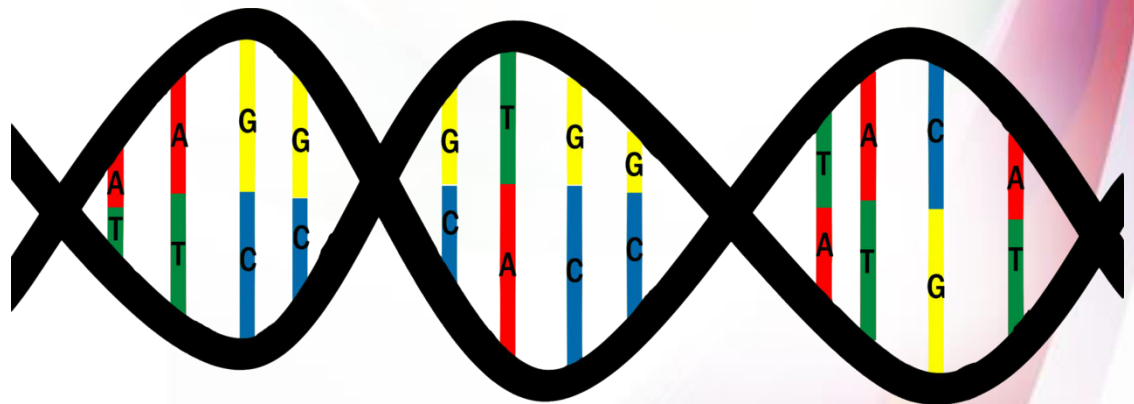
RADIATION REASON

- Ultraviolet rays
- Ionizing radiation



Virology

Types of Mutant Viruses



Types of Mutant Viruses

TYPES

- Biochemical Markers
- Deletions
- Host Range
- Non-Sense
- Plaque morphology
- Temperature sensitive
- Cold Sensitive
- Reverents

Types of Mutant Viruses

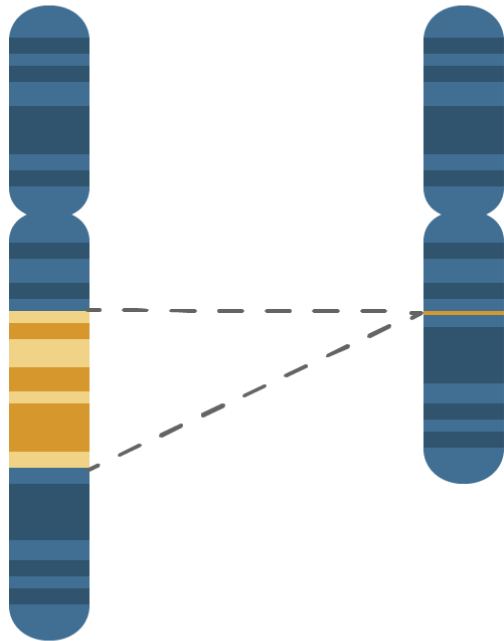
BIOCHEMICAL MARKERS



- Drug resistance mutations
- Altered virulence
- Polymorphism
- Altered proteins or nucleic acids
- Sensitivity to inactivating agents

Types of Mutant Viruses

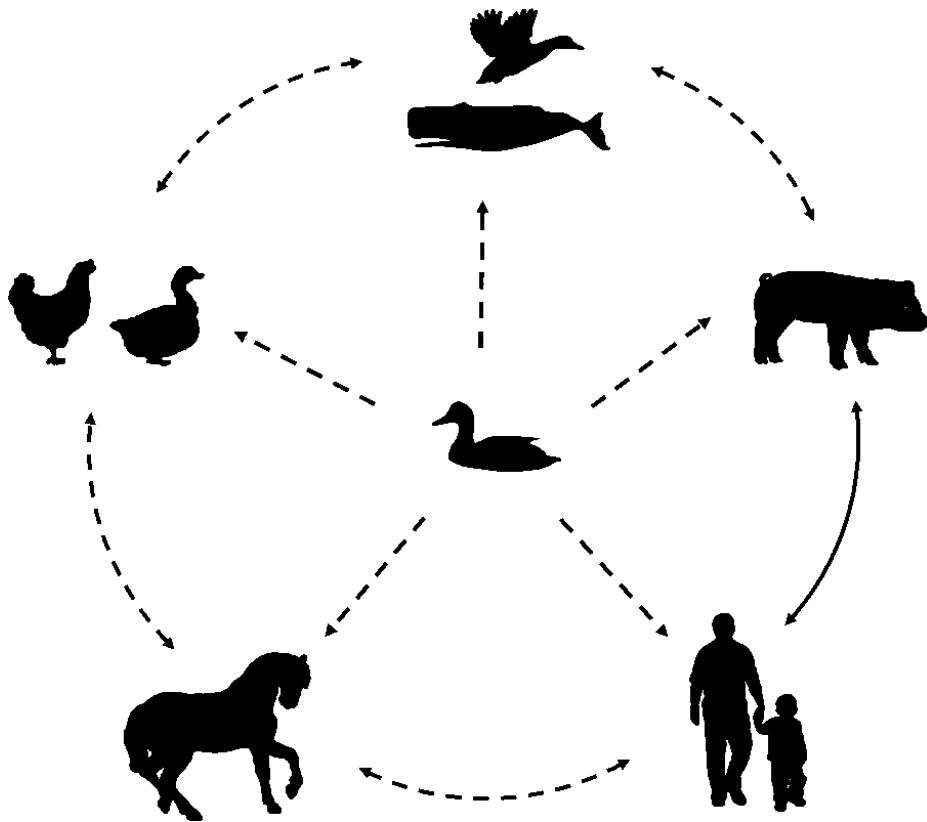
DELETIONS



- Similar to nonsense mutation
- May include one or more viral genes
- Involve non-coding regions of genome
- Defective-interfering particles (DIP)
- Can only revert to wild-type by recombination

Types of Mutant Viruses

HOST RANGE

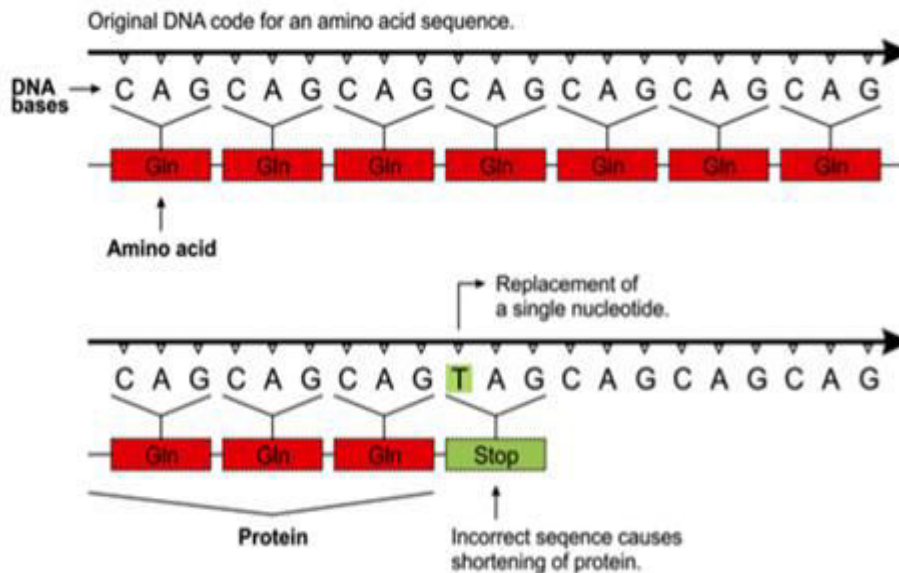


- Refers to whole host animal or cell type in vitro
- Conditional mutants

Types of Mutant Viruses

NON-SENSE

- Alteration of coding sequence of one of the three stop codons (UAG, UAA, UGA)
- Termination of translation
- Low reversion frequency

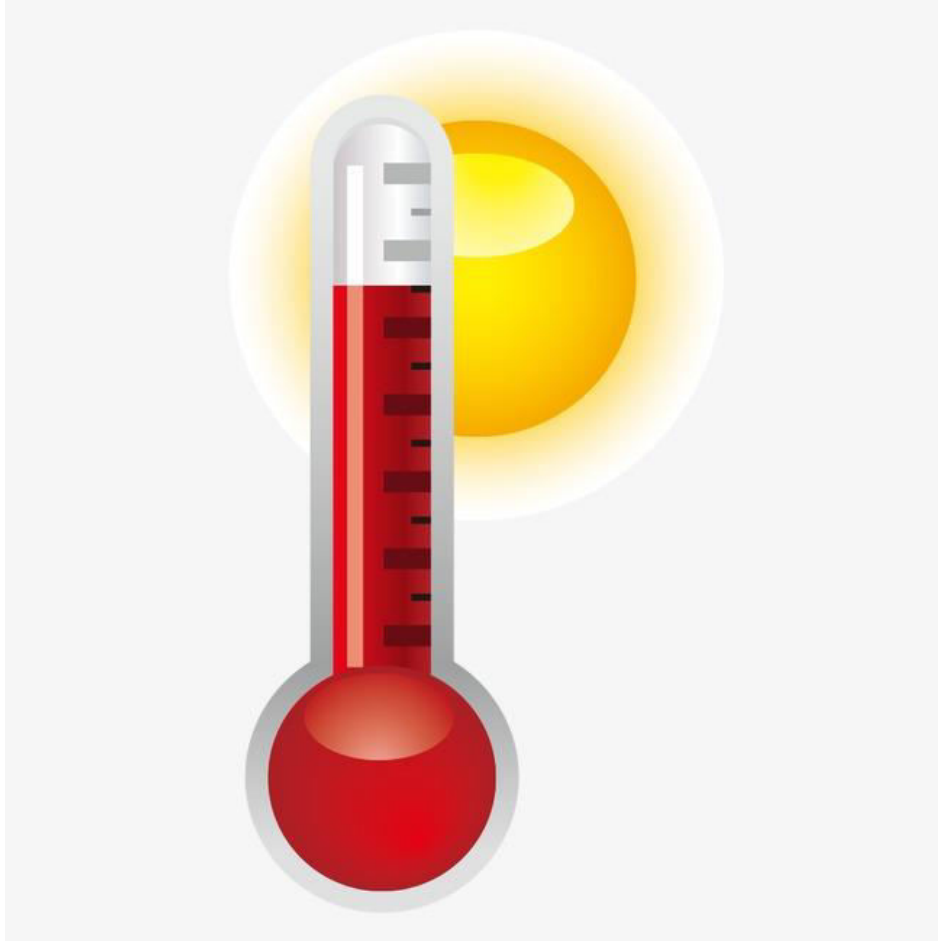


Types of Mutant Viruses

PLAQUE MORPHOLOGY

- Large plaque mutants or small plaque mutants
- Plaque size referred to temp. sensitivity
- Useful as unselected markers in multi-factorial crosses

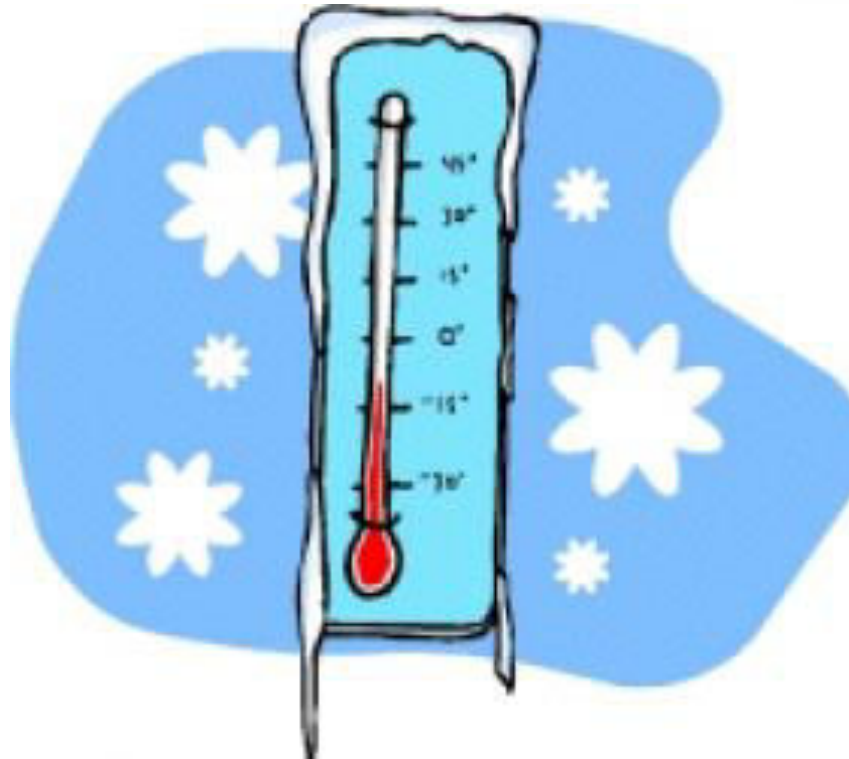
Types of Mutant Viruses



TEMPERATURE SENSITIVE

- Useful for isolation of conditional-lethal mutations
- Usually occur by mis-sense mutation in proteins
- Some activities work on non-permissive temperatures

Types of Mutant Viruses



COLD SENSITIVE

- Opposite to temperature sensitive
- Useful in bacteriophages and plant viruses as their cells can be propagated at low temperatures

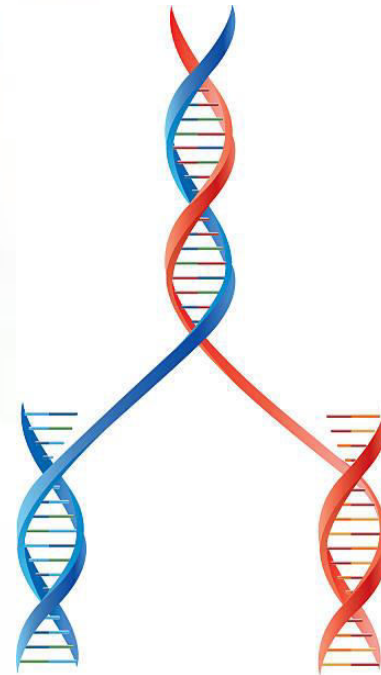
Types of Mutant Viruses

REVERTANTS

- Valid type of mutation
- Reverse mutations
- 'back mutation' or
'compensatory
mutation'

Virology

VIRAL RECOMBINATION



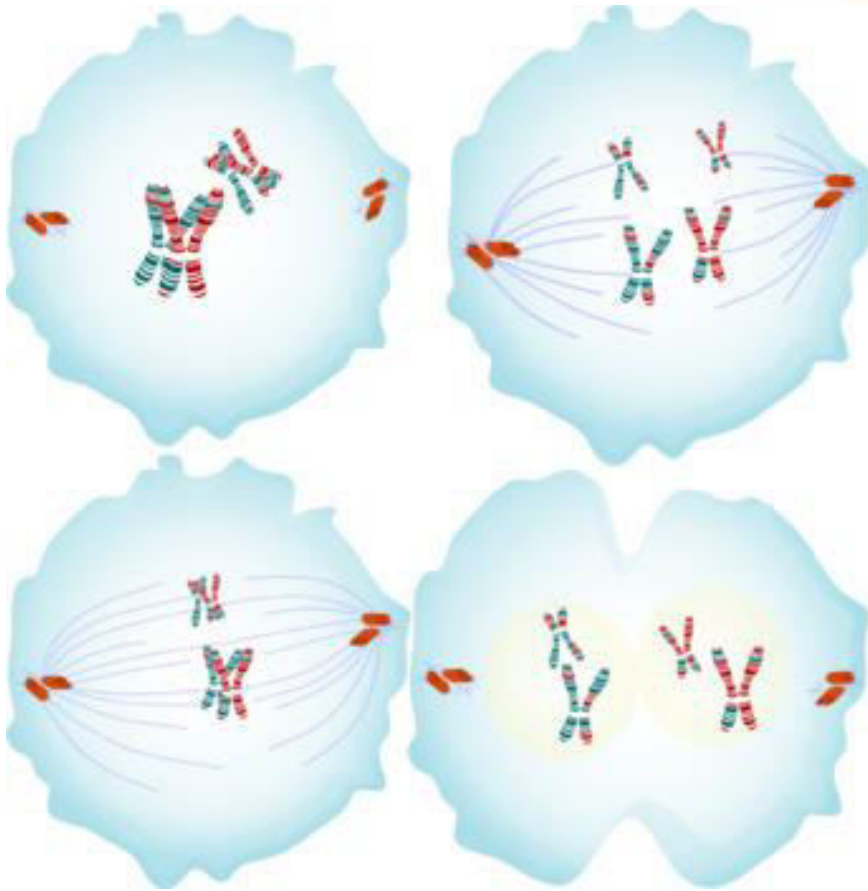
Viral Recombination

DEFINITION

- The exchange or transfer of genetic material between different but closely related viruses infecting the same cell

Viral Recombination

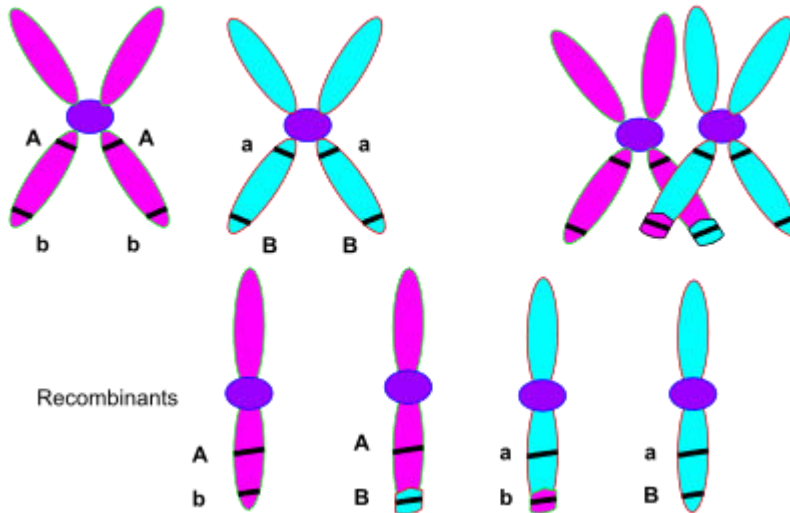
HOW?



- Can also occur between virus and host
- Recombinants contain new genetic information

Viral Recombination

INTRAMOLECULAR RECOMBINATION



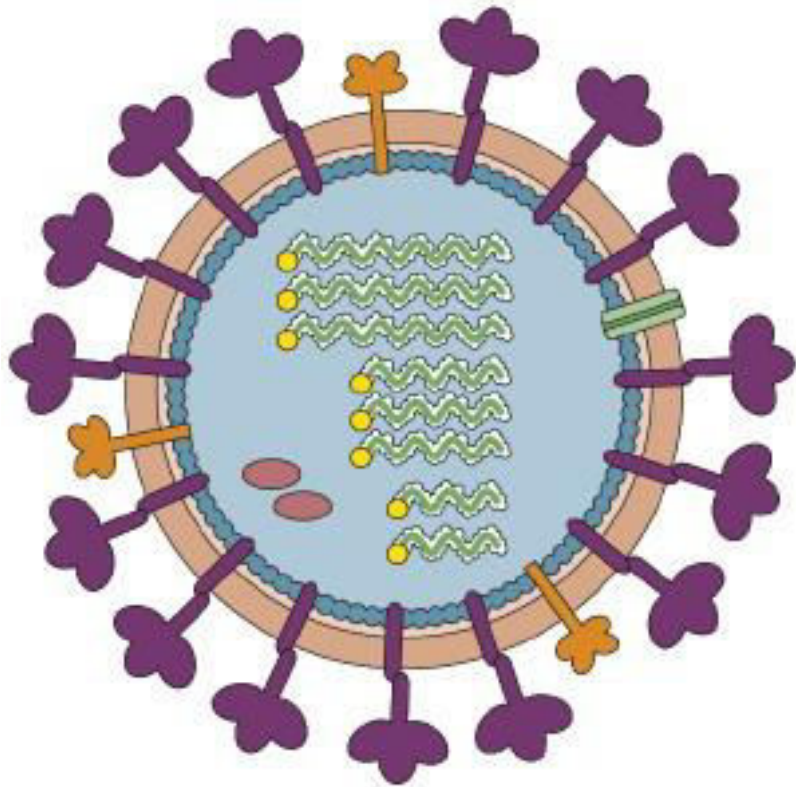
- Usually in DNA viruses
- Dissociation and re-establishment of covalent bonds with nucleic acids
- Template switching mechanism

Viral Recombination

EXAMPLES

- Picornavirus
- Togavirus
- Corona virus
- Influenza A virus

Viral Recombination



RNA VIRUSES

- Randomly with segmented genome
- E.g orthomyxovirus, reovirus, bunyavirus
- Genome segments show recombinations

Viral Recombination

GENETIC REACTIVATION

- Infectious progeny produced from parents, in which one or both are noninfectious
- Mixed infection of cell



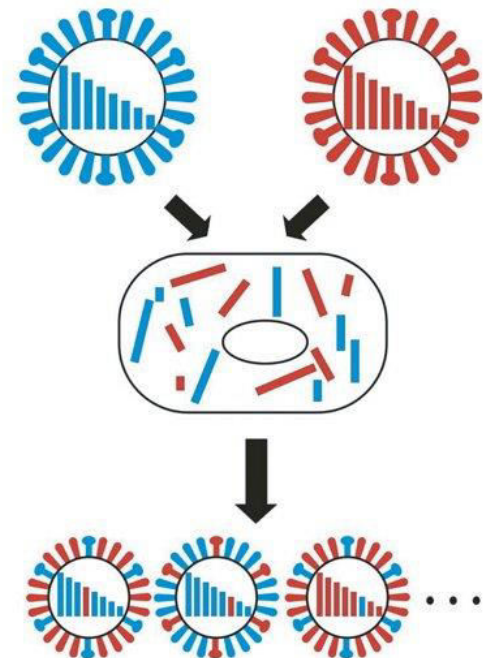
Viral Recombination

ADVANTAGE

- Genetic variability in nature
- Rapid adaptation of viruses
- Virulence changes
- Epidemiology of viruses (e.g. Influenza A)

Virology

GENETIC INTERACTIONS BETWEEN VIRUSES

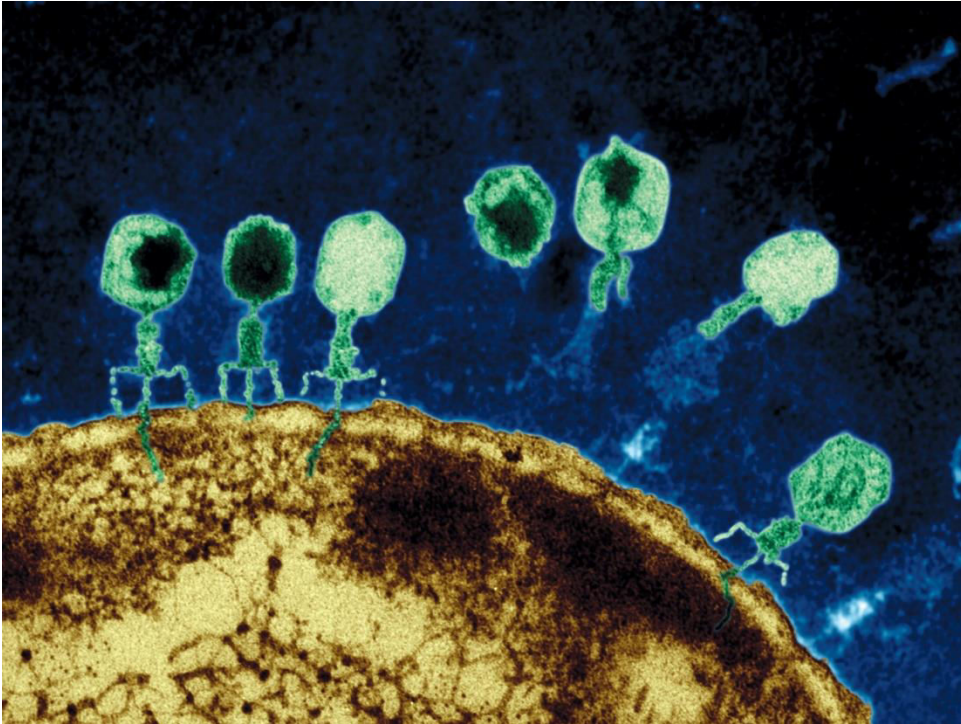


Genetic Interactions between viruses

WHEN?

- Naturally
- More than one infection in host

Genetic Interactions between viruses



Super-infection

- Mixed infection in cell culture
- Two purposes:
 - **COMPLEMENTATION**
 - **RECOMBINATION**

Genetic Interactions between viruses

COMPLEMENTATION

- Assignment of mutants to functional groups
- Resulting production of one or both parent viruses
- One of the viruses provides functional gene product
- Allows functional analysis of unknown mutations

Genetic Interactions between viruses

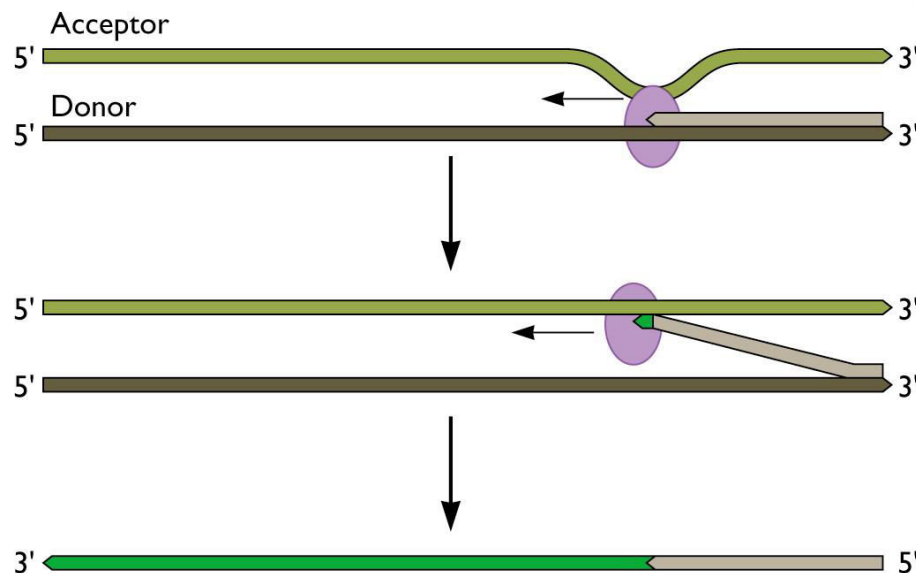
TYPES OF COMPLEMENTATION

- Allelic (intragenic) complementation where different mutants have defects in same protein
- Non-allelic (intergenic) results from mutants with defects in different genes

Genetic Interactions between viruses

RECOMBINATION

- Physical interaction of viral genomes resulting gene combinations not present in either parents
- Three mechanisms: Intra-molecular recombination by strand breakage and re-ligation, intra-molecular recombination by copy-choice and Reassortment



Genetic Interactions between viruses

INTRA-RECOMBINATION BY STRAND BREAKAGE AND LIGATION

- Occurs in all DNA and RNA viruses that replicate via DNA intermediate
- Mediated by cellular enzymes

Genetic Interactions between viruses

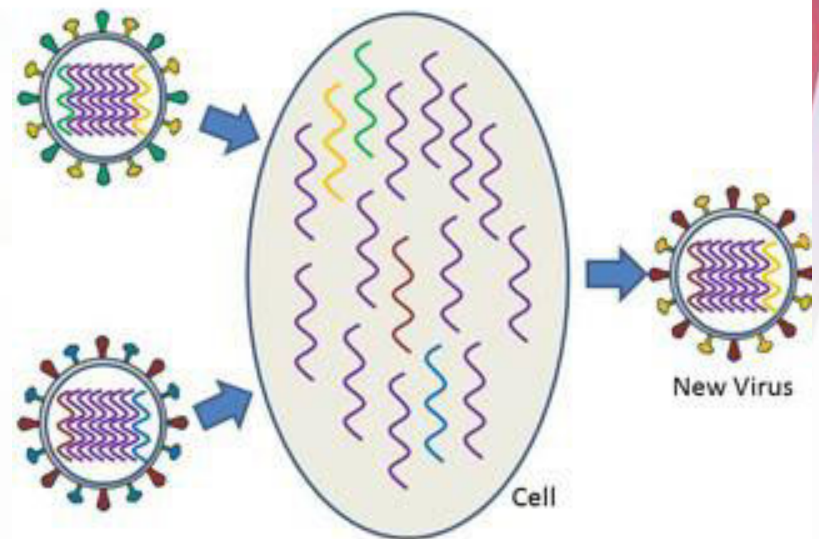
INTRA-MOLECULAR RECOMBINATION BY COPY CHOICE

- In RNA viruses
- During switching of viral polymerase
- Cellular enzymes involved
- Defective Interfering

Genetic Interactions between viruses

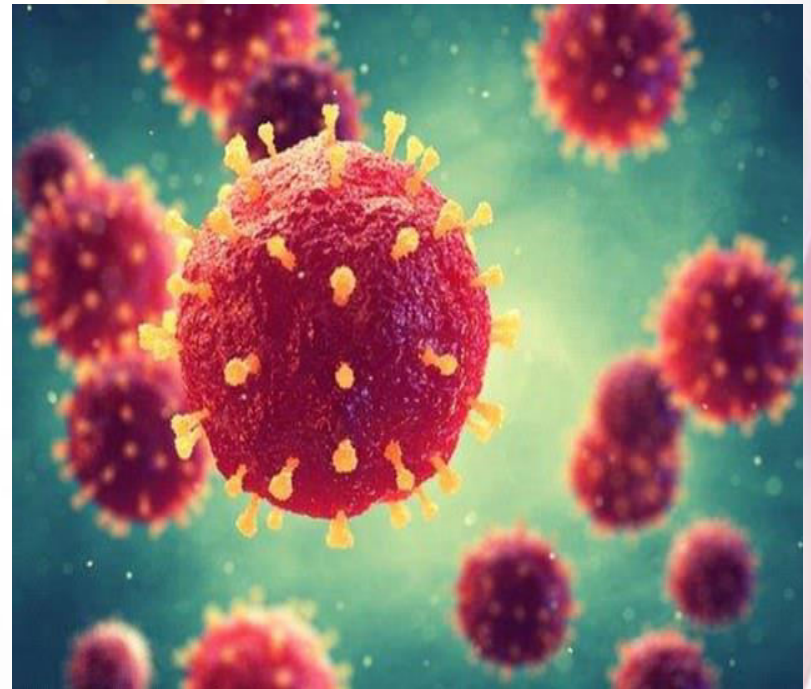
REASSORTMENT

- Random shuffling
- Progeny viruses receive at least one of the parent gene segment
- Packaging mechanism not understood



Virology

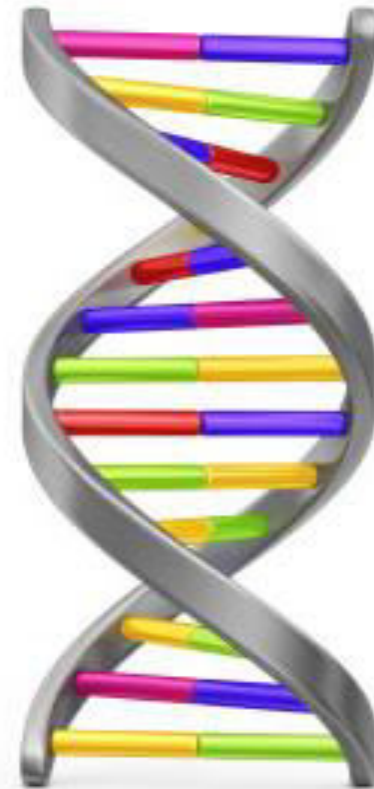
NON-GENETIC INTERACTIONS BETWEEN VIRUSES



Non-Genetic Interactions between viruses

EUKARYOTIC CELLS

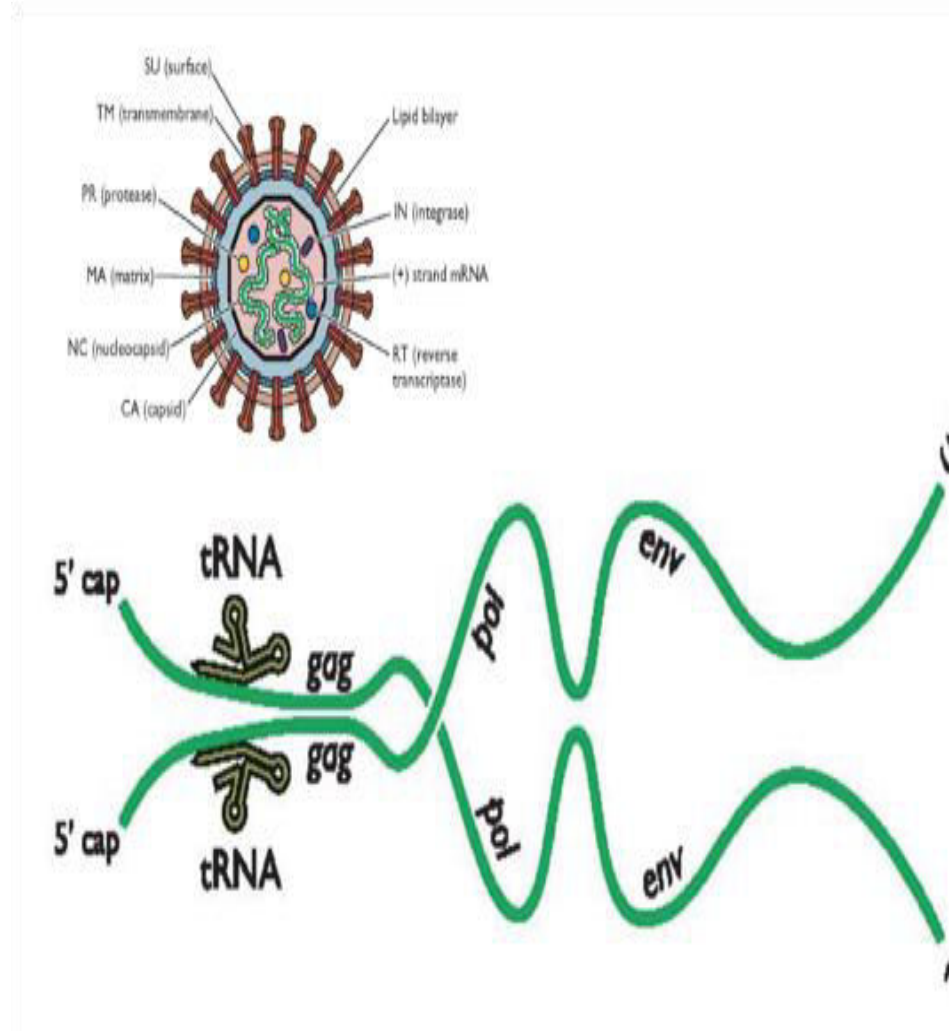
- Diploid genome
- Two copies of each chromosome
- Each having own allele of same gene
- Difference in allelic markers at loci



Non-Genetic Interactions between viruses

RETROVIRUS

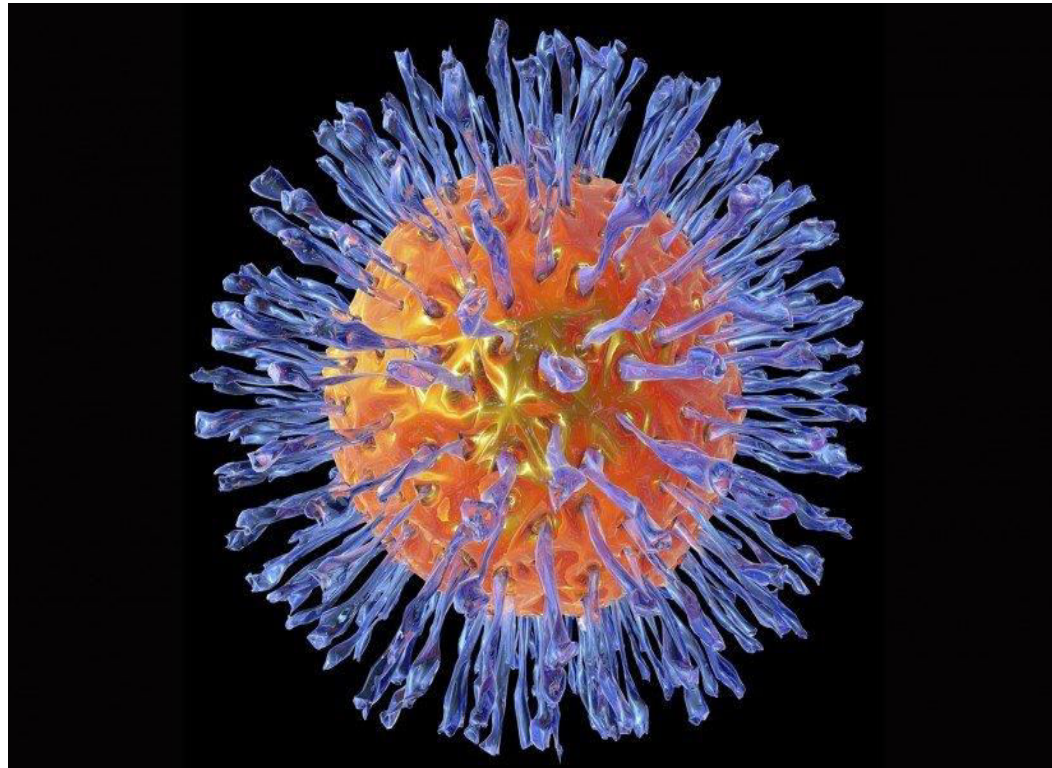
- Truly diploid
- Two copies of entire genome



Non-Genetic Interactions between viruses

HERPESVIRUS

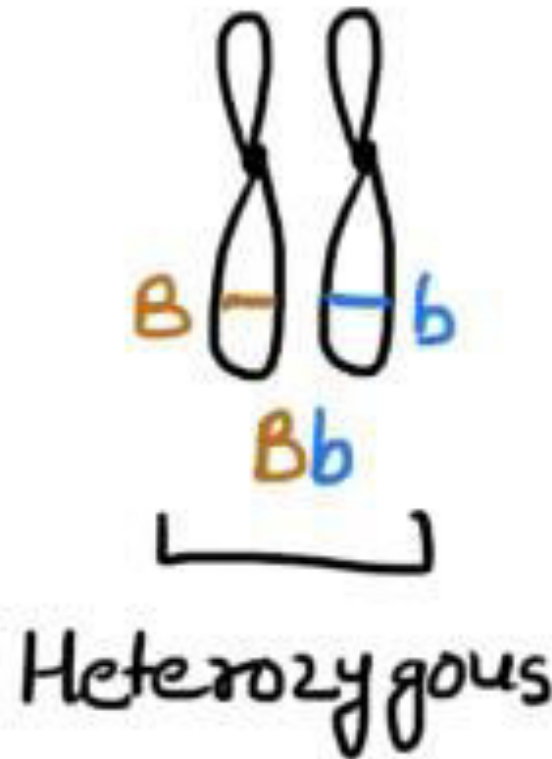
- Partially heterozygous
- Repeated sequences



Non-Genetic Interactions between viruses

HETEROZYGOSIS

- Aberrant packaging of multiple genomes
- May result in multiploid particles
- genetic complexity of population



Non-Genetic Interactions between viruses

SUPERINFECTION

- Interference as a result of superinfection
- Homologous interference against same virus from presence of particles which blocks replication
- Interference by mutation
- Interference by sequestration of virus receptors due to virus-attachment protein

Non-Genetic Interactions between viruses

PHENOTYPIC MIXING

Variable

Genome maybe enclosed in capsid/envelope of another (pseudo-typing)

capsid contains genomic particles of both viruses

Non-specific incorporation of different viral glycoproteins

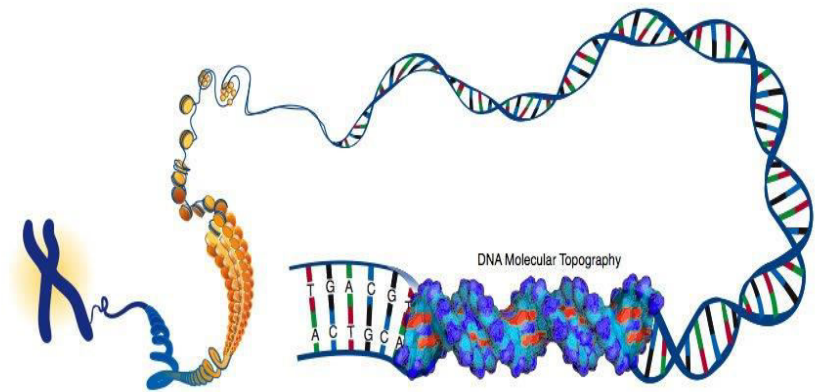
Non-Genetic Interactions between viruses

PHENOTYPIC MIXING AS A TOOL

- Examining biological properties of viruses
- Vesicular stomatitis virus forms pseudotypes containing retrovirus envelope glycoproteins
- Plaque forming properties of VSV, cell tropism properties of retrovirus
- Used for HIV and Retrovirus

Virology

LARGE DNA GENOMES



Large DNA genomes

- Viral genome could be considerably big e.g

DS DNA genomes

- Generally, Genetically similar to host cells e.g

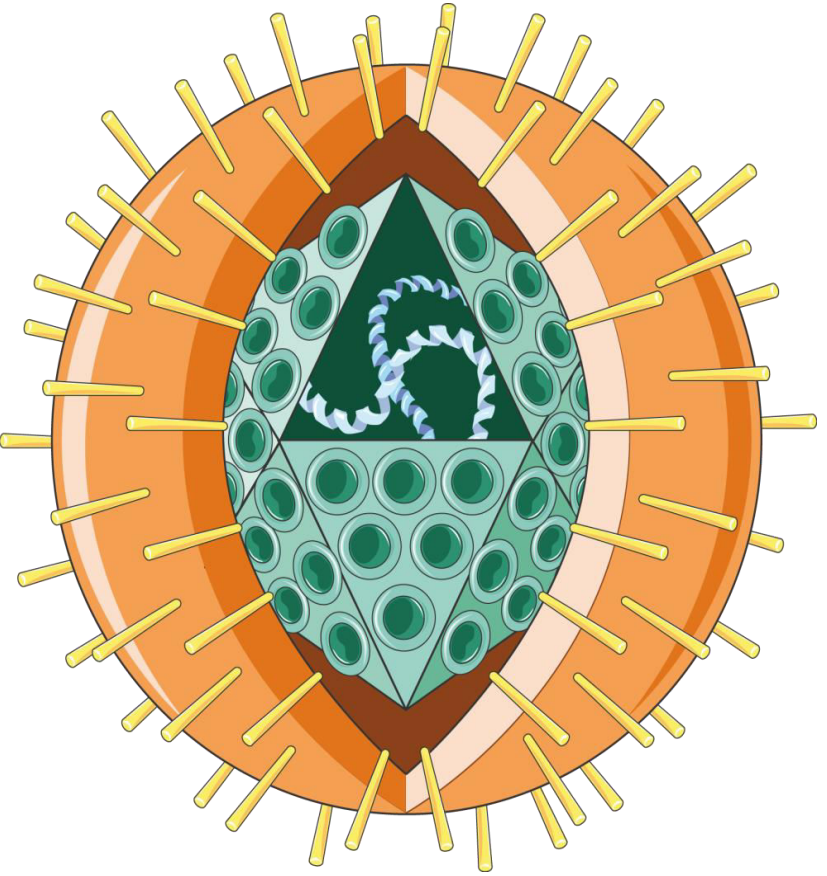
Adenovirus, herpes

ctc.

Large DNA genomes

HERPESVIRIDAE

- Large family
- More than 100 members
- Eight human viruses, with similar overall structure
- Subdivided into 3 families



Large DNA genomes

SUBFAMILIES

Alphaherpesvirinae

- Latent infections in sensory ganglion, genome size 120-180kbp
- Simplexvirus, Varicellovirus, Human herpes virus-I and II,
- Human herpes virus-III

Betaherpesvirinae

- Restricted host range, genome size 140-253 kbp
- Cytomegalovirus, Muromegalovirus, Roseolovirus, Human herpes virus-V, Mouse cytomegalovirus-I, Human herpes virus-VI and VII

Gammaherpesvirinae

- Infection of lymphoblastic cells, genome size 105-175 kbp
- Lymphocryptovirus, Rhadinovirus, Human herpesvirus-IV, Human herpes virus-VIII

Large DNA genomes

HERPESVIRUS

- large linear double stranded DNA genome containing 35 viral polypeptides encoding variety of enzymes
- Two covalently joined sections, Unique long (U_L) and Unique short regions (U_S) by inverted repeats
- Allows structural rearrangements of unique regions

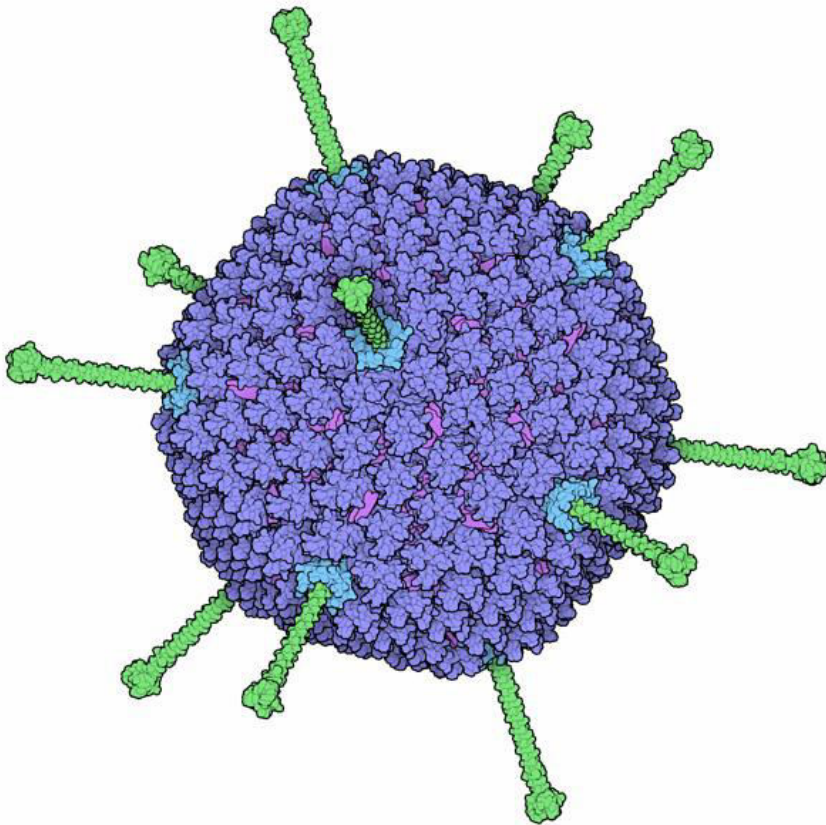
Large DNA genomes

HERPES SIMPLEX VIRUS

- 152 kbps ds DNA, 80 genes
- Densely packed with overlapping reading frames
- Each gene expressed by own promoter
- Extensively mapped by conventional genetic analysis
- Most intensively studied virus

Large DNA genomes

ADENOVIRUS

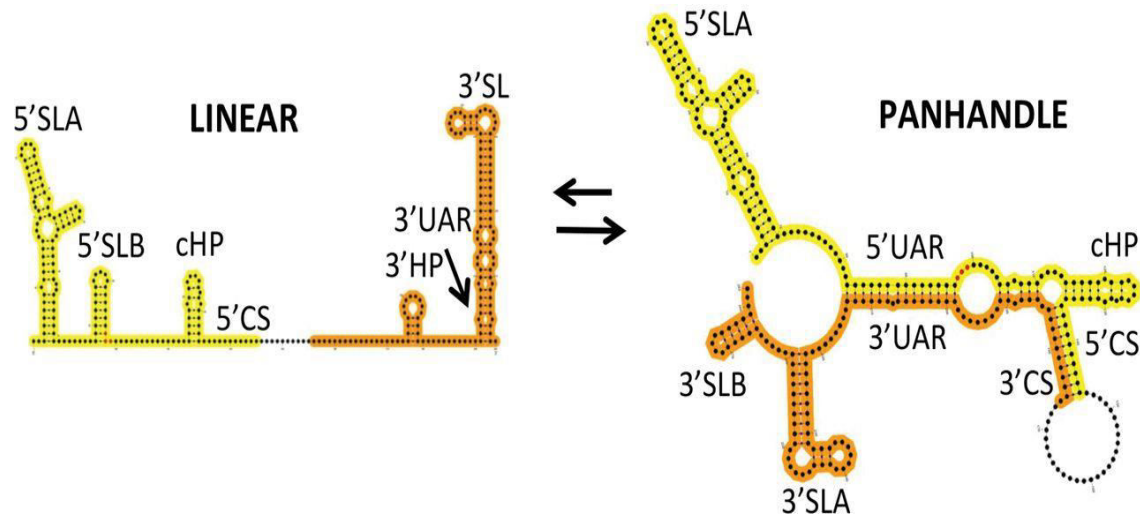


- Linear double stranded DNA genome of 30 to 38 kbp, 30-40 genes
- Terminal sequence is inverted repeat of 100 to 140bp
- Denatured single strands form panhandled structure

Large DNA genomes

PANHANDLED STRUCTURES

- Important in replication
- 55kDA protein known as terminal protein
- Attached to 5' end of each strand
- Acts as primer and initiates synthesis of new strands



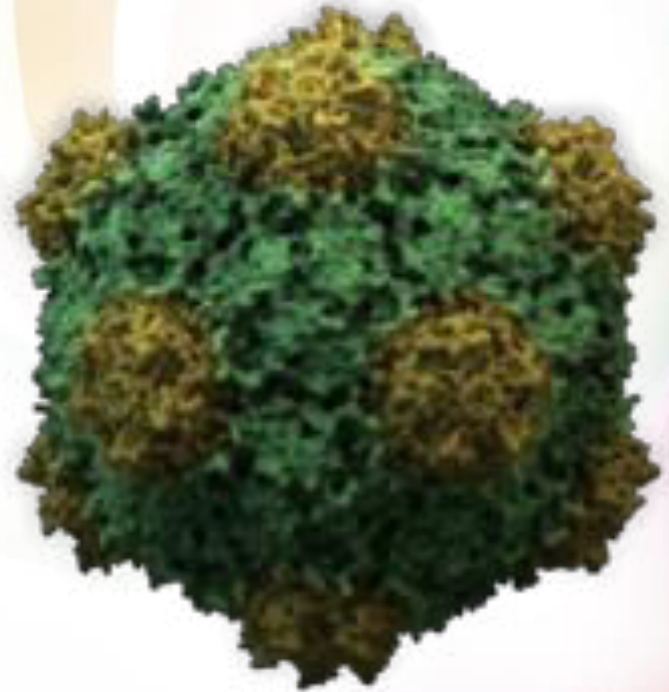
Large DNA genomes

EXPRESSION

- Promoters
- Splicing patterns

Virology

VIRUSES WITH SMALL DNA GENOMES



Small DNA genomes

M-13 PHAGE

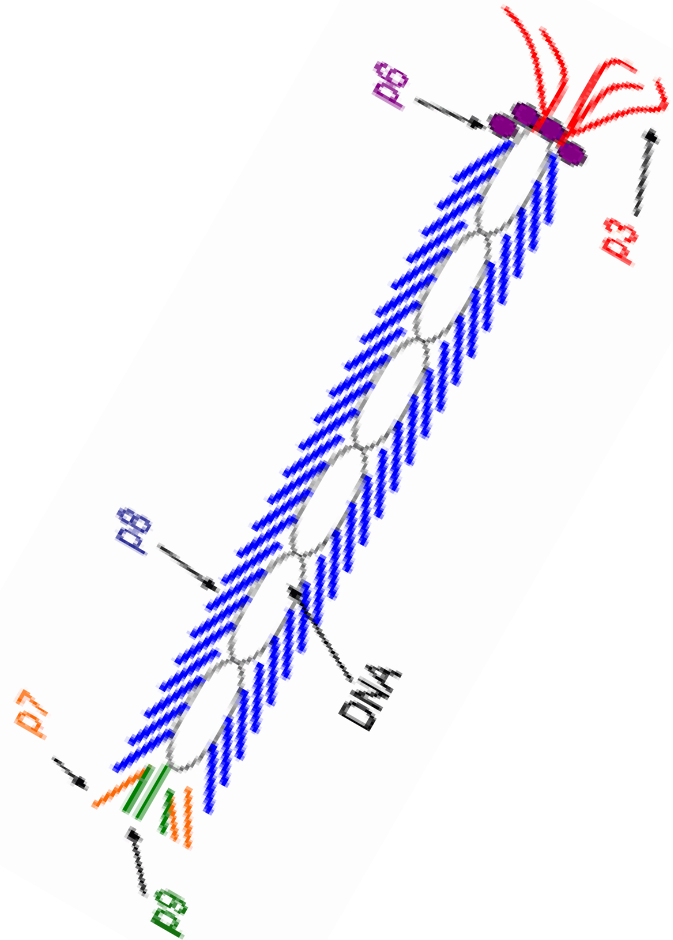
- Simple genome
- 6.4kb
- Single stranded
- Positive sense
- Circular DNA

encoding ten genes

Small DNA genomes

M-13 PHAGE

- Icosahedral symmetry
- Capsid can be expanded by protein subunits
- Hence, genomic size can also be increased



Small DNA genomes

LAMBDA PHAGE

- Rigid packaging restraints
- Only DNA of between 95 and 110% of normal genome size can be packaged into the capsid
- 49kbp
- Linear genome

Small DNA genomes

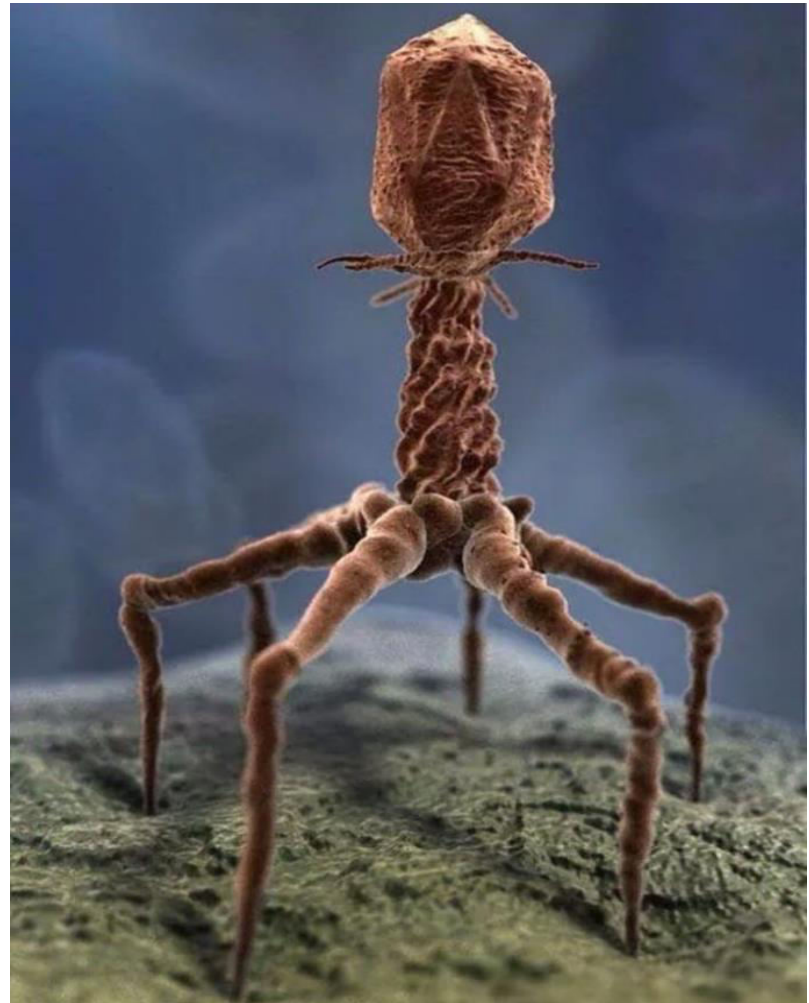
LAMBDA PHAGE

- Substrate packaged into phage heads during assembly consist of long concatemers of phage DNA
- DNA apparently whirled by phage head
- Complete genome cleaved by phage coded endonuclease
- *Cos-site*
- DNA ligase

Small DNA genomes

T-4 PHAGE

- 160 kbp
- Double stranded DNA
- Linear genome



Small DNA genomes

T-4 PHAGE

- Terminal redundancy
- Long Contactmers of
DNA genome
- Longer than complete
genome

Small DNA genomes

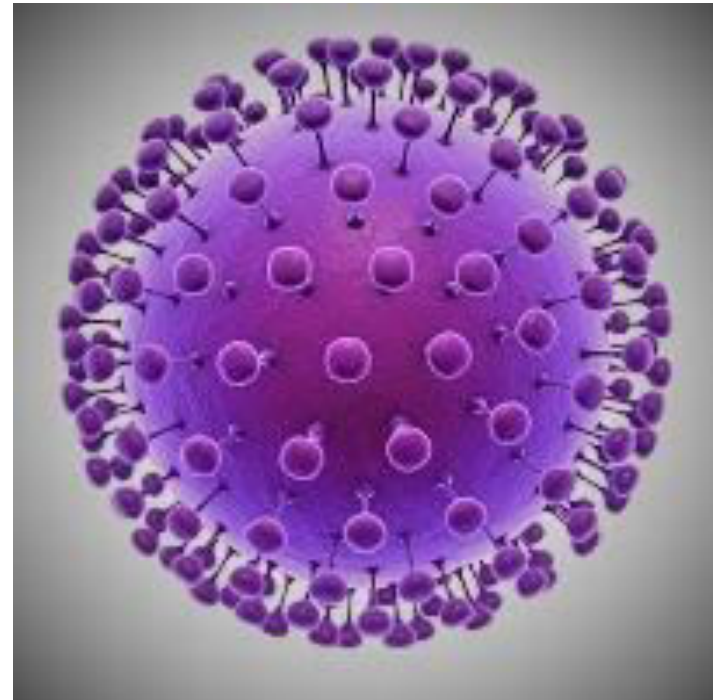
PARVOVIRUS

- Linear, non-segmented, single stranded negative sense genome, 5kb
- Most packaged as negative sense
- A portion of positive sense strands , very small
- Two genes *cap* and *rep*

Small DNA genomes

PARVOVIRUS

- Splicing pattern for each gene
- Pallindromic sequences for hair-pin structures



Small DNA genomes

POLYOMAVIRUS

- Double stranded, circular DNA, 5 kb
- DNA in super-coiled form
- Four cellular histones
- Maximum information in minimum space

Virology

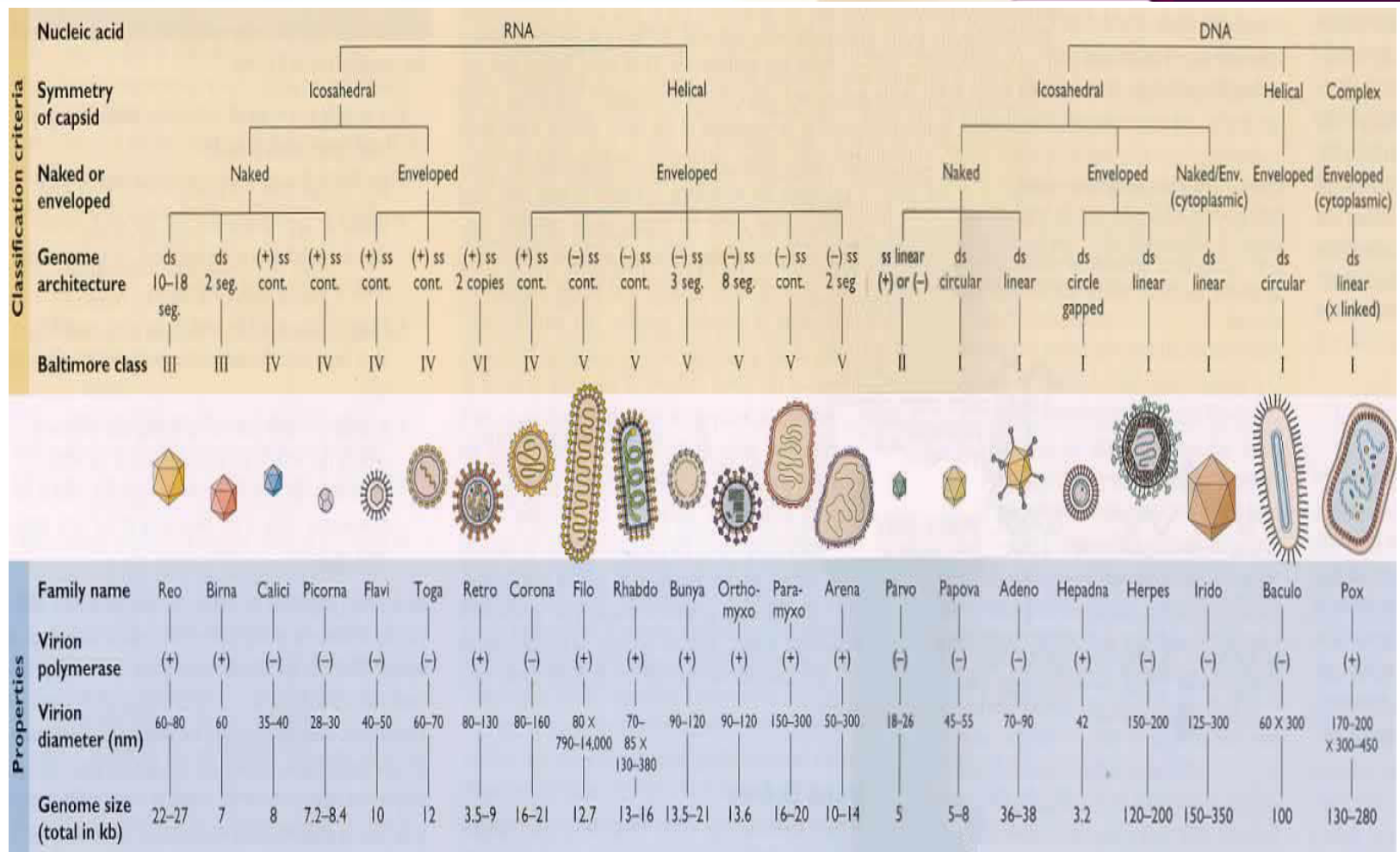
GENERAL CLASSIFICATION OF VIRUSES

Viral Classification

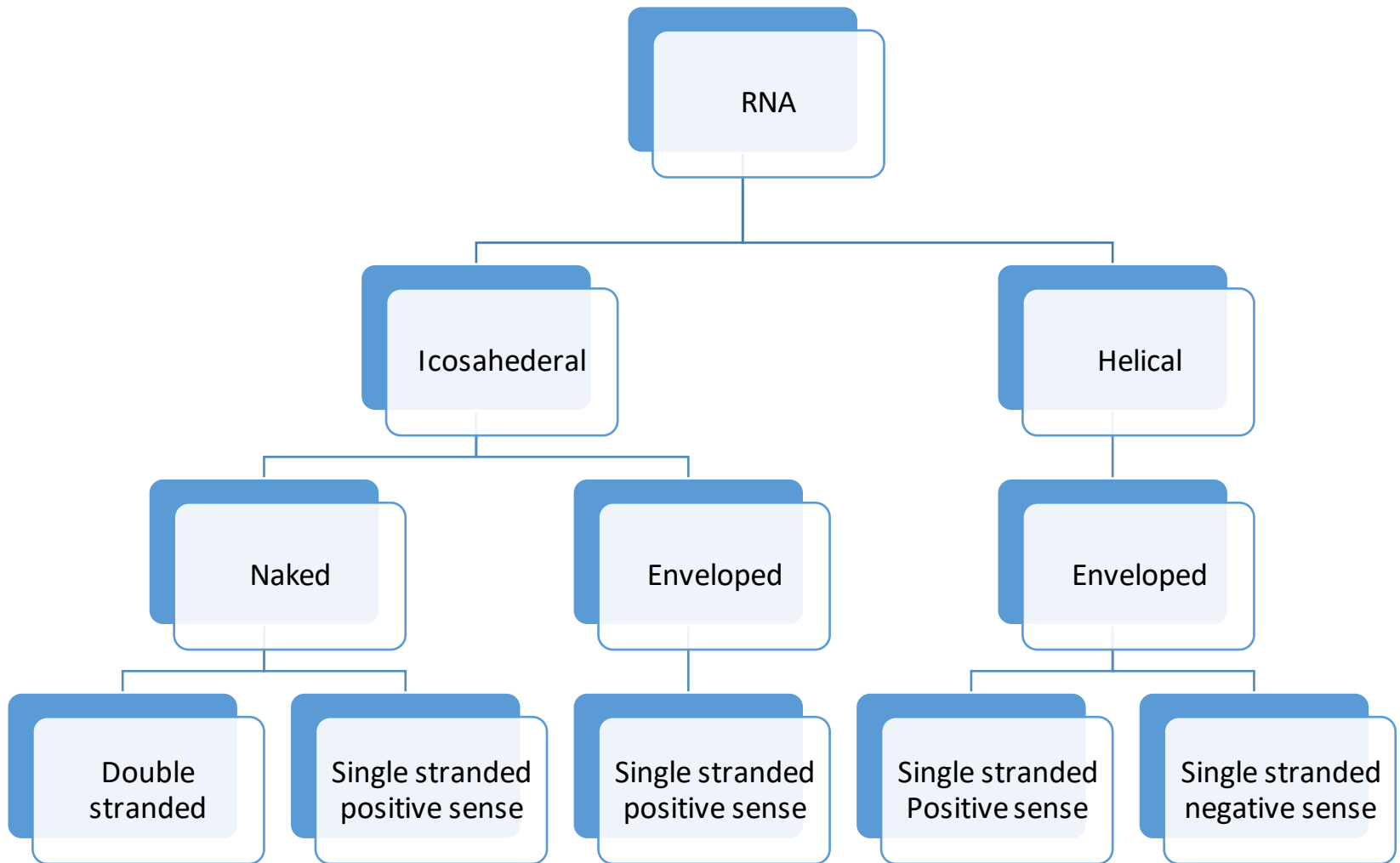
- Naming of organisms and categorizing them
- Based on a certain factors like phenotype, nucleic acid type, replication etc



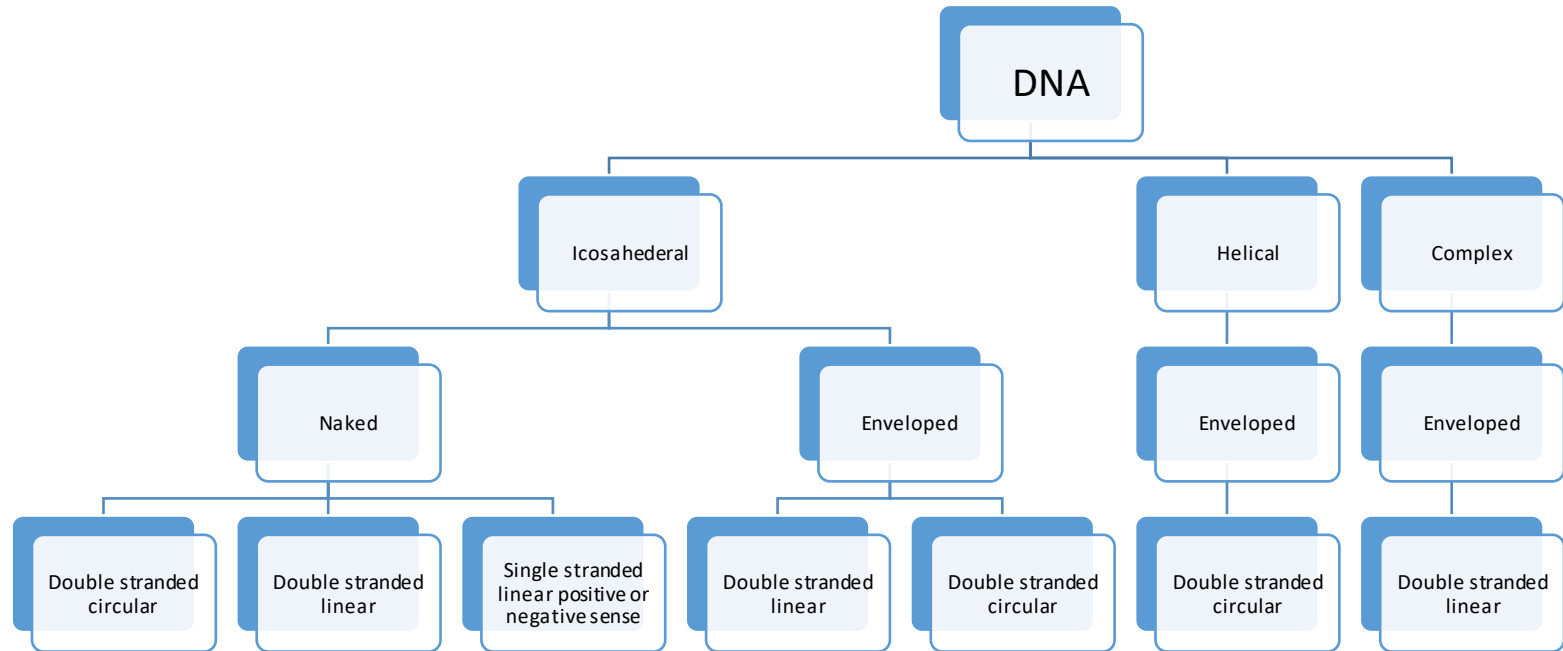
Viral Classification



Viral Classification



Viral Classification



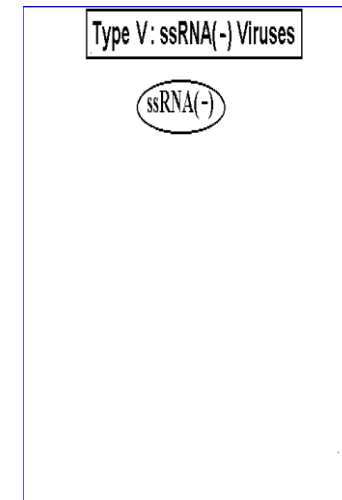
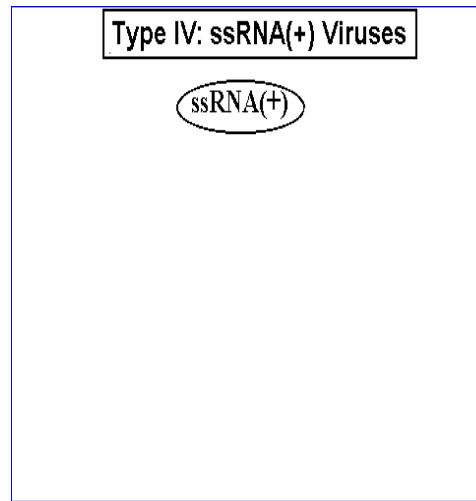
Virology

CLASSIFICATION OF POSITIVE SENSE RNA VIRUSES

Classification of RNA viruses

RNA VIRUS

- Ribonucleic acid as genome
- Single stranded or double stranded
- Belongs to Group III, Group IV or Group V of Baltimore Classification system



Classification of RNA viruses

DOUBLE STRANDED RNA VIRUSES

FAMILY	GENUS	TYPE SPECIES
Birnaviridae	Avibirnavirus Aquabirnavirus	Infectious bursal disease virus Infectious pancreatic necrosis virus
Reoviridae	Orthoreovirus Orbivirus Rotavirus Coltivirus Aquareovirus	Mammalian orthoreovirus Bluetongue virus-1 Rotavirus A Colorado tick fever virus Aquareovirus A

Classification of RNA viruses

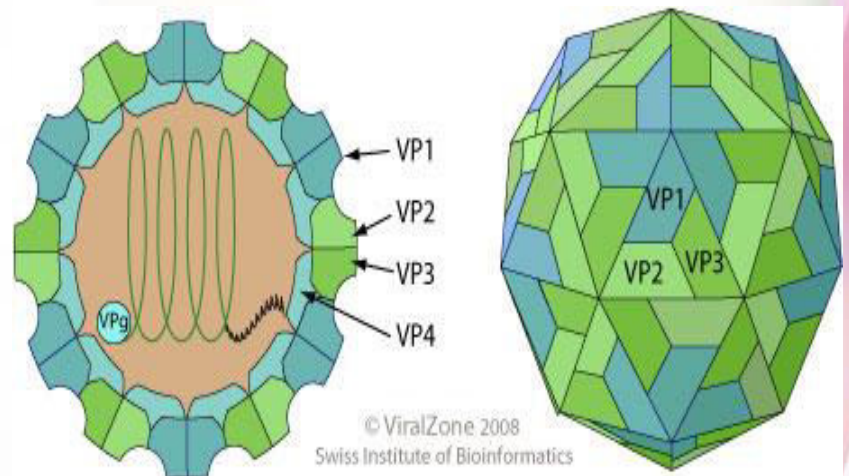
SINGLE STRANDED RNA VIRUSES

FAMILY	GENUS	TYPE SPECIES
Arteriviridae	Arterivirus	Equine arterivirus
Astroviridae	Astrovirus	Human astrovirus-1
Calciviridae	Vesivirus Lagovirus Hepatitis E like virus	Swine vesicular exanthema virus Hepatitis E virus
Coronaviridae	Coronavirus Torovirus	Infectious bronchitis virus Equine torovirus
Flaviviridae	Flavivirus Hepacivirus pestivirus	Yellow fever virus Hepatitis C virus Bovine viral diarrhea
Picornaviridae	Enterovirus Rhinovirus Hepatovirus	Poliovirus Human rhinovirus A Hepatitis A virus
Togaviridae	Alphavirus Rubivirus	Sindbis virus Rubella virus

Virology

PICORNAVIRUS

Properties



Picornavirus

PROPERTIES

- Non-enveloped
- Icosahedral symmetry

Biochemically,

- Ether- resistant
- Acid stable

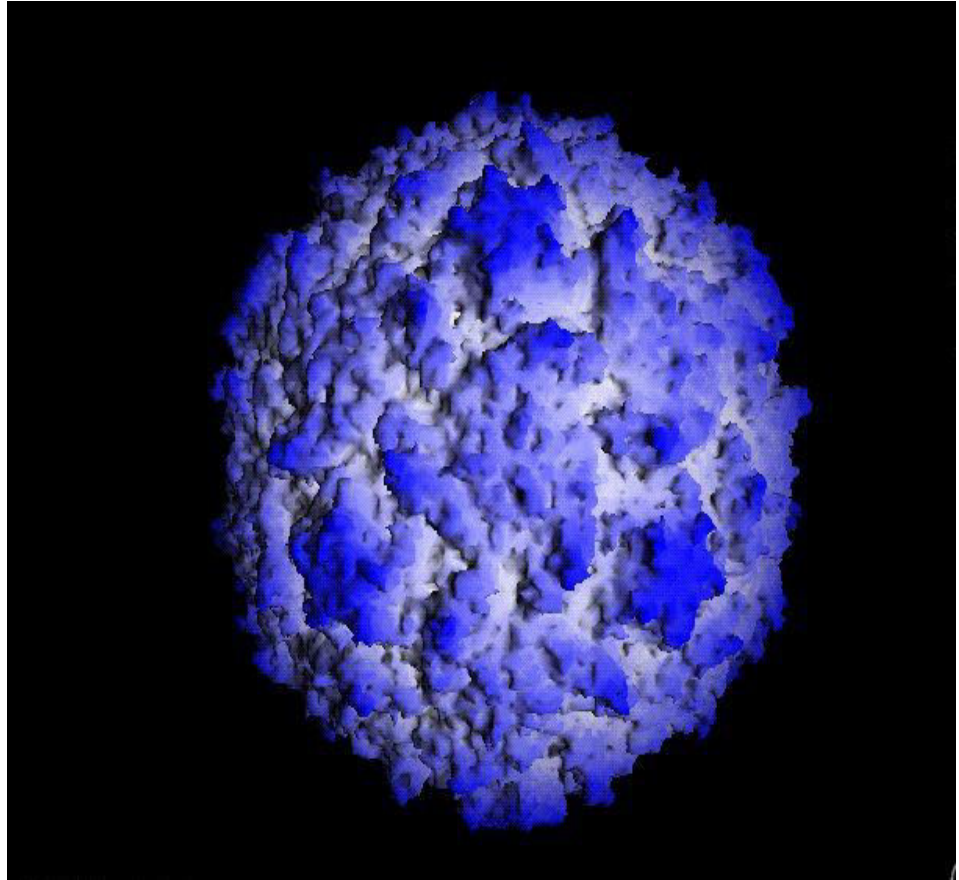
Picornavirus

- One of the largest families
- PICO RNA



Small sized RNA genome

- 9 genera
- 230 members



Picornavirus

- A picornavirus is a virus belonging to the family *Picornaviridae*, and order Picornavirales.
- Vertebrates, including humans, serve as natural hosts.
- Birds also serve as host
- Picornaviruses are nonenveloped viruses that represent a large family of small, cytoplasmic, plus-strand RNA (~7.5kb) viruses .
- 30-nm icosahedral capsid.

Picornavirus

- There are currently 80 species in this family, divided among 35 genera.
- These include the *Enterovirus*, *Aphthovirus*, *Cardiovirus*, *Rhinovirus* and *Hepatovirus* genera.
- The viruses in this family can cause a range of diseases including paralysis, meningitis, hepatitis and poliomyelitis.

Picornavirus

Enteroviruses

- poliomyelitis
- Echovirus
- Bovine enterovirus
- Simian Enterovirus
- Avian Enterovirus

Rhinoviruses

- Human cold virus
- Equine rhinovirus
- Bovine rhinovirus
- Foot and mouth disease
- Vesicular exanthema

Other picornaviruses

- Avian encephalomyelitis
- Mouse encephalitis
- Feline picornavirus

Virology

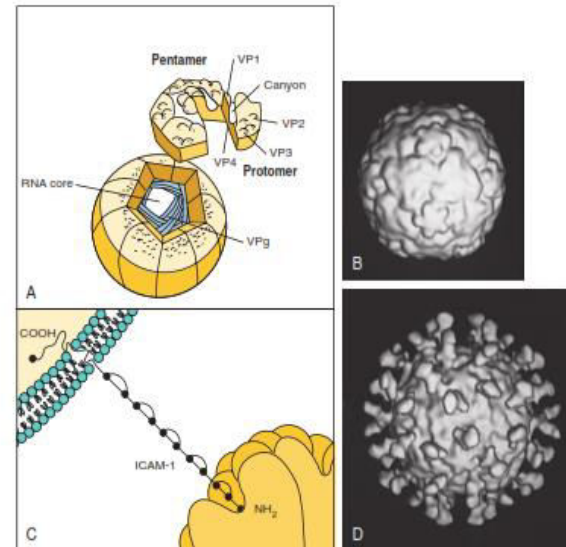
PICORNAVIRUS

Genome

Picornavirus

GENOMIC PROPERTIES

- Resembles mRNA
- Positive sense
- Single stranded RNA
- 7200 to 8450 bases



Picornavirus

GENOME

- The genome is non-segmented and positive-sense.
- The genome RNA is unusual because it has a protein on the 5' end that is used as a primer for transcription by RNA polymerase.
- This primer is called VPg genome range between 2–3 kb.
- VPg contain tyrosine residue at the 3' end.
- Tyrosine as a –OH source for covalently linked to 5' end of RNA.

Picornavirus

- Picornaviruses are non-enveloped, with an Icosahedral capsid.
- The capsid consists of 60 protomers.
- Each protomer consists of 4 polypeptides
- These peptides are called VP (viral protein) 1, 2, 3 and 4.
- VP2 and VP4 polypeptides originate from one protomer known as VP0
- VP0 is cleaved to give the different capsid components.

Picornavirus

REPLICATION

- The mRNA encodes RNA dependent RNA polymerase.
- This polymerase makes complementary minus strands of RNA.
- RNA will act as template to make more plus strands.

Picornavirus

REPLICATION

an overview of the steps in picornavirus replication
attachment,

- entry
- translation
- transcription/genome replication
- Assemble
- exit

Virology

PICORNAVIRUS

Important diseases

PICORNAVIRUS

POLIOVIRUS

- Causes polio
- Deadly neurological conditions
- Mostly subclinical
- 1% suffer from paralytic legs
- Deadly respiratory failure



PICORNAVIRUS

RHINOVIRUS

- Most common virus
- Common flu in humans
- Rhinovirus infection proliferates in temperatures between 33–35 °C (91–95 °F)

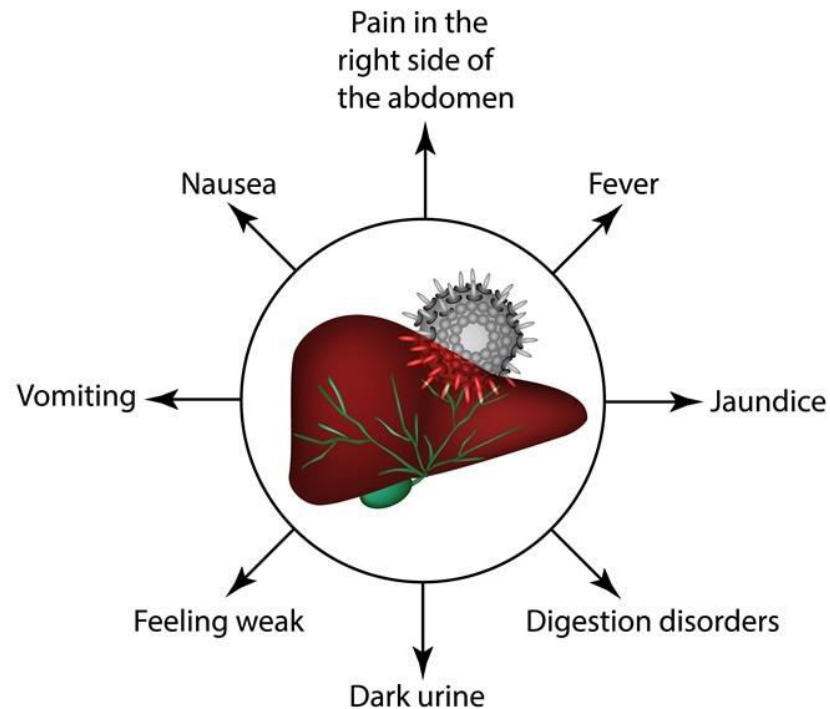


PICORNAVIRUS

HEPATITIS A

- Loss of appetite
- Jaundice
- Dark urine
- No long term consequences

SYMPTOMS OF HEPATITIS A



PICORNAVIRUS

FOOT AND MOUTH DISEASE

- In hooved animals
- Lesions around foot and mouth
- Inability to walk
- Fast spread
- Significant economic damage

PICORNAVIRUS

Lesions



Virology

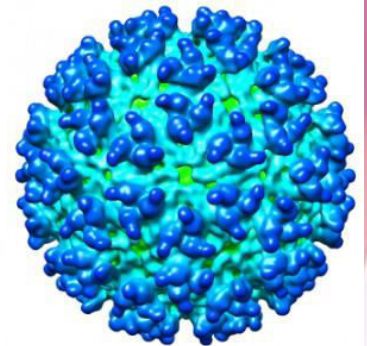
TOGAVIRUS

General properties

TOGAVIRUS

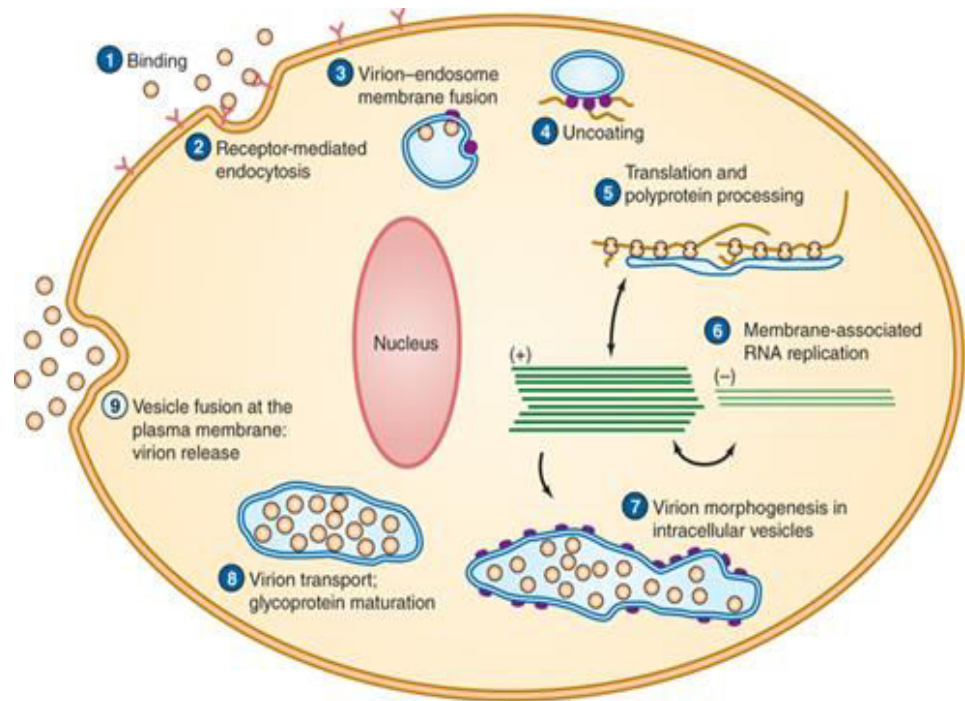
PROPERTIES

- Enveloped
- Positive single stranded RNA
- Icosahedral
- 2 genera having 32 species
 - Arbovirus
 - Rubula virus



TOGAVIRUS

- Early nonstructural and late structural protein synthesis
- Cytoplasmic replication
- Budding at plasma membrane



TOGAVIRUS

GENOME

- Linear, non-segmented, single-stranded, positive sense RNA
- 10,000–12,000 Nucleotides
- The virus is enveloped and forms spherical particles (65–70 nm diameter)
- The capsid is icosahedral, constructed of 240 monomers, having a triangulation number of 4.

TOGAVIRUS

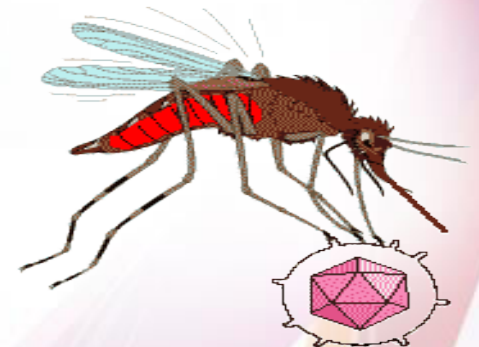
REPLICATION

- Entry into the host cell is achieved by attachment of the viral E glycoprotein to host receptors
- which mediates clathrin-mediated endocytosis.
- The receptors for binding are unknown
- Glycoprotein petal-like spikes act as attachment proteins.
- After virus attachment and entry into the cell, gene expression and replication takes place within the cytoplasm.

TOGAVIRUS

ARBOVIRUS

- Arthropod born virus (mosquitoes)
- Broad host range
- (Humans and plants)
- Zoonotic



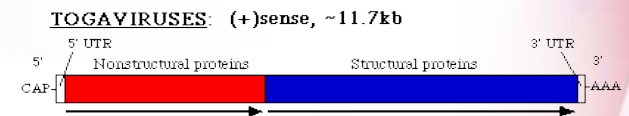
Virology

TOGAVIRUS Genome

TOGAVIRUS

GENETIC CHARACTERS

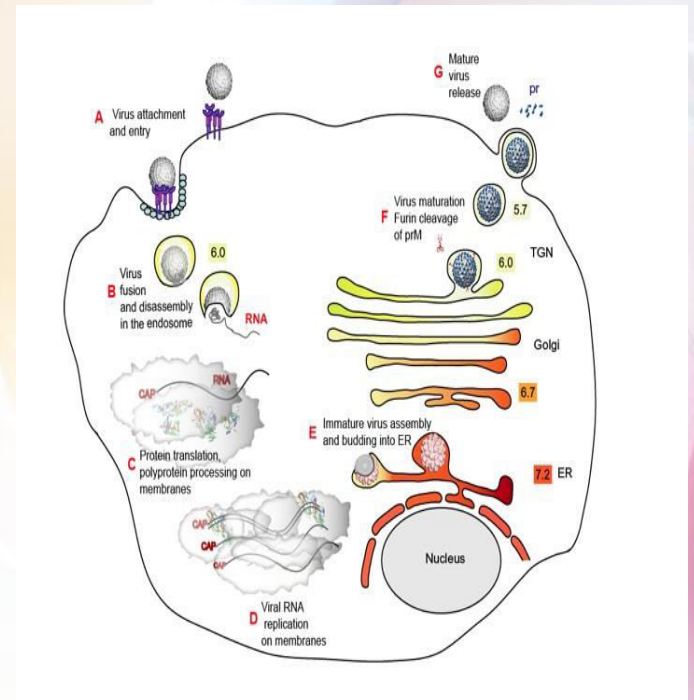
- Positive sense
- Single stranded RNA
- Around 11.7kb
- The 5'-terminus has a methylated nucleotide cap.



TOGAVIRUS

REPLICATION

- Viral RNA acts as both genome as well as mRNA
- After fusion of viral capsid with cell receptor
- RNA released in cytoplasm



TOGAVIRUS

VIRION STRUCTURE

- Togavirus virions consist of an envelope and a nucleocapsid.
- The virions are spherical to pleomorphic in shape, and about 70 nm in diameter.
- Glycoprotein spikes evenly cover the surface of the virion.
- The genome is contained in the T=4 icosahedral capsid, which measures 40 nm in diameter.

TOGAVIRUS

REPLICATION

- Nonstructural proteins formed first
- Replication within cytoplasm

Virology

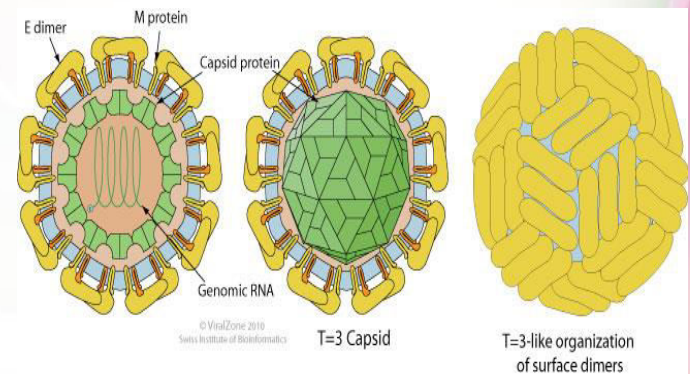
FLAVIVIRUS

Properties

FLAVIVIRUS

CHARACTERISTICS

- RNA virus
- Enveloped



FLAVIVIRUS

DENGUE FEVER VIRUS

- Mosquito borne
 - *Aedes Egyptei*
- Infects bone marrow (break bone fever)
- Low platelet level
- Hemorrhagic fever
- Can lead to renal failure
- Septicemia and death

FLAVIVIRUS

YELLOW FEVER

- *Aedes* mosquito
- Jaundice
- Backache
- Bloody diarrhea
- Bloody vomiting
- Vaccine available



FLAVIVIRUS

WEST NILE VIRUS

- Birds are reservoir
- Mosquitoes Vector
 - Culex
- Encephalitis
- Meningitis
- Flaccid paralysis
- Seizures and coma



FLAVIVIRUS

HEPATITIS C

- Transmission: Exposure to infected blood
- Highly antigenic variable
- Jaundice
- 60-80% become chronic cases
- Cirrhosis
- Primary cause of hepatocellular carcinoma
- Interferon treatment

Virology

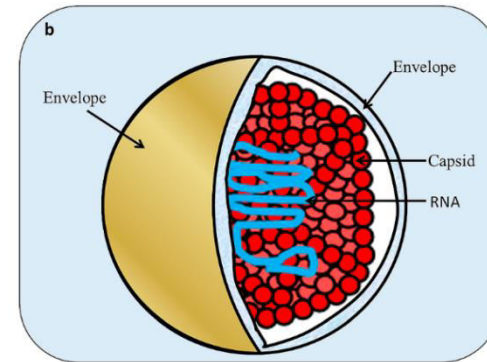
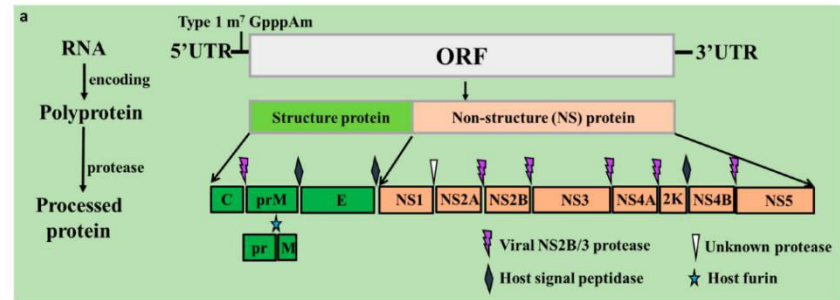
FLAVIVIRUS

Genome

FLAVIVIRUS

GENOME

- RNA genome
- Positive sense
- Non-segmented



FLAVIVIRUS

TWO GROUP HYPOTHESIS

On the basis of
morphological
information

Obtained using
electron microscopy
supported the
hypothesis of the
existence of at least
two virus groups

FLAVIVIRUS

- One group includes non-enveloped virus
- Family Reoviridae (genera Orbivirus, Coltivirus and Seadornavirus)
- Viruses with an overall diameter of 60–80 nm
- Icosahedral symmetry
- Several concentric capsid layers that surround a segmented double-stranded RNA (dsRNA) genome

FLAVIVIRUS

- Second group includes enveloped viruses (inactivated by ether and deoxycholate)
- 50–60 nm in diameter
- Infectious ssRNA genome of positive polarity.
- Two antigenically distinct subgroups i.e.
- Group A and Group B.

FLAVIVIRUS

MEDICAL IMPORTANCE

- Mostly serologically related to each other
- Antibodies of one useful against others
- Antigenic variability of Hepatitis C proteins
- The 17D vaccine is one of the most efficient vaccines ever developed and was derived from a strain of Flavivirus isolated from a man who recovered from infection by the virus.

Virology

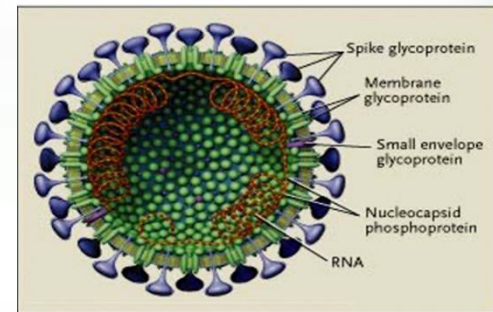
CORONAVIRUS

Structure

Coronavirus

STRUCTURE

- Enveloped/ encapsulated
- Club shaped projections on envelope
- 80-160nm diameter
- Helical shaped



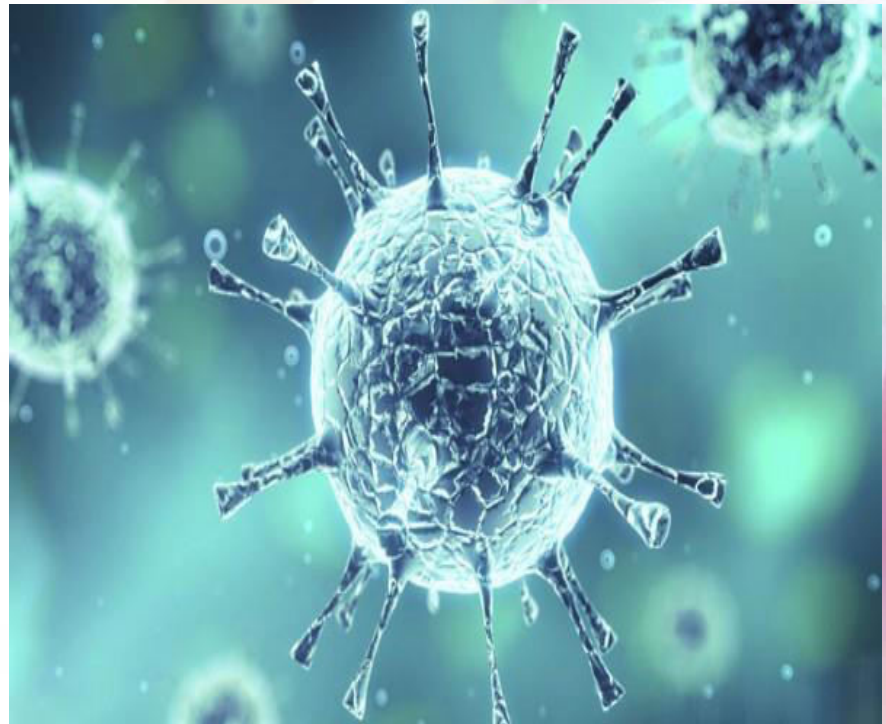
Coronavirus

- *Coronaviruses* are species belonging to the subfamily Coronavirinae in the family Coronaviridae, in the order Nidovirales.
- Coronaviruses are enveloped viruses with a positive sense single-stranded RNA genome and with a nucleocapsid of helical symmetry.
- The Genomic size of coronaviruses ranges from approximately 26 to 32 Kb, the largest for an RNA virus

Coronavirus

COMMON COLD

- Second most common cause of common cold
- SARS (Severe Acute Respiratory syndrome)
- GI tract infections in kids



Coronavirus

SARS

- Zoonotic origin
- Acute respiratory form
- Outbreaks leads to enormous number of deaths
- Flu and Fever
- Lethargy
- Cough
- Pneumonia

Coronavirus

MERS

- Middle east respiratory syndrome
- Fever
- Cough
- Shortness of breath
- Sometimes diarrhea
- Mortality rate: 35%
- Camels are major reservoir hosts

Virology

CORONAVIRUS

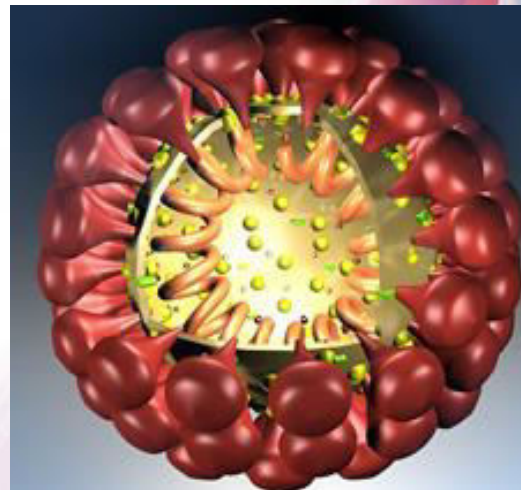
Genome

Coronavirus

GENOME

- Positive sense
- Single stranded
- RNA
- 27,000 -30,000 kb

genome



Coronavirus

- Enveloped
- Spherical
- About 120 nm in diameter
- The RNA genome is associated with the N protein to form the nucleocapsid (helical for the genus coronavirus, and tubular for the genus torovirus)
- Linear ssRNA

Coronavirus

- 27-32kb in size (the largest of all RNA virus genomes)
- Capped
- Polyadenylated
- The leader RNA (65-89 bp) at the 5' end of the genome is also present at the end of each subgenomic RNAs.

Coronavirus

REPLICATION

CYTOPLASMIC

- Attachment of the viral S protein receptors
- Endocytosis of the virus into the host cell
- Fusion of virus membrane with endosomal membrane
- Genome is released into the cytoplasm.
- Synthesis and proteolytic cleavage of the replicase polyprotein.

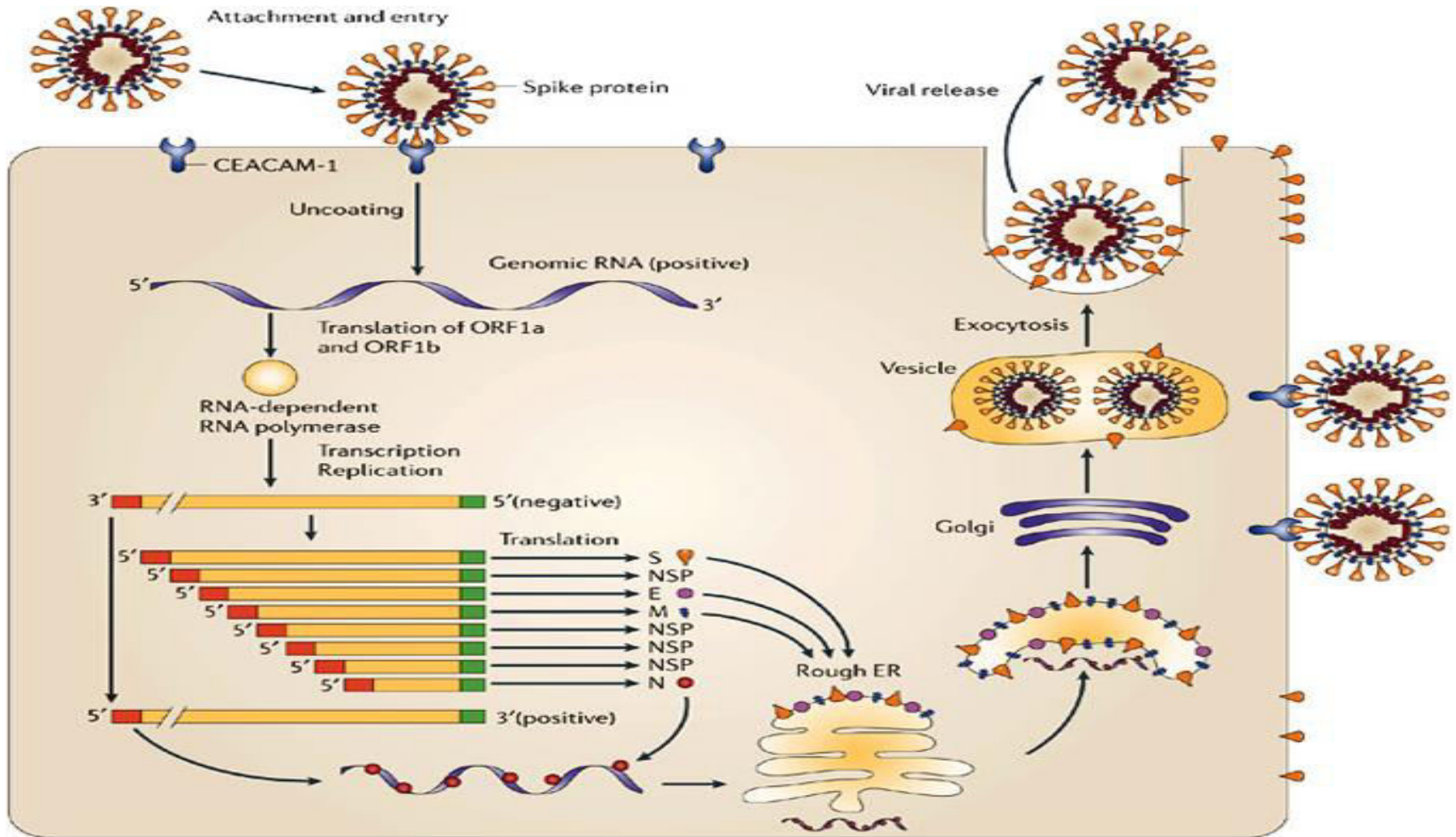
Coronavirus

REPLICATION

- A dsRNA genome is synthesized from the genomic ssRNA(+).
- The dsRNA genome is transcribed by providing viral mRNAs/new ssRNA(+) genomes.
- Synthesis of structural proteins encoded by subgenomic mRNAs.
- Assembly and budding at membranes of the endoplasmic reticulum (ER), the intermediate compartments, and/or the Golgi complex.
- Release of new virions by exocytosis.

Coronavirus

REPLICATION IN CYTOPLASM



Virology

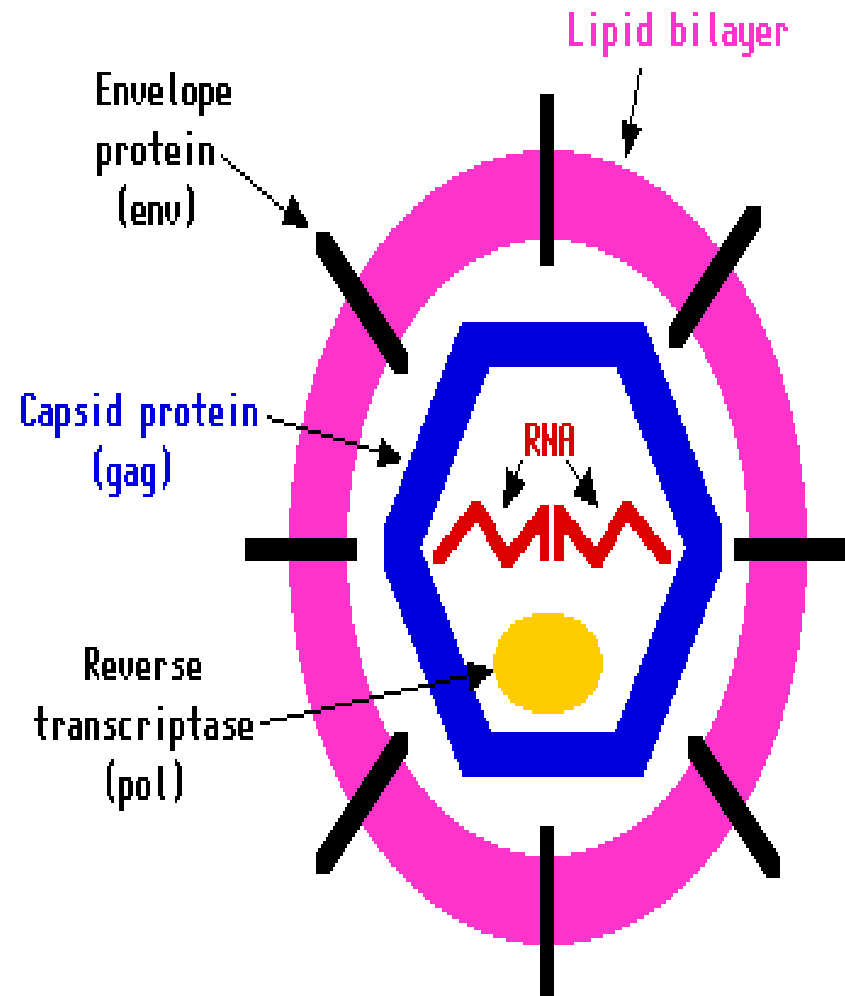
RETROVIRUS



RETROVIRUS

BASICS

- An envelope
- Envelope protein
- A capsid
- Two molecules of RNA
- Reverse transcriptase



RETROVIRUS

REPLICATION PATTERN

Once inside the host cell cytoplasm, the virus uses its own RT enzyme to produce DNA from its RNA genome

The reverse of the usual pattern, thus *retro* (backwards)

The new DNA is then incorporated into the host cell genome by an integrase enzyme referred to as a provirus

RETROVIRUS

REPLICATION PATTERN

The host cell then treats the viral DNA as part of its own genome, transcribing and translating the viral genes along with the cell's own genes

The proteins are produced which are required to assemble new copies of the virus.

It is difficult to detect the virus until it has infected the host.

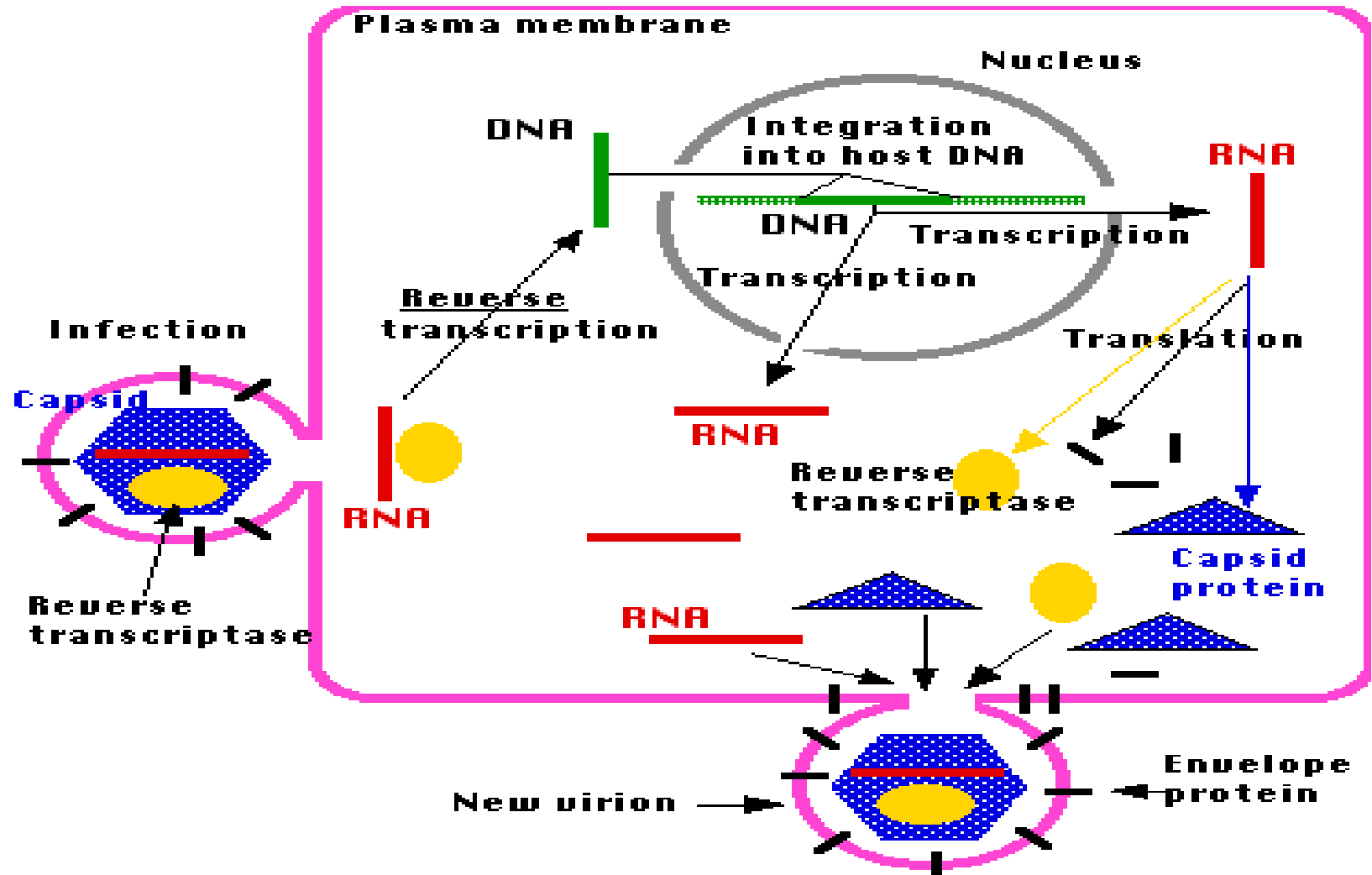
Infection will persist indefinitely

RETROVIRUS

REPLICATION PATTERN

- The gag gene ---
capsid protein
- The pol gene ---
molecules of reverse
transcriptase
- The env gene ---
molecules of
the envelope protein

RETROVIRUS



RETROVIRUS

HUMAN IMMUNODEFICIENCY VIRUS

- Most important immunosuppressing virus
- Causes immunosuppression
- HIV-1 and HIV-2
- AIDS



RETROVIRUS

DISCOVERY

- In February 1997
- Parkinson's disease
- Transplant of pig tissue to human beings
- Recipients immune system need to suppress to avoid rejection
- HIV-1 and HIV-2, retroviruses of primates

RETROVIRUS

TRANSMISSION

- blood
- Semen
- pre-seminal fluid
- rectal fluids
- vaginal fluids
- breast milk
- **Blood transfusion**
- **Syringes**

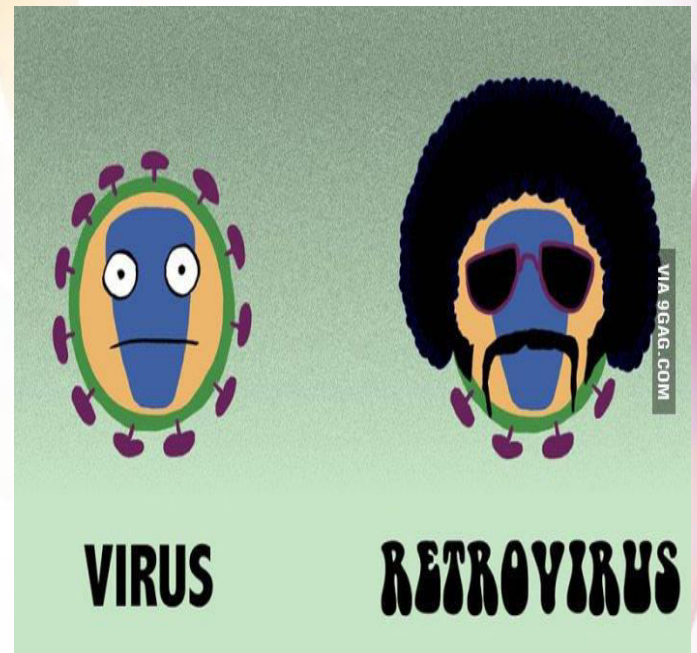
Virology

RETROVIRUS

Characteristics

RETROVIRUS

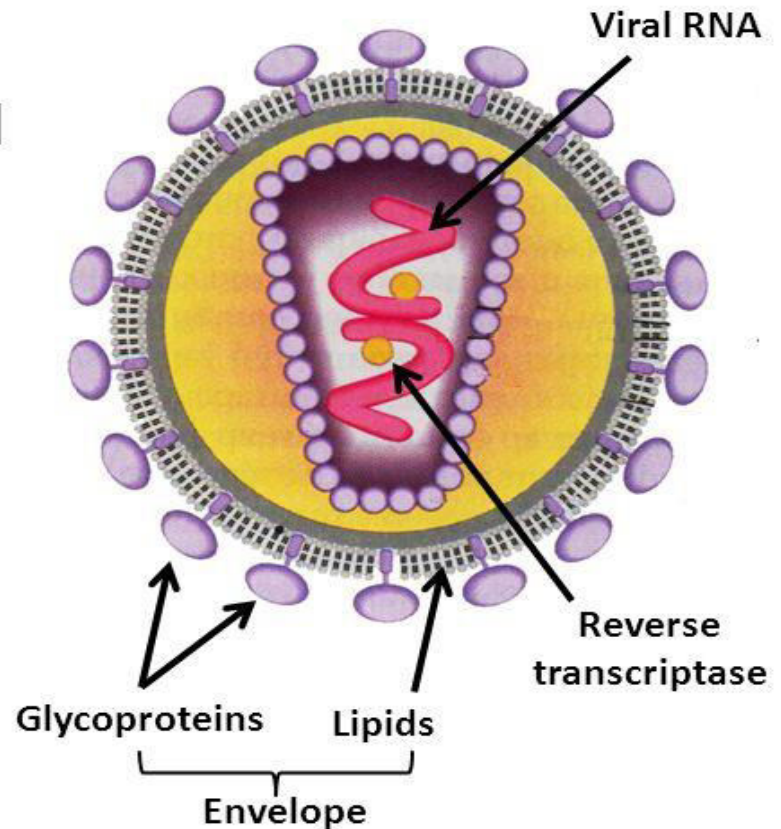
**Most challenging virus for
human beings so far.....**



RETROVIRUS

Retrovirus Structure

- All retroviruses:
 - Are surrounded by a viral envelope
 - Are icosahedral in shape
 - Contain 2 identical molecules of RNA
 - Contain the enzyme *reverse transcriptase*



RETROVIRUS

- Virions of retroviruses consist of enveloped particles about 100 nm in diameter.
- The virions also contain two identical single-stranded RNA molecules 7–10kb in length.
- Although virions of different retroviruses do not have the same morphology or biology
- All the virion components are very similar.

RETROVIRUS

Virion components

- Envelope
- Nucleic acid (RNA)
- Proteins

RETROVIRUS

ENVELOPE

- composed of lipids
- obtained from the host plasma membrane
- glycoprotein encoded by the env gene.

Functions

- protection from the extracellular environment via the lipid bilayer
- enabling the retrovirus to enter/exit host cells through endosomal membrane trafficking
- the ability to directly enter cells by fusing with their membranes.

RETROVIRUS

NUCLEIC ACID

- dimer RNA
- cap at the 5' end and a poly(A) tail at the 3' end
- terminal noncoding regions for replication
- internal regions that encode virion proteins
- The 5' end includes four regions, which are R, U5, PBS, and L.

PROTEINS

- consisting of gag proteins, protease, pol proteins, and env proteins

RETROVIRUS

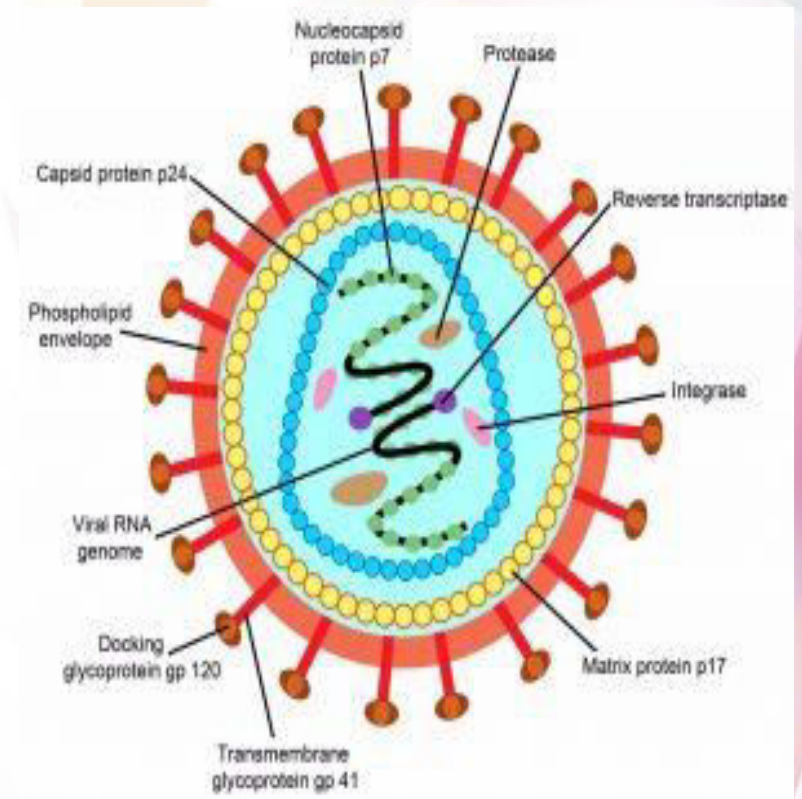
SALIENT CHARACTERISTICS

- Retroviruses are the only animal viruses that integrate into the host cell's genome
- The retroviral genomes can accommodate changes to its configuration.
- The retroviral life cycle begins in the nucleus of an infected cell.
- Retroviruses offer gene therapy researchers aid for delivering genes to target cells at high efficiency that allows for long-term, stable expression of introduced genetic elements

Virology

RETROVIRUS

Genome



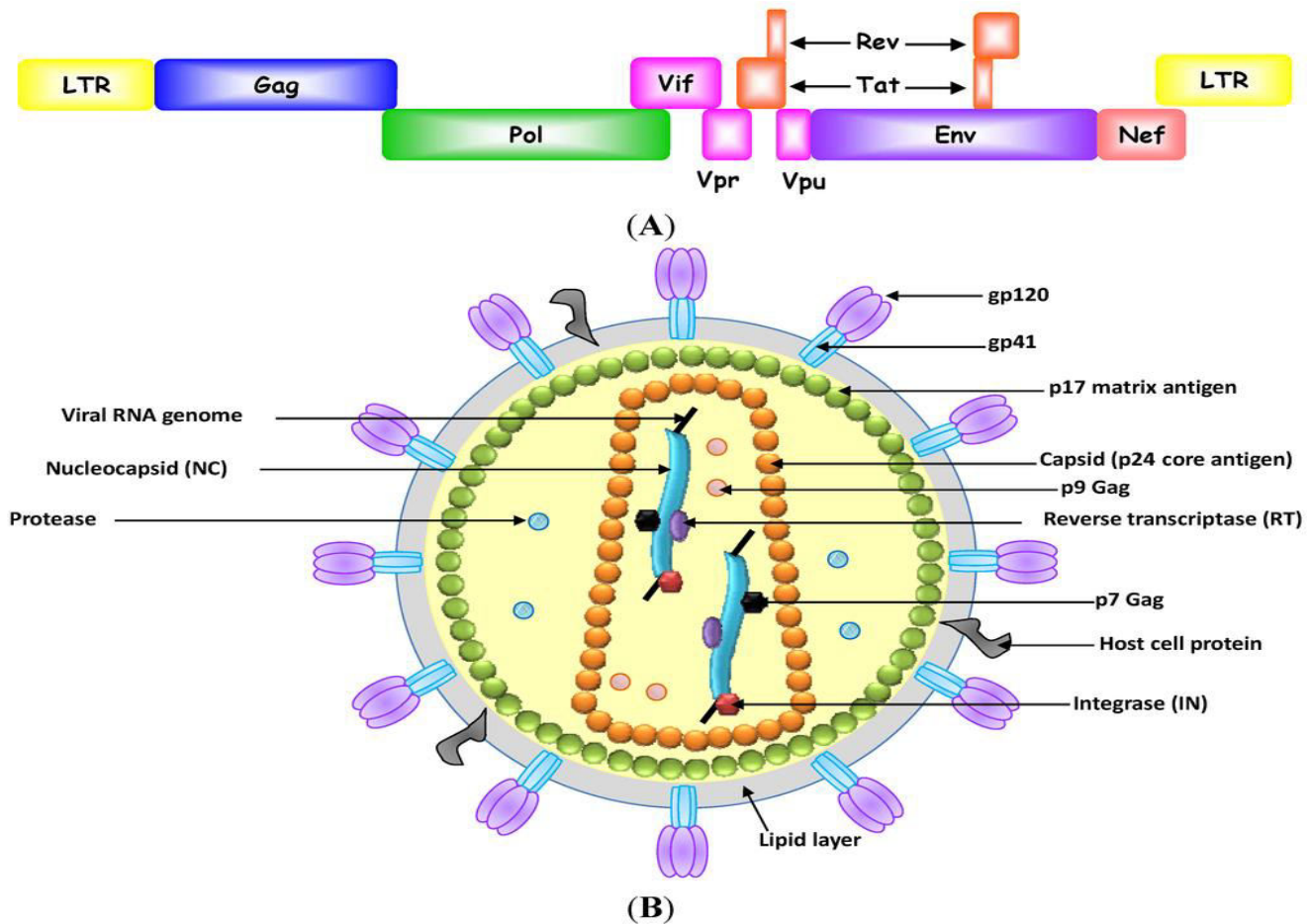
RETROVIRUS

GENETIC CHARACTERS

- Single stranded RNA viruses
- Positive sense
- enveloped
- Changes to DNA by RT enzyme

RETROVIRUS

Genomic Structure



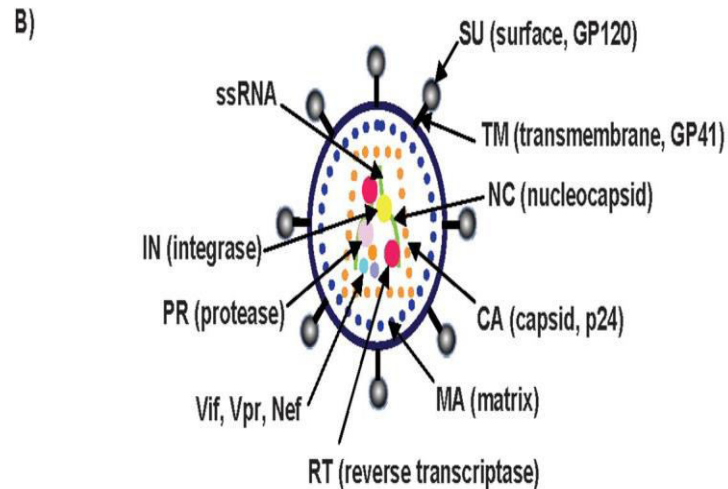
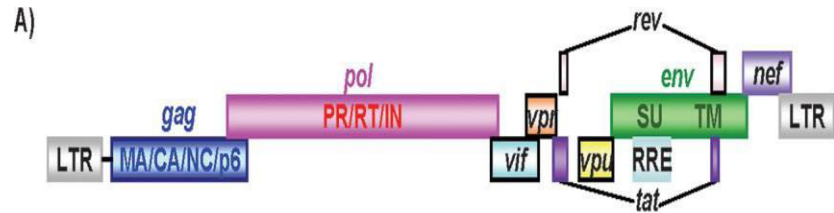
RETROVIRUS

Genetic makeup

Diploid nature of virus

3 very imp. genes:

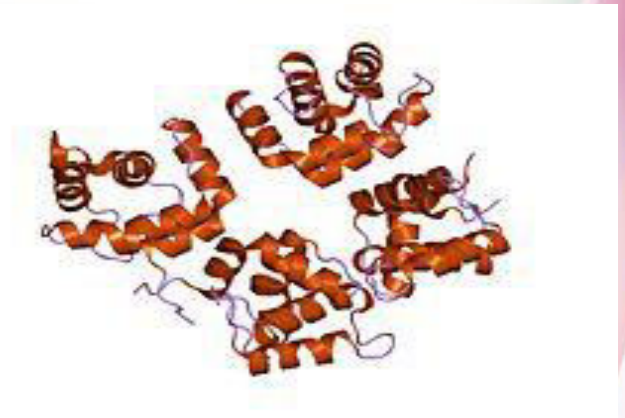
- GAG
- ENV
- POL



RETROVIRUS

GAG

- Gag gene makes protein p24
- Which serves as capsule around RNA
- Cone shaped



RETROVIRUS

ENV

- Envelope glycoproteins
- gp41 and gp120
- Gp41: transmembrane protein
- Gp120: outer glycoprotein
- Host receptor

RETROVIRUS

POL

- Reverse transcriptase
Polymerase
- Integrase activity
- DNA from RNA

Virology

RETROVIRUS Lifecycle

Virology

RETROVIRAL REPLICATION CYCLE

1- early steps

from the cell surface to the nucleus

- Retroviral particles must bind specifically to their target cells
- cross the plasma membrane
- reverse-transcribe their RNA genome
- Uncoat the cores
- find their way to the nuclear membrane
- penetrate into the nucleus

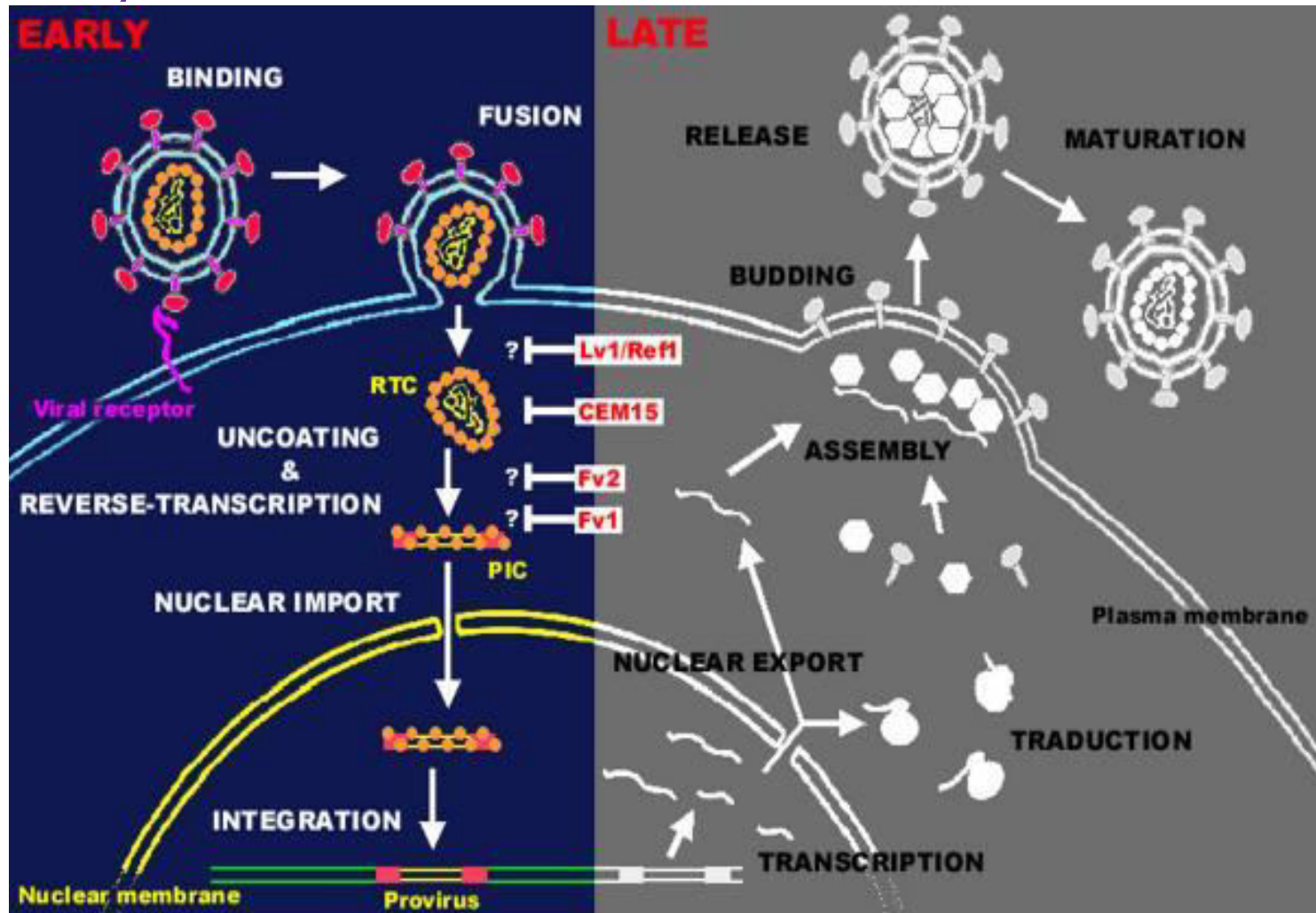
Hijack the cellular machinery

Virology

2- LATER STAGE

- Transcription
- Transduction
- Assembly
- Maturation
- Budding
- Release

- Lifecycle





Life Cycle of a Retrovirus - YouTube.MP4

Virology

POSITIVE SENSE RNA
PLANT VIRUS

Positive Sense RNA Plant Virus

LUTEOVIRIDAE

- Sobemovirus



Positive Sense RNA Plant Virus

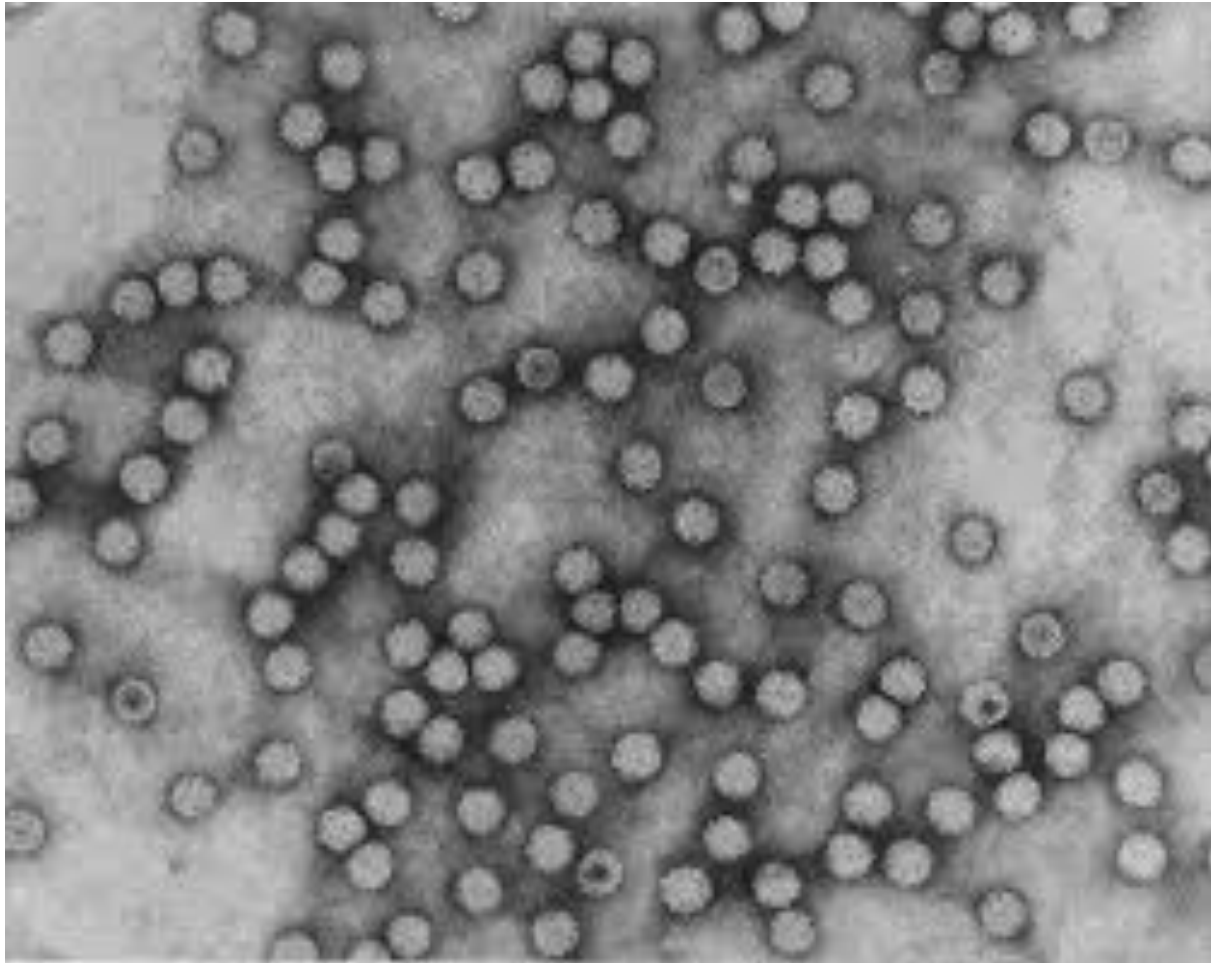
SEQUIVIRIDAE

- Sadwavirus
- Chevavirus



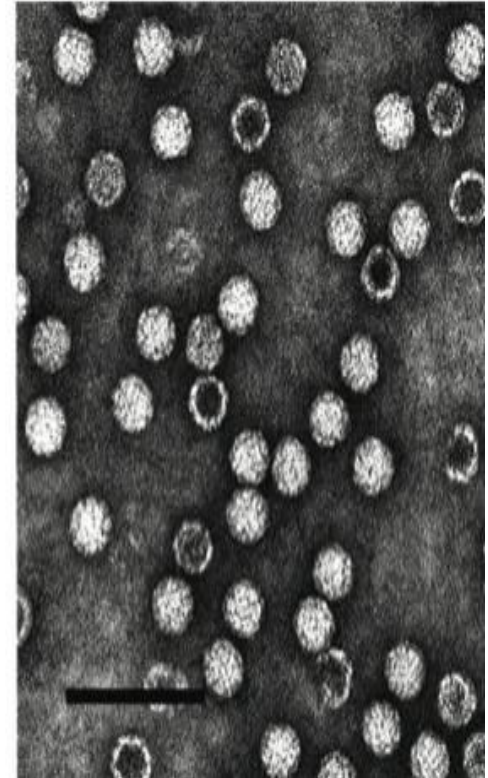
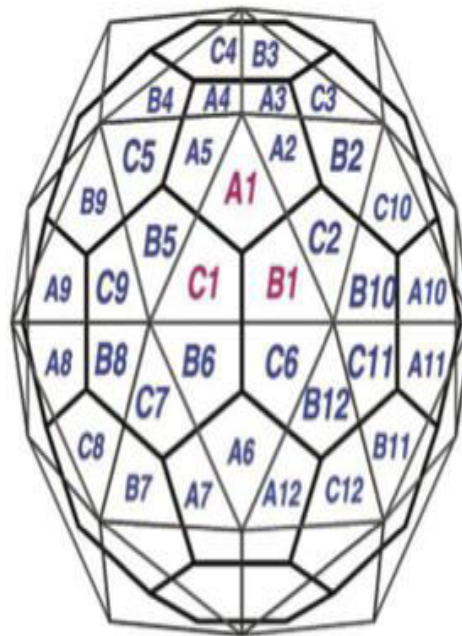
Positive Sense RNA Plant Virus

TOMBUSVIRIDAE



Positive Sense RNA Plant Virus

TYMOVIRIDAE



Positive Sense RNA Plant Virus

COMOVIRIDAE

- Idaeovirus



Positive Sense RNA Plant Virus

BROMOVIRIDAE

- Cucumovirus
- Bromovirus
- Ilavirus
- Alfamovirus
- Ourmiavirus
- Tobamovirus
- Hordievirus
- Pecluvirus
- Furovirus
- Pomovirus
- Bemivirus



Positive Sense RNA Plant Virus

OTHERS...

- Flexiviridae
- Potyviridae
- Closteroviridae

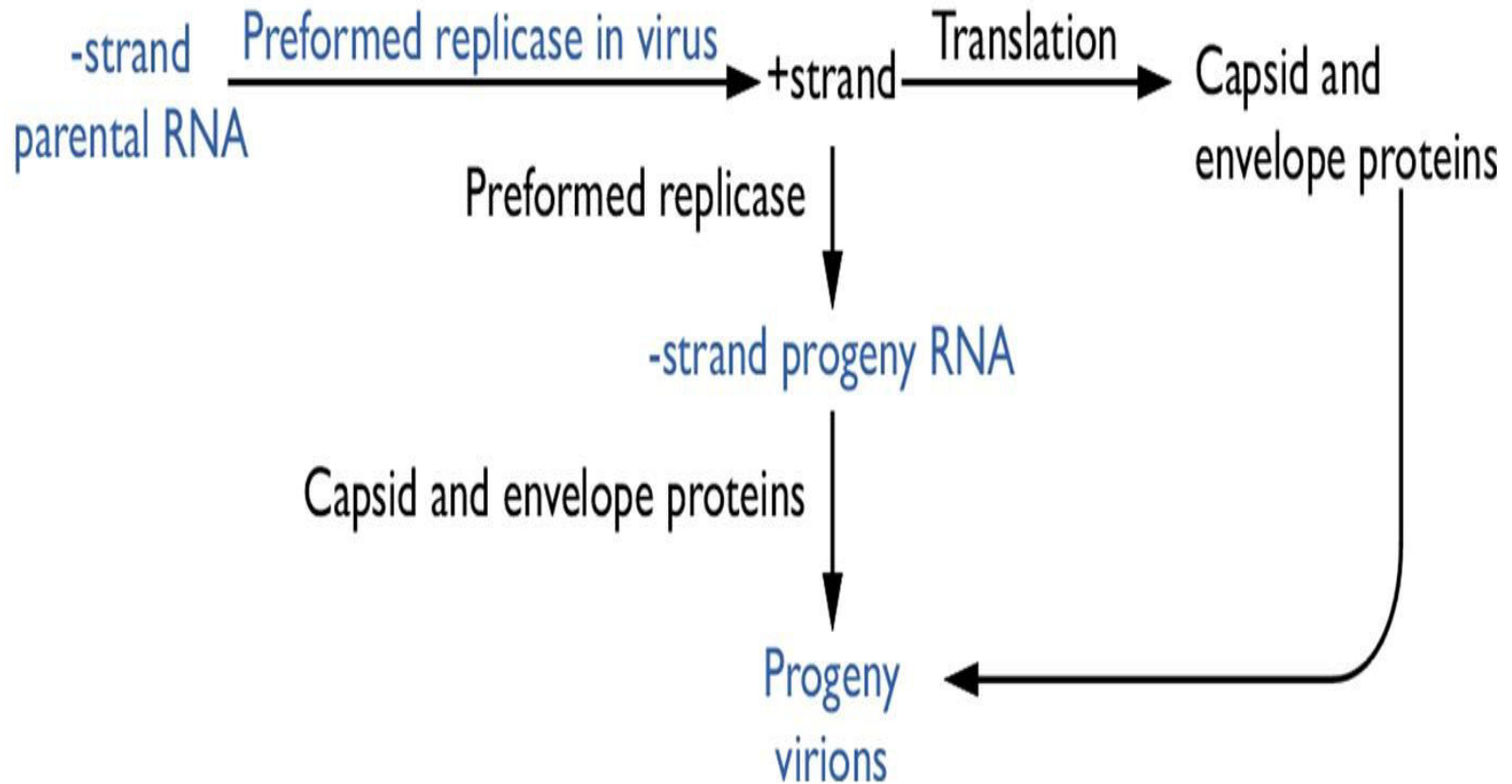
Virology

NEGATIVE SENSE RNA VIRUSES

Negative Sense RNA viruses

REPLICATION

ssRNA (-strand) viruses



Negative Sense RNA viruses

Family	Subfamily	Genus	Species
<i>Bornaviridae</i>		<i>Bornavirus</i>	Borna disease virus
<i>Bunyaviridae</i>		<i>Bunyavirus</i> <i>Hantavirus</i> <i>Phlebovirus</i>	Bunyamweravirus Hantaanvirus Rift valley fever
<i>Orthoviridae</i>		<i>Influenza virus A,B,C</i> <i>Thogotovirus</i>	Influenza A,B,C Thogotovirus
<i>Paraviridae</i>	<i>Paramyxovirinae</i> <i>Pneumovirinae</i>	<i>Respirovirus</i> <i>Morbillivirus</i> <i>Rubulavirus</i> <i>Pneumovirus</i> <i>Metapneumovirus</i>	Sendai virus Measles Virus Mumps virus Human respiratory syncytial virus
<i>Rhabdoviridae</i>		<i>Lyssavirus</i> <i>Vesiculovirus</i> <i>Ephemerovirus</i> <i>Novirhabdovirus</i>	Rabies virus Vesicular Stomatitis Virus Bovine Ephemeral fever virus

Virology

BUNYAVIRUS

Characteristics

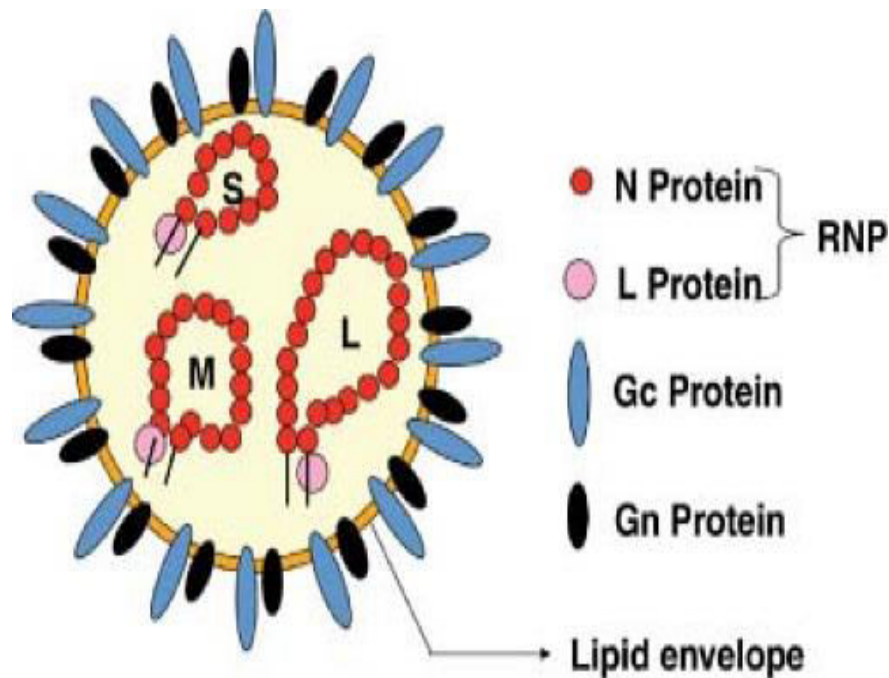
Bunyavirus

GENERAL PROPERTIES

- Enveloped RNA viruses
- Found in Arthropods and rodents ; occasionally infecting humans
- Some of them also infect plants
- Vector-borne except Hantaviruses
- Vectors (mosquitos, tick, or sandfly)
- Hantaviruses are transmitted through contact with deer mice feces
- Vector activity, for example, mosquito-borne viruses are more common in the summer

Bunyavirus

MORPHOLOGY



- Spherical
- 90-100nm
- Lipid envelope
- Spikes
- Lack matrix proteins

Bunyavirus

CLASSIFICATION

- Largest family
- 5 genera:
 1. Orthobunyavirus
 2. Hantavirus
 3. Nairovirus
 4. Phleovirus
 5. Tospovirus

Bunyavirus

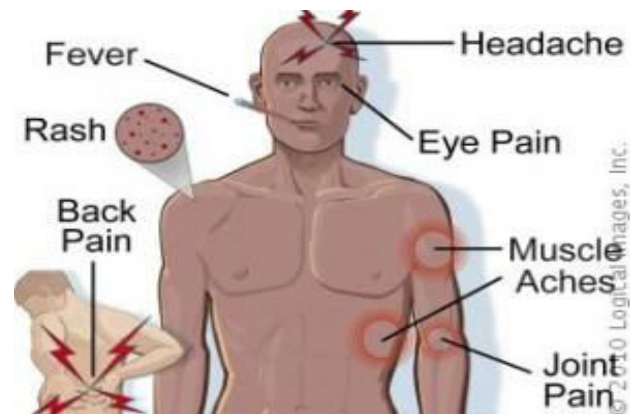
DISEASES

Family: Peribunyaviridae

California encephalitis
virus, La Crosse
encephalitis virus,
Jamestown Canyon virus
and Snowshoe hare virus
vector: mosquitoes

Family: Nairoviridae

Crimmean Hemorrhagic Congo
fever
Vector: ticks



Bunyavirus

DISEASES

Family:Phenuiviridae

- Rift valley fever
- Vector: mosquitoes

Family:Hantaviridae

- Hantavirus
- Vector: aerosolized excreta from these mammals

Virology

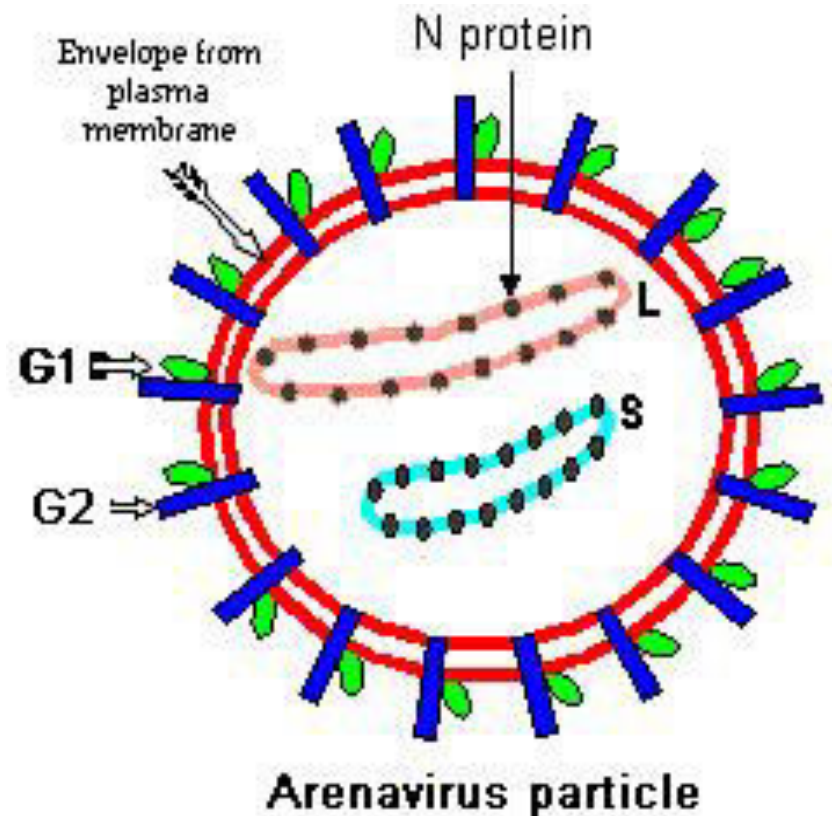
ARENNAVIRUS

Characteristics

Arenavirus

GENERAL PROPERTIES

- Name from Latin
“arenosus” meaning sand
- Sandy appearance of ribosomes
- two single-stranded RNA segments
- Virion Pleomorphic or spherical
- 60 to 300nm (variable)



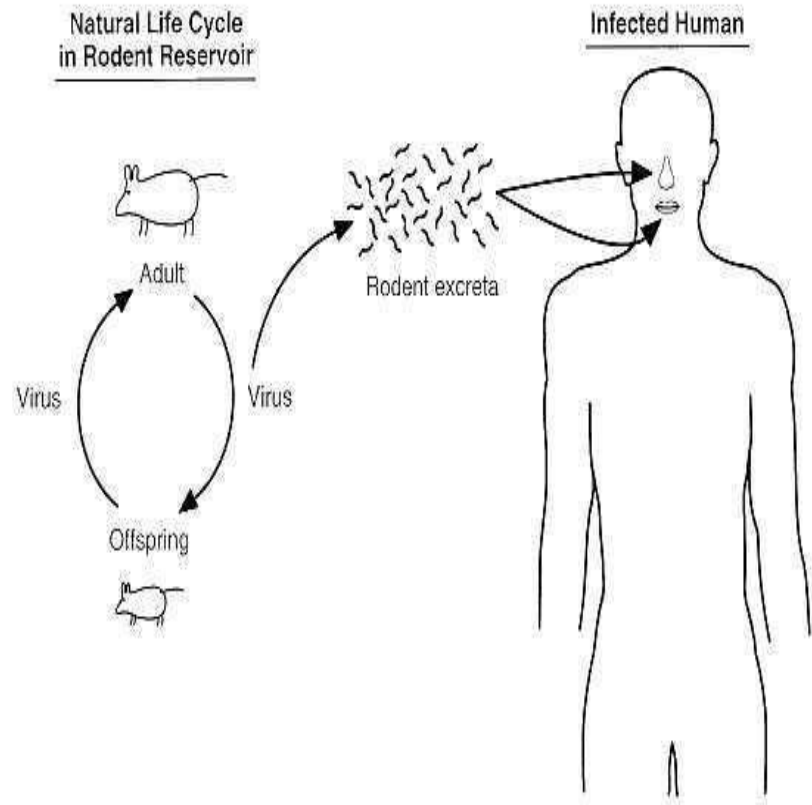
Arenavirus

- family *Arenaviridae*
- Hosts; rodents and occasionally humans
- Snakes
- At least eight arenaviruses are known to cause human disease
- The diseases derived from arenaviruses range in severity

Arenavirus

DISEASES

- Transferred from rodents to humans
- Aseptic meningitis
- Viral hemorrhagic fever



Arenavirus

DISEASE

- Lymphocytic
Choriomeningitis
Virus infection
- Hemorrhagic fever
from a number of
virus belonging to this
family

Virology

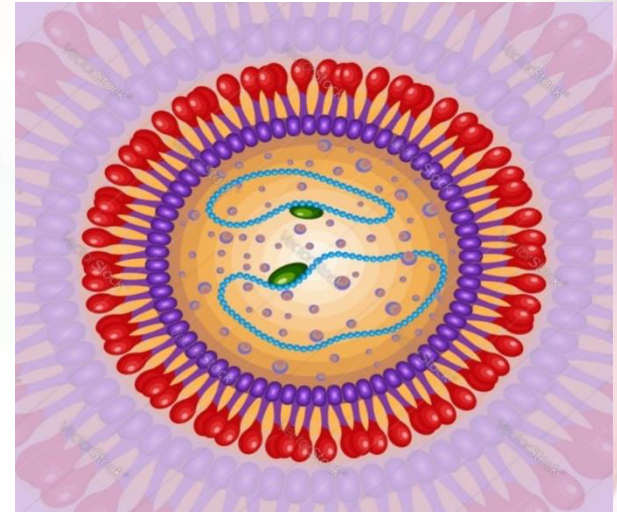
ARENAVIRUS

Genome

Arenavirus

GENOMIC CHARACTERS

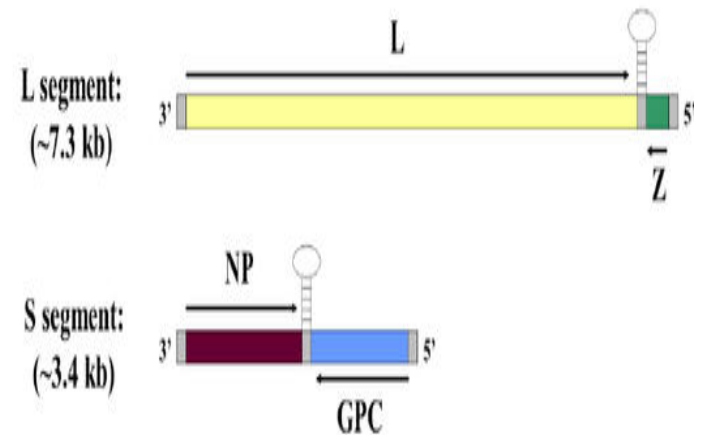
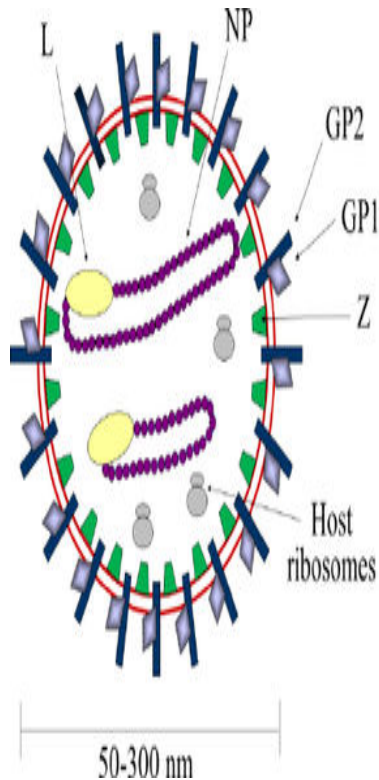
- Two single stranded RNA strands
- Ambisense RNAs
- Negative sense



Arenavirus

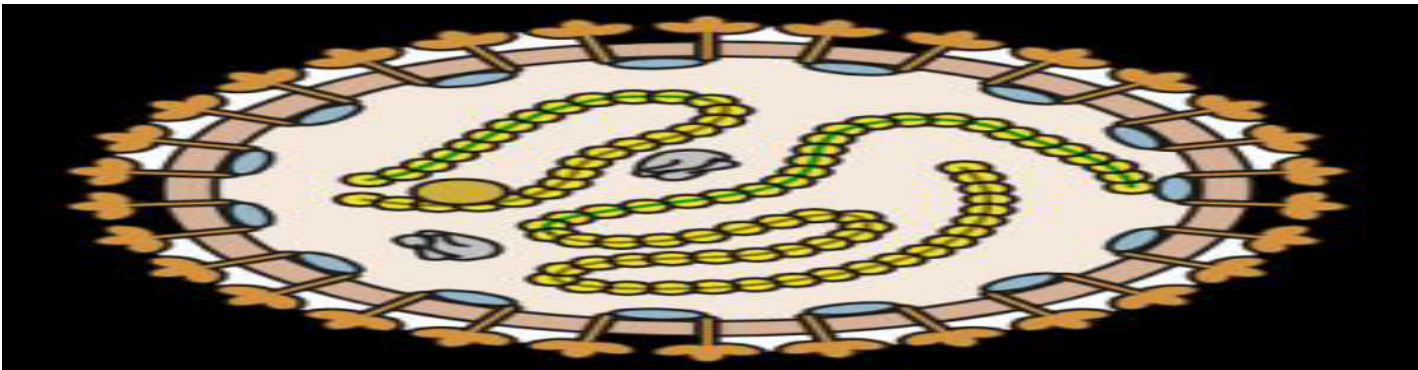
SEGMENTED GENOME

- Segmented genome
 - Small S
 - Large L



Arenavirus

- The S-segment RNA is approximately 3.5 kb
- Encodes the viral nucleocapsid protein (NP) and glycoprotein (GPC).
- The L-segment RNA is approximately 7.2 kb
- Encodes the viral RNA-dependent RNA-polymerase (L) and a small RING-domain containing protein (Z).



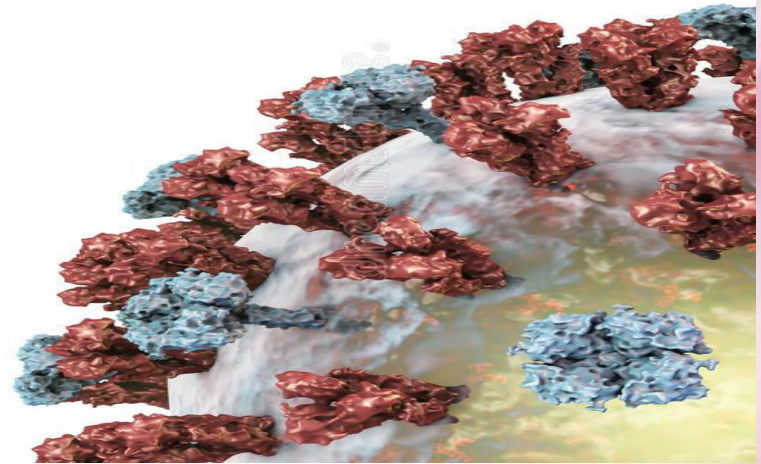
Arenavirus

- The Z protein forms homo oligomers and a structural component of the virions.
- The formation of these oligomers is an essential step for particle assembly and budding. Binding between Z and the viral envelope glycoprotein complex is required for virion infectivity.
- Z also interacts with the L and NP proteins.
- Polymerase activity appears to be modulated by the association between the L and Z proteins.
- Interaction between the Z and NP proteins is critical for genome packaging.

Virology

ORTHOMYXOVIRIDAE

Characteristics



Orthomyxoviridae

NOMENCLATURE

- Greek word
- Orthos meaning proper
- Myx meaning mucus

Orthomyxoviridae

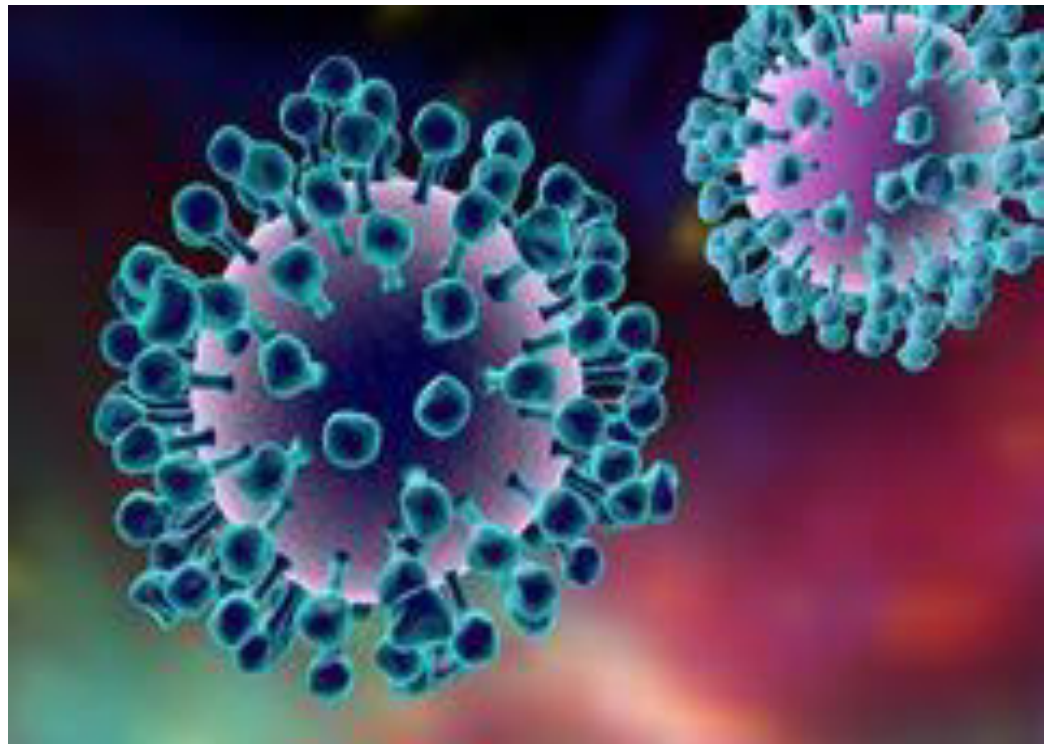
MORPHOLOGY

- Spherical/ pleomorphic
- Enveloped
- 80-120nm
- Spikes on Influenza A and B
- Labile to heat, pH variations, detergents

Orthomyxoviridae

PROTEINS

- The [hemagglutinin](#) (HA)
- [The neuraminidase](#) (NA)



Orthomyxoviridae

GENERA

- Influenza virus A
- Influenza virus B
- Influenza virus C
- Influenza virus D
- Isavirus
- Thogotovirus
- Quaranjavirus

Orthomyxoviridae

DISEASES

- Influenza A infects humans, other mammals, and birds, and causes all flu pandemics
- Influenza B infects humans and seals
- Influenza virus C infects humans, pigs, and dogs
- Influenza virus D infects cattle and pigs

Virology

ORTHOMYXOVIRIDAE

Genome

Orthomyxoviridae

GENOMIC PROPERTIES

- Helical nucleocapsid
- Segmented genome
- Negative sense
- Single stranded RNA
- 10 – 13.6kb
- 12000-15000kb

Orthomyxoviridae

RNPs

- The segmented genome is encapsidated with each segment in a separate nucleocapsid called as *Ribonucleoprotein complexes (RNPs)*

Orthomyxoviridae

- terminal sequences of RNA bound to trimeric RdRp.
- All these RNPs are surrounded by one envelope

Orthomyxoviridae

segment	Protein
PB1	Polymerase
PB2	Polymerase
PA	Polymerase
HA	Hemagglutinin
NP	Nucleoprotein
NA	Neuraminidase
M	Membrane protein(s)
NS	non-structural protein(s)

Orthomyxoviridae

REPLICATION

- Entrance through body fluids
- Virus binds through hemagglutinin-glycoprotein and sialic acid receptors
- Virus enters by endocytosis
- Proteins and polymerase forms complex
- New protein synthesis
- Replication in the nucleus
- Eight segments of genome formed

Virology

PARAMYXOVIRIDAE

Characteristics

Paramyxoviridae

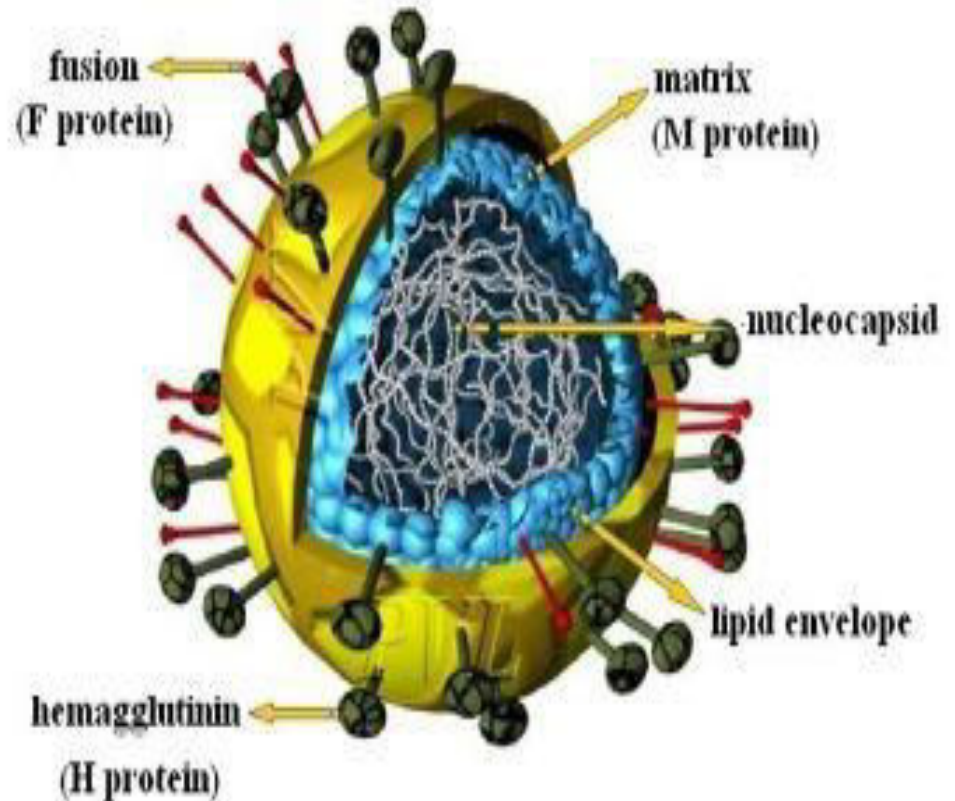
NOMENCLATURE

- **Para** means beyond
- **Myxo** means mucus
- Paramyxo and orthomyxo were formerly known as myxoviruses

Paramyxoviridae

PROPERTIES

- Pleomorphic
- 150-300nm diameter
- Enveloped
- Single stranded RNA



Paramyxoviridae

- Spikes on envelope
- Sensitive to heat, chemicals and radiations
- Host: Vertebrates
- Genera: 7
- Species: 49

Paramyxoviridae

PROTEINS

- N – the nucleocapsid protein associates with genomic RNA and protects the RNA from nuclease digestion
- P – the phosphoprotein binds to the N and L proteins and forms part of the RNA polymerase complex
- M – the matrix protein assembles between the envelope and the nucleocapsid core, it organizes and maintains virion structure

Paramyxoviridae

PROTEINS

- F – the fusion protein projects from the envelope surface as a trimer, and mediates entry
- H/HN/G – the cell attachment proteins span the viral envelope and project from the surface as spikes.
- L – the large protein is the catalytic subunit of RNA dependant RNA polymerase (RDRP)
- Accessory proteins – a mechanism known as RNA editing allows multiple proteins to be produced from the P gene.

Virology

PARAMYXOVIRUS

Genome

Paramyxoviridae

GENOMIC CHARACTERS

- Nucleocapsid: helical
- 13-18nm
- Non Segmented linear
- Negative sense
- Single stranded RNA
- 15-16kb
- Contain 6-10 genes

Paramyxoviridae

Extracistronic regions

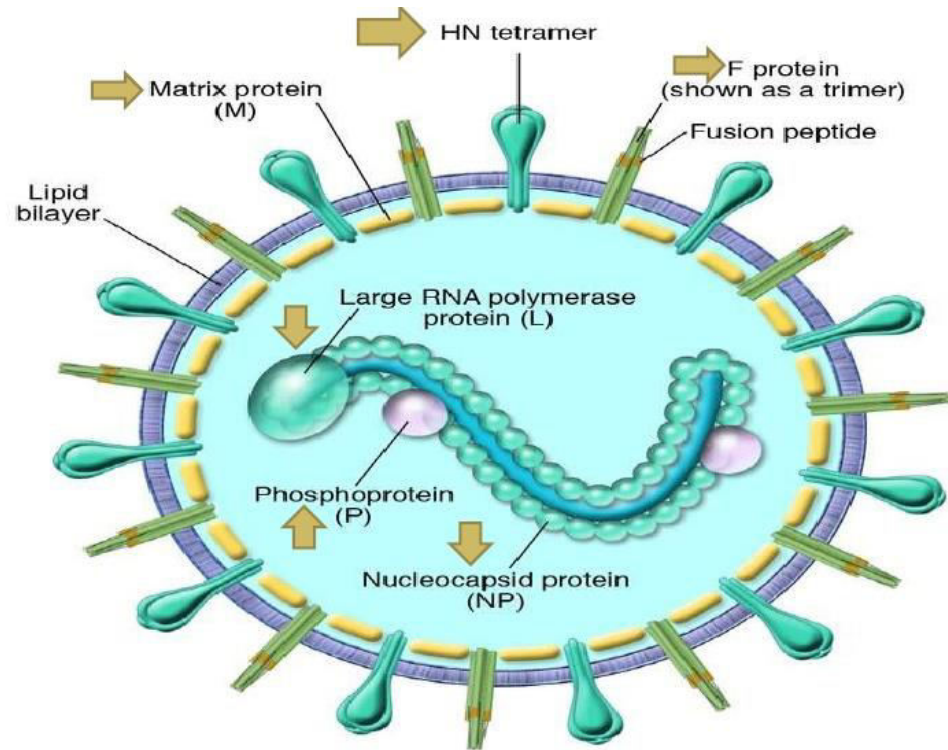
- A 3' leader sequence, , which acts as a transcriptional promoter.
- A 5' trailer sequence, 50–161 nucleotides long

Intergenomic regions

- between each gene
- which are three nucleotides long for morbilliviruses, respiroviruses and henipaviruses
- variable length (1-56 nucleotides) for rubulaviruses

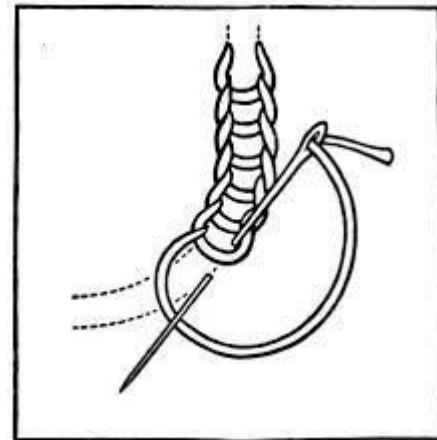
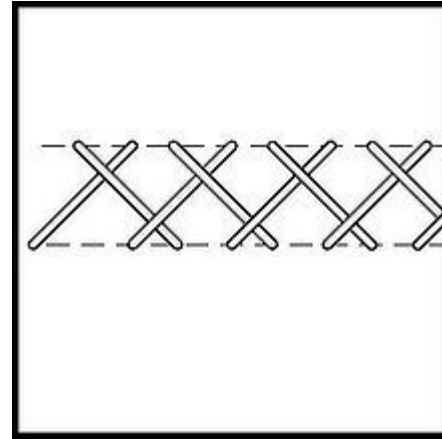
Paramyxoviridae

- Fusion Protein
- Haemagglutinin-Neuraminidase
- Matrix protein
- Phosphoprotein
- Nucleoprotein
- Polymerase protein



Paramyxoviridae

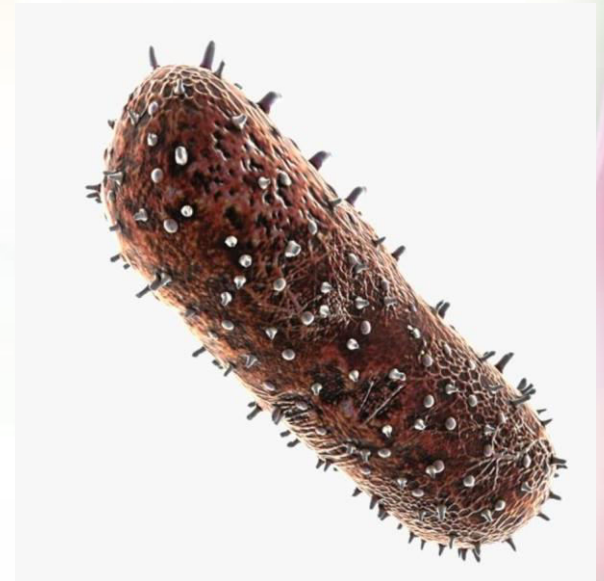
- Characteristic herring bone appearance
- Nucleocapsid core contains:
 - Genomic RNA
 - Nucleocapsid protein
 - Polymerase protein



Virology

RHABDOVIRIDAE

General Properties



Rhabdoviridae

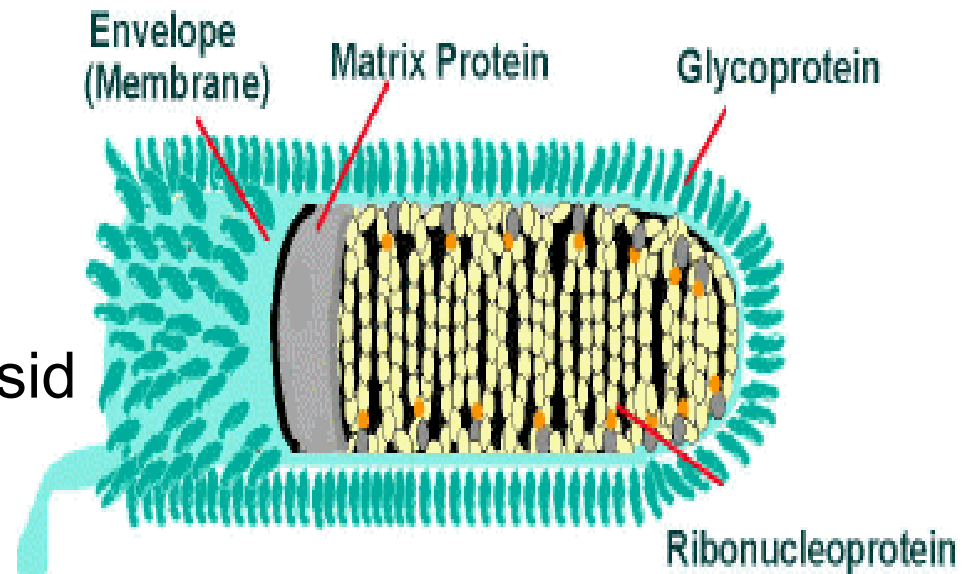
MEANING

- Greek word
- Rhabdos meaning rod
- Referring to shape of virus
- Mononegavirales

Rhabdoviridae

MORPHOLOGY

- Rod or bullet shaped
- Enveloped
- Bacilliform
- Helical nucleocapsid
- 75-180nm long
- RNA genome
- 18 genera



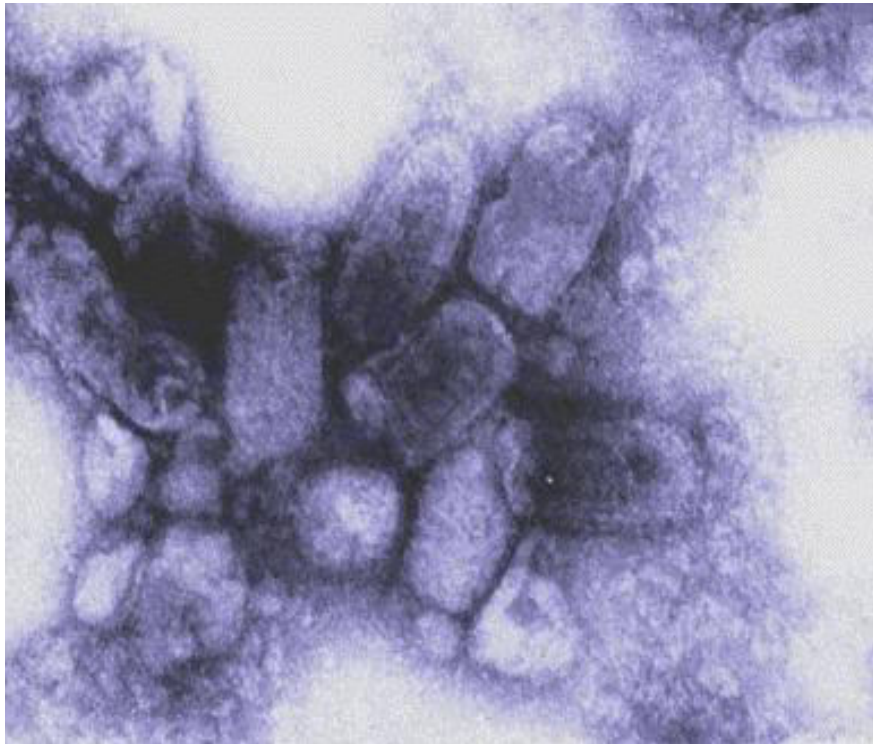
Rhabdoviridae

PROTEINS

- large protein (L)
- glycoprotein (G)
- nucleoprotein (N)
- phosphoprotein (P)
- matrix protein (M)

Rhabdoviridae

MEMBERS



- Vesicular stomatitis
Indiana Virus
- Lyssavirus
- Ephemerovirus
- Cytorhabdovirus
- Nucleorhabdovirus

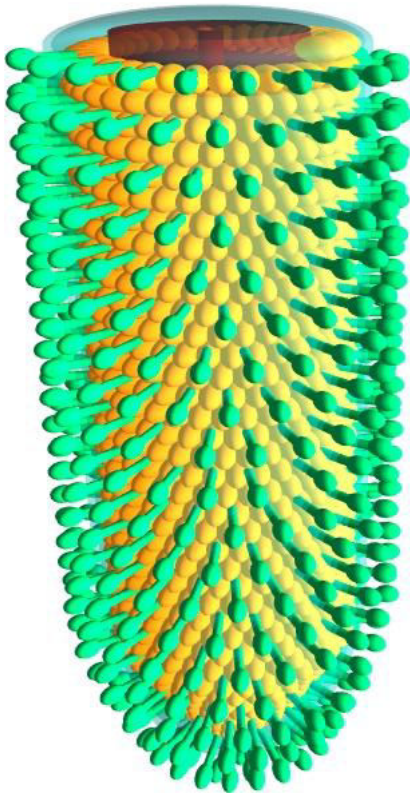
Rhabdoviridae

VESICULOVIRUS

- Infects human, cattle, horses
- Flu-like illness
- Fever
- Seizures
- Diarrhea
- Vomiting
- Coma
- Death

Rhabdoviridae

LYSSAVIRUS



- Infects human
- Bat and rodent carrier
- Throughout world
- Causes rabies
- Anxiety, confusion, agitation, paranoia, hallucinations
- Death

Rhabdoviridae



EPHEMEROVIRUS

- Infects cattle primarily
- Causes Bovine Ephemeral Fever
- Fever, stiffness, lameness
- Nasal and ocular discharges
- High morbidity and low mortality



Rhabdoviridae

CYTORHABDOVIRUS



- Plant viruses
- Bacilliform
- Mature in cell cytoplasm
- lettuce necrotic yellow virus
- barley yellow striate mosaic virus

Rhabdoviridae

NUCLEORHABDOVIRUS

- Plant viruses
- Bacilliform
- Linear genomes
encoding 5 to 6 proteins
- Nuclear viral replication
transmitted by insect bite

Virology

RHABDOVIRIDAE

GENOME



Rhabdoviridae

Genomic Properties

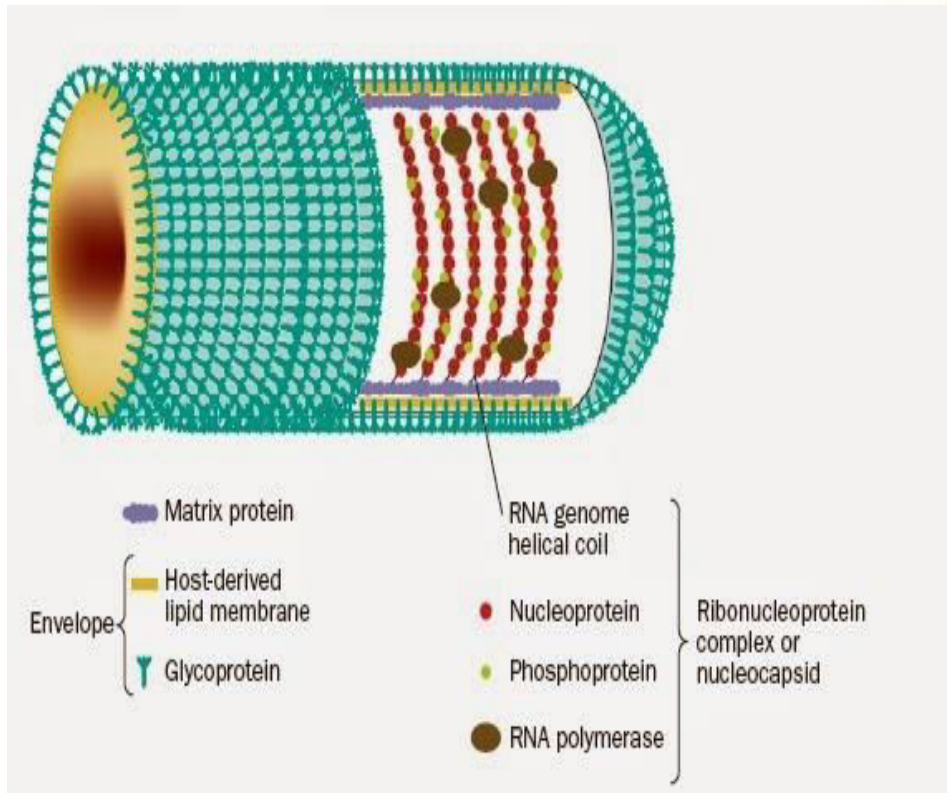
- Single stranded
- RNA



Rhabdoviridae

GENES

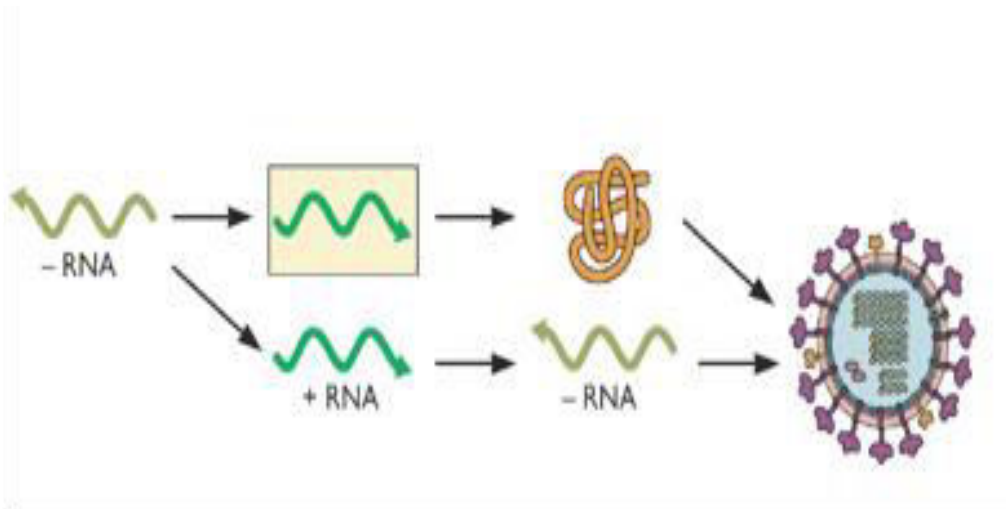
- Five proteins:
- Large protein (L)
- Glycoprotein (G)
- Nucleoprotein (N)
- Phosphoprotein (P)
- Matrix protein (M)



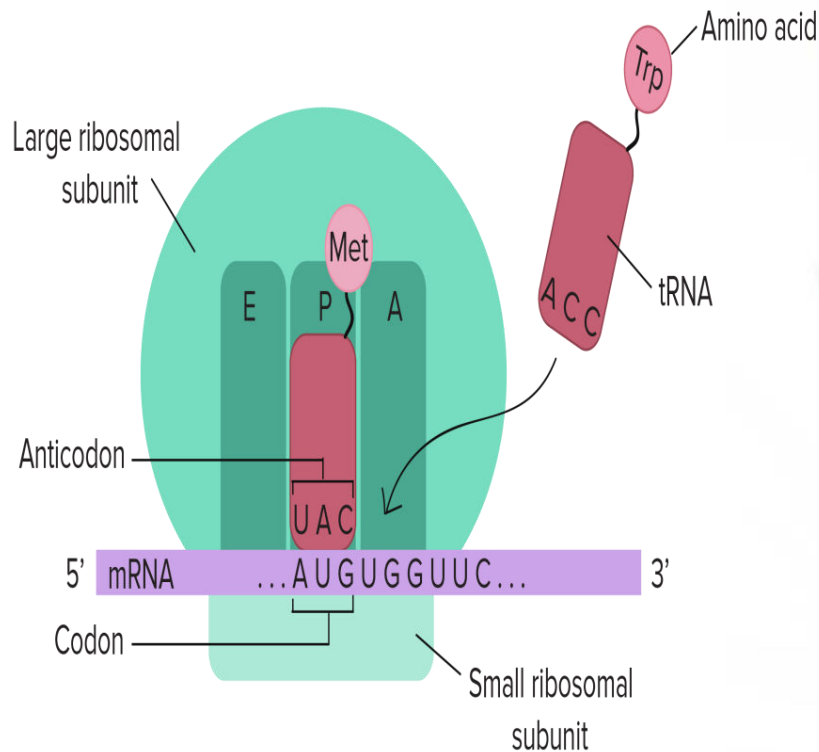
Rhabdoviridae

TRANSCRIPTION

- 1 L and 3 P proteins
- Right after entry
- From 3' to 5'
- Produces mRNA



Rhabdoviridae



TRANSLATION

- Translated on free ribosomes
- G protein translated on RER
- Single peptides of G protein

Rhabdoviridae

REPLICATION

- Cytoplasmic
- Attachment of G protein
- Endocytosis
- Negative strand of RNA transcribed
- Virus leaves by budding

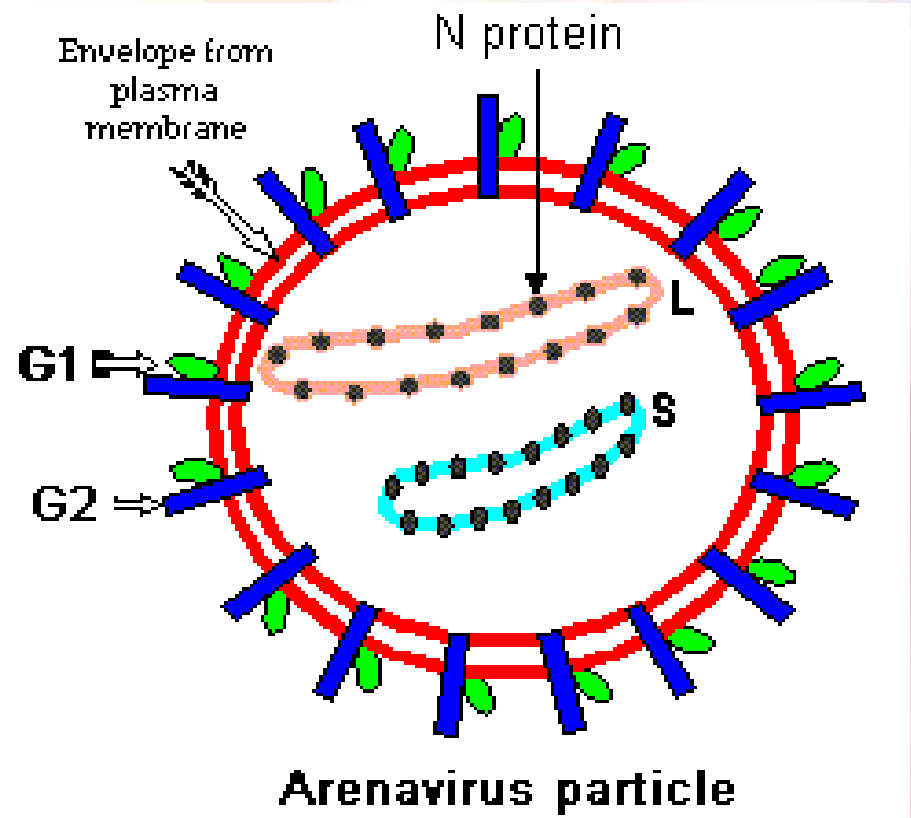
Virology

SEGMENTED AND MULTIPARTITE GENOME

Multipartite and Segmented Genome

ARENAVIRUS

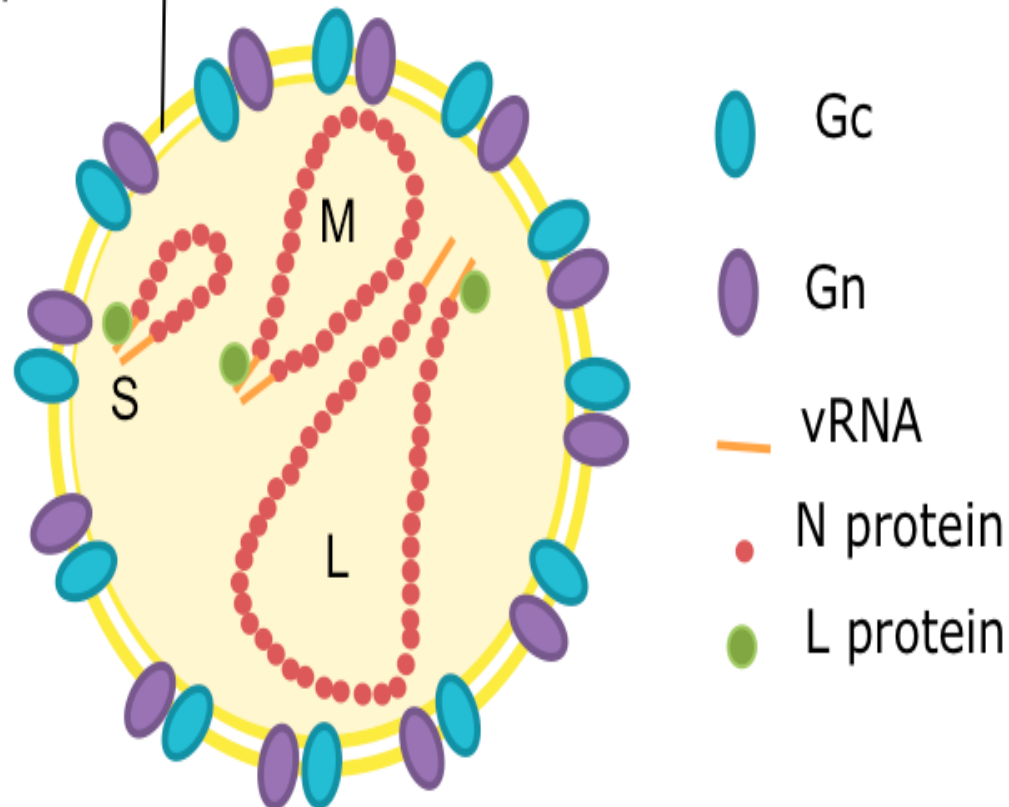
2 segments



Multipartite and Segmented Genome

BUNYAVIRUS 3 segments

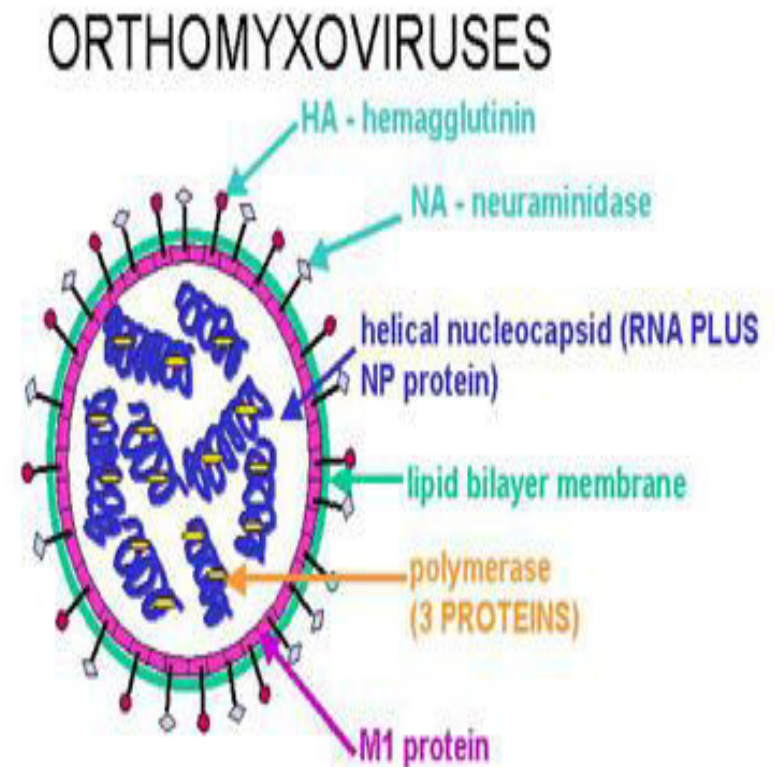
Lipid envelope



Multipartite and Segmented Genome

ORTHOMYXOVIRUS

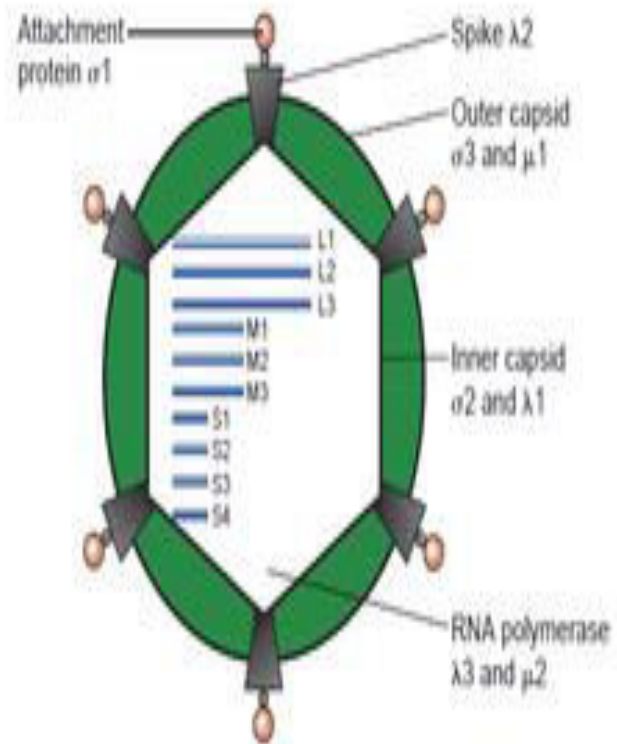
8 segments



Multipartite and Segmented Genome

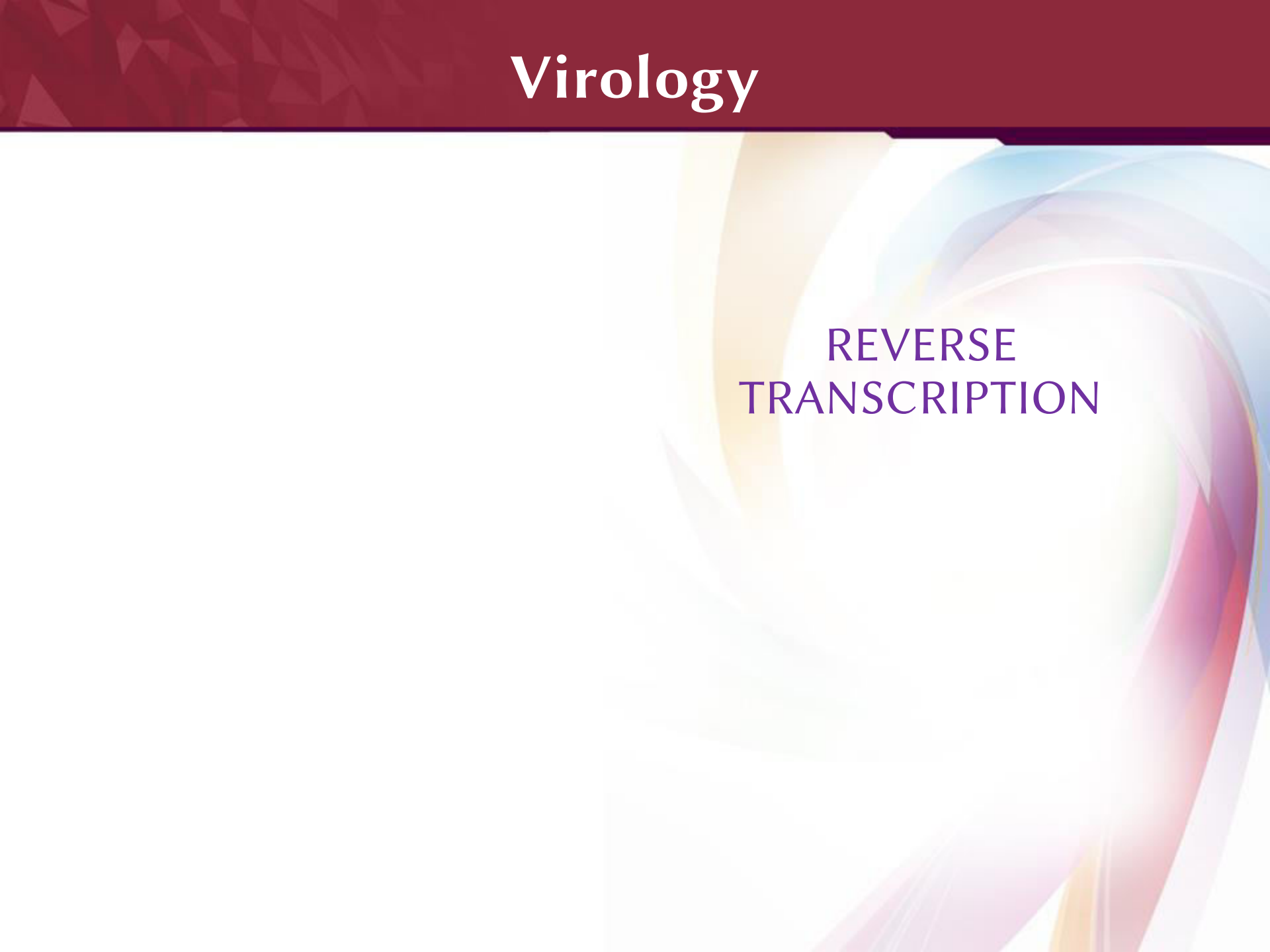
REOVIRUS

10 segments



Virology

REVERSE
TRANSCRIPTION



Reverse Transcription

Reverse transcription is a process in which a double standard DNA molecules are made from a single stranded RNA

The name of this method originated by its opposite direction to transcription process

Reverse Transcription

It involves the presence of

- Reverse transcriptase,
an enzyme

- A primer
some bases
of nucleotides
segments

- Reverse transcriptase
inhibitor

leucine rich repeat
protein

Reverse Transcription

Reverse Transcription

Virology

TRANSPOSONS

Importance

Transposons and its Important

TRANSPOSONS

A sequence of DNA that can change its position, from place to place within a genome

Transposons and its Important



DISCOVERY

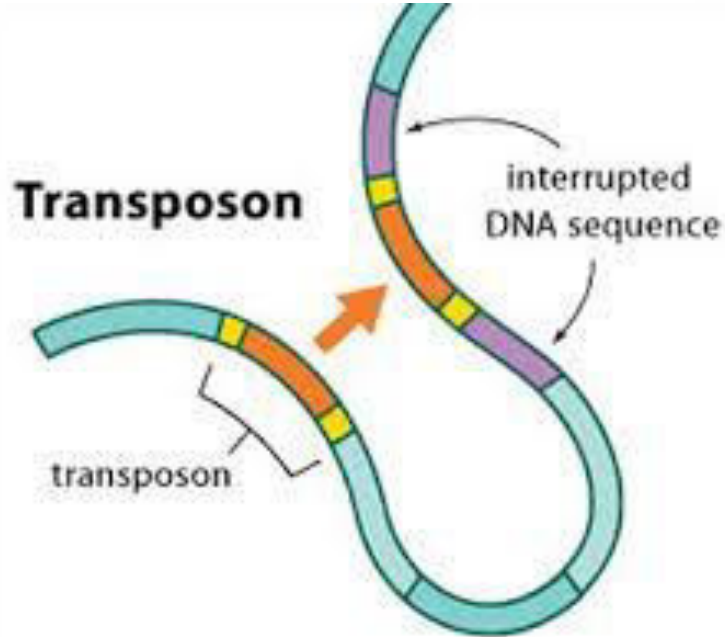
- 1940s
- Barbera McClintock

Transposons and its Important

TYPES

Two groups:

- Simple transposons
- Reterotransposons



Transposons and its Important

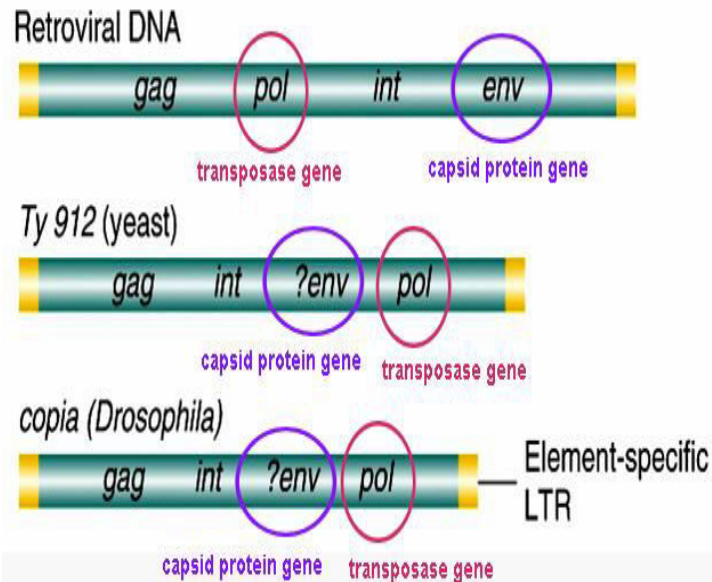
SIMPLE TRANSPOSONS

- In prokaryotes
- Which do not undergo reverse transcription
- e.g. genome of enterobacteria phage Mu

Transposons and its Important

RETROTRANSPOSONS

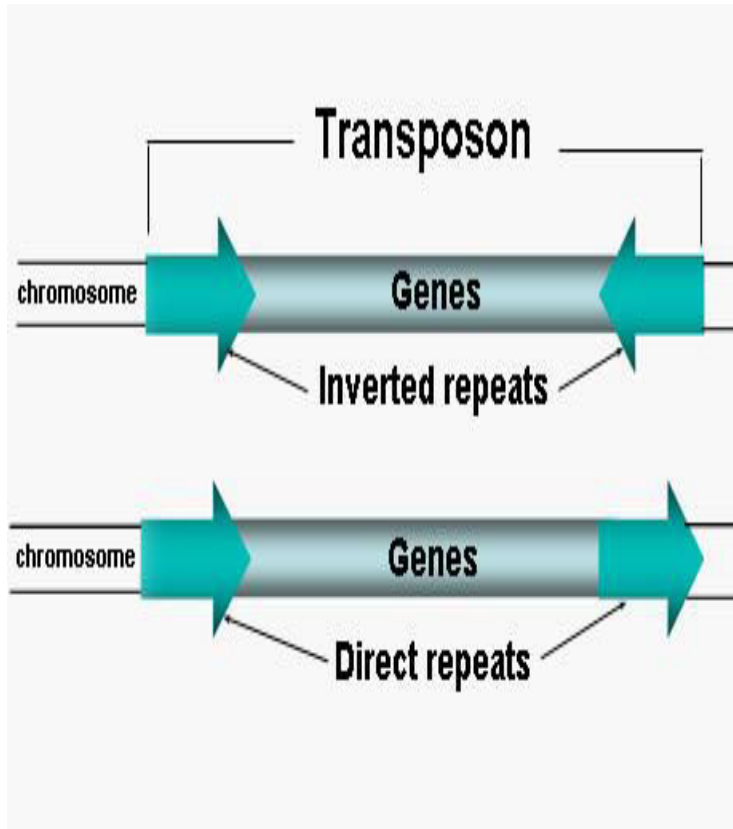
Genes in retroviral DNA and viral retrotransposons



- Closely resemble retroviral genome
- Bound by long repeat units (LTRs)
- Move by means of transcription/ reverse transcription/ integration mechanism
- Found in eukaryotes
- In *Metaviridae* and *Pseudoviridae*

Transposons and its Important

PROPERTIES



- Responsible for high mutation
- Promote genetic rearrangements like deletion, inversion, duplication etc
- Often accompanied by replication
- Control their own transposition functions

Virology

Isolation, Cultivation and Identification of Viruses

Isolation, Cultivation and Identification

ISOLATION

➤ Centrifugation

- Differential centrifugation- high vs low speed to separate cells from viruses
- Gradient centrifugation
 - separate by size or density

Isolation, Cultivation and Identification

CULTIVATION

- To cultivate the virus in specific host



Isolation, Cultivation and Identification

CULTIVATION



Animal



Plant



Bacteria



Insect

Isolation, Cultivation and Identification

METHOD OF CULTIVATION

Cell culture

Organized tissue

Live animal

Embryonated egg

Isolation, Cultivation and Identification

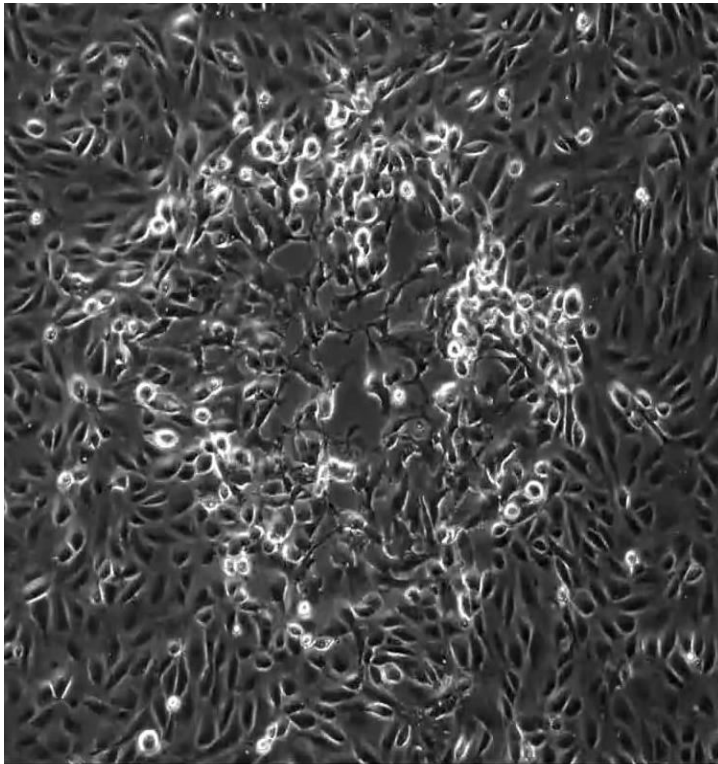
IDENTIFICATION



- Plaque formation
- Cytopathic Effects
- Fluorescent Antibody technique
- Nucleic Acid hybridization
- Electron Microscopy

Isolation, Cultivation and Identification

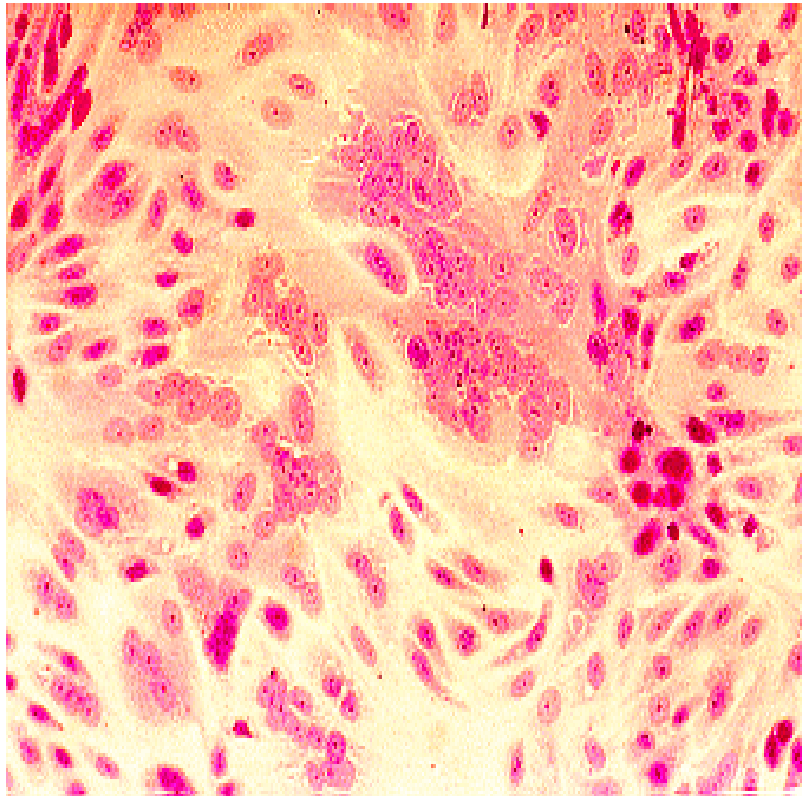
PLAQUE FORMATION



- a visible structure formed within a cell culture
- The bacteriophage viruses replicate and spread, thus generating regions of cell destructions known as plaques

Isolation, Cultivation and Identification

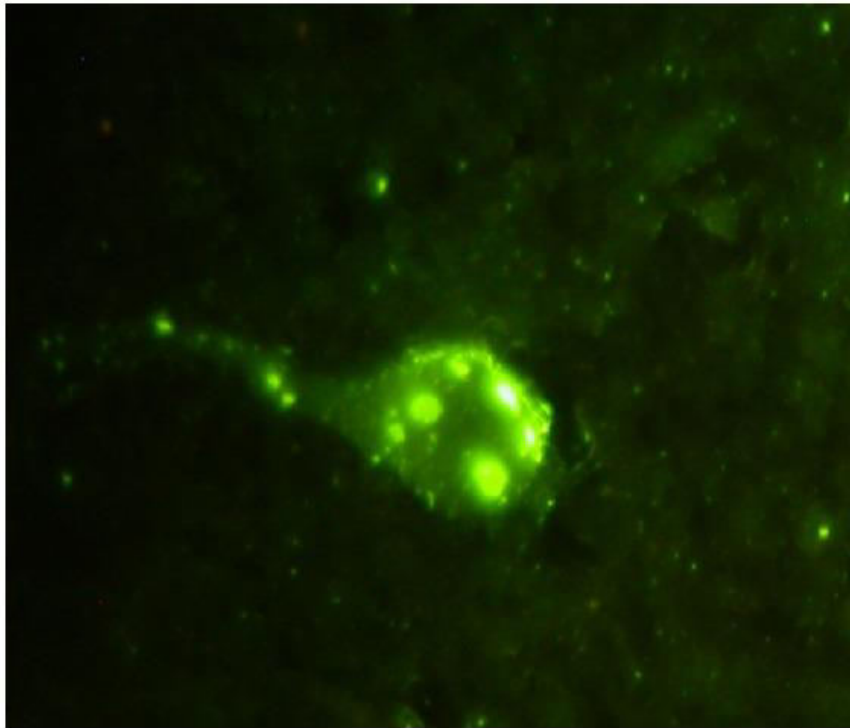
CYTOPATHIC EFFECTS



- Syncytia
- Cell rounding
- Membrane proliferations
- Vacuolization
- Inclusion bodies
- Focus (foci)
- Hemadsorption

Isolation, Cultivation and Identification

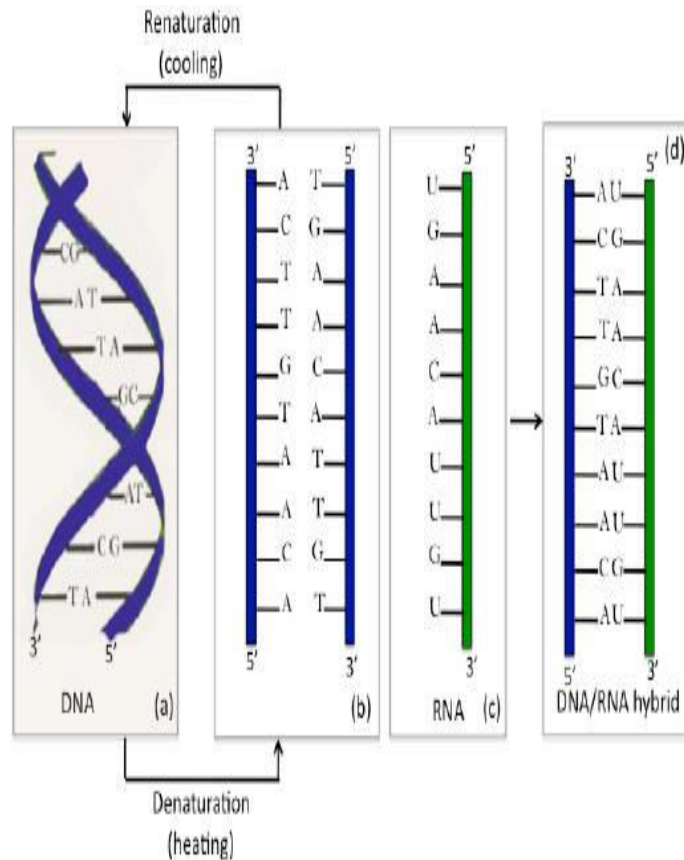
FLUORESCENT ANTIBODY TECHNIQUE



- based on the observation that virus proteins are (antigen) present in the tissues
- e.g rabies virus present in nervous tissues

Isolation, Cultivation and Identification

NUCLEIC ACID HYBRIDIZATION

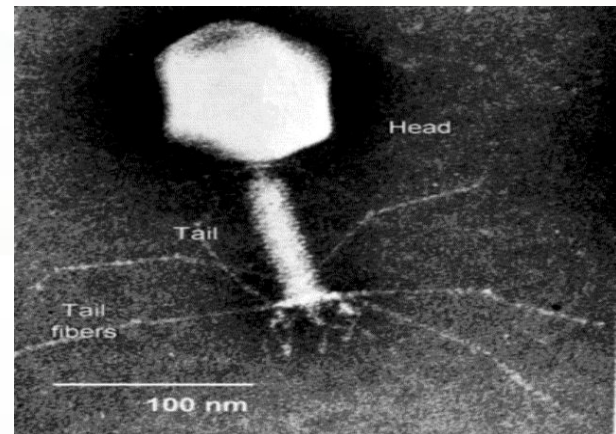


A technique in which single-stranded nucleic acids (DNA or RNA) are allowed to interact so that complexes called hybrids are formed by molecules with similar, complementary sequences

Isolation, Cultivation and Identification

ELECTRON MICROSCOPY

- uses a beam of **electrons** to create an image of the specimen
- Inclusion bodies formed



END

Virology

Growing Bacteriophages in Laboratory



Growing Bacteriophages

METHODS

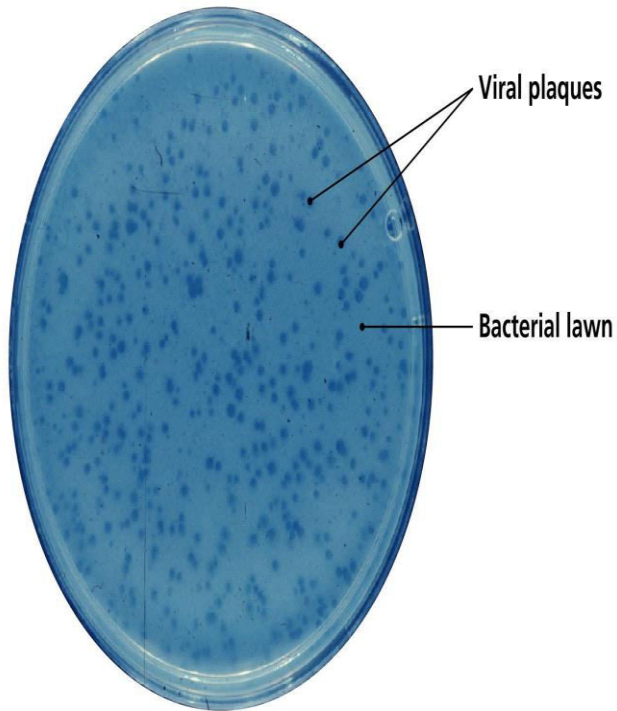
Two methods

- In liquid media
- On solid media



Growing Bacteriophages

ON SOLID MEDIA



- Like bacteria
- Helps in plaque test to detect and count viruses
- Bacteriophage sample mixed with host bacteria and melted agar

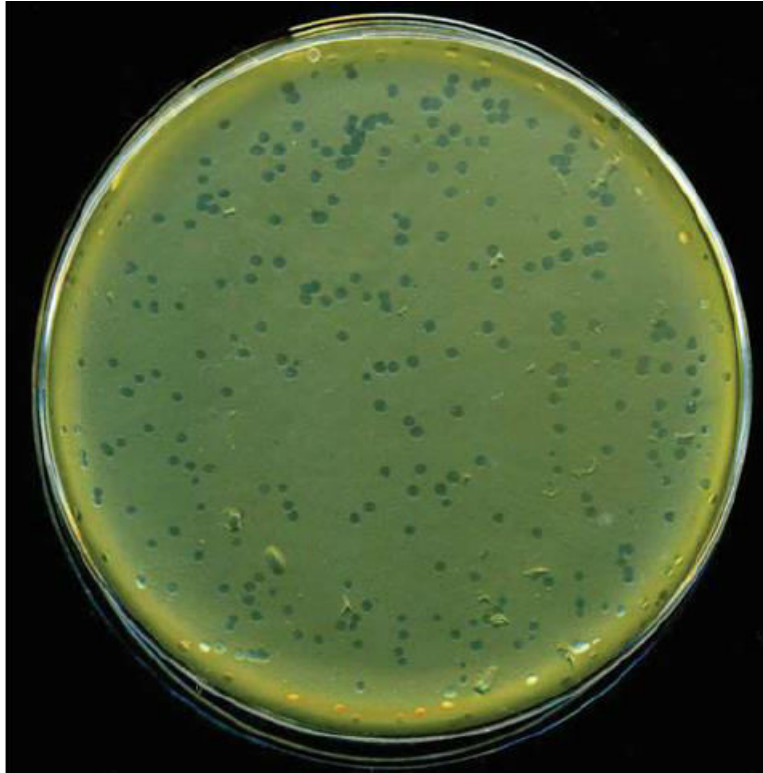
Growing Bacteriophages

INNOCULATION



- Mixed agar poured into petri plate
- Petri plate already containing hardened layer of agar media
- Mixture solidifies into top thin layer

Growing Bacteriophages

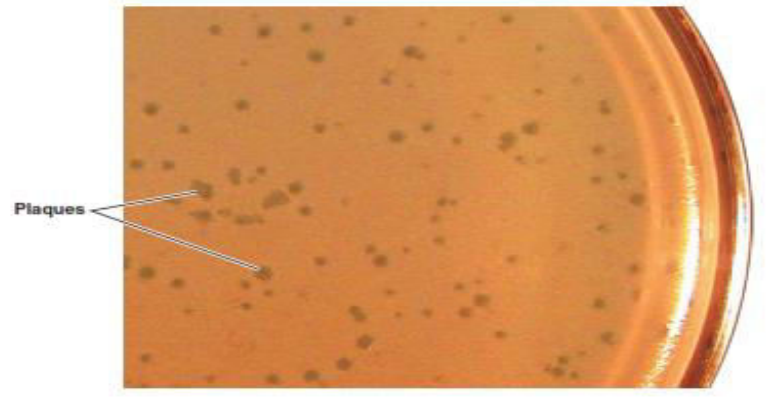


- ❖ Each virus infects bacteria
- ❖ Multiplies
- ❖ Releases new viruses
- ❖ New viruses infect other bacteria
- ❖ All bacteria get destroyed

Growing Bacteriophages

PLAQUE FORMATION

- Produces a number of clearings or plaques
- Visible by naked eye
- Uninfected bacteria form



END

Virology

Growing Animal Viruses in Embryonated Eggs

Growing Animal viruses

METHODS

1. Animals
2. Cell culture
- 3. Embryonated eggs**

Growing Animal viruses

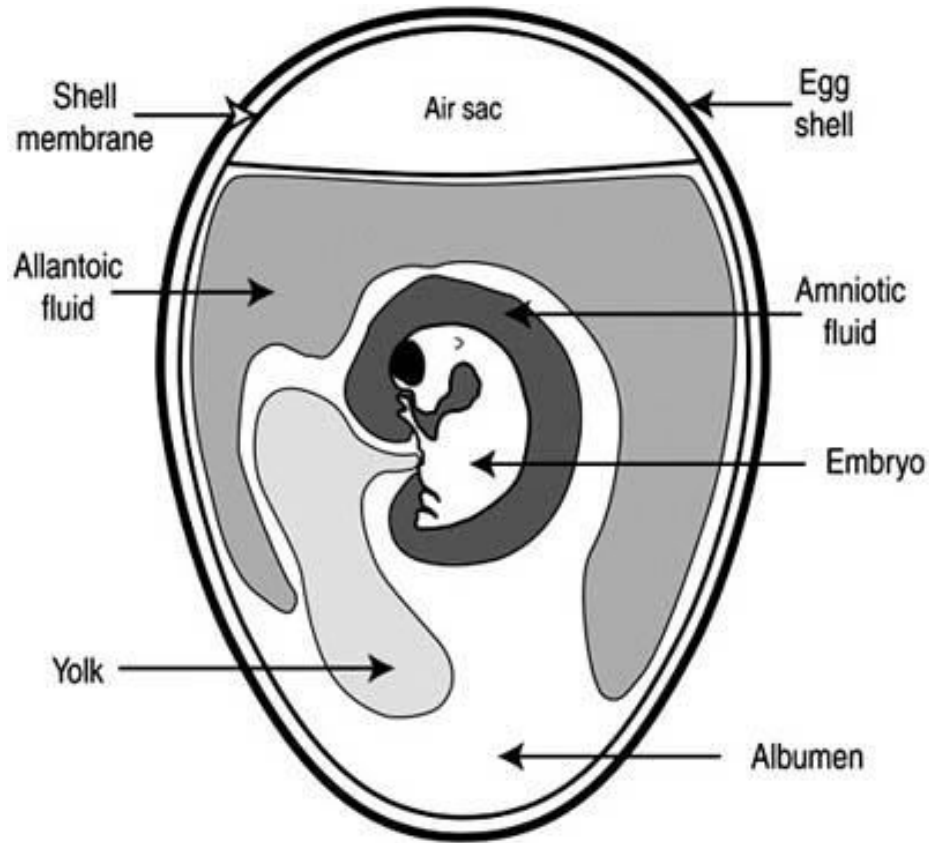
ADVANTAGE



- Convenient
- Inexpensive
- Host for many viruses

Growing Animal viruses

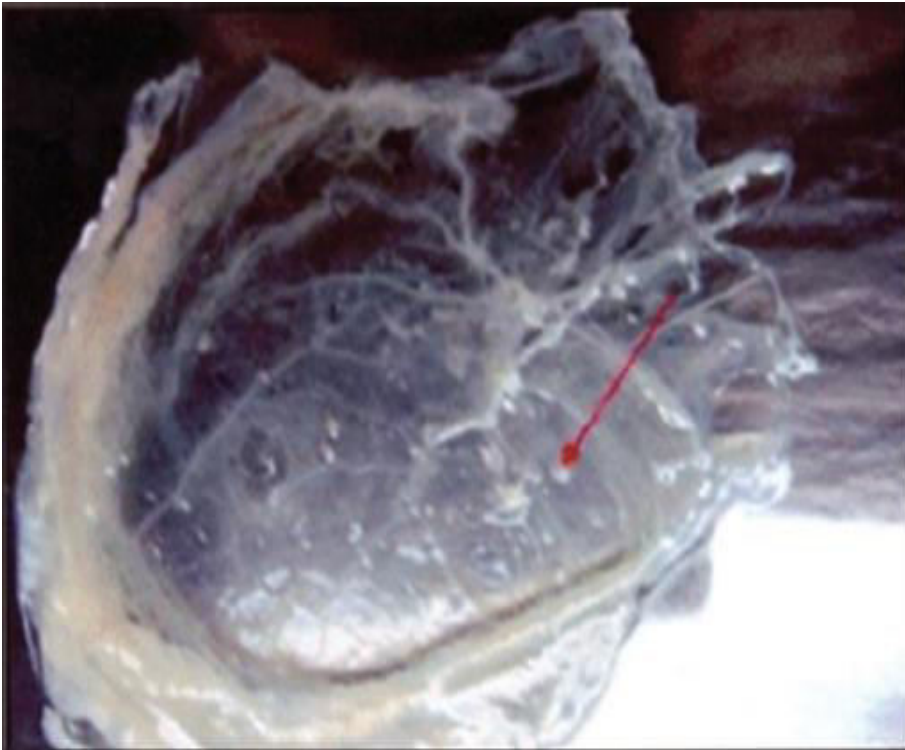
PROCEDURE



- Hole is drilled in the shell
- Viral suspension inoculated
- Many routes

Growing Animal viruses

VIRAL GROWTH

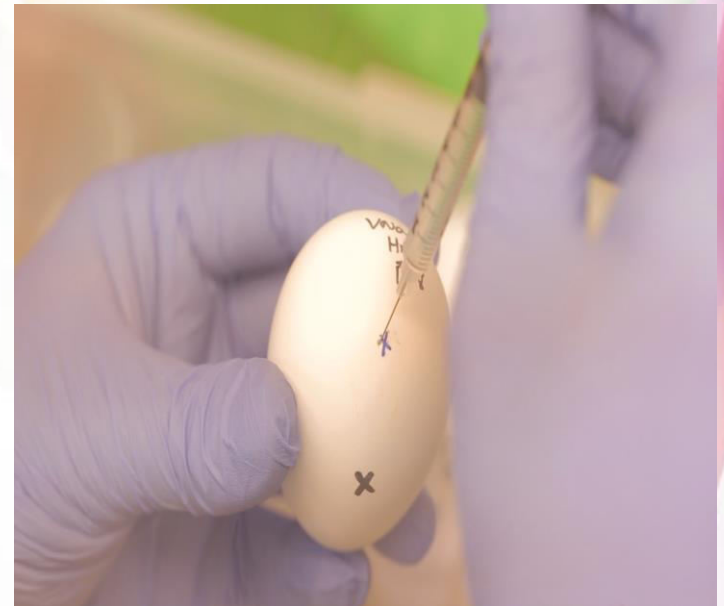


- Death of embryo
- Damage to embryo
- Pocks or lesions on egg membrane

Growing Animal viruses

PURPOSE

- Isolation and growth of virus
- Growth for vaccination



Growing Animal viruses

ROUTES

1. Chorioallantoic membrane inoculation
2. Amniotic inoculation
3. Allantoic inoculation
4. Yolk sac inoculation

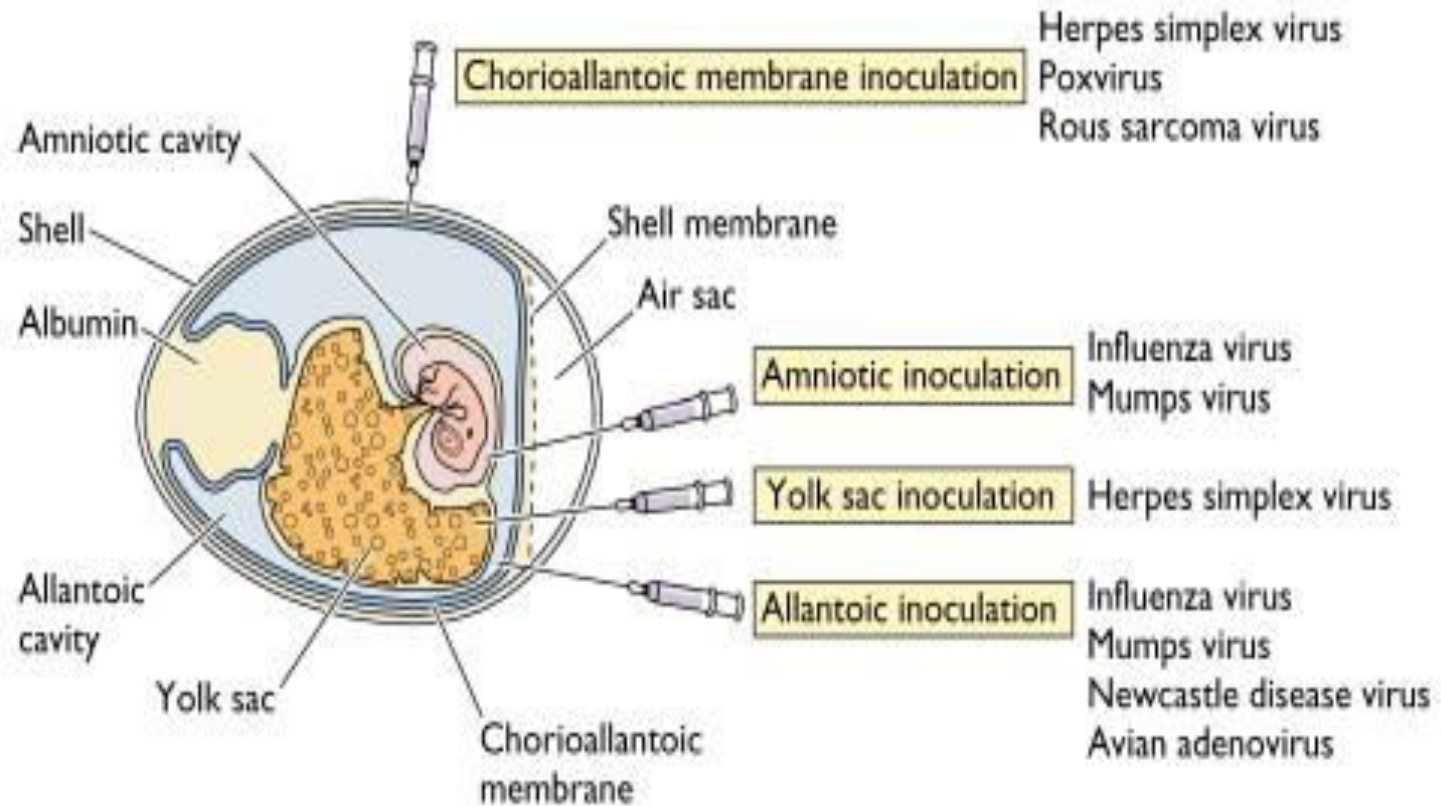
END

Virology

Chorioallantoic Membrane Inoculation

Chorioallantoic Membrane Inoculation

ROUTES OF INOCULATION

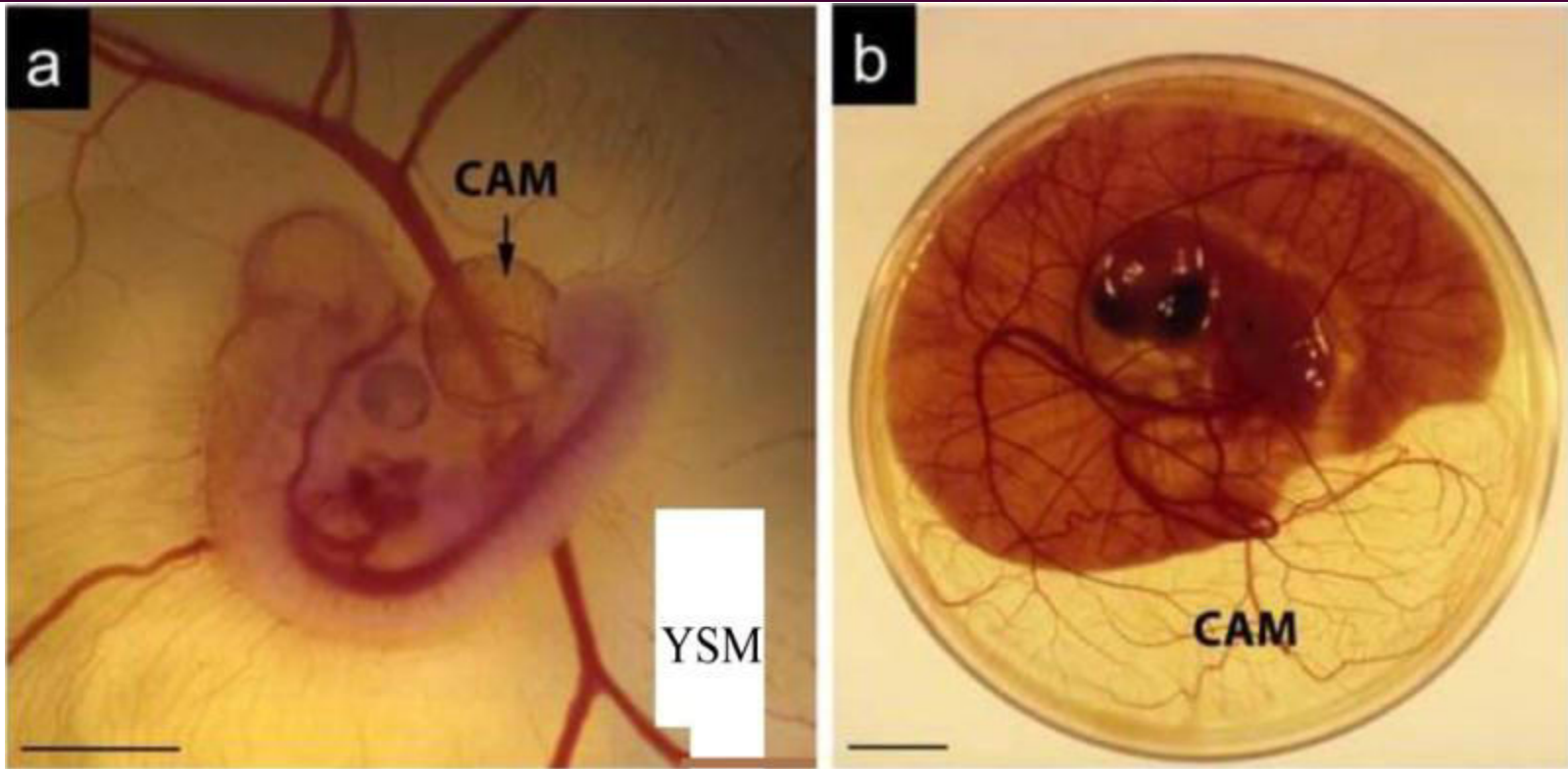


The different routes of inoculation into the egg are shown, as well as the different compartments in which viruses replicate.

Chorioallantoic Membrane Inoculation

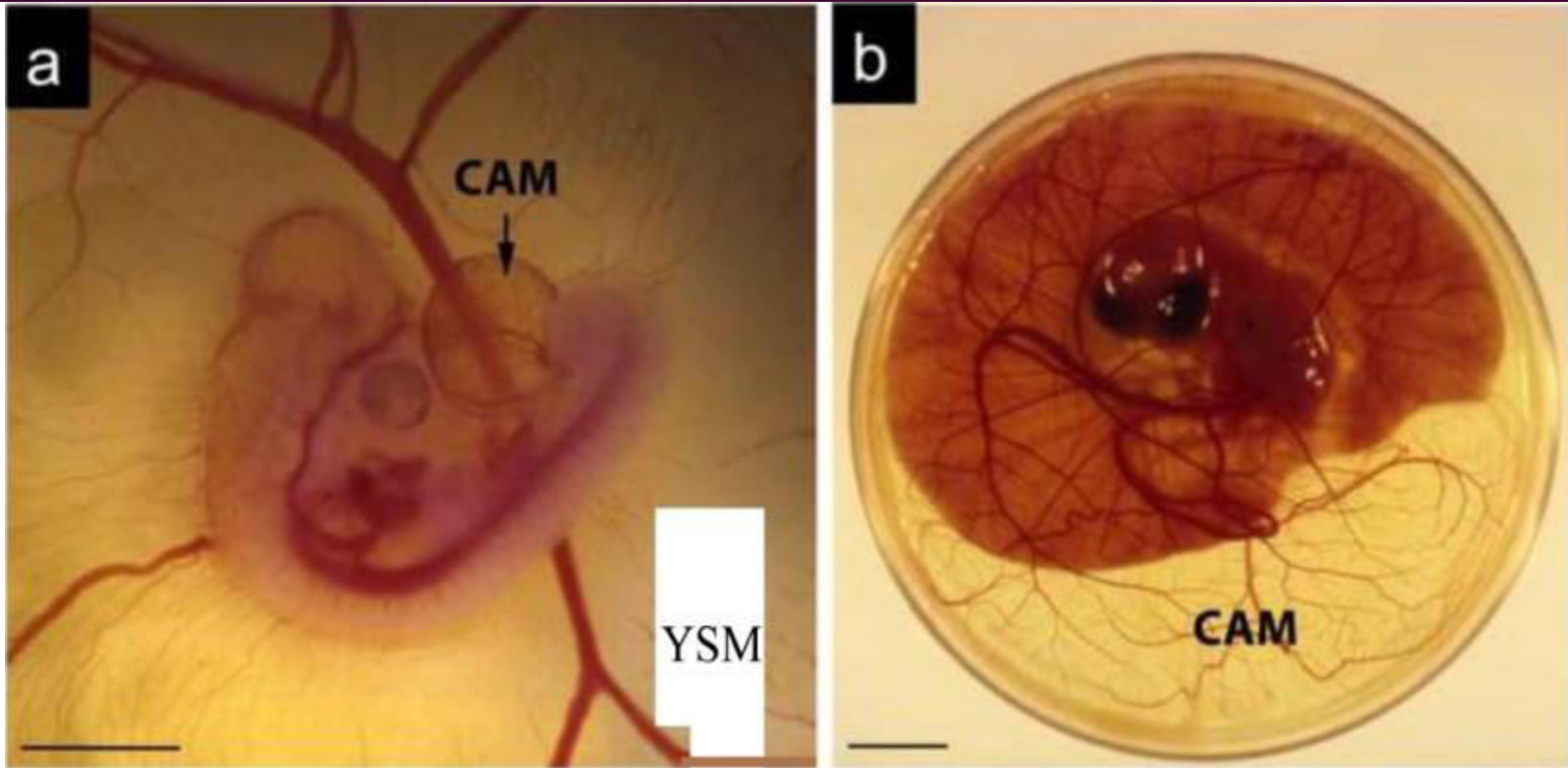
- The chicken chorioallantoic membrane (CAM) is a simple, highly vascularized extraembryonic membrane, which performs multiple functions during embryonic development. Over the last two decades, interest in the CAM as a robust experimental platform to study blood vessels has been shared by specialists working in bioengineering, development, morphology, biochemistry, transplant biology, cancer research and drug development

Chorioallantoic Membrane Inoculation



Chicken embryo and associated extra-embryonic structures. a, Chicken embryo at day 4 post-fertilization the highly vascularized chorioallantoic membrane (CAM) that expands from the hind-gut. The yolk sac membrane (YSM), also highly vascularized, is seen in the background; b Embryo at day 12 post-fertilization into a 10 cm petri dish.

Chorioallantoic Membrane Inoculation



Chicken embryo and associated extra-embryonic structures. a Chicken embryo at day 4 post-fertilization. Note the highly vascularized chorioallantoic membrane (CAM) that expands from the hind-gut. The yolk sac membrane (YSM), also highly vascularized, is seen in the background; b Embryo at day 12 post-fertilization into a 10 cm petri dish.

Chorioallantoic Membrane Inoculation



Virology

Allantoic Cavity Inoculation

Allantoic Cavity inoculation

- Taking Flu virus as an example
- Virus inoculated in allantoic cavity

Chorioallantoic Membrane Inoculation



Virology

**Intraembryonic, Intracranial and
Intravenous Inoculation**

I/E, I/C and I/V inoculation

- Inoculated
- Incubated for few days
- Lesions or death observed

I/E, I/C and I/V inoculation

- These inoculations of embryos are not widely used.
- Embryonic inoculation:
- Embryos are inoculated and incubated for a period of one to six days during which they are observed twice a day for the evidence of death of embryo. Some viruses, such as equine encephalomyelitis virus kill the embryo within 24 hours while others kill the embryos in 2 to 3 days. Some viruses readily multiply within the embryonic tissue and are seldom lethal.

I/E, I/C and I/V inoculation

- Embryo death is accompanied by collapse of the large blood vessels of the chorioallantois so that they are difficult to see upon candling the egg, whereas they are quite distinct while the embryos are alive. Normal movements of the embryo are no longer observed if the embryo is moribund or dead.
- If the embryo is not dead 5 or 6 days of incubation, it is killed or examined for lesions.
-

I/E, I/C and I/V inoculation



II. Preparation and Setup for Stereotaxic Delivery of Viral Vectors

Virology

SEROLOGICAL METHODS

Agglutination and
Precipitation

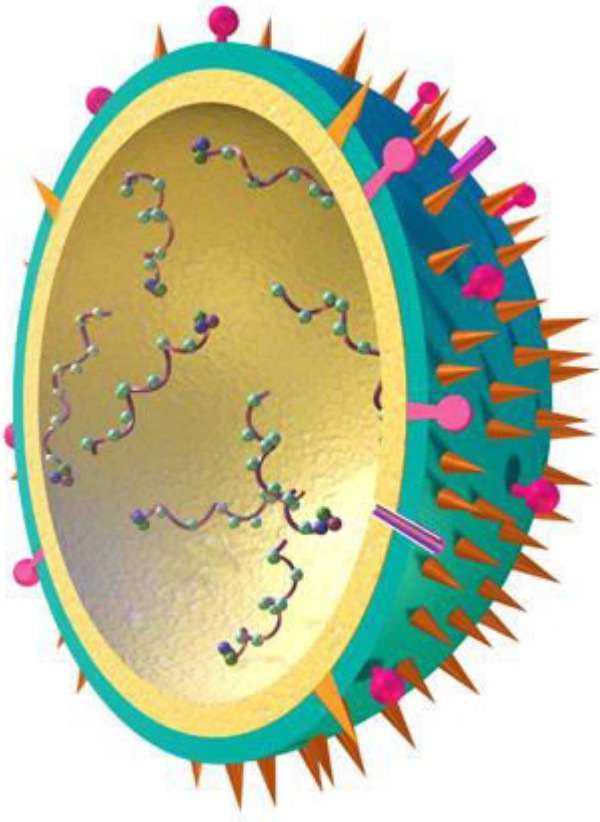
Agglutination and Precipitation

PRINICPILE

- Viruses maybe sensitized by specific antibody
- In the presence of electrolyte the virus particles cling to each other forming aggregates
- Observed through examinations
 - microscopic
 - macroscopic

Agglutination and Precipitation

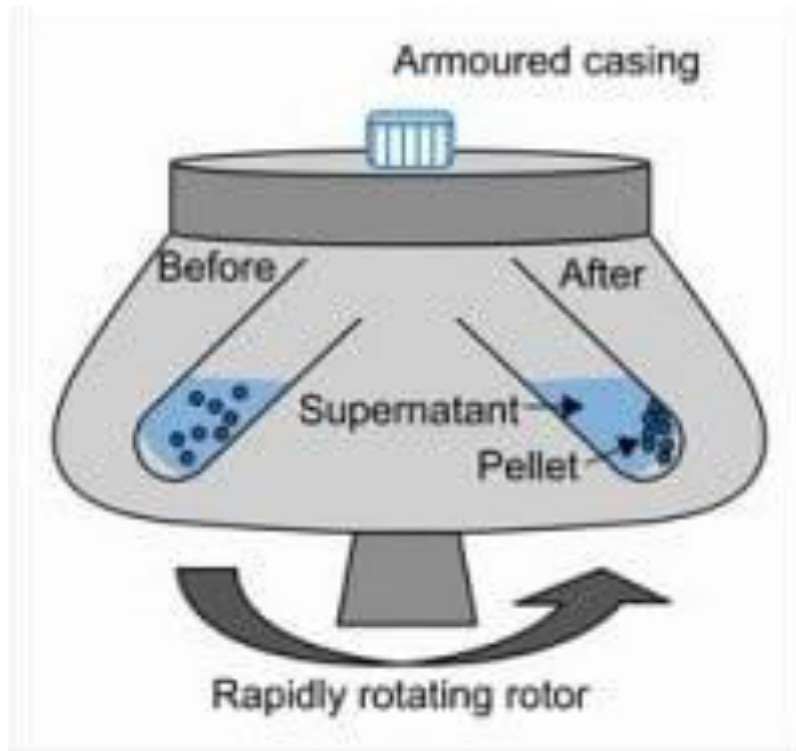
FLOCCULATION



- Size of virus is intermediate between bacteria and protein molecule
- Hence the term, Flocculation is more appropriate

Agglutination and Precipitation

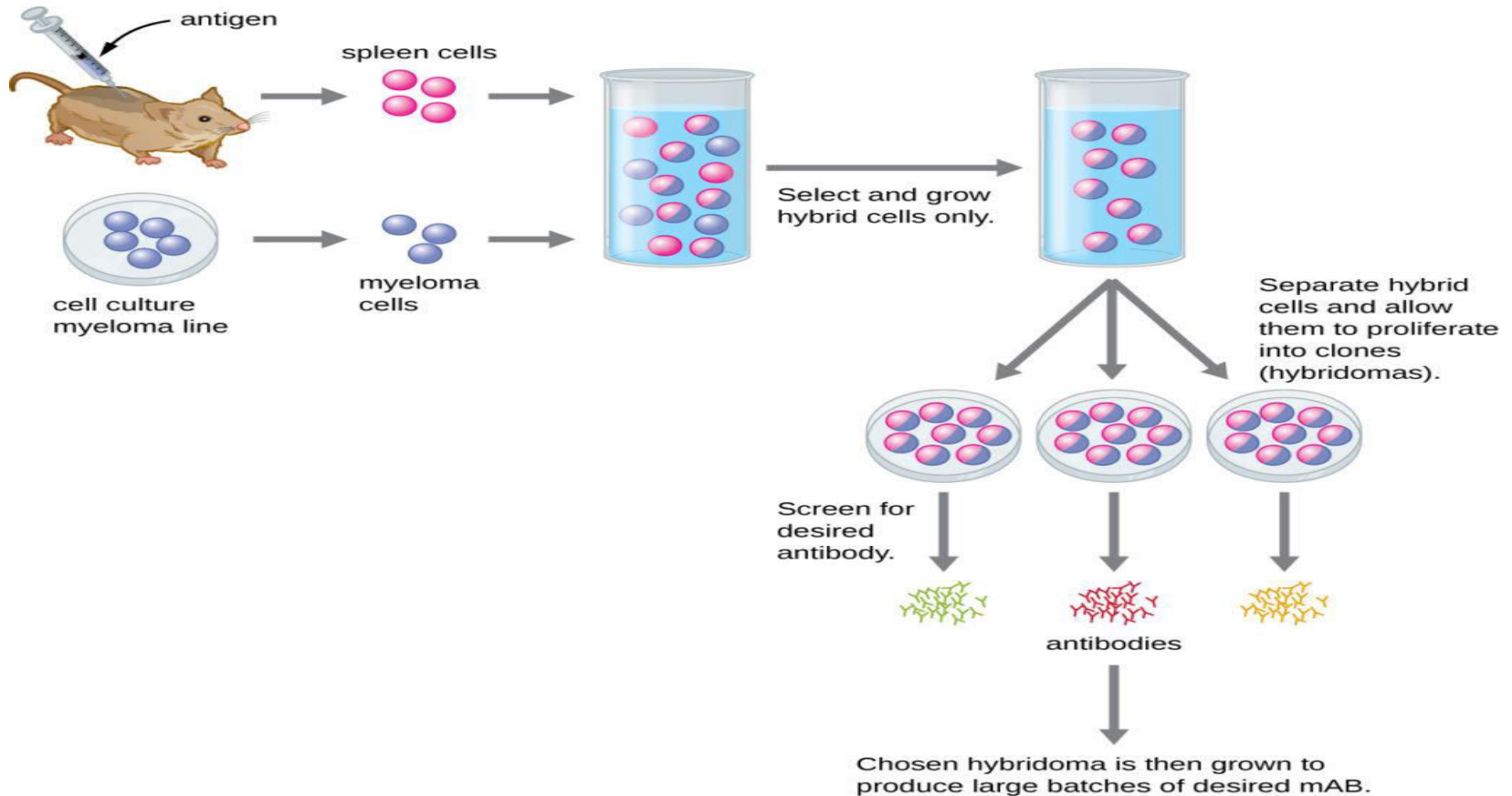
ANTIGEN PREPARATION



- Centrifugation
- Washing in buffered solution
- Recentrifugation

Agglutination and Precipitation

LINE OF ACTION



Agglutination and Precipitation

APPLICATIONS

- Two ways:
 1. If virus is known, antibodies can be detected
 2. If known serum is available, an unknown virus can be detected

Agglutination and Precipitation

CHALLENGES

- Preparation of antigen is difficult
- Viruses must be grown in the animal tissues

END

Virology

SEROLOGICAL METHODS

Complement Fixation

Complement Fixation

- The complement fixation test is an immunological test used to detect the presence of either specific antibody or specific antigen on the basis of complement fixation
- The complement fixation is a system of serum proteins that react with antigen-antibody complexes.
- If this reaction occurs on a cell surface, it will result in the formation of trans-membrane pores and therefore destruction of the cell.

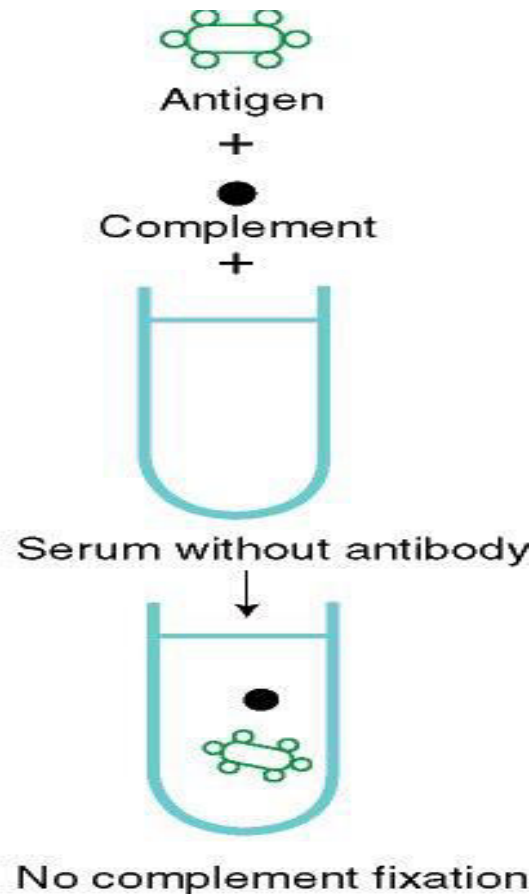
Complement Fixation

HOW IT WORKS

- Viruses as antigen
- Virus antigen for this test obtained from infected animal tissues
- Tissues such as, lungs or brain

Complement Fixation

APPLICATIONS



- Detection of presence of antibodies in serum
- Identification of unknown virus by using known antibody

Complement Fixation

SHORT PROCEDURE

- Serum is separated from the patient.
- Patients naturally have different levels of complement proteins in their serum. To negate any effects this might have on the test, the complement proteins in the patient's serum must be destroyed and replaced by a known amount of standardized complement proteins.
 - The serum is heated in such a way that all of the complement proteins—but none of the antibodies—within it are destroyed.
 - A known amount of standard complement proteins are added to the serum. (These proteins are frequently obtained from guinea pig serum.)
- The antigen of interest is added to the serum.
- Sheep red blood cells (sRBCs) which have been pre-bound to anti-sRBC antibodies are added to the serum.

Complement Fixation

RESULT INTERPRETATION

- If the patient's serum contains antibodies against the antigen of interest, they will bind to the antigen in step 3 to form antigen-antibody complexes.
- The complement proteins will react with these complexes and be depleted.
- Thus when the sRBC-antibody complexes are added in step 4, there will be no complement left in the serum.
- However, if no antibodies against the antigen of interest are present, the complement will not be depleted and it will react with the sRBC-antibody complexes added in step 4, lysing the sRBCs and spilling their contents into the solution, thereby turning the solution pink

Complement Fixation



Virology

SEROLOGICAL METHODS

Serum Neutralization Test

Serum Neutralization Test

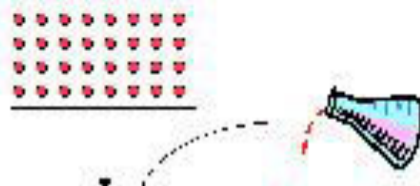
HOW IT WORKS

- Used to confirm identity of an unknown viral isolate
- Use of known dilution of specific antiserum mixed with multiple dilution
- Observation of virus to produce Cytopathic effects

Serum Neutralization Test

NECESSITIES

Serial dilutions of test sera



Addition of virus



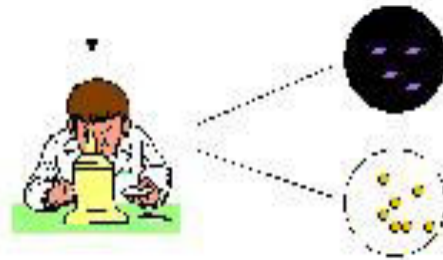
First incubation

Addition of cells



Second incubation

Plate reading



- Amount of virus
- Amount of serum
- Test animal
- Route of inoculation

Serum Neutralization Test

BASIC PRINCIPLES

- Wide range of dilutions used
- Equal volumes of viral dilution
- Equal volume of virus dilution and immune serum used
- Inoculation of 5 to 6 animals
- Equal number of controls
- Observation of animals for signs, lesions or deaths

Serum Neutralization Test

For comparisons

- L.D50 (Lethal dose 50)
- Viral dilutions checked for causing deaths and infectivity
- Titer noted

Serum Neutralization Test

CREATED USING
POWTOON

Virology

SEROLOGICAL METHODS

Virus Neutralization Test

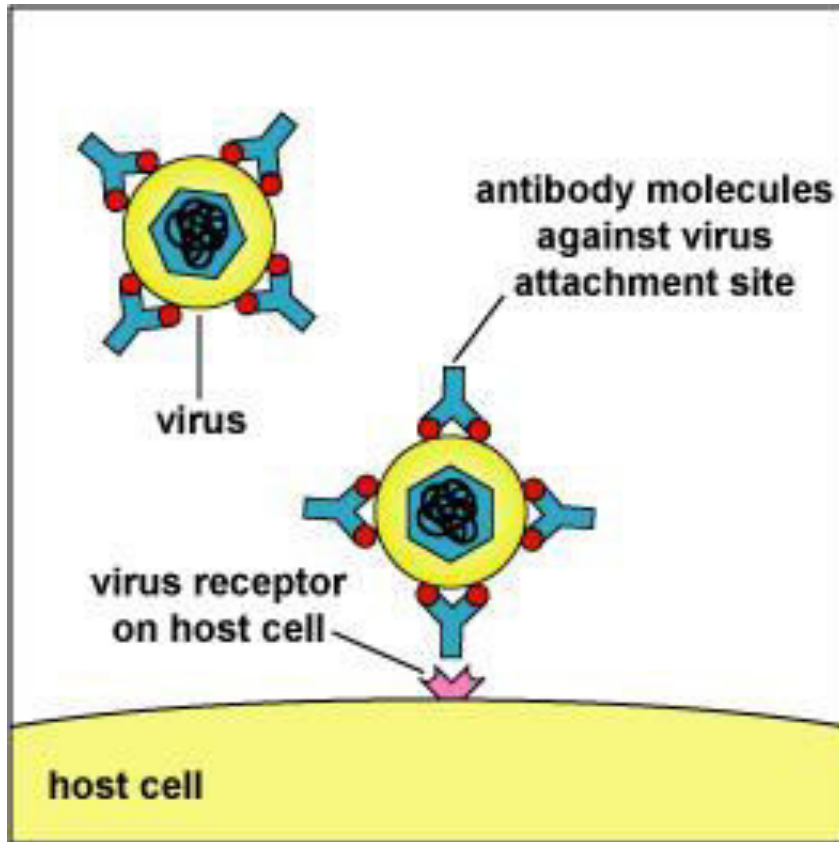
Virus Neutralization Test

INTRODUCTION

- Serological method that detects the presence of viral neutralizing antibodies
- The antibodies bind to viral particles
 - Monoclonal and polyclonal antibodies
- Block viral infectivity – no cell infection

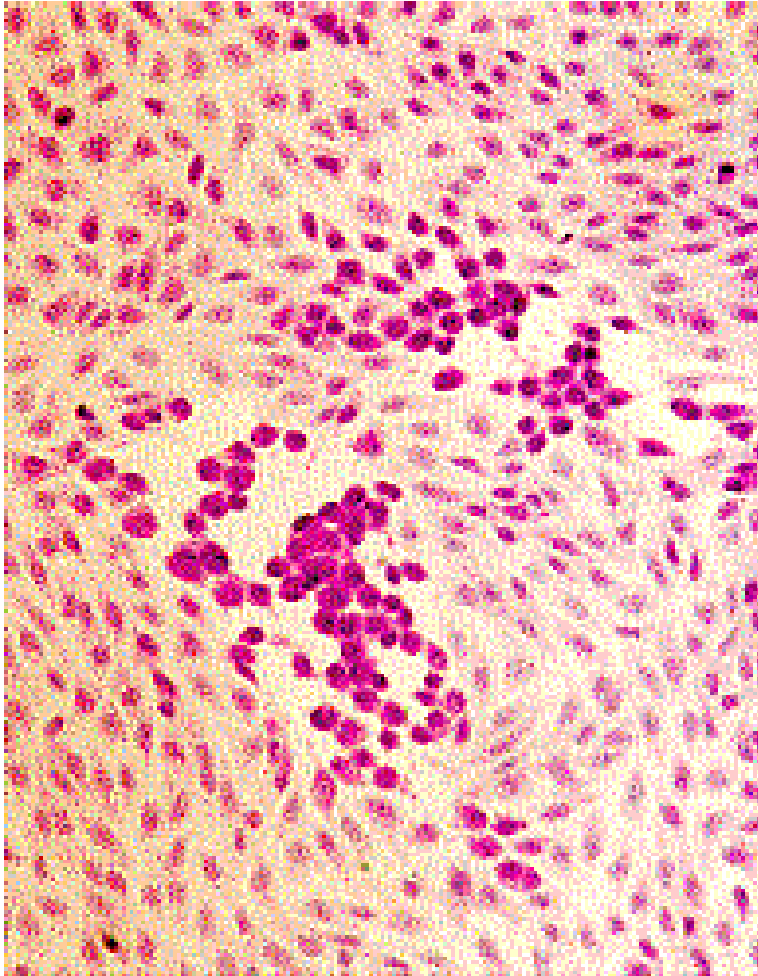
Virus Neutralization Test

- Conducted by:



- Mixing dilutions of antibodies with standardized amount of virus , incubating them and cultured into cells, eggs or animals to have clear cytopathic observation

Virus Neutralization Test



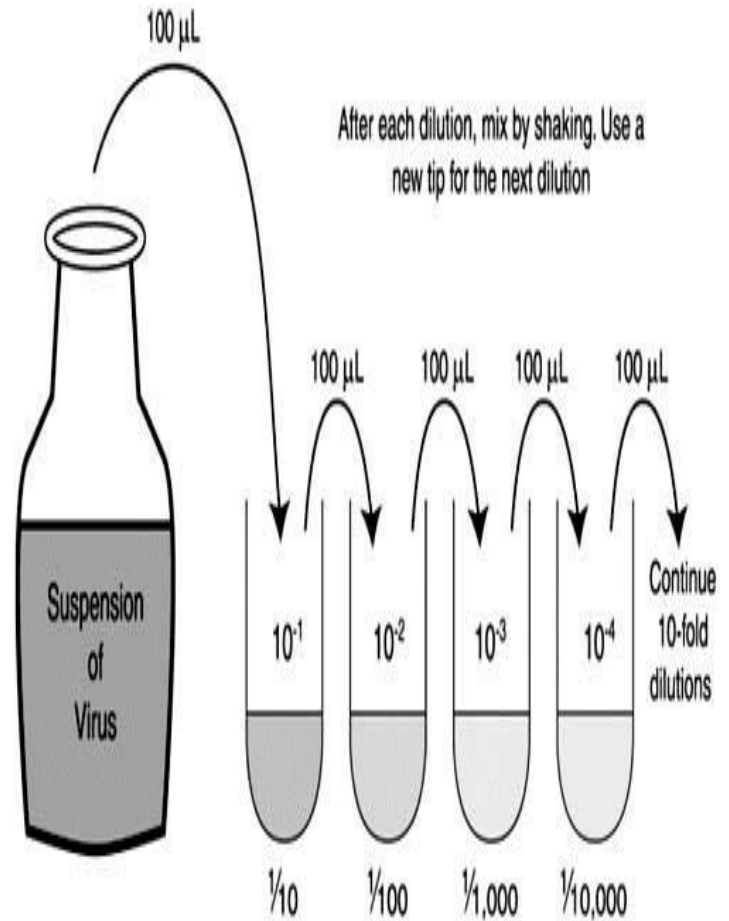
CYTOPATHIC EFFECTS

- Rounding of cells
- Change in texture
 - Granular
 - Hyaline
 - Glassy
- Formation of syncytium

Virus Neutralization Test

STEPS

- First step:
 - Serial dilution of serum
 - Known virus
 - Incubated for 1-2 hours at 37°C
- Second step
 - Cell culture inoculated with mixture
 - Incubated at 37°C
 - Direct microscope



Virus Neutralization Test

PROS

- Can be specific
- Sensitive
- Vaccine production/
immunological status

Virus Neutralization Test

CONS

- Very slow
- Intensive
- Required skilled people
- Depends on cell lines

END

Virology

SEROLOGICAL METHODS

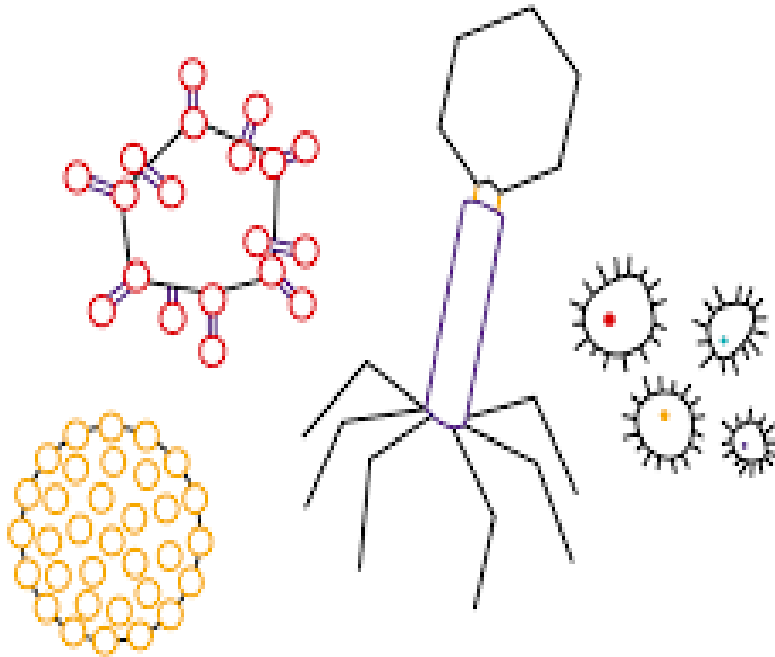
Protection Tests

Protection Test

INTRODUCTION

- Not strictly serological
- For identification of those viruses
- Used to produce successful vaccines
- Procedure also employed for testing commercial serum and vaccine products

Protection Test



- Production of either active or passive immunity in an animal followed by a challenge dose of virulent virus
- It is also called in-vivo neutralization test

Protection Test



- Vaccine prepared for all types of immunity to different lab animals
- When satisfactory immunity is produced, all are challenged with a known virus

Protection Test

EXAMPLES

- Used commercially for Rabies and hog cholera vaccines
- Used by injecting in graded doses

END

Virology

SEROLOGICAL METHODS

**Result interpretation of
Protection Tests**

Protection Test

HYPOTHETICAL TEST

- Suppose we have three types of vaccines
- We inject them in three different guinea pigs
- Observe for immunity

Protection Test

Immunized guinea pigs	10-3	10-4	10-5	10-6	10-7
Eastern type vaccine	5	5	2	1	0
Western type vaccine	1	0	0	0	0
Venezuelan type vaccine	5	5	3	0	0
Controls	5	5	3	1	0

Protection Test

- Disease: equine encephalomyelitis
- Control also set



Protection Test

- Vaccine that provides best immunity and no death proves to be the best

END

Virology

SEROLOGICAL METHODS

**Haemagglutination
/Haemagglutination
inhibition Test**

HA/HI Test

INTRODUCTION

- Few viruses possess haemagglutinin protein that binds to Sialic Acid receptors on RBCs

HA/HI Test

SAMPLE

- Blood
- Using anticoagulant like EDTA or Heparin



HA/HI Test

1% RBC'S PREPARATION

- Preparation of 1% RBCs
- Equal amount of blood and Normal Saline (NS) centrifuged in Falcon tube
- Remove supernatant and add the same volume of NS again and centrifuge
- Repeat the procedure until clear supernatant
- Remove the supernatant and add ten times of its volume of NS

HA/HI Test

HAEMAGGLUTINATION TEST

- A ninety six well plate marked according to sample
- Each row assigned to one sample
- Add 50uL NS to all wells
- Add antigen to first well and make two fold dilution till second last well
- Leave last well as control
- Add 25uL RBCs solution to all wells

HA/HI Test

HAEMAGGLUTINATION HAEMINHIBITION TEST

- 96 well plate labelled according to samples
- 25uL antiserum added to first well and make two fold dilution till end
- 0.5uL RBCs added to all wells
- Incubation for 30 minutes

Virology

SEROLOGICAL METHODS

**Result interpretation of
HA/HI test**

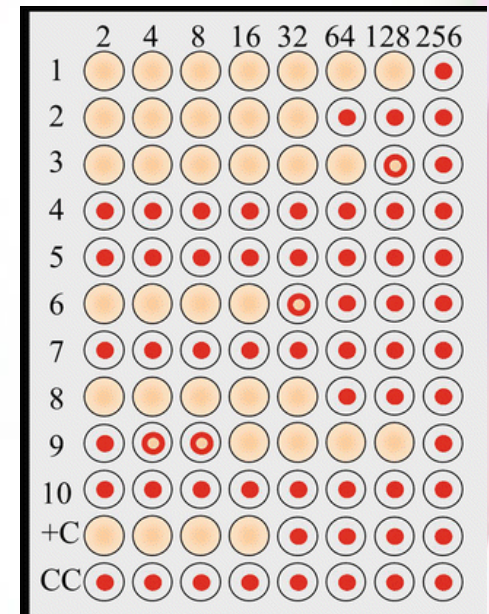
HA/HI Test

INTERPRETATION OF HA TEST

- Button formation showed no hemagglutination
- Hazy wells showed agglutination

HA/HI Test

- One HA calculates on the basis of last well showing button formation
- Button formation at controls



HA/HI Test

- Dilution value shows us 4HA unit value
- To determine one HA unit the value is divided by four

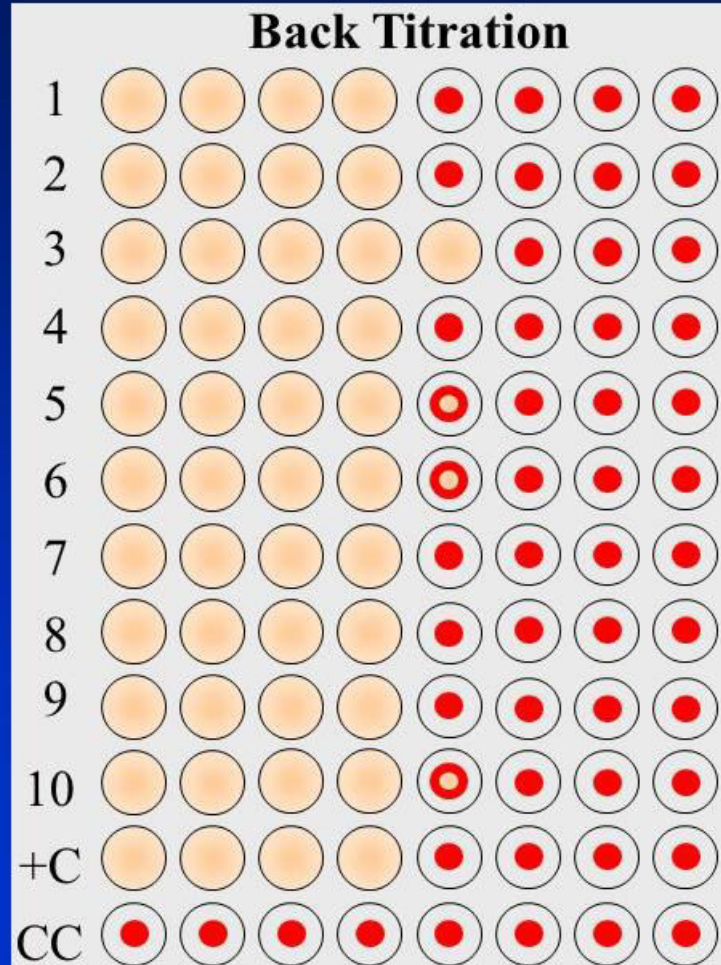
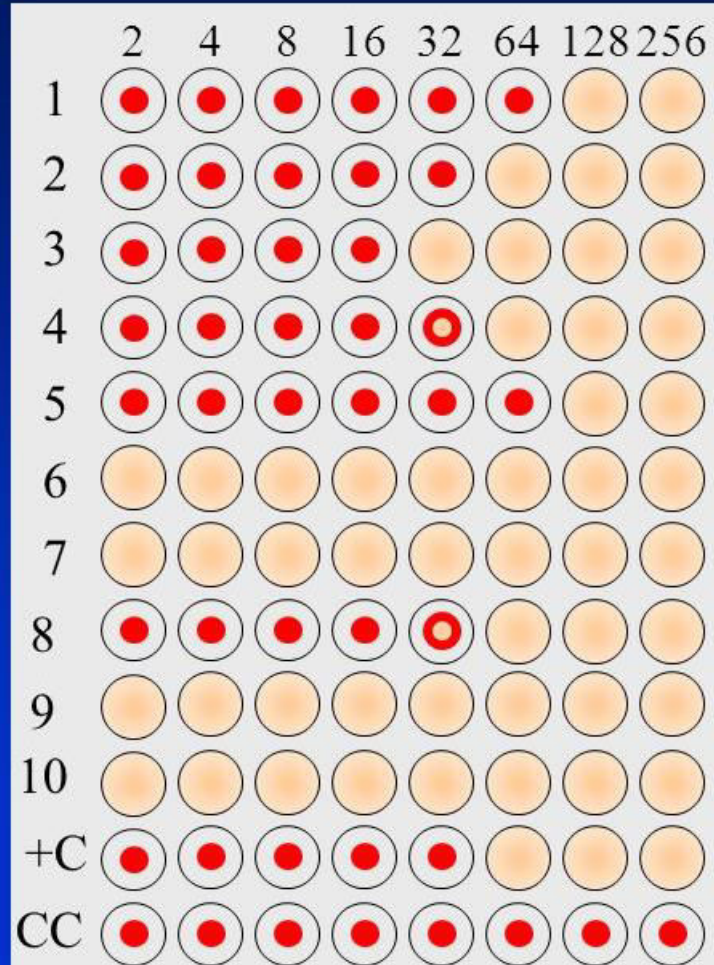
Well	1	2	3	4	5	6	7	8	9	10	11	12
Dilution	1/10	1/20	1/40	1/80	1/160	1/320	1/640	1/1280	1/2560	1/5120	1/10240	Control
Pattern												

HA/HI Test

INTERPRETATION OF HI TEST

- Shows button formation at the start
- Hazy wells at the end
- 8HA units were calculated per 25uL
- By dividing endpoint by 16

HA/HI Test



Virology

CYTOPATHOGENECITY IN TISSUE CULTURE

Tissue Culture

Basic Ingredients

1. Tissue cells
2. Lactalbumin hydrolysate
3. Serum
4. Balanced salt solution

Tissue Culture

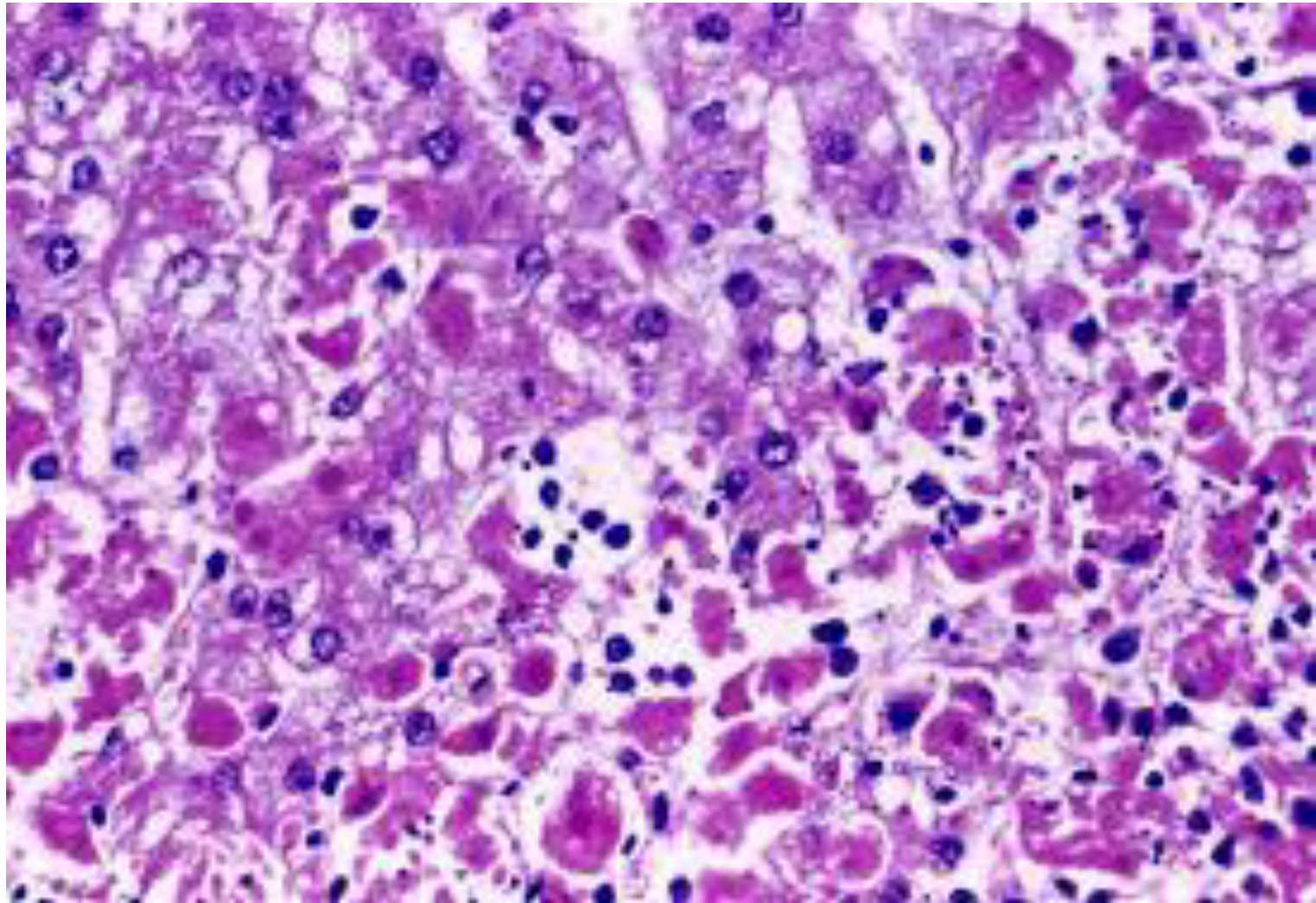
Other Ingredients

1. Ascitic fluid
2. Amniotic fluid
3. Serum ultrafilterate
4. Amino acids
5. Vitamins
6. Glucose
7. Inorganic salts
8. Penicillin and streptomycin

Tissue Culture

- Glassware used must be absolutely clean and free film and toxic substances
- Chemicals and distil water must be free from toxic ions

Tissue Culture



Tissue Culture

Cytopathogenicity

- Abnormalities observed in cells
- Inclusion bodies observed
- Syncytia formation

Virology

SEROLOGICAL METHODS

Haemadsorption Test

Haemadsorption

Introduction

- Technique similar to haemagglutination test
- Used for detection of certain viruses in tissue culture

Haemadsorption

Haemagglutinin

- Viruses producing haemagglutinin will adsorb erythrocytes to the surface of cells in which the virus has grown



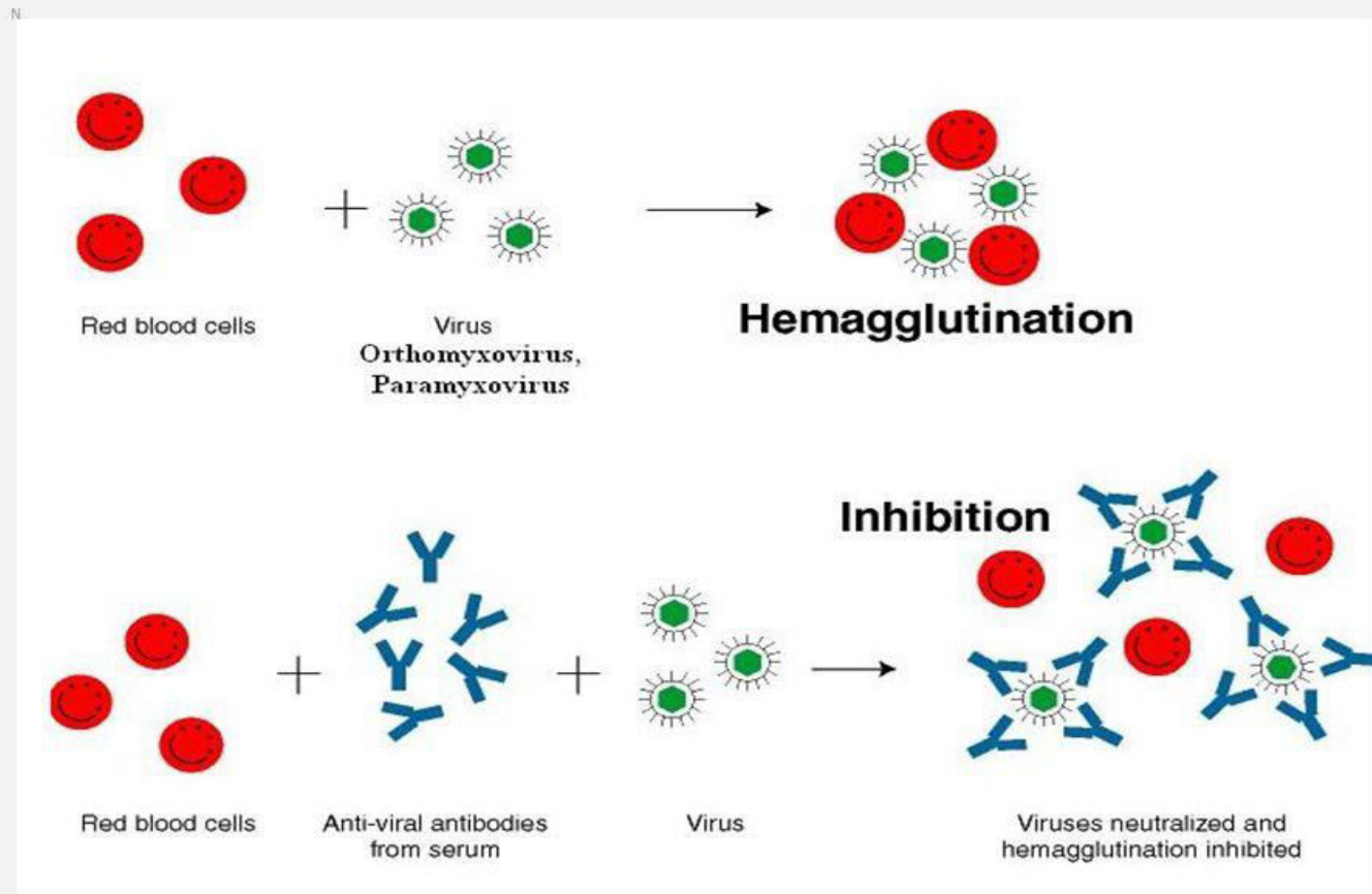
Haemadsorption

PROCEDURE

- A suspension of washed RBCs is added to tissue culture tubes
- Presences of haeadsorbing virus, results in clustering of RBCs
- Rosettes around the infected cells of the culture
- Can be observed microscopically

Haemadsorption

Haemadsorption and agglutination inhibition:



Haemadsorption

USE

- Useful for detection of viruses producing haemagglutinins

Haemadsorption

Inhibition

- Can be inhibited by specific antibody
- Hence
haemadsorption-inhibition test is
designed for antibody
titer

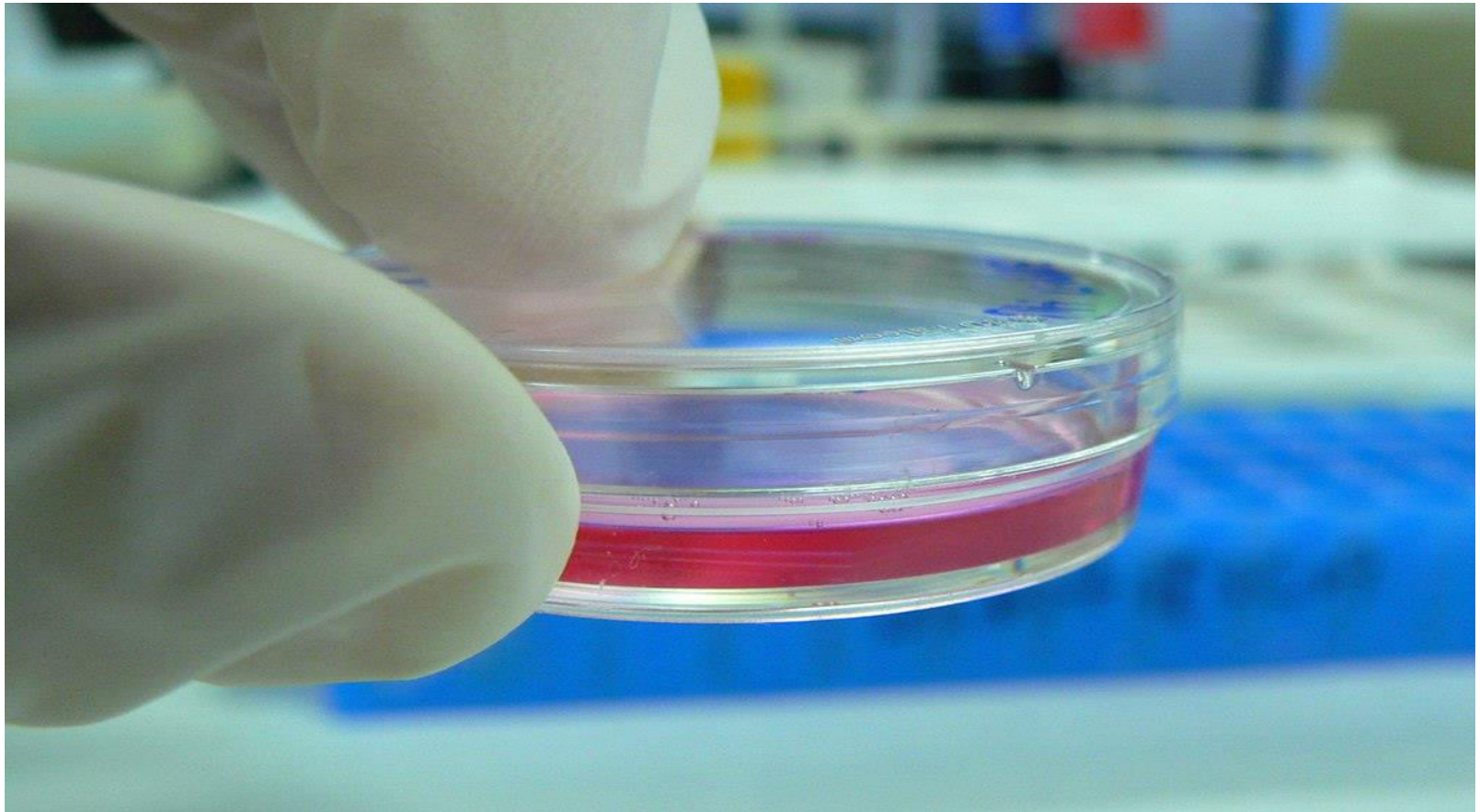
END

Virology

Growing Animal Viruses in Cell Culture

Cell Culture

THE PROCESS BY WHICH CELLS ARE GROWN UNDER CONTROLLED CONDITIONS



Cell Culture

CONTROLLED CONDITIONS

Vary for each cell type

- Essential nutrients (amino acids, carbohydrates etc.
- Growth factors
- Hormones
- Gases (CO_2 or O_2)
- Physio-chemical environment regulators (pH buffer, osmotic pressure, temperature etc.

Cell Culture

History

Cell culture was first successfully undertaken by Ross Harrison in 1907

- Roux in 1885 for the first time maintained embryonic chick cells in a cell culture
- 1878: Claude Bernard proposed that physiological systems of an organism can be maintained in a living system after the death.
- 1885: Roux maintained embryonic chick cells in a saline culture.
- 1897: Loeb demonstrated the survival of cells isolated from blood and connective tissue in serum and plasma.
- 1903: Jolly observed cell division of salamander leucocytes *in vitro*.

1907

Harrison cultivated frog nerve cells in a lymph clot held by the hanging drop' method and observed the growth of nerve fibers *in vitro* for several weeks. He was considered by some as the father of cell culture.

Cell Culture

APPLICATIONS

Areas where cell culture technology is currently playing a major role

- Model systems for
Studying basic cell biology, interactions between disease causing agents and cells, effects of drugs on cells, process and triggering of aging & nutritional studies
- Toxicity testing
Study the effects of new drugs
- Cancer research
Study the function of various chemicals, virus & radiation to convert normal cultured cells to cancerous cells

Cell Culture

APPLICATIONS

Areas where cell culture technology is currently playing a major role

- Virology

Cultivation of virus for vaccine production, also used to study their infectious cycle.

- Genetic Engineering

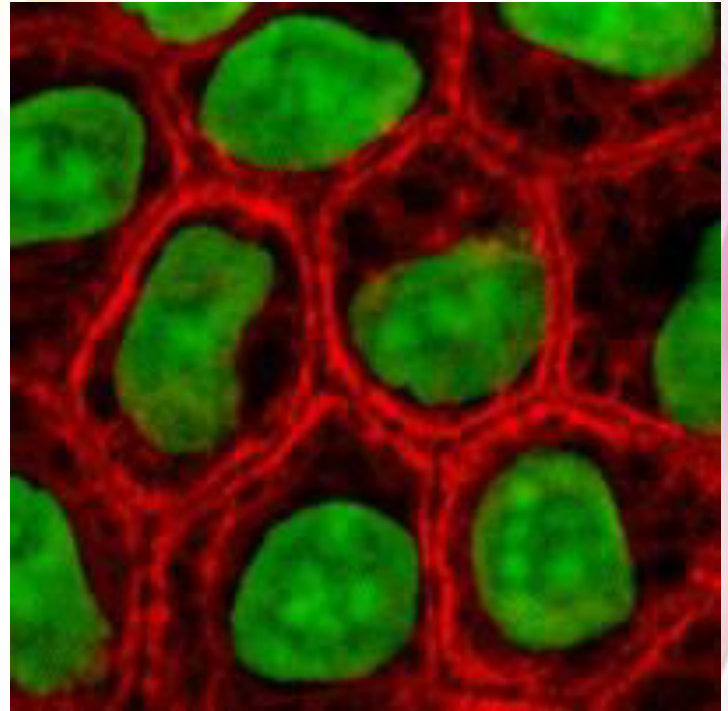
Production of commercial proteins, large scale production of viruses for use in vaccine production e.g. polio, rabies, chicken pox, hepatitis B & measles

- Gene therapy

Cells having a functional gene can be replaced to cells which are having non-functional gene

Cell Culture

Culture of Mammalian Cells
Culture of Non-mammalian Cells



MIDTERM SYLLABUS

Effort By

Amaan Khan